14 A liquid is evaporating. The liquid is not boiling.

Which statement about the liquid is correct?

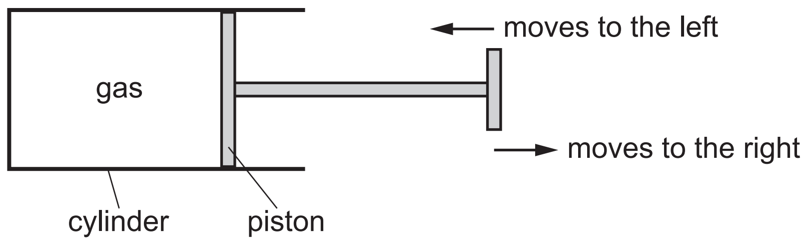
A Any molecule can escape, and from any part of the liquid.

B Any molecule can escape, but only from the liquid surface.

C Only molecules with enough energy can escape, and only from the liquid surface.

D Only molecules with enough energy can escape, but from any part of the liquid.

15 The diagram shows a quantity of gas enclosed in a cylinder by a piston.



The piston is moved to the left or to the right. The temperature of the gas is kept constant.

Which row describes the effect of moving the piston slowly in the direction shown in the table?

	movement of piston	speed of gas molecules	pressure of gas
A	to the left	increases	decreases
B	to the left	no change	increases
C	to the right	increases	decreases
D	to the right	no change	increases

16 Which statement describes what happens as ice at 0 °C starts to melt to become water?

A Energy is absorbed and the temperature remains constant.

B Energy is absorbed and the temperature rises.

C Energy is released and the temperature remains constant.

D Energy is released and the temperature rises.

17 What is meant by the *fixed points* of the scale of a liquid-in-glass thermometer?

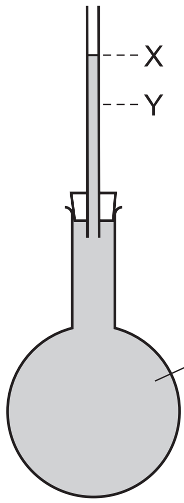
A the distance between one scale division and the next

B the highest and lowest temperatures that the thermometer can record

C the maximum and minimum depth to which the thermometer should be submerged in a liquid

D the two agreed temperatures used for marking the temperature scale

18 A liquid at room temperature fills a flask and a glass tube to level X.



The flask is now placed in ice, and the liquid level in the tube falls to level Y.

Why does the level fall?

A The flask contracts.

B The flask expands.

C The liquid contracts.

D The liquid expands.

19 Which process involves convection?

A bread toasting under a grill

B energy from the Sun warming a road surface

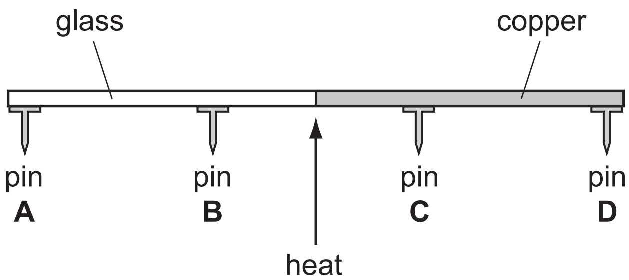
C hot air rising to the top of a cool room

D thermal energy transfer through a copper bar

20 A rod is made half of glass and half of copper. Four pins **A**, **B**, **C** and **D** are attached to the rod by wax. The rod is heated in the centre as shown.

The pins fall off when the wax melts.

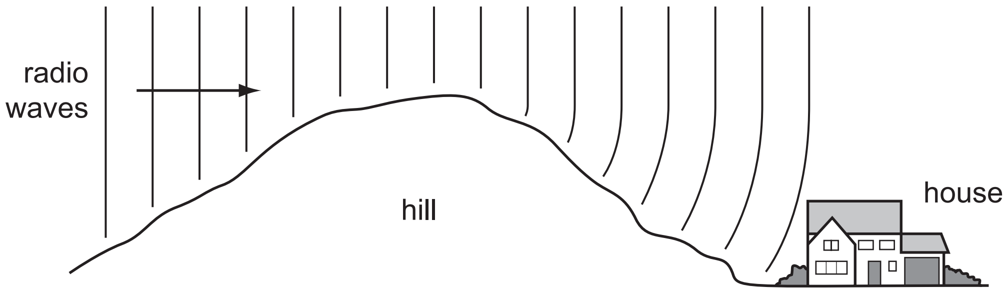
Which pin falls off first?



21 Which row shows the natures of light waves, sound waves and X-rays?

	light waves	sound waves	X-rays
A	longitudinal	longitudinal	transverse
B	longitudinal	transverse	longitudinal
C	transverse	longitudinal	transverse
D	transverse	transverse	longitudinal

22 Radio waves are received at a house at the bottom of a hill.



The waves reach the house because the hill has caused them to be

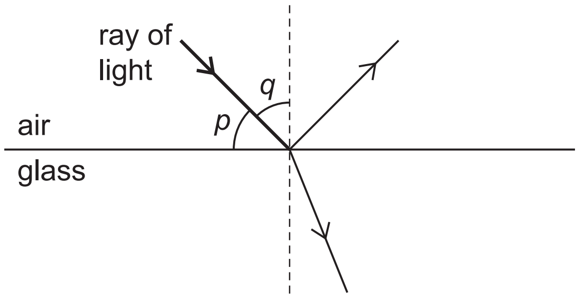
A diffracted.

B radiated.

C reflected.

D refracted.

23 The diagram shows a ray of light in air incident on a glass block. Some of the light is refracted, and some of the light is reflected. Two angles *p* and *q* are marked on the diagram.



Which row gives the angle of incidence and shows whether the ray undergoes total internal reflection?

	angle of incidence	total internal reflection
A	*p*	no
B	*p*	yes
C	*q*	no
D	*q*	yes

24 Visible light has a frequency of approximately 5.0×10^{14} Hz.

M and N are two other types of electromagnetic radiation.

The frequency of M is 5.0×10^{6} Hz.

The frequency of N is 5.0×10^{15} Hz.

Which types of radiation are M and N?

	M	N
A	radio waves	infra-red
B	radio waves	ultraviolet
C	ultraviolet	X-rays
D	X-rays	infra-red

25 What is the approximate range of hearing of a healthy human ear?

A 2.0 Hz to 2.0 kHz

B 2.0 Hz to 20 kHz

C 20 Hz to 2.0 kHz

D 20 Hz to 20 kHz

26 A singer sings two notes. The first note is louder and lower in pitch than the second note.

Which statement about the two notes is correct?

A The first note has a larger amplitude and a larger frequency than the second note.

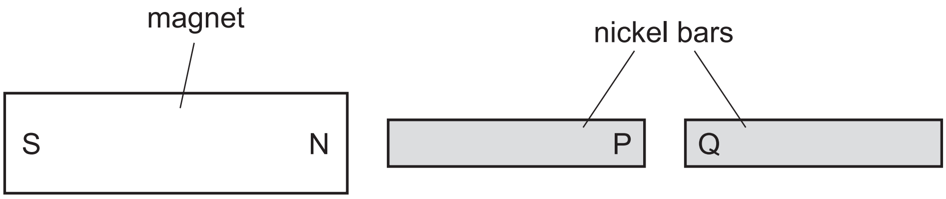
B The first note has a larger amplitude and a smaller frequency than the second note.

C The first note has a smaller amplitude and a larger frequency than the second note.

D The first note has a smaller amplitude and a smaller frequency than the second note.

27 Two nickel bars are placed close to the N-pole of a bar magnet.

The nickel bars become magnetised.



Which row states the pole induced at P, the pole induced at Q, and the type of magnetic force between P and Q?

	pole induced at P	pole induced at Q	force between P and Q
A	N	S	attraction
B	N	S	repulsion
C	S	N	attraction
D	S	N	repulsion

28 A student wishes to make a permanent magnet. She has an iron rod and a steel rod.

Which rod should she use to make the permanent magnet, and is this rod a hard magnetic material or a soft magnetic material?

	rod	type of magnetic material
A	iron	hard
B	iron	soft
C	steel	hard
D	steel	soft

29 Two meters are connected in a circuit to measure the current in a component and the potential difference across the component.

Which meters are used and how are they connected to the component?

A an ammeter in parallel for current, a voltmeter in series for potential difference

B an ammeter in series for current, a voltmeter in parallel for potential difference

C a voltmeter in parallel for current, an ammeter in series for potential difference

D a voltmeter in series for current, an ammeter in parallel for potential difference

30 A wire has a certain electrical resistance.

The diameter and length of the wire may be changed.

Which pair of changes **must** cause the resistance of the wire to increase?

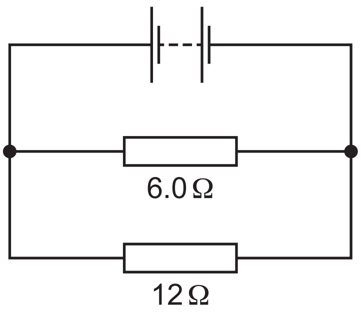
	change of diameter	change of length
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

31 P and Q are the circuit symbols for two electrical components.

Which components are represented by P and by Q?

	P	Q
A	thermistor	fuse
B	thermistor	relay
C	variable resistor	fuse
D	variable resistor	relay

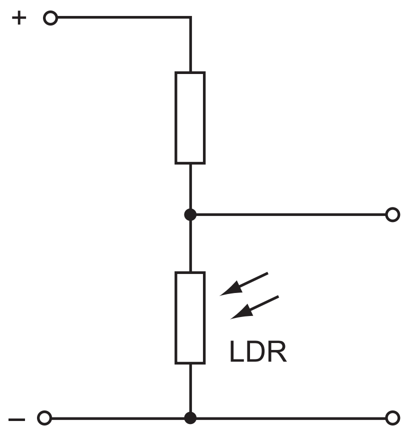
32 The diagram shows two resistors connected in a circuit.



What could be the combined resistance of this arrangement of resistors?

A 4.0 Ω **B** 6.0 Ω **C** 9.0 Ω **D** 18 Ω

33 The diagram shows part of a circuit used to switch street lamps on and off automatically.



In the evening it gets dark.

Which row shows the effect on the resistance of the light-dependent resistor (LDR) and on the potential difference (p.d.) across it?

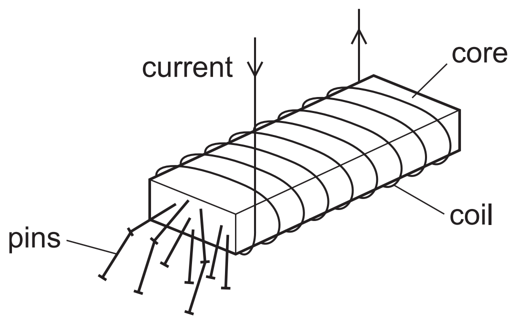
	resistance of LDR	p.d. across LDR
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

34 A domestic circuit includes a 30 A fuse. This protects the wiring if there is too much current in the circuit.

In which wire is the 30 A fuse positioned, and what does it do when it operates?

	position	operation
A	live wire	disconnects the circuit
B	live wire	reduces the current to 30 A
C	neutral wire	disconnects the circuit
D	neutral wire	reduces the current to 30 A

35 A strong electromagnet is used to attract pins.



What happens when the current in the coil is halved?

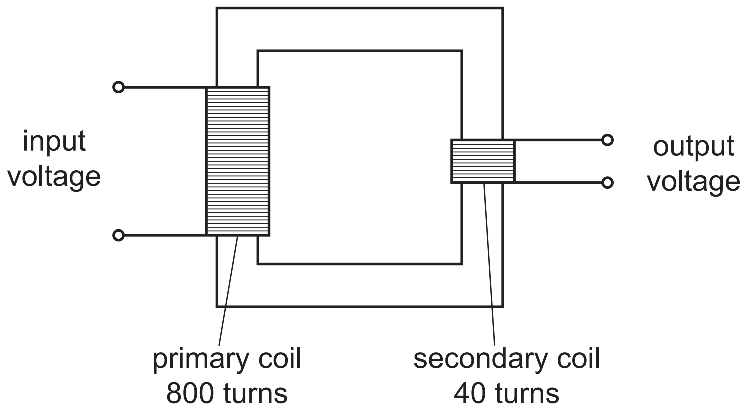
A No pins are attracted.

B Some pins are attracted, but not as many.

C The same number of pins is attracted.

D More pins are attracted.

36 The diagram shows a transformer.



The input voltage is 240 V.

What is the output voltage?

A 6.0 V **B** 12 V **C** 20 V **D** 40 V

37 A neutral atom consists of electrons orbiting a nucleus. The nucleus contains protons and neutrons.

Which statement about the atom **must** be correct?

A The number of electrons is equal to the number of neutrons.

B The number of electrons is equal to the number of protons.

C The number of neutrons is equal to the number of protons.

D The number of electrons, neutrons and protons are all different.

38 Below are the symbols for five different nuclides.

$^{35}_{17}X$	$^{37}_{17}X$	$^{38}_{18}X$	$^{81}_{35}X$	$^{81}_{37}X$
nuclide 1	nuclide 2	nuclide 3	nuclide 4	nuclide 5

Which two nuclides are isotopes of the same element?

A nuclide 1 and nuclide 2

B nuclide 2 and nuclide 3

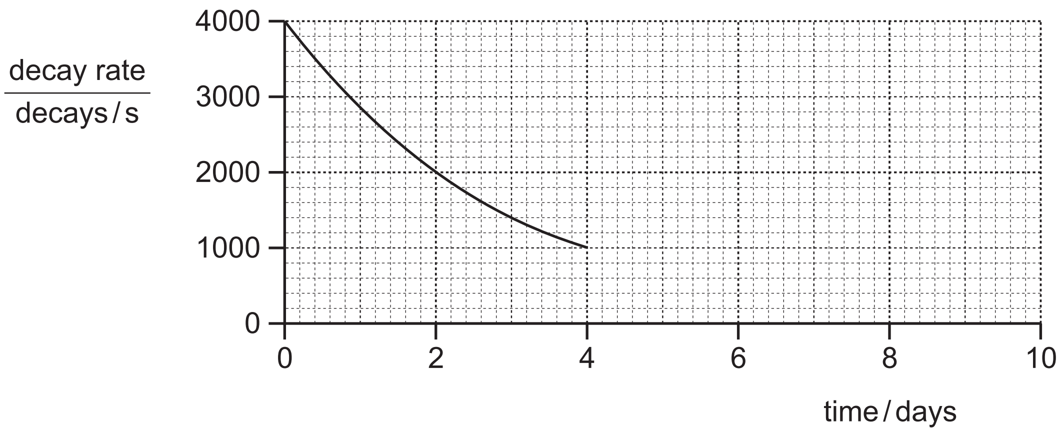
C nuclide 2 and nuclide 5

D nuclide 4 and nuclide 5

39 Which row describes the nature and a property of all β-particles?

	nature	property
A	electrons	can travel through a vacuum
B	electrons	stopped by a thin sheet of paper
C	helium nuclei	can travel through a vacuum
D	helium nuclei	stopped by a thin sheet of paper

40 The graph shows how the decay rate of a radioactive source changes with time.



What will be the activity at 8 days?

A 0 decays / s

B 125 decays / s

C 250 decays / s

D 500 decays / s

Cambridge IGCSE

Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **May/June 2016**

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

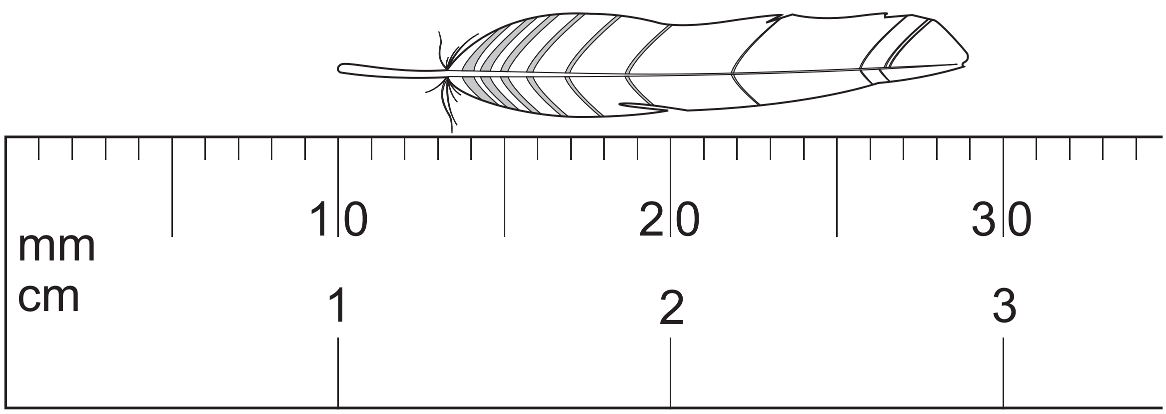
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Electronic calculators may be used.
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m / s^2).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **17** printed pages and **3** blank pages.

CAMBRIDGE International Examinations

1 The diagram shows an enlarged drawing of the end of a metre rule. It is being used to measure the length of a small feather.



What is the length of the feather?

A 19 mm **B** 29 mm **C** 19 cm **D** 29 cm

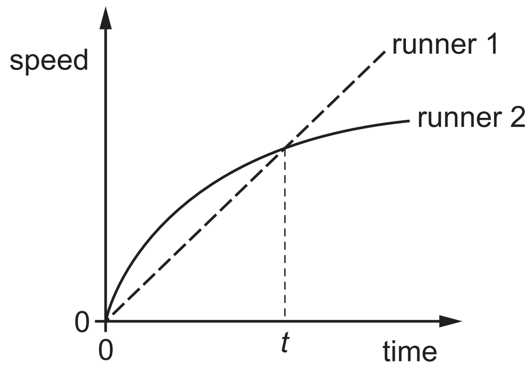
2 A train begins a journey from a station and travels 60 km in a time of 20 minutes.

What is the average speed of the train?

A 3.0 m / s **B** 5.0 m / s **C** 50 m / s **D** 60 m / s

3 Two runners take part in a race.

The graph shows how the speed of each runner changes with time.



What does the graph show about the runners at time *t*?

A Both runners are moving at the same speed.

B Runner 1 has zero acceleration.

C Runner 1 is overtaking runner 2.

D Runner 2 is slowing down.

4 A cup contains hot liquid.

Some of the liquid evaporates.

What happens to the mass and what happens to the weight of the liquid in the cup?

	mass	weight
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

5 An object has a mass of 50 kg.

The gravitational field strength on Earth is 10.0 N / kg.

The gravitational field strength on a distant planet is 4.0 N / kg.

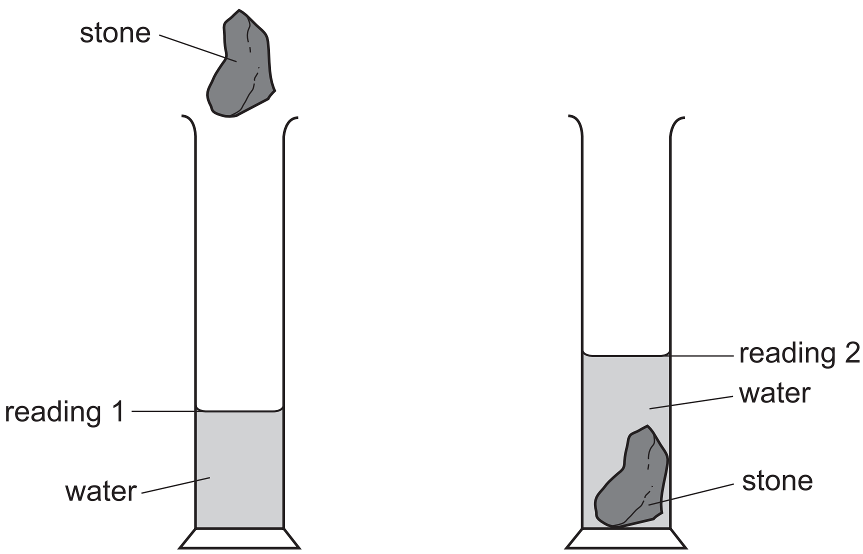
What is the weight of the object on Earth, and what is its weight on the distant planet?

	on Earth	on the distant planet
A	5.0 kg	12.5 kg
B	5.0 N	12.5 N
C	500 kg	200 kg
D	500 N	200 N

6 A student wishes to determine the density of an irregularly-shaped stone.

First he finds the mass of the stone. Next he lowers the stone into a measuring cylinder containing water.

The diagrams show the measuring cylinder before and after the stone is lowered into it.



How should the student calculate the density of the stone?

A mass of stone × reading 2

B mass of stone × (reading 2 – reading 1)

C mass of stone ÷ reading 2

D mass of stone ÷ (reading 2 – reading 1)

7 Which is an example of a force?

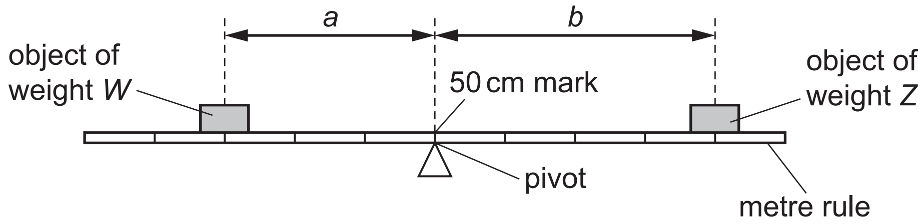
A energy

B power

C pressure

D weight

8 The diagram shows an object of weight W and an object of weight Z balanced on a uniform metre rule.



Which equation relating to W, Z, a and b is correct?

A $\dfrac{W}{a} = \dfrac{Z}{b}$

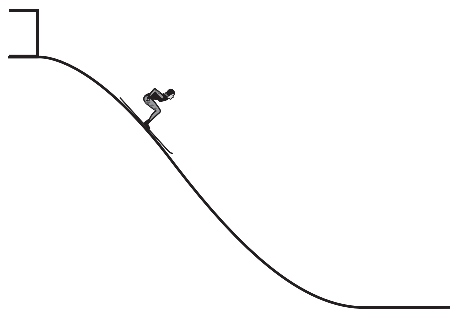
B $W \times Z = a \times b$

C $W \times a = Z \times b$

D $W \times (a + b) = Z$

9 A skier walks from the bottom of a ski slope to the top and gains 10 000 J of gravitational potential energy.

She skis down the slope. At the bottom of the slope, her kinetic energy is 2000 J.



How much energy is dissipated in overcoming friction and air resistance as the skier moves down the slope?

A 2000 J **B** 8000 J **C** 10 000 J **D** 12 000 J

10 A coal-fired power station generates electricity. Coal is burnt and the energy released is used to boil water. The steam from the water makes the generator move and this produces electricity.

Which words are used to describe the energy stored in the coal and the energy of the moving generator?

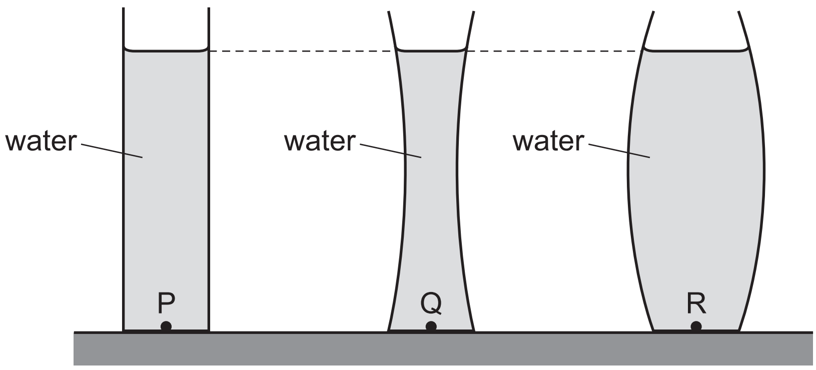
	coal	generator
A	chemical	hydroelectric
B	chemical	kinetic
C	geothermal	hydroelectric
D	geothermal	kinetic

11 Four different children run up the same set of stairs.

For which child is the useful power to climb the stairs the greatest?

	mass of child / kg	time taken / s
A	40	15
B	50	25
C	60	25
D	70	15

12 The diagram shows three vases each with the same base area. Each vase contains water of the same depth.



Which statement about the water pressures at points P, Q and R is correct?

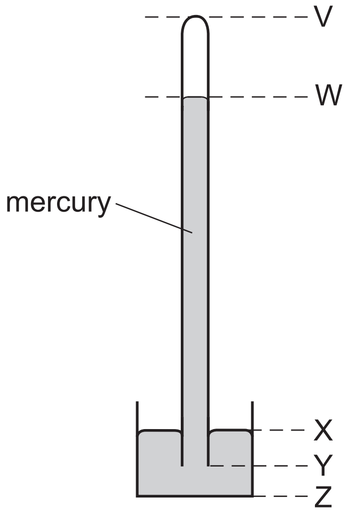
A The pressure at point P is the greatest.

B The pressure at point Q is the least.

C The pressure at point R is the greatest.

D The pressures at points P, Q and R are the same.

13 The diagram shows a simple mercury barometer.



The atmospheric pressure increases.

Which distance increases?

A VW **B** WY **C** XY **D** XZ

14 Which statement about evaporation is correct?

A Evaporation causes the temperature of the remaining liquid to decrease.

B Evaporation does not occur from a cold liquid near its freezing point.

C Evaporation does not occur from a dense liquid, such as mercury.

D Evaporation occurs from all parts of a liquid.

15 A gas is stored in a sealed container of constant volume. The temperature of the gas increases. This causes the pressure of the gas to increase.

What happens to the gas molecules during this pressure increase?

A The average kinetic energy of the molecules increases.

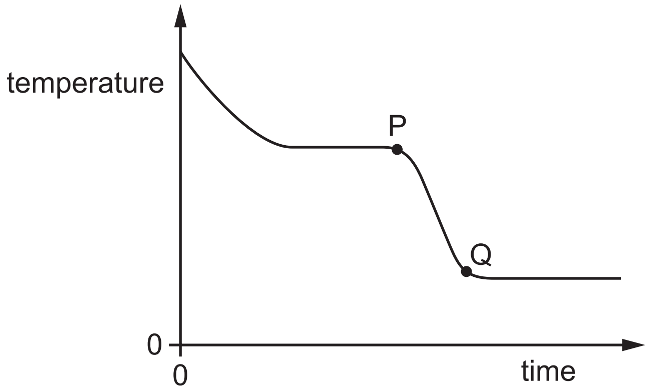
B The average separation of the molecules decreases.

C The average separation of the molecules increases.

D The volume of each molecule increases.

16 A substance loses thermal energy (heat) to the surroundings at a steady rate.

The graph shows how the temperature of the substance changes with time.



What could the portion PQ of the graph represent?

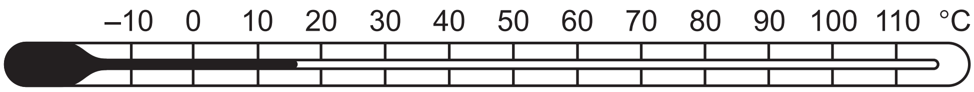
A gas condensing

B gas cooling

C liquid cooling

D liquid solidifying

17 A student wishes to check the upper and the lower fixed points on a Celsius scale thermometer.



She has four beakers P, Q, R and S.

Beaker P contains a mixture of ice and salt.

Beaker Q contains a mixture of ice and water.

Beaker R contains boiling salt solution.

Beaker S contains boiling water.

Which two beakers should she use to check the fixed points?

A P and R **B** P and S **C** Q and R **D** Q and S

18 The same quantity of thermal energy is supplied to two solid objects X and Y. The temperature increase of object X is greater than the temperature increase of object Y.

Which statement explains this?

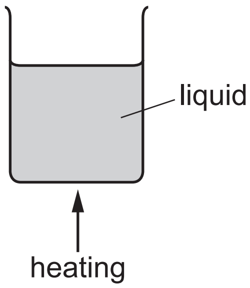
A X has a lower melting point than Y.

B X has a lower density than Y.

C X has a lower thermal capacity than Y.

D X is a better thermal conductor than Y.

19 A liquid is heated in a beaker.



The density of the liquid changes as its temperature increases. This causes energy to be transferred throughout the liquid.

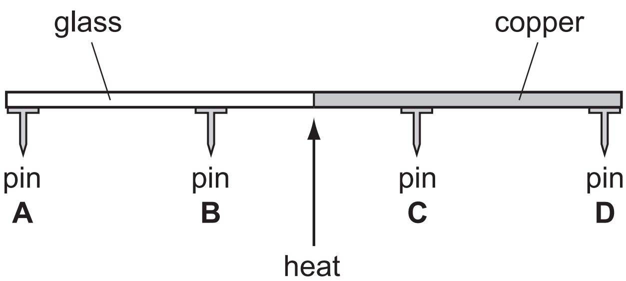
How does the density change and what is this energy transfer process?

	density	energy transfer process
A	decreases	conduction
B	decreases	convection
C	increases	conduction
D	increases	convection

20 A rod is made half of glass and half of copper. Four pins **A**, **B**, **C** and **D** are attached to the rod by wax. The rod is heated in the centre as shown.

The pins fall off when the wax melts.

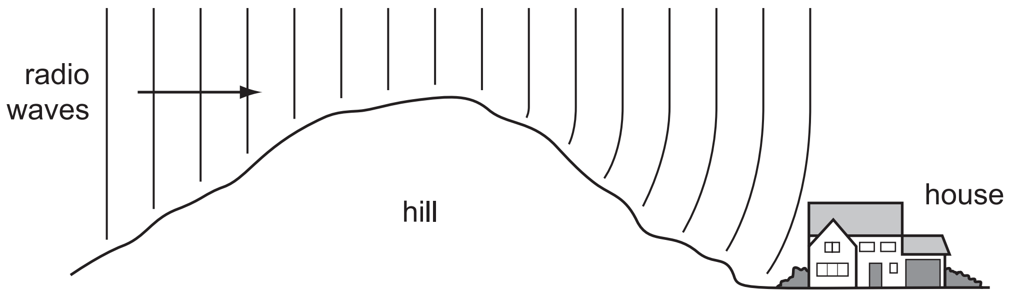
Which pin falls off first?



21 Which row shows the natures of light waves, sound waves and X-rays?

	light waves	sound waves	X-rays
A	longitudinal	longitudinal	transverse
B	longitudinal	transverse	longitudinal
C	transverse	longitudinal	transverse
D	transverse	transverse	longitudinal

22 Radio waves are received at a house at the bottom of a hill.



The waves reach the house because the hill has caused them to be

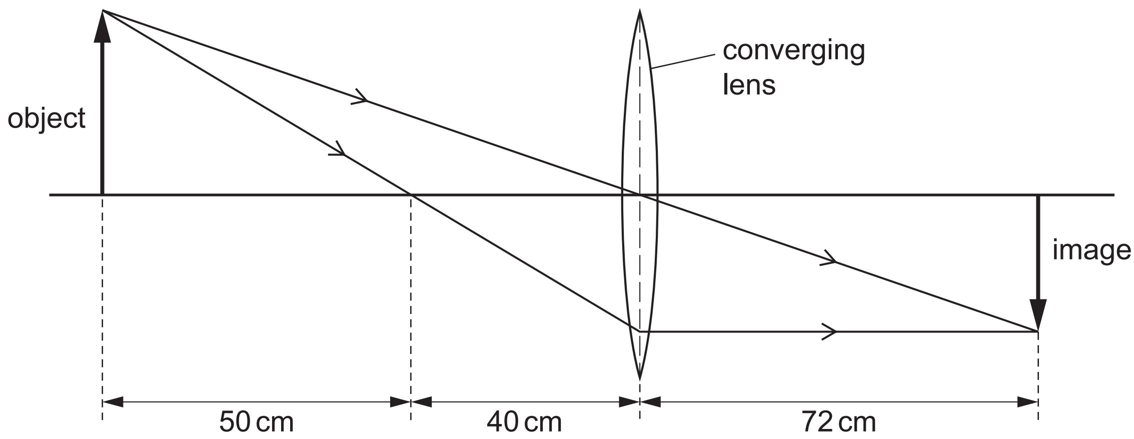
A diffracted.

B radiated.

C reflected.

D refracted.

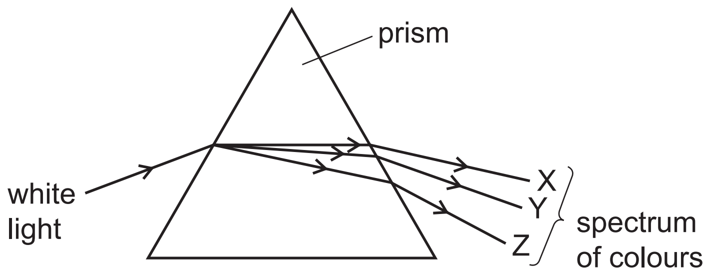
23 The ray diagram shows the image of an object formed by a converging lens.



What is the focal length of the lens?

A 40 cm **B** 50 cm **C** 72 cm **D** 90 cm

24 The diagram shows the dispersion of white light by a prism.



Which row could be correct for the colours seen at X, at Y and at Z?

	colour at X	colour at Y	colour at Z
A	red	violet	yellow
B	red	yellow	violet
C	violet	yellow	red
D	yellow	red	violet

25 Why can ultrasound **not** be heard by humans?

A The amplitude is too great.

B The frequency is too great.

C The speed is too great.

D The wavelength is too great.

26 A sound wave has a certain amplitude and a certain frequency.

A second sound wave is quieter and lower in pitch than the first sound wave.

The second wave has

A a larger amplitude and a greater frequency.

B a larger amplitude and a smaller frequency.

C a smaller amplitude and a greater frequency.

D a smaller amplitude and a smaller frequency.

27 Which statement about a magnet is correct?

A A magnet attracts a gold rod.

B A magnet does not attract a plastic rod.

C A magnet never repels another magnet.

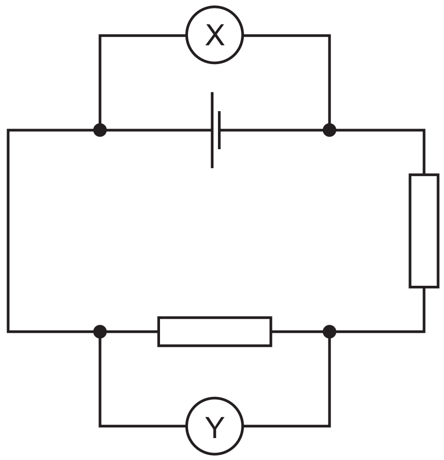
D A magnet sometimes repels an unmagnetised nickel rod.

28 A student wishes to make a permanent magnet. She has an iron rod and a steel rod.

Which rod should she use to make the permanent magnet, and is this rod a hard magnetic material or a soft magnetic material?

	rod	type of magnetic material
A	iron	hard
B	iron	soft
C	steel	hard
D	steel	soft

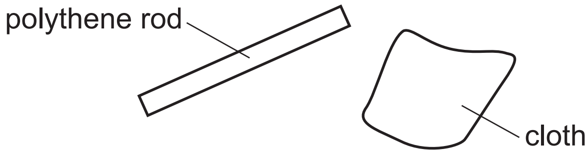
29 The circuit shown includes two meters X and Y, connected correctly.



Which row gives the unit of the quantity measured by X and the unit of the quantity measured by Y?

	meter X	meter Y
A	ampere	ampere
B	ampere	volt
C	volt	ampere
D	volt	volt

30 A polythene rod is rubbed with a cloth.

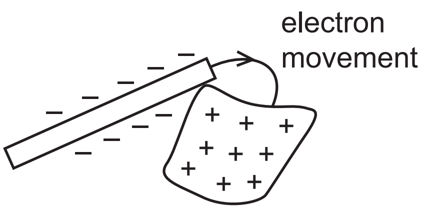


The rod and the cloth both become charged as electrons move between them.

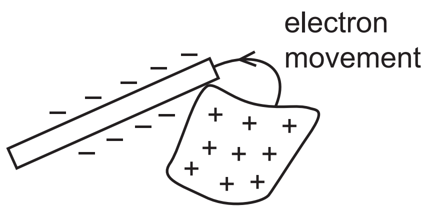
The rod becomes negatively charged.

Which diagram shows how the rod becomes negatively charged, and the final charge on the cloth?

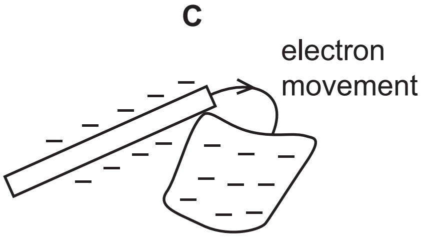
A



B



C



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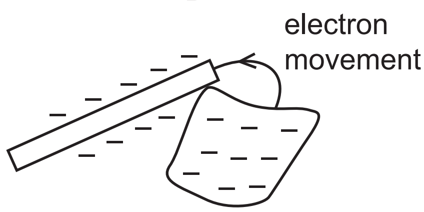


31 What is the function of a relay?

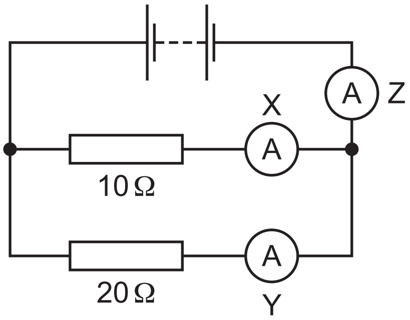
A to allow a current in one circuit to operate a switch in another circuit

B to prevent an electric shock by earthing a metal case

C to protect a circuit by melting if the current becomes too large

D to transform a d.c. voltage to a different value

32 The circuit shown contains three ammeters X, Y and Z.



Which ammeter has the largest reading?

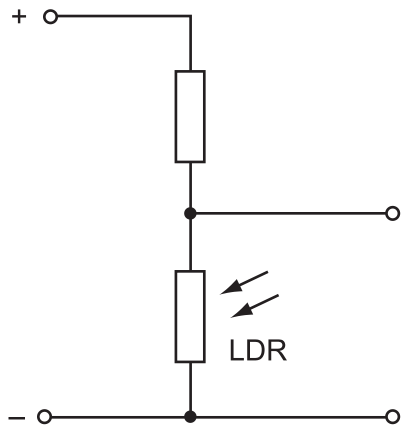
A X

B Y

C Z

D They all have the same reading.

33 The diagram shows part of a circuit used to switch street lamps on and off automatically.



In the evening it gets dark.

Which row shows the effect on the resistance of the light-dependent resistor (LDR) and on the potential difference (p.d.) across it?

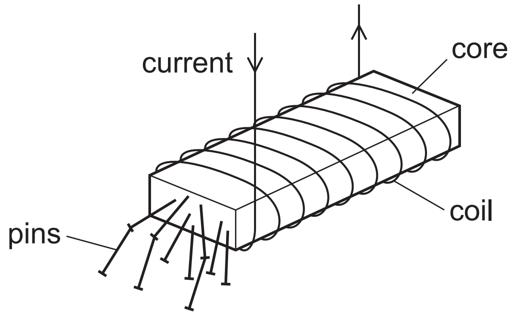
	resistance of LDR	p.d. across LDR
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

34 A domestic circuit includes a 30 A fuse. This protects the wiring if there is too much current in the circuit.

In which wire is the 30 A fuse positioned, and what does it do when it operates?

	position	operation
A	live wire	disconnects the circuit
B	live wire	reduces the current to 30 A
C	neutral wire	disconnects the circuit
D	neutral wire	reduces the current to 30 A

35 A strong electromagnet is used to attract pins.



What happens when the current in the coil is halved?

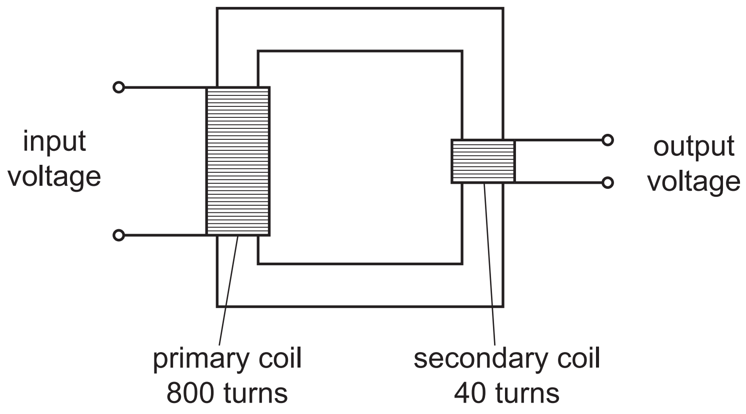
A No pins are attracted.

B Some pins are attracted, but not as many.

C The same number of pins is attracted.

D More pins are attracted.

36 The diagram shows a transformer.



The input voltage is 240 V.

What is the output voltage?

A 6.0 V **B** 12 V **C** 20 V **D** 40 V

37 How many neutrons are in a nucleus of the nuclide $^{37}_{17}\mathrm{C}l$?

A 17 **B** 20 **C** 37 **D** 54

38 A certain element has several isotopes.

Which statement about these isotopes is correct?

A They must have different numbers of electrons orbiting their nuclei.

B They must have the same number of neutrons in their nuclei.

C They must have the same number of nucleons in their nuclei.

D They must have the same number of protons in their nuclei.

39 A radioactive nucleus emits either an α-particle or a β-particle.

What are the products of these two types of radioactive emission?

	product after α-emission	product after β-emission
A	a nucleus of a different element	a nucleus of a different element
B	a nucleus of a different element	a nucleus of the same element
C	a nucleus of the same element	a nucleus of a different element
D	a nucleus of the same element	a nucleus of the same element

40 A reading is taken every 10 minutes of the number of emissions per second from a radioactive source. The table shows the readings.

time / min	number of emissions per second
0	800
10	560
20	400
30	280
40	200
50	140
60	100

What is the half-life of the source?

A 10 min **B** 20 min **C** 40 min **D** 60 min

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Cambridge International General Certificate of Secondary Education

PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **May/June 2016**

MARK SCHEME

Maximum Mark: 40

Published

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Question Number	*Key*	*Question Number*	*Key*
1	**A**	21	**C**
2	**C**	22	**A**
3	**B**	23	**B**
4	**C**	24	**C**
5	**A**	25	**B**
6	**C**	26	**C**
7	**C**	27	**A**
8	**C**	28	**C**
9	**B**	29	**A**
10	**B**	30	**D**
11	**B**	31	**C**
12	**A**	32	**D**
13	**B**	33	**D**
14	**A**	34	**A**
15	**D**	35	**B**
16	**D**	36	**B**
17	**B**	37	**B**
18	**B**	38	**D**
19	**D**	39	**B**
20	**C**	40	**C**

Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **May/June 2016**

MARK SCHEME

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Question Number	*Key*	*Question Number*	*Key*
1	**A**	21	**C**
2	**B**	22	**A**
3	**B**	23	**C**
4	**A**	24	**B**
5	**A**	25	**D**
6	**C**	26	**B**
7	**B**	27	**A**
8	**C**	28	**C**
9	**B**	29	**B**
10	**D**	30	**B**
11	**D**	31	**A**
12	**C**	32	**A**
13	**B**	33	**D**
14	**C**	34	**A**
15	**B**	35	**B**
16	**A**	36	**B**
17	**D**	37	**B**
18	**C**	38	**A**
19	**C**	39	**A**
20	**C**	40	**C**

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Cambridge International General Certificate of Secondary Education

PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **May/June 2016**

MARK SCHEME

Maximum Mark: 40

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Question Number	*Key*	*Question Number*	*Key*
1	**A**	21	**C**
2	**C**	22	**A**
3	**A**	23	**A**
4	**A**	24	**B**
5	**D**	25	**B**
6	**D**	26	**D**
7	**D**	27	**B**
8	**C**	28	**C**
9	**B**	29	**D**
10	**B**	30	**B**
11	**D**	31	**A**
12	**D**	32	**C**
13	**B**	33	**D**
14	**A**	34	**A**
15	**A**	35	**B**
16	**C**	36	**B**
17	**D**	37	**B**
18	**C**	38	**D**
19	**B**	39	**A**
20	**C**	40	**B**

Cambridge IGCSE

Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/13**

Paper 1 Multiple Choice **May/June 2015**

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

0951356906

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Electronic calculators may be used.

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **18** printed pages and **2** blank pages.

IB15 06_0625_13/2RP

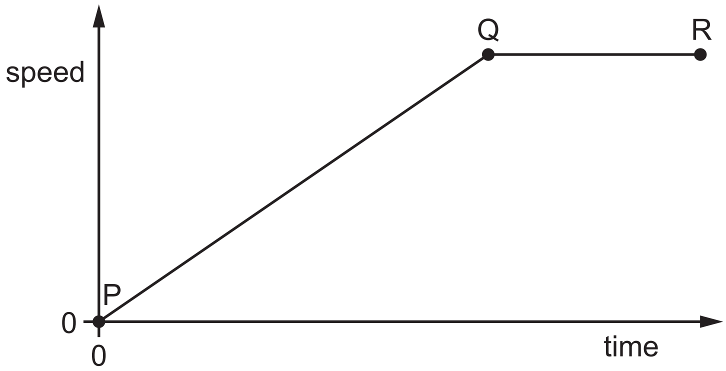
CAMBRIDGE
International Examinations

1 A cook wants to prepare some food to be cooked by 1.15 p.m. He uses an oven with an automatic timer that can be set to switch on and off at certain times. The oven needs to be switched on for 2 hours 10 minutes.

At which time does the oven need to switch on?

A 11.05 a.m. **B** 11.25 a.m. **C** 3.05 p.m. **D** 3.25 p.m.

2 The speed-time graph shows the motion of a car.



Which row describes the motion?

	between P and Q	between Q and R
A	accelerating	moving at constant speed
B	accelerating	not moving
C	moving at constant speed	decelerating
D	moving at constant speed	not moving

3 A car travels 100 km. The journey takes two hours. The highest speed of the car is 80 km/h, and the lowest speed is 40 km/h.

What is the average speed for the journey?

A 40 km/h **B** 50 km/h **C** 60 km/h **D** 120 km/h

4 Weight is an example of which quantity?

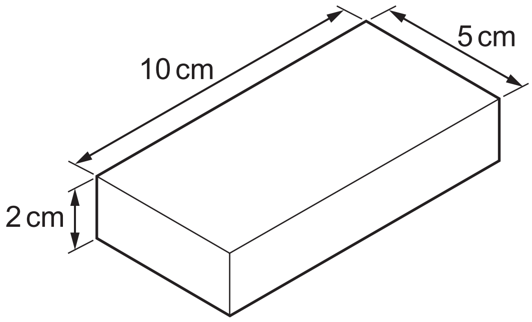
A acceleration

B force

C mass

D pressure

5 A metal block has the dimensions shown. Its mass is 1000 g.



What is the density of the metal?

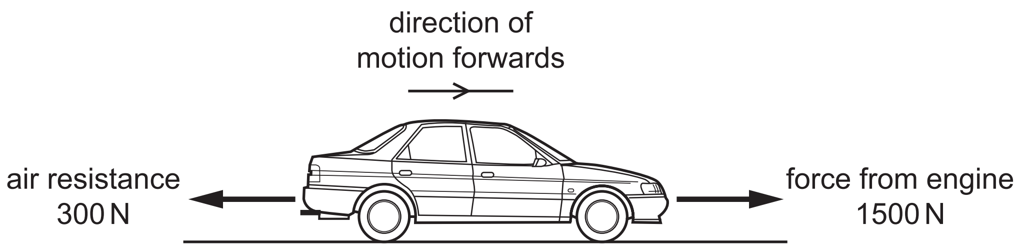
A $\left(\dfrac{5 \times 10}{1000 \times 2}\right)$ g/cm^3

B $\left(\dfrac{2 \times 5 \times 10}{1000}\right)$ g/cm^3

C $\left(\dfrac{1000 \times 2}{5 \times 10}\right)$ g/cm^3

D $\left(\dfrac{1000}{2 \times 5 \times 10}\right)$ g/cm^3

6 A car travels along a horizontal road at a constant speed. Three horizontal forces act on the car. The diagram shows two of these three forces.



What is the size and the direction of the third horizontal force acting on the car?

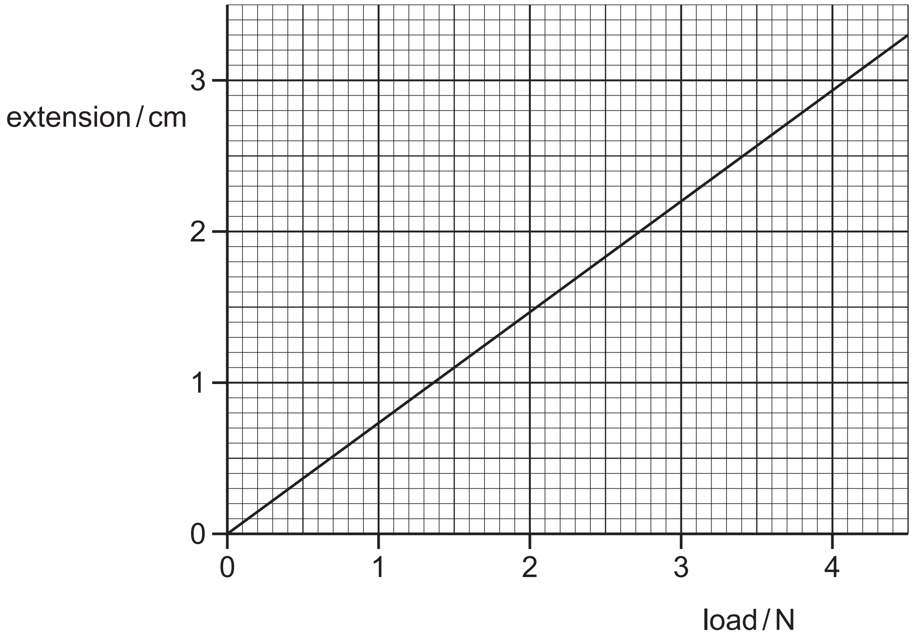
A 1200 N backwards

B 1200 N forwards

C 1800 N backwards

D 1800 N forwards

7 The extension-load graph for a spring is shown. The unstretched length of the spring is 17.0 cm.



When an object is hung from the spring, the length of the spring is 19.2 cm.

What is the weight of the object?

A 1.4 N **B** 1.6 N **C** 2.6 N **D** 3.0 N

8 To calculate the power produced by a force, the size of the force must be known.

What else needs to be known to calculate the power?

	the distance the force moves the object	the time for which the force acts on the object
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

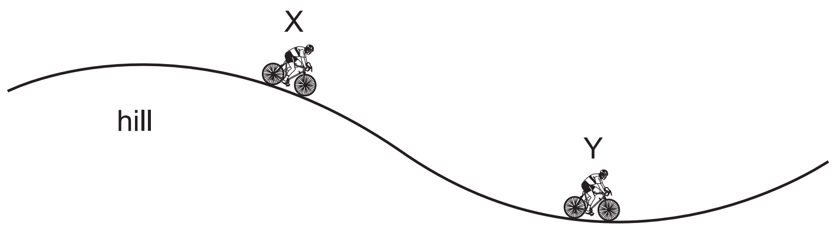
key

✓ = needed

✗ = not needed

9 A cyclist travels down a hill from rest at point X, without pedalling.

The cyclist applies his brakes and the cycle stops at point Y.



Which energy changes have taken place between X and Y?

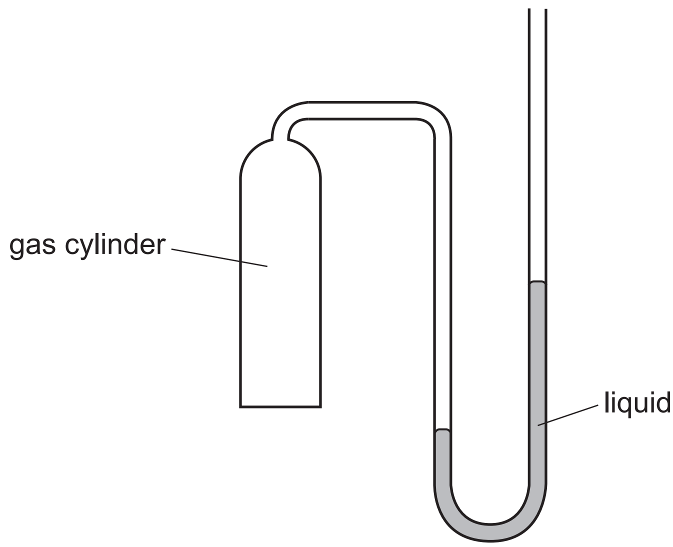
A gravitational potential → kinetic → thermal (heat)

B gravitational potential → thermal (heat) → kinetic

C kinetic → gravitational potential → thermal (heat)

D kinetic → thermal (heat) → gravitational potential

10 The diagram shows a manometer with one side connected to a gas cylinder and the other side open to the atmosphere.



Which conclusion can be made using only the information from liquid levels in the manometer?

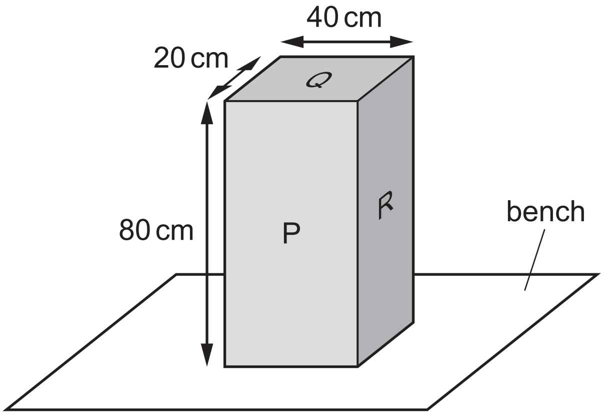
A The density of the gas is less than the density of air.

B The density of the gas is greater than the density of air.

C The pressure of the gas is less than atmospheric pressure.

D The pressure of the gas is greater than atmospheric pressure.

11 The diagram shows a solid block resting on a bench. The dimensions of the block are shown.



On which labelled surface should the block rest to produce the smallest pressure on the bench?

A P

B Q

C R

D any of P, Q or R

12 Small smoke particles suspended in air are viewed through a microscope.

The smoke particles move randomly.

What does this show?

A The air consists of fast-moving molecules.

B The pressure of the air is increasing.

C There are convection currents in the air.

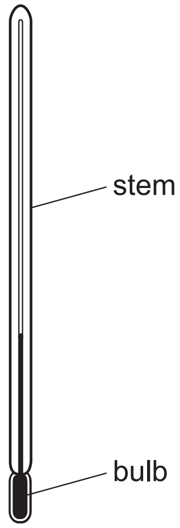
D The temperature of the air is increasing.

13 Molecules escape from a liquid during evaporation. The temperature of the remaining liquid changes.

Which molecules escape and how does the temperature change?

	molecules escaping	temperature of remaining liquid
A	least energetic	decreases
B	least energetic	increases
C	most energetic	decreases
D	most energetic	increases

14 The thermometer in the diagram has no scale.



Where must the bulb be placed so that 0 °C can be marked on the stem?

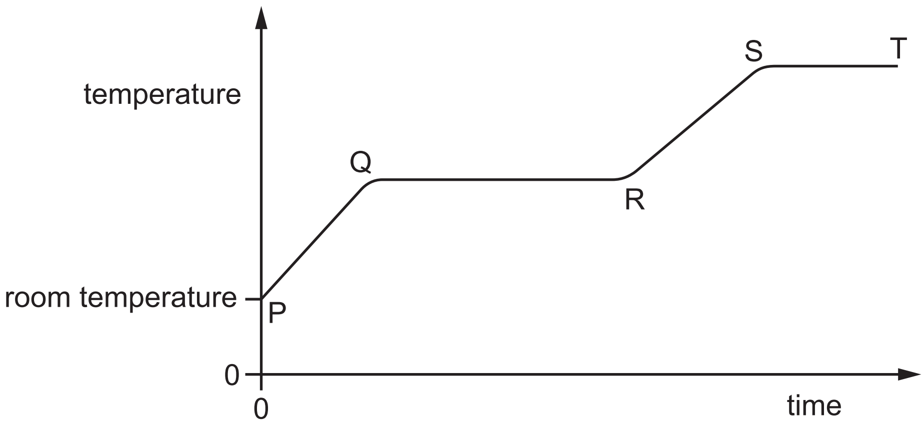
A in a freezer

B in pure boiling water

C in pure cold water

D in pure melting ice

15 A solid is heated from room temperature. The graph shows how its temperature changes with time as it is heated constantly.



Between which labelled points on the graph is the substance partly solid and partly liquid?

A between P and Q

B between Q and R

C between R and S

D between S and T

16 One method of heat transfer involves the energy travelling at a much greater speed than in other methods.

What is the name of this method?

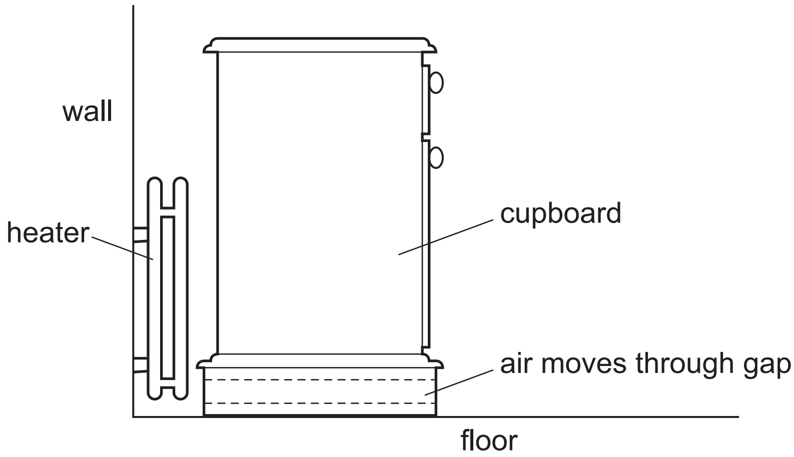
A conduction

B convection

C evaporation

D radiation

17 A cupboard is placed in front of a heater. Air can move through a gap under the cupboard.



Which row describes the temperature, and the direction of movement, of the air in the gap?

	air temperature	air direction
A	cool	away from the heater
B	cool	towards the heater
C	warm	away from the heater
D	warm	towards the heater

18 What is the frequency of a wave?

A the distance that a wavefront travels every second

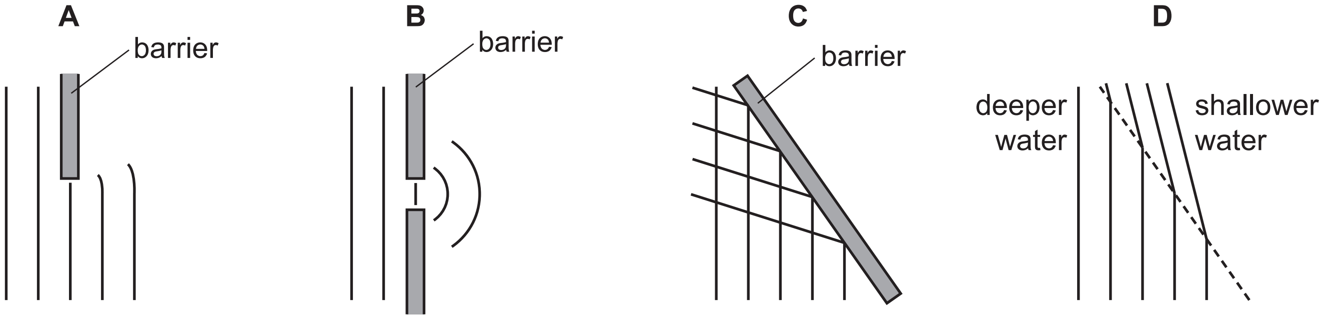
B the distance from one wavefront to the next

C the number of wavefronts produced per second

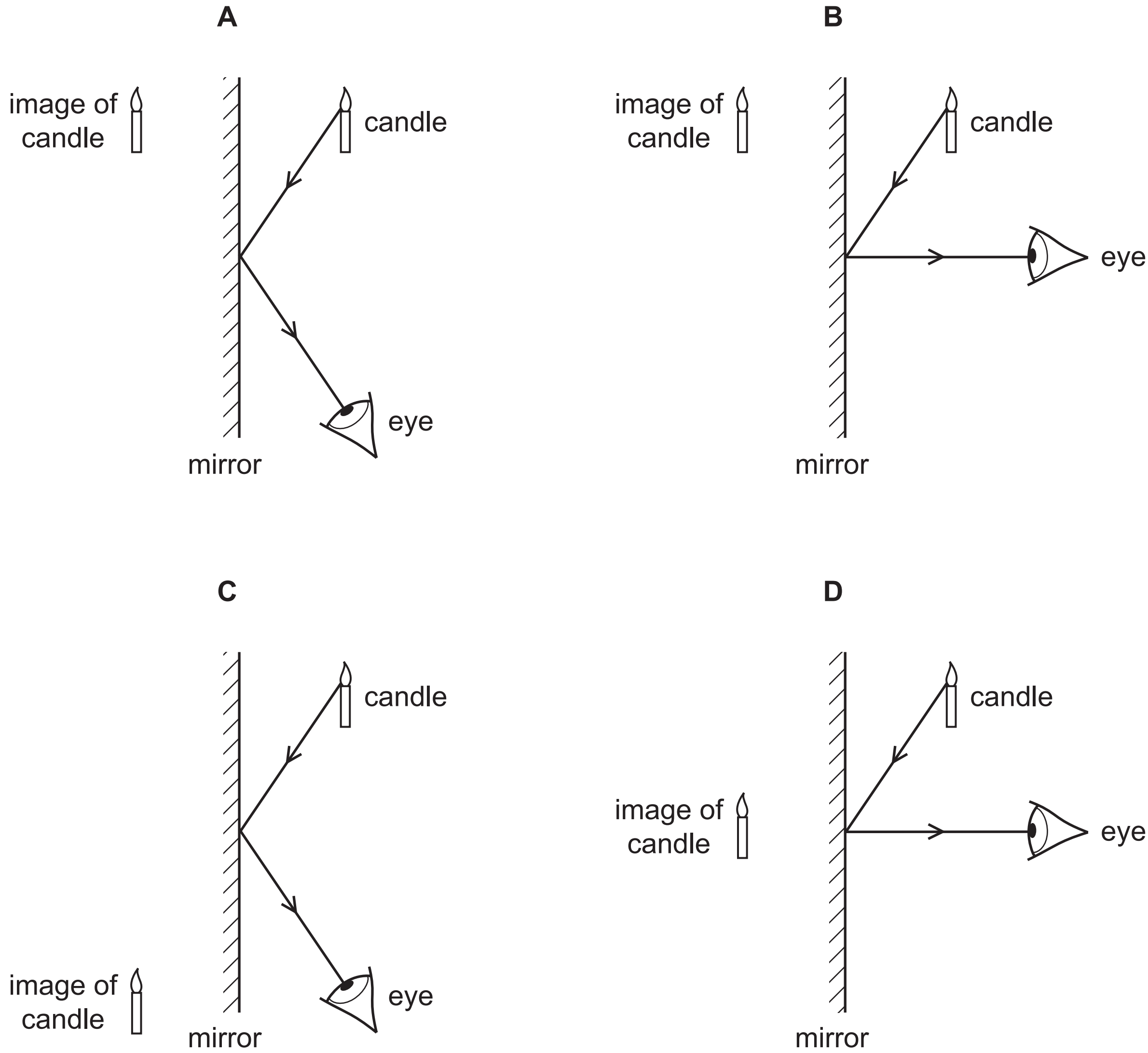
D the time taken for a wavefront to pass a certain point

19 The diagrams represent water waves in a tank.

Which diagram represents a wave that changes speed?

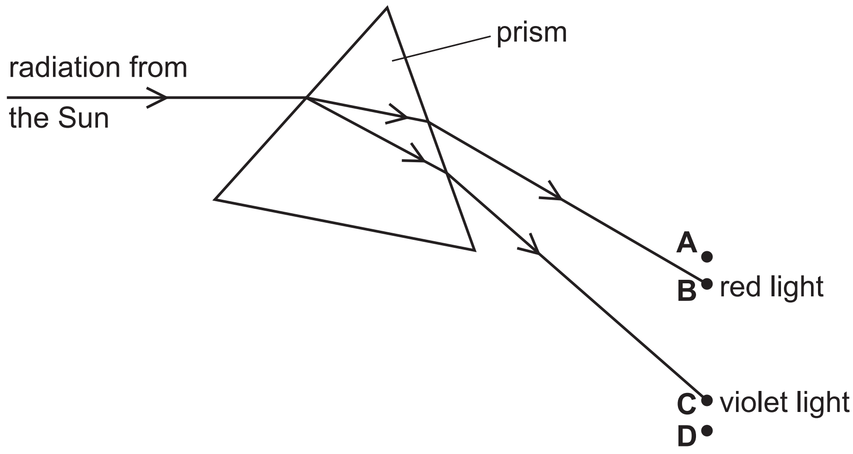


20 Which diagram shows how the light from a candle is reflected by a mirror, and shows the position of the image formed?



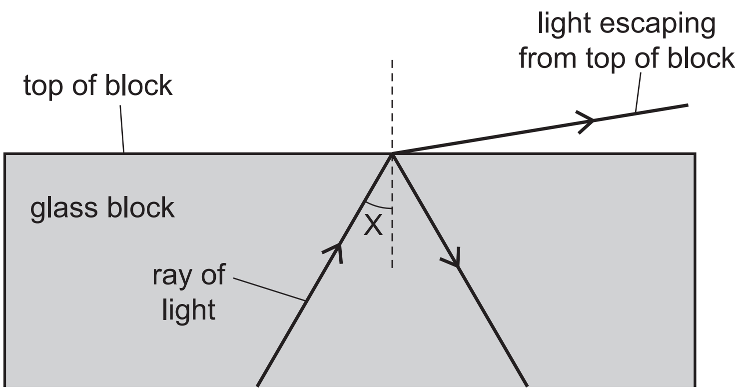
21 Radiation from the Sun is dispersed by a prism. The prism does not absorb any of the radiation. Four identical thermometers are placed, one at each of the labelled positions.

In which position does the thermometer show the greatest rise in temperature?



22 A scientist tries to direct a ray of light in a glass block so that no light escapes from the top of the block.

However, some light does escape.



The scientist changes angle X and stops the light escaping from the top.

Which row in the table describes the change to angle X and the name of the effect produced?

	change to angle X	name of effect produced
A	decrease	total internal reflection
B	decrease	total internal refraction
C	increase	total internal reflection
D	increase	total internal refraction

23 The frequency of a musical note is increased.

A student hearing the sound detects an increase in which property?

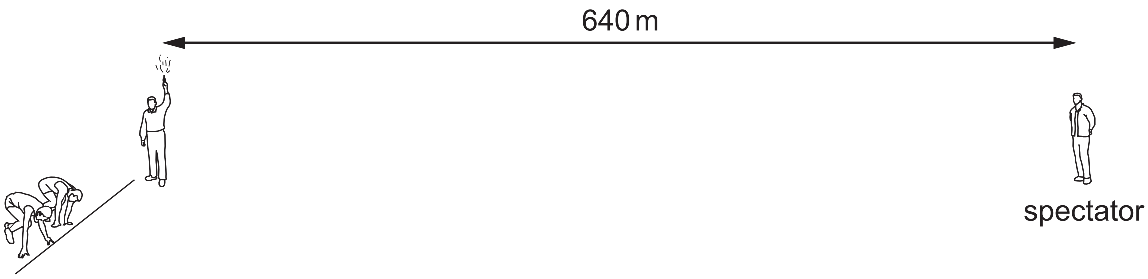
A loudness of the sound

B pitch of the sound

C speed of the sound wave

D wavelength of the sound wave

24 A man holding a starting pistol stands 640 m away from a spectator.



The spectator hears the sound of the starting pistol 2.0 s after seeing the flash from the pistol.

Using this information, what is the speed of sound in air?

A 160 m/s **B** 320 m/s **C** 640 m/s **D** 1280 m/s

25 Which action will demagnetise a magnetised piece of steel?

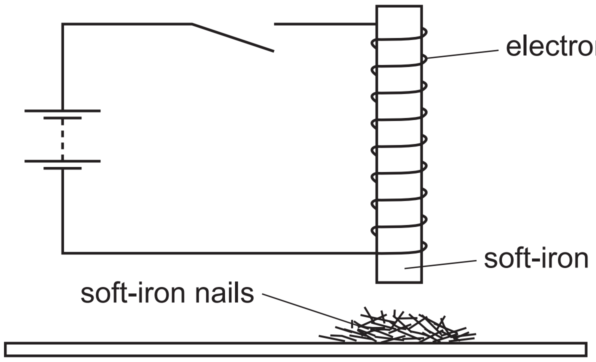
A Cool it in a freezer for several hours.

B Hit it repeatedly with a hammer.

C Put it in a coil carrying a direct current (d.c.).

D Put it near an unmagnetised piece of iron.

26 An electromagnet with a soft-iron core is connected to a battery and an open switch. The soft-iron core is just above some small soft-iron nails.



The switch is now closed, left closed for a few seconds, and then opened.

What do the soft-iron nails do as the switch is closed, and what do they do when the switch is then opened?

	as switch is closed	as switch is opened
A	nails jump up	nails fall down
B	nails jump up	nails stay up
C	nails stay down	nails jump up
D	nails stay down	nails stay down

27 What is the unit of electromotive force (e.m.f.)?

A ampere

B newton

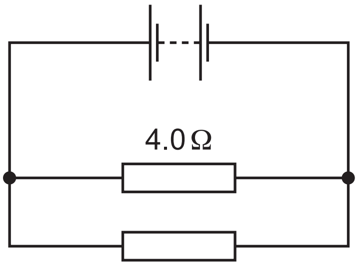
C ohm

D volt

28 Which sample of copper wire has the greatest electrical resistance?

	length of wire / m	diameter of wire / mm
A	1.0	2.0
B	1.0	4.0
C	10	2.0
D	10	4.0

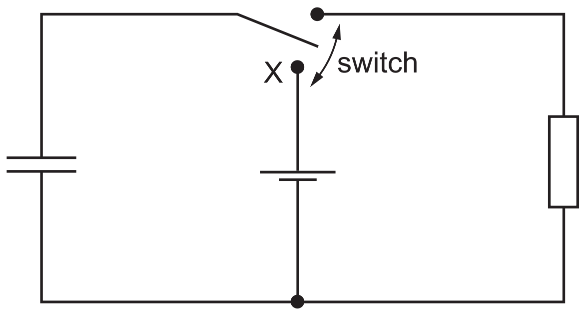
29 The circuit diagram shows a 4.0 Ω resistor and an 8.0 Ω resistor connected to a 6.0 V battery.



What is the current in the 8.0 Ω resistor?

A 0 A **B** 0.50 A **C** 0.75 A **D** 1.0 A

30 The diagram shows a circuit which includes an uncharged capacitor and a switch.

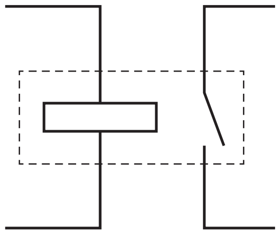


The switch can be moved between position X and position Y.

What happens to the capacitor when the switch is moved to position X, and what happens when the switch is then moved to position Y?

	switch at X	switch at Y
A	capacitor charges	capacitor charges
B	capacitor charges	capacitor discharges
C	capacitor discharges	capacitor charges
D	capacitor discharges	capacitor discharges

31 Which component is represented by this circuit symbol?



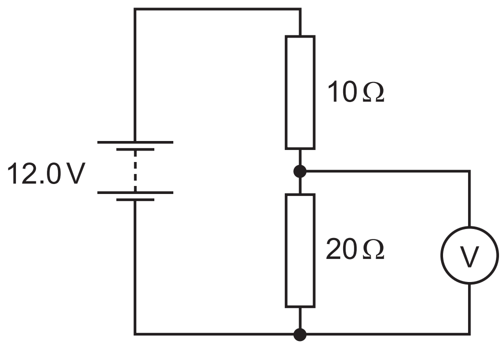
A a bell

B a fuse

C a relay

D a transformer

32 The diagram shows a 10 Ω resistor and a 20 Ω resistor connected in a potential divider circuit.

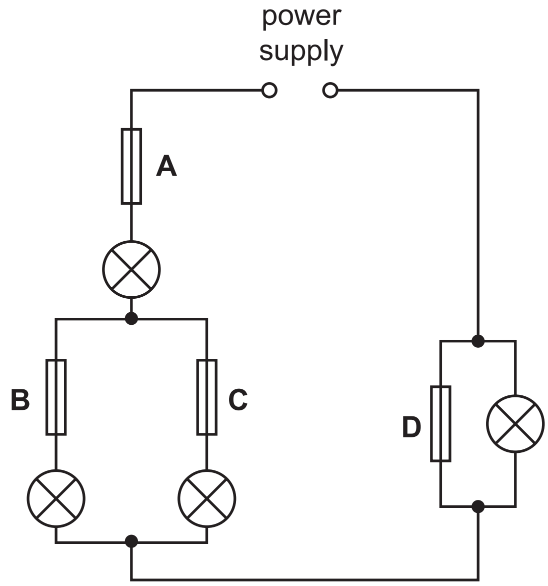


What is the reading on the voltmeter?

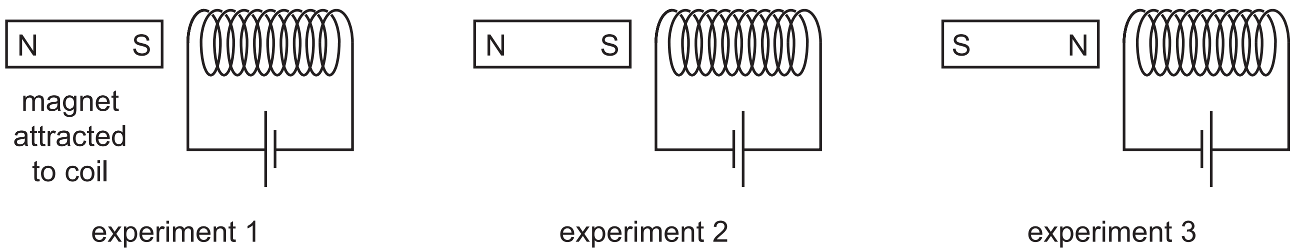
A 4.0 V **B** 6.0 V **C** 8.0 V **D** 12.0 V

33 In the circuit shown, only one of the fuses has blown, but none of the lamps is lit.

Which fuse has blown?



34 A student investigates the force on a bar magnet placed near a current-carrying coil. She carries out three different experiments.

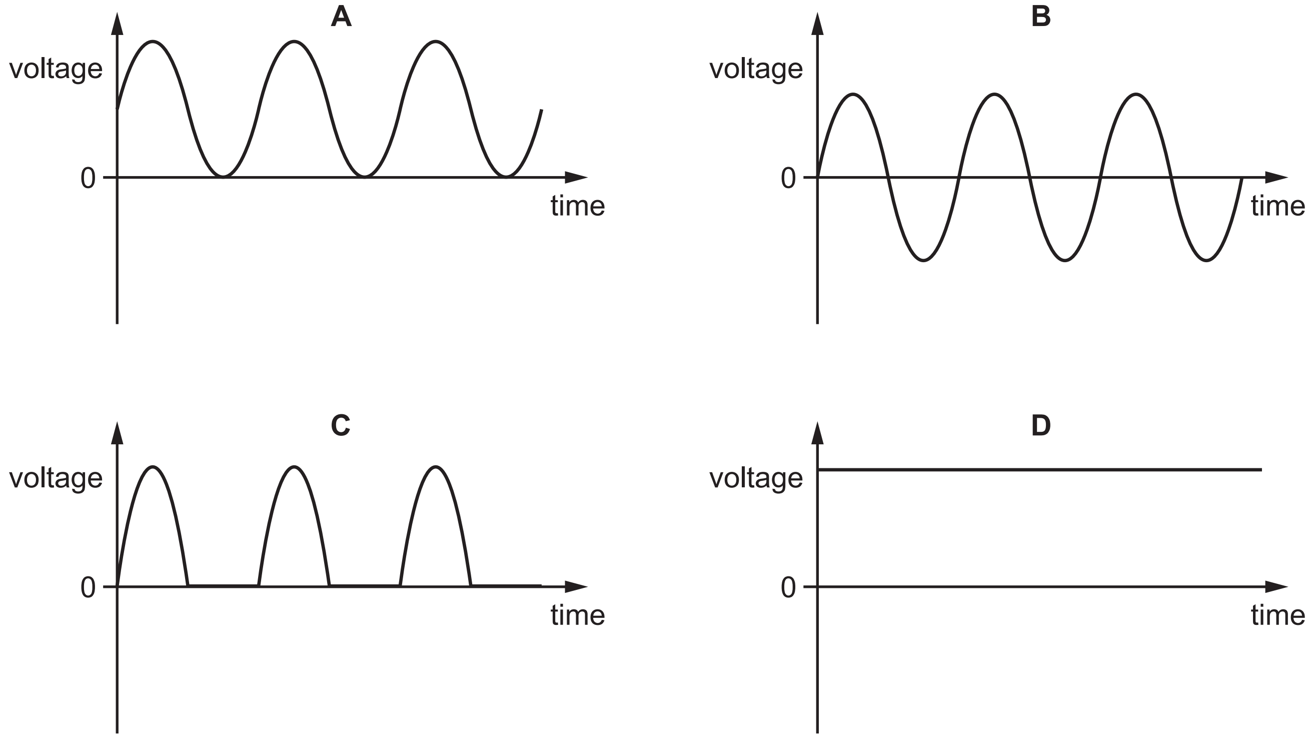


In experiment 1, the magnet is attracted to the coil.

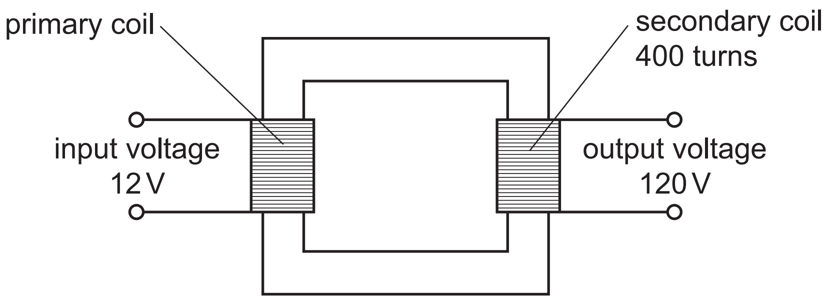
Which row shows what happens in the other two experiments?

	experiment 2	experiment 3
A	magnet attracted	magnet attracted
B	magnet attracted	magnet repelled
C	magnet repelled	magnet attracted
D	magnet repelled	magnet repelled

35 Which diagram represents the voltage output of a simple a.c. generator?



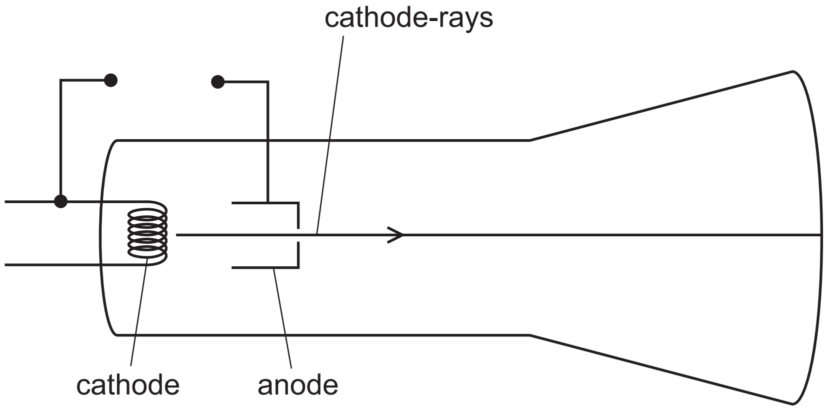
36 A transformer has 400 turns on its secondary coil. An input voltage of 12 V is applied to the primary coil, and an output voltage of 120 V is induced across the secondary coil.



How many turns are on the primary coil of the transformer?

A 12 **B** 40 **C** 400 **D** 4000

37 The diagram shows a cathode-ray tube.



Which electrode should be heated, and which electrode should be positive?

	heated electrode	positive electrode
A	anode	anode
B	anode	cathode
C	cathode	anode
D	cathode	cathode

38 α, β and γ-radiations are emitted by radioactive substances.

Which statement is correct?

A α-radiation consists of charged particles and is the most highly ionising radiation.

B β-radiation consists of charged particles and is the most penetrating radiation.

C β-radiation consists of uncharged particles and is the least highly ionising radiation.

D γ-radiation consists of uncharged particles and is the least penetrating radiation.

39 The nucleus of an americium atom contains 146 neutrons and 95 protons. It decays by emitting an α-particle.

How many neutrons and how many protons remain in the nucleus when this form of americium decays?

	number of neutrons remaining	number of protons remaining
A	142	93
B	142	95
C	144	93
D	144	95

40 Which statement is correct for the nucleus of **any** atom?

A The nucleus contains electrons, neutrons and protons.

B The nucleus contains the same number of protons as neutrons.

C The nucleus has a total charge of zero.

D The nucleus is very small compared with the size of the atom.

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Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/12**

Paper 1 Multiple Choice **May/June 2015**

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

1041620377

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Electronic calculators may be used.

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **19** printed pages and **1** blank page.

IB15 06_0625_12/3RP

CAMBRIDGE
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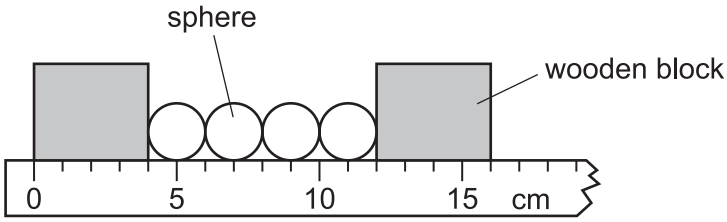
1 The diagram shows four identical spheres placed between two wooden blocks on a ruler.



What is the diameter of one sphere?

A 1.0 cm **B** 2.0 cm **C** 3.0 cm **D** 4.0 cm

2 What does the area under a speed-time graph represent?

A acceleration

B average speed

C deceleration

D distance travelled

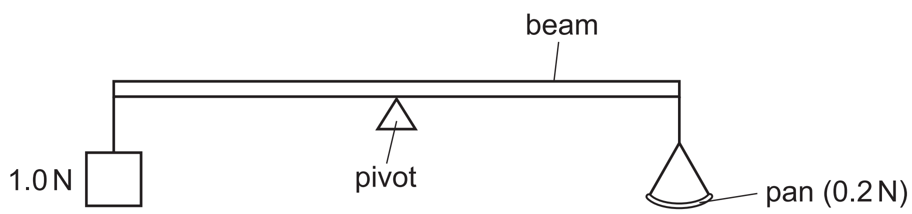
3 A car travels 100 km. The journey takes two hours. The highest speed of the car is 80 km / h, and the lowest speed is 40 km / h.

What is the average speed for the journey?

A 40 km / h **B** 50 km / h **C** 60 km / h **D** 120 km / h

4 The diagram shows a uniform beam being used as a balance. The beam is pivoted at its centre.

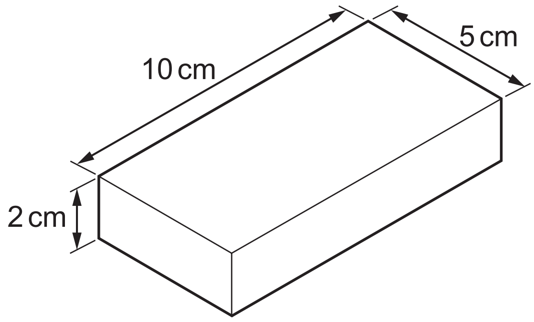
A 1.0 N weight is attached to one end of the beam. An empty pan weighing 0.2 N is attached to the other end of the beam.



How many 0.1 N weights must be placed on the pan in order to balance the beam?

A 5 **B** 8 **C** 10 **D** 12

5 A metal block has the dimensions shown. Its mass is 1000 g.



What is the density of the metal?

A $\left(\dfrac{5 \times 10}{1000 \times 2}\right) \text{g/cm}^3$

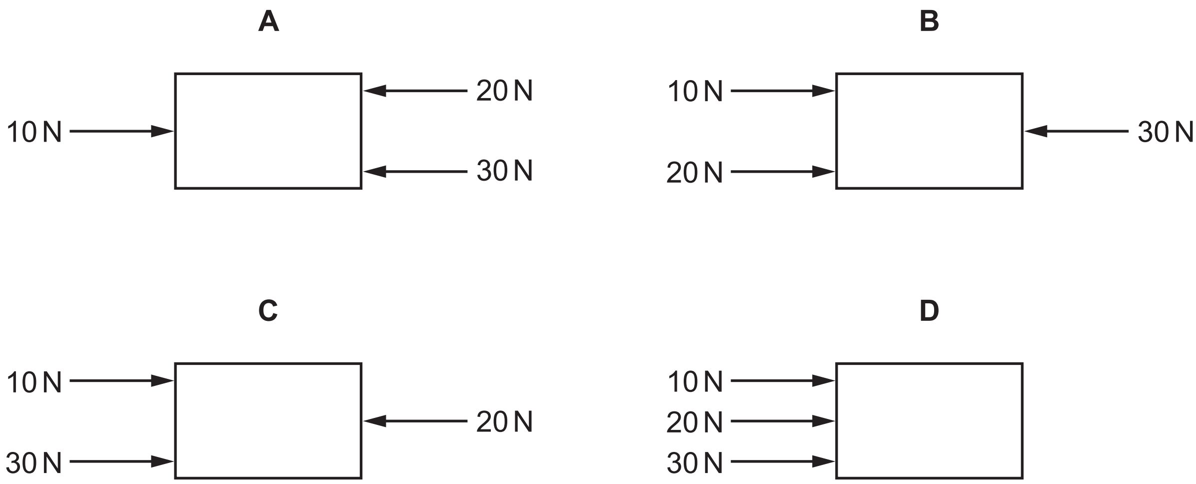
B $\left(\dfrac{2 \times 5 \times 10}{1000}\right) \text{g/cm}^3$

C $\left(\dfrac{1000 \times 2}{5 \times 10}\right) \text{g/cm}^3$

D $\left(\dfrac{1000}{2 \times 5 \times 10}\right) \text{g/cm}^3$

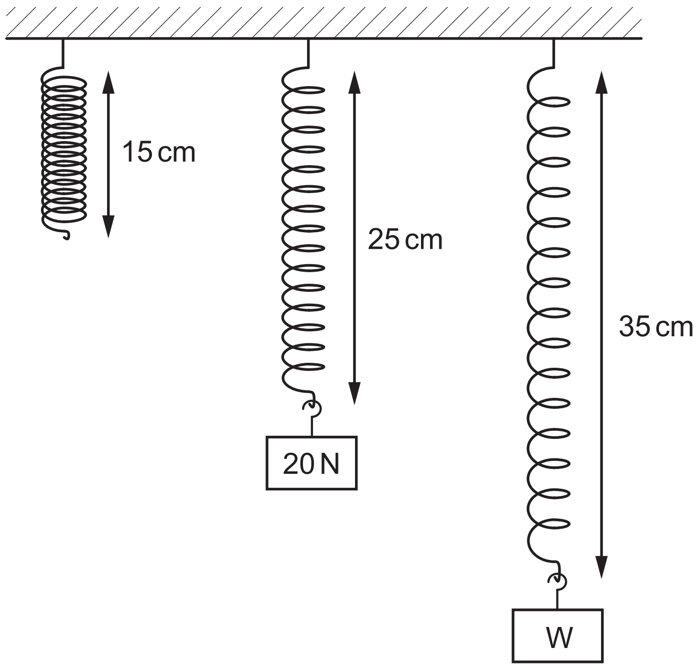
6 The diagrams show four identical objects. Each object is acted on by only the three forces shown.

Which object accelerates to the right, with the **smallest** acceleration?



[Turn over

7 Different weights are hung from a spring. The diagram shows the original length of the spring, and the lengths when different weights are added.



The extension of the spring is directly proportional to the weight hung from it.

What is the weight of W?

A 30 N **B** 35 N **C** 40 N **D** 45 N

8 Which source of energy involves the splitting of heavy atoms?

A chemical energy

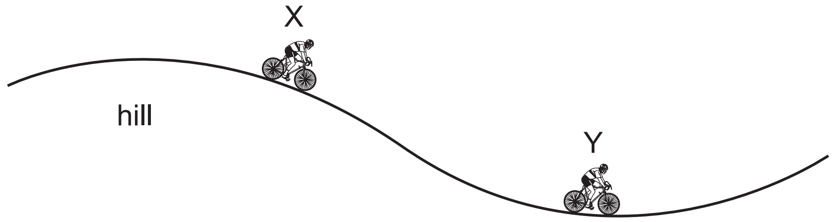
B geothermal energy

C hydroelectric energy

D nuclear energy

9 A cyclist travels down a hill from rest at point X, without pedalling.

The cyclist applies his brakes and the cycle stops at point Y.



Which energy changes have taken place between X and Y?

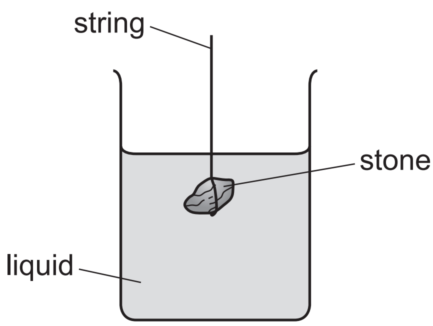
A gravitational potential → kinetic → thermal (heat)

B gravitational potential → thermal (heat) → kinetic

C kinetic → gravitational potential → thermal (heat)

D kinetic → thermal (heat) → gravitational potential

10 The diagram shows a stone suspended under the surface of a liquid from a string. The stone experiences a pressure caused by the liquid.



What would increase the pressure on the stone?

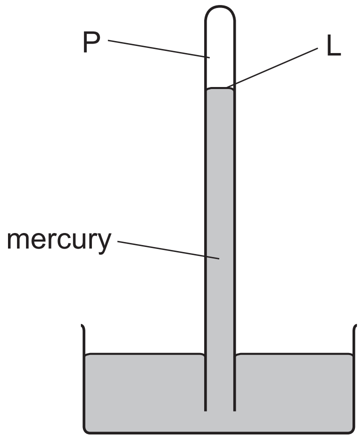
A decreasing the surface area of the stone

B increasing the mass of the stone

C lowering the stone deeper into the liquid

D using a liquid with a lower density

11 The diagram shows a simple mercury barometer, used to measure atmospheric pressure.



Atmospheric pressure decreases.

Which row states what happens to the pressure at point P and what happens to the level L?

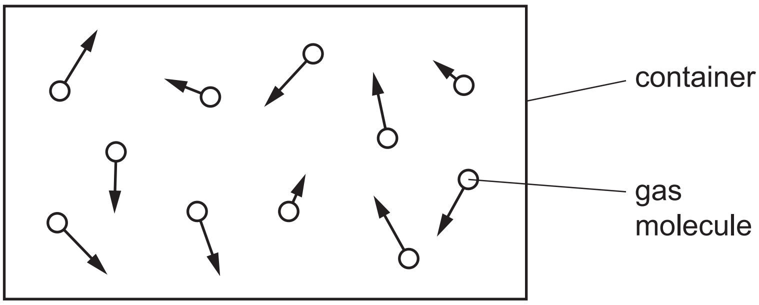
	pressure at P	level L
A	decreases	falls
B	decreases	rises
C	stays the same	falls
D	stays the same	rises

12 Puddles of rain water remain after a storm. The water in the puddles gradually evaporates.

How does the evaporation affect the temperature of the water remaining in the puddle, and how does it affect the average speed of the remaining water molecules in the puddle?

	temperature of water in puddle	average speed of water molecules in puddle
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

13 The diagram represents moving gas molecules in a sealed container of fixed volume.

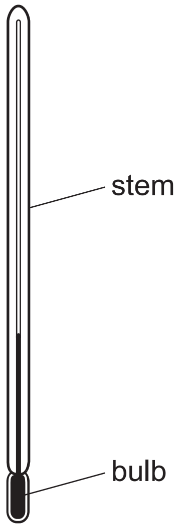


The temperature of the gas is now increased.

What happens to the pressure of the gas, and what happens to the speed of the gas molecules?

	pressure of gas	speed of molecules
A	increases	increases
B	increases	unchanged
C	unchanged	increases
D	unchanged	unchanged

14 The thermometer in the diagram has no scale.



Where must the bulb be placed so that 0 °C can be marked on the stem?

A in a freezer

B in pure boiling water

C in pure cold water

D in pure melting ice

15 Two metal blocks X and Y are at room temperature. Each block is heated so that its temperature rises by 10 °C.

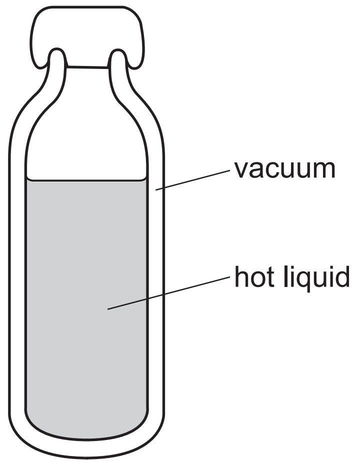
The blocks are now allowed to cool back to room temperature.

Block Y has a greater thermal capacity than block X.

Which block needs more thermal (heat) energy to heat it up by 10 °C and which block loses more thermal (heat) energy as it cools back to room temperature?

	more energy	
	heating	cooling
A	X	X
B	X	Y
C	Y	X
D	Y	Y

16 The diagram shows a vacuum flask used to keep liquid hot.



How does thermal energy pass through the vacuum?

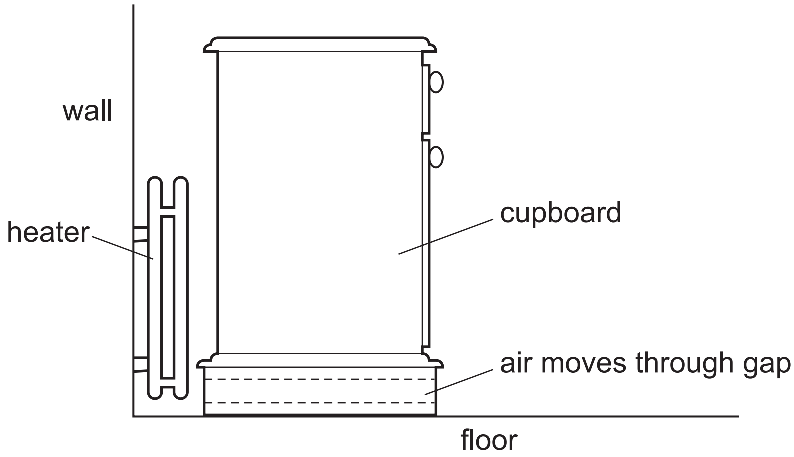
A conduction only

B convection only

C radiation

D conduction and convection

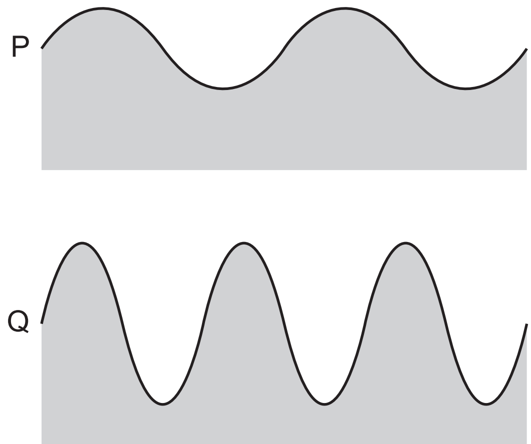
17 A cupboard is placed in front of a heater. Air can move through a gap under the cupboard.



Which row describes the temperature, and the direction of movement, of the air in the gap?

	air temperature	air direction
A	cool	away from the heater
B	cool	towards the heater
C	warm	away from the heater
D	warm	towards the heater

18 The diagrams show two water waves P and Q that are travelling at the same speed on the surface of a pond. The diagrams are to the same scale.

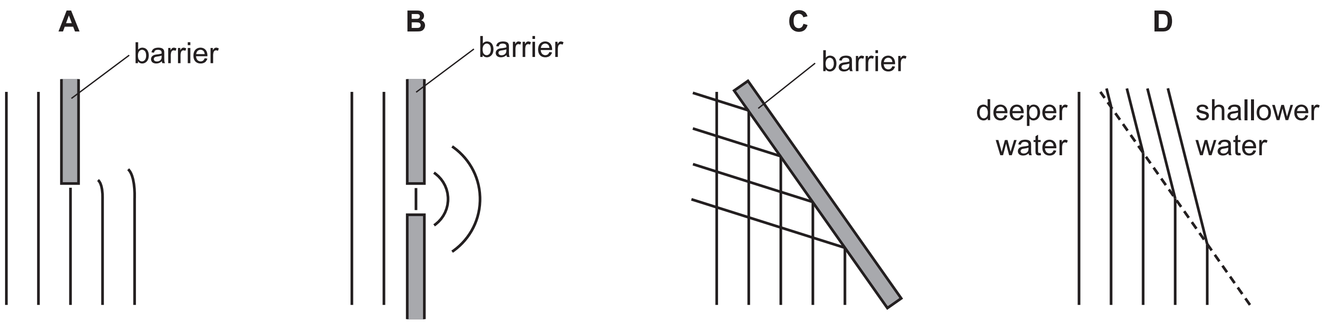


Which wave has the greater amplitude and which wave has the greater frequency?

	greater amplitude	greater frequency
A	P	P
B	P	Q
C	Q	P
D	Q	Q

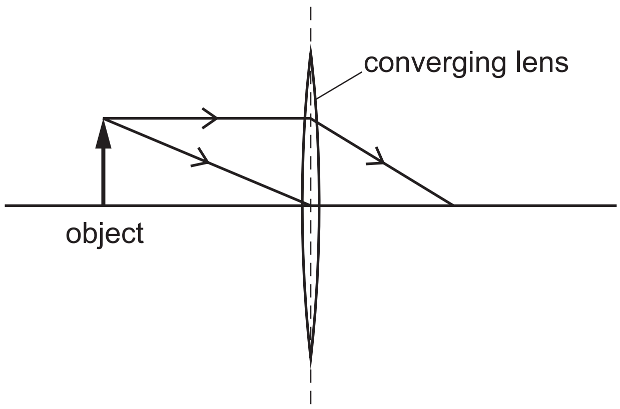
19 The diagrams represent water waves in a tank.

Which diagram represents a wave that changes speed?



20 An object is placed in front of a thin converging lens.

The diagram shows the paths of two rays from the top of the object.



An image of the object is formed on a screen to the right of the lens.

How does this image compare with the object?

A It is larger and inverted.

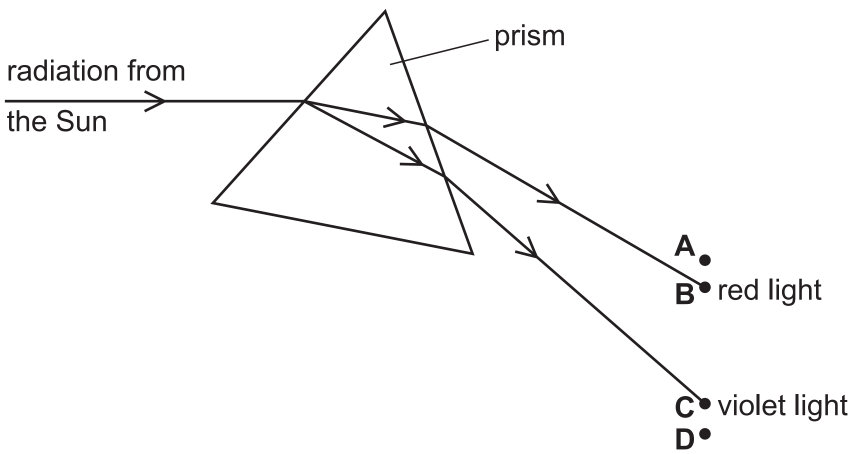
B It is larger and the same way up.

C It is smaller and inverted.

D It is smaller and the same way up.

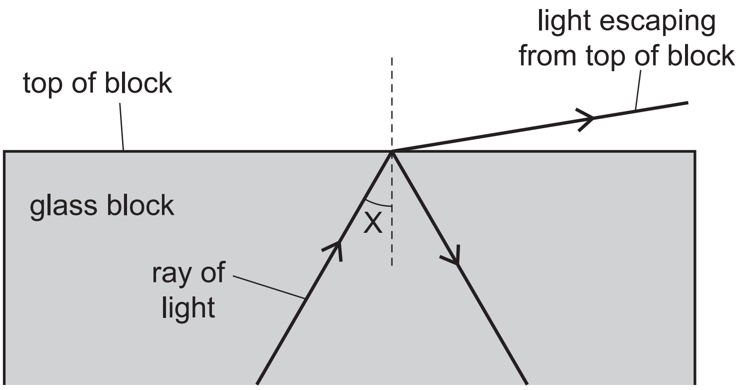
21 Radiation from the Sun is dispersed by a prism. The prism does not absorb any of the radiation. Four identical thermometers are placed, one at each of the labelled positions.

In which position does the thermometer show the greatest rise in temperature?



22 A scientist tries to direct a ray of light in a glass block so that no light escapes from the top of the block.

However, some light does escape.



The scientist changes angle X and stops the light escaping from the top.

Which row in the table describes the change to angle X and the name of the effect produced?

	change to angle X	name of effect produced
A	decrease	total internal reflection
B	decrease	total internal refraction
C	increase	total internal reflection
D	increase	total internal refraction

23 A quiet sound is produced by a loudspeaker. The loudness of the sound is increased.

Which property of the sound wave is increased?

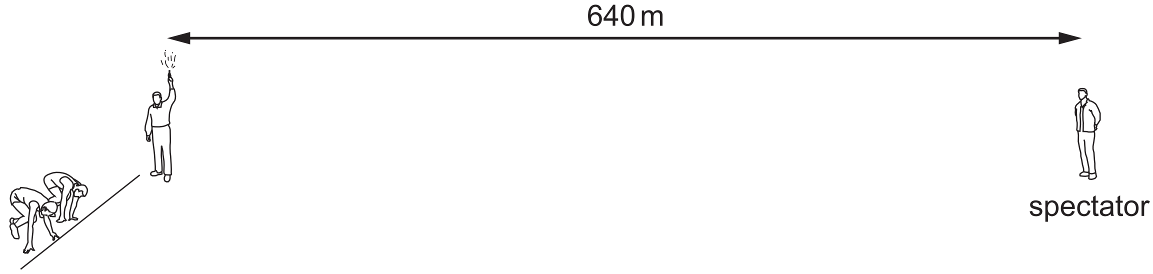
A amplitude

B frequency

C speed

D wavelength

24 A man holding a starting pistol stands 640 m away from a spectator.



The spectator hears the sound of the starting pistol 2.0 s after seeing the flash from the pistol.

Using this information, what is the speed of sound in air?

A 160 m / s **B** 320 m / s **C** 640 m / s **D** 1280 m / s

25 Which group contains only non-ferrous metals?

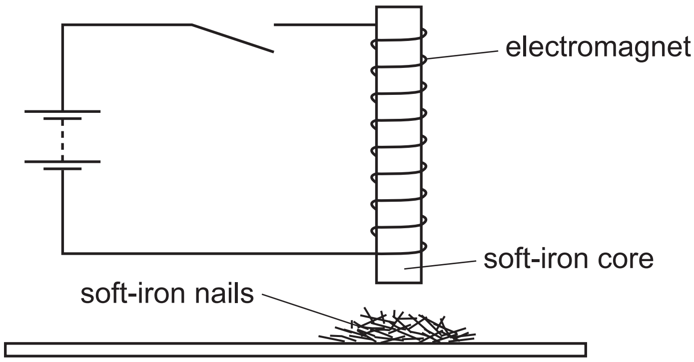
A aluminium, brass, iron

B brass, copper, lead

C copper, iron, steel

D copper, lead, steel

26 An electromagnet with a soft-iron core is connected to a battery and an open switch. The soft-iron core is just above some small soft-iron nails.



The switch is now closed, left closed for a few seconds, and then opened.

What do the soft-iron nails do as the switch is closed, and what do they do when the switch is then opened?

	as switch is closed	as switch is opened
A	nails jump up	nails fall down
B	nails jump up	nails stay up
C	nails stay down	nails jump up
D	nails stay down	nails stay down

27 A student has wires of different lengths and different diameters. The wires are all made of the same metal.

The student measures the resistance of one wire.

Which wire has a greater resistance than the wire he has measured?

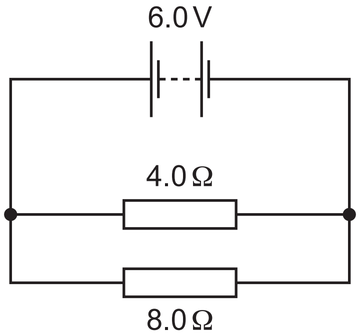
A a shorter wire with a larger diameter

B a shorter wire with the same diameter

C a wire of the same length with a larger diameter

D a wire of the same length with a smaller diameter

28 The circuit diagram shows a 4.0 Ω resistor and an 8.0 Ω resistor connected to a 6.0 V battery.

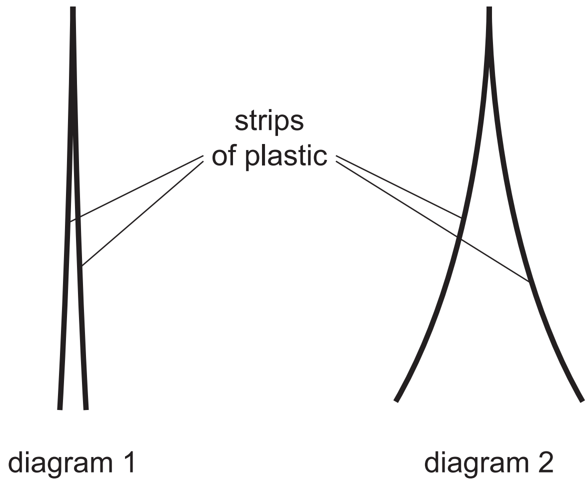


What is the potential difference (p.d.) across the 4.0 Ω resistor?

A 0.5 V **B** 2.0 V **C** 4.0 V **D** 6.0 V

29 Diagram 1 shows two thin, uncharged strips of plastic.

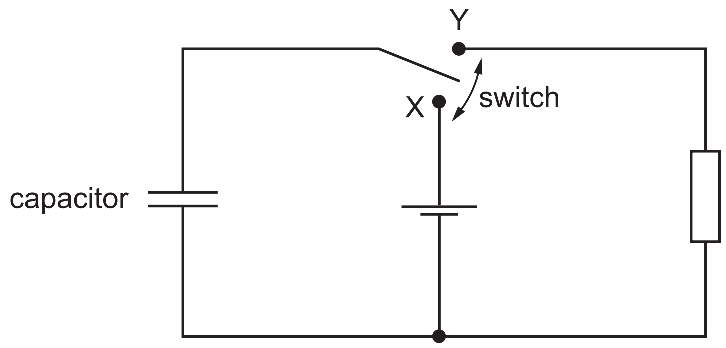
Diagram 2 shows the same strips after they have been rubbed with a dry cloth.



Which row describes the charge on the strips after rubbing, and the force between the strips after rubbing?

	charge on strips	force between strips
A	opposite	attraction
B	opposite	repulsion
C	the same	attraction
D	the same	repulsion

30 The diagram shows a circuit which includes an uncharged capacitor and a switch.

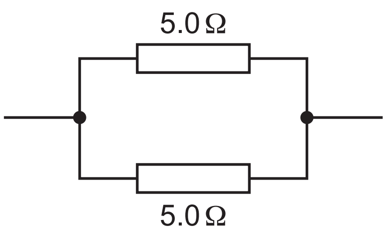


The switch can be moved between position X and position Y.

What happens to the capacitor when the switch is moved to position X, and what happens when the switch is then moved to position Y?

	switch at X	switch at Y
A	capacitor charges	capacitor charges
B	capacitor charges	capacitor discharges
C	capacitor discharges	capacitor charges
D	capacitor discharges	capacitor discharges

31 Two $5.0\,\Omega$ resistors are connected as shown in the diagram.



What is the total resistance of this combination?

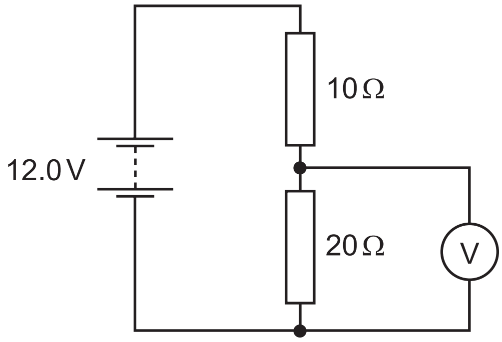
A less than $5.0\,\Omega$

B $5.0\,\Omega$

C more than $5.0\,\Omega$ but less than $10.0\,\Omega$

D $10.0\,\Omega$

32 The diagram shows a 10 Ω resistor and a 20 Ω resistor connected in a potential divider circuit.

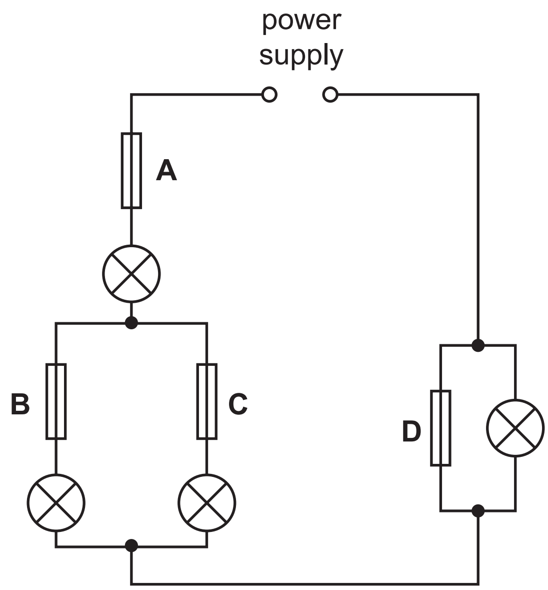


What is the reading on the voltmeter?

A 4.0 V **B** 6.0 V **C** 8.0 V **D** 12.0 V

33 In the circuit shown, only one of the fuses has blown, but none of the lamps is lit.

Which fuse has blown?



34 An e.m.f. is induced across a wire when it moves through the magnetic field between the poles of a magnet.

Which electrical device operates because of this effect?

A a battery

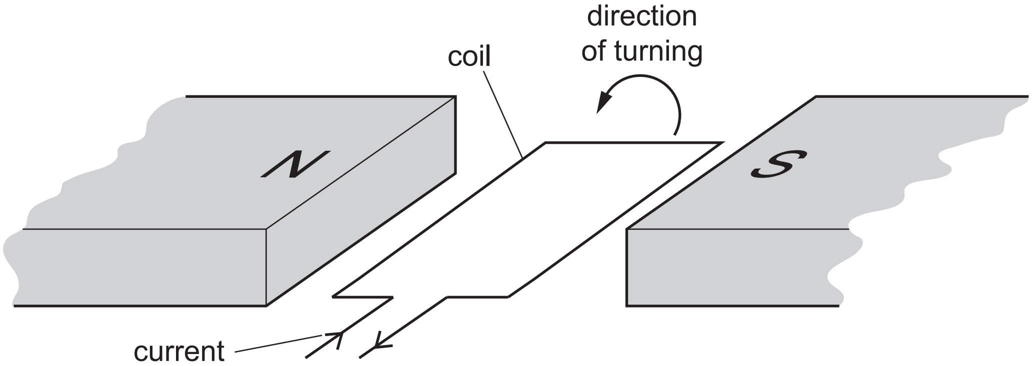
B a cathode-ray tube

C a generator

D a motor

35 The diagram shows a flat, rectangular coil placed between the poles of a magnet.

There is a current in the coil that makes it turn in the direction shown in the diagram.



Which change would make the coil turn in the opposite direction?

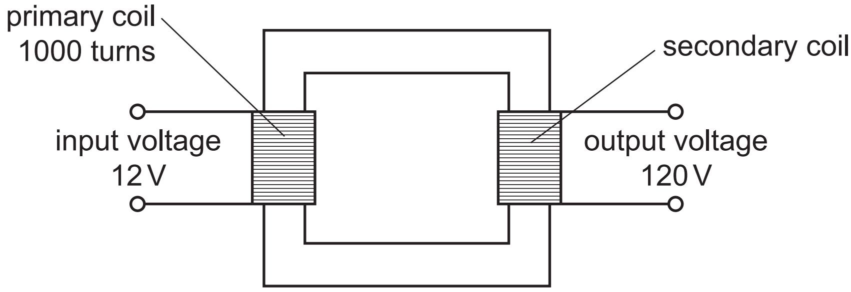
A decreasing the current in the coil

B increasing the number of turns on the coil

C reversing both the direction of the current in the coil and the poles of the magnet

D reversing only the direction of the current in the coil

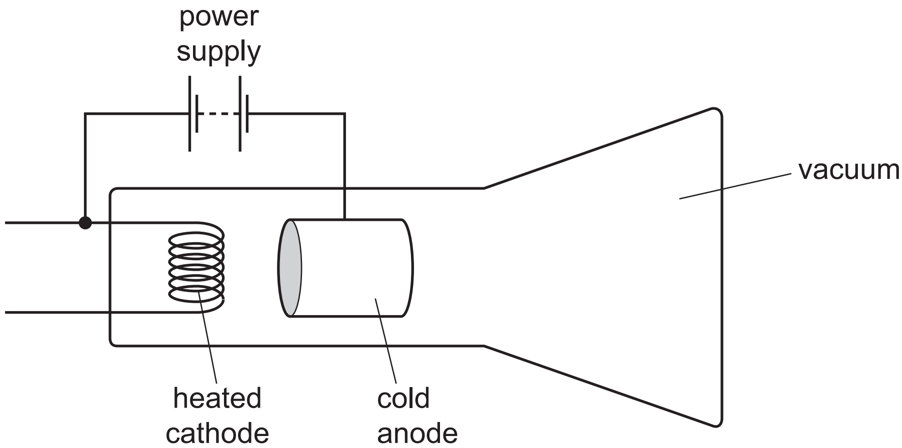
36 A transformer has 1000 turns on its primary coil. An input voltage of 12 V is applied to the primary coil, and an output voltage of 120 V is induced across the secondary coil.



How many turns are on the secondary coil of the transformer?

A 100 **B** 120 **C** 1000 **D** 10 000

37 The diagram shows a cathode-ray tube.



The tube is not working properly.

Which change should be made so that the tube works properly to produce a continuous beam of cathode rays?

A Heat the anode instead of the cathode.

B Reverse the connections of the power supply.

C Use an a.c. power supply instead of a d.c. power supply.

D Use air in the tube instead of a vacuum.

38 Which statement about α-radiation is correct?

A It is a stream of fast-moving electrons.

B It is a form of electromagnetic radiation.

C It is more highly ionising than γ-radiation.

D It is more penetrating than β-radiation.

39 A radioactive source produces a count rate on a detector of 1600 counts / s.

After 32 hours the count rate has fallen to 100 counts / s.

Both count rates have been corrected for background radiation.

What is the half-life of the source?

A 2.0 hours **B** 6.4 hours **C** 8.0 hours **D** 16 hours

40 A nuclide has the symbol $^{22}_{10}\mathrm{Ne}$.

What is the proton number of a nucleus of this nuclide?

A 10 **B** 12 **C** 22 **D** 32

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MARK SCHEME for the May/June 2015 series

0625 PHYSICS

0625/13 Paper 1 (Multiple Choice), maximum raw mark 40

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Question Number	*Key*	*Question Number*	*Key*
1	**A**	21	**A**
2	**A**	22	**C**
3	**B**	23	**B**
4	**B**	24	**B**
5	**D**	25	**B**
6	**A**	26	**A**
7	**D**	27	**D**
8	**A**	28	**C**
9	**A**	29	**C**
10	**D**	30	**B**
11	**A**	31	**C**
12	**A**	32	**C**
13	**C**	33	**A**
14	**D**	34	**C**
15	**B**	35	**B**
16	**D**	36	**B**
17	**B**	37	**C**
18	**C**	38	**A**
19	**D**	39	**C**
20	**A**	40	**D**

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Cambridge International General Certificate of Secondary Education

MARK SCHEME for the May/June 2015 series

0625 PHYSICS

0625/12 Paper 1 (Multiple Choice), maximum raw mark 40

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Question Number	*Key*	*Question Number*	*Key*
1	**B**	21	**A**
2	**D**	22	**C**
3	**B**	23	**A**
4	**B**	24	**B**
5	**D**	25	**B**
6	**C**	26	**A**
7	**C**	27	**D**
8	**D**	28	**D**
9	**A**	29	**D**
10	**C**	30	**B**
11	**C**	31	**A**
12	**A**	32	**C**
13	**A**	33	**A**
14	**D**	34	**C**
15	**D**	35	**D**
16	**C**	36	**D**
17	**B**	37	**B**
18	**D**	38	**C**
19	**D**	39	**C**
20	**A**	40	**A**

CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International General Certificate of Secondary Education

MARK SCHEME for the May/June 2015 series

0625 PHYSICS

0625/11 Paper 1 (Multiple Choice), maximum raw mark 40

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2015 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

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Question Number	*Key*	*Question Number*	*Key*
1	**B**	21	**A**
2	**C**	22	**C**
3	**B**	23	**C**
4	**D**	24	**B**
5	**D**	25	**A**
6	**B**	26	**A**
7	**D**	27	**A**
8	**D**	28	**C**
9	**A**	29	**A**
10	**C**	30	**B**
11	**C**	31	**A**
12	**A**	32	**C**
13	**C**	33	**A**
14	**D**	34	**D**
15	**C**	35	**C**
16	**C**	36	**C**
17	**B**	37	**A**
18	**B**	38	**C**
19	**D**	39	**B**
20	**C**	40	**C**

Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2025**

45 minutes

You must answer on the multiple choice answer sheet.

* 4 5 7 3 5 6 4 8 7 6 *

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m / s^2).

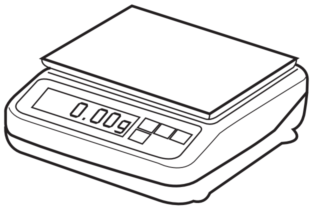
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

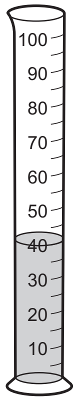
This document has **16** pages. Any blank pages are indicated.

1 A student is asked to find the volume of a small irregularly shaped piece of rock.

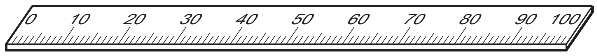
The student has the following apparatus available.



balance



measuring cylinder containing water



ruler

Which apparatus must the student use to find the volume of the small piece of rock?

A balance and ruler

B ruler only

C balance and measuring cylinder

D measuring cylinder only

2 A digital stop-clock measures time in minutes and seconds.

The stop-clock reads 00:50 when it is started (i.e. 00 minutes 50 seconds).

It reads 02:10 when it is stopped.

What is the shortest possible time that has passed between starting and stopping the stop-clock?

A 1 minute 06 seconds

B 1 minute 20 seconds

C 2 minutes 10 seconds

D 3 minutes 00 seconds

3 Four students are instructed to find the time it takes for a ball bearing to fall through a vertical height of 1.0 m.

Each student uses a different procedure.

Which student follows the correct procedure?

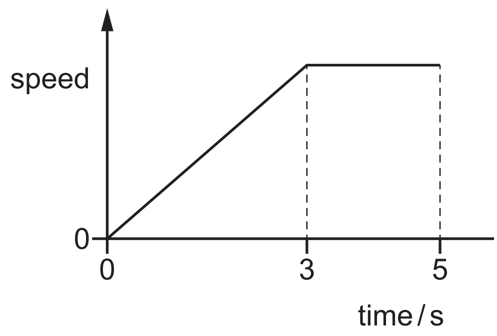
A The student measures the time taken for the ball to fall 1.0 m five times. He then adds the times taken together and divides by five.

B The student measures the time taken for the ball to fall 1.0 m five times. He then multiplies the times taken together and divides by five.

C The student measures the time taken for the ball to fall 2.0 m. He then divides the time taken by two.

D The student measures the time taken for the ball to fall 2.0 m five times. He then adds the times taken together and divides by ten.

4 The graph shows the motion of a car for a 5-second period.



Which row shows a time when the car is at rest and a time when it is moving at a constant speed?

	the car is at rest at	the car is moving at a constant speed at
A	0.0 s	2.0 s
B	0.0 s	4.0 s
C	4.0 s	0.0 s
D	4.0 s	2.0 s

5 Which property of an object is a gravitational force acting on its mass?

A density

B surface area

C volume

D weight

6 A stone has a weight of 5.7 N.

The gravitational field strength g is 9.8 N/kg.

What is the mass of the stone?

A 0.58 kg **B** 5.7 kg **C** 56 kg **D** 550 kg

7 Which equation is correct?

A mass = $\dfrac{\text{acceleration}}{\text{force}}$

B mass = density × volume

C mass = $\dfrac{\text{force}}{\text{impulse}}$

D mass = $\dfrac{\text{kinetic energy}}{\text{velocity}^2}$

8 What is meant by the moment of a force?

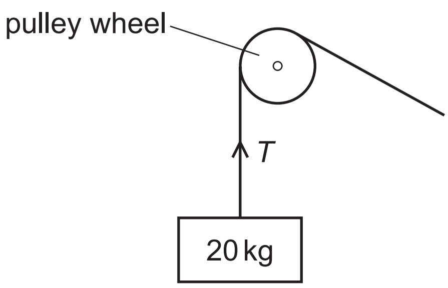
A the magnitude of the force

B the work done by the force

C the time for which the force acts

D the turning effect of the force

9 A mass of 20 kg is held stationary by a rope passing over a frictionless pulley.



What is the force T in the rope?

A 10 kg **B** 20 kg **C** 100 N **D** 200 N

10 Which equation relates the work done W on an object by a force F when it moves a distance d?

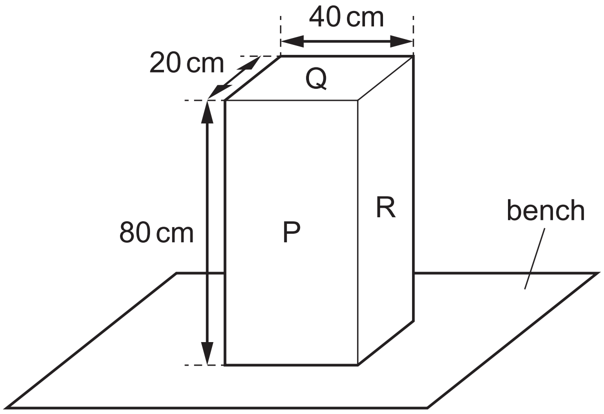
A $d = WF$ **B** $d = \dfrac{F}{W}$ **C** $F = \dfrac{d}{W}$ **D** $F = \dfrac{W}{d}$

11 A boy produces an average useful power output of 60 W as he rides his bicycle for 2.0 minutes.

How much useful energy does he transfer?

A 0.50 J **B** 30 J **C** 120 J **D** 7200 J

12 The diagram shows a solid block resting on a bench. The dimensions of the block are shown.



On which labelled surface should the block rest to produce the smallest pressure on the bench?

A P

B Q

C R

D P, Q and R produce the same pressure

13 A skier is standing still on a flat area of snow.

The total weight of the skier is 550 N. The total area of his skis in contact with the ground is $0.015\,m^2$.

What is the pressure exerted on the ground by the skier?

A $0.83\,N/m^2$ **B** $8.3\,N/m^2$ **C** $3700\,N/m^2$ **D** $37\,000\,N/m^2$

14 A force acts on an area to produce a pressure.

Which changes produce the same pressure?

A doubling the area and doubling the force

B doubling the area and halving the force

C doubling the area and making the force four times bigger

D halving the area and doubling the force

15 The diagram shows a cylinder containing oxygen under high pressure.

The temperature of the cylinder and the oxygen decreases.

Which statement describes what happens to the pressure in the cylinder and gives the reason why the pressure changes?

A The pressure decreases because the oxygen particles move faster.

B The pressure decreases because the oxygen particles move more slowly.

C The pressure increases because the oxygen particles move faster.

D The pressure increases because the oxygen particles move more slowly.

16 Which statement describes what happens during solidification?

A Particles begin to move around each other easily.

B Particles stop moving around each other.

C The temperature of the particles continues to decrease during solidification.

D The temperature of the particles continues to increase during solidification.

17 Solids expand the ......1...... and the average separation between molecules is the ......2...... .

Which words complete the sentence about the expansion of solids when they are heated, relative to liquids and gases?

	1	2
A	least	greatest
B	least	smallest
C	most	greatest
D	most	smallest

18 A girl blows some air into a party balloon and ties a knot to seal the air in the balloon.

The girl holds the sealed balloon above a hot convector heater.

She observes that the volume of the balloon increases significantly.

Which statement identifies the cause of this increase in volume?

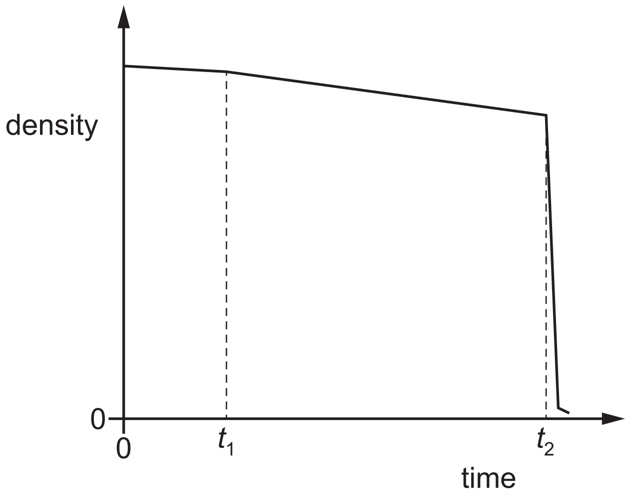
A The air in the balloon contracts.

B The air in the balloon expands.

C The material of the balloon contracts.

D The material of the balloon expands.

19 The graph shows what happens to the density of a substance as its temperature changes over time.



Which term describes the change of state taking place at time t_2?

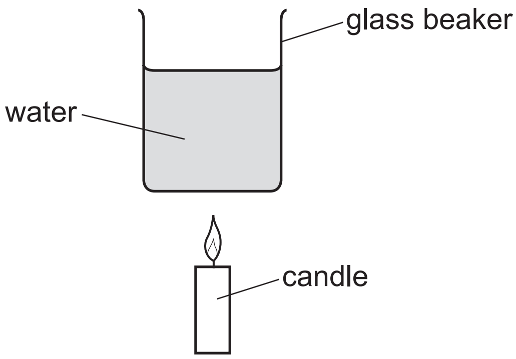
A melting

B freezing

C condensation

D boiling

20 A candle is used to heat some water in a glass beaker.



What is the **main** method of heat transfer through the glass and within the water?

	through the glass	within the water
A	conduction	conduction
B	conduction	convection
C	convection	conduction
D	convection	convection

21 Which equation enables the speed of a wave to be calculated?

A speed = amplitude × frequency

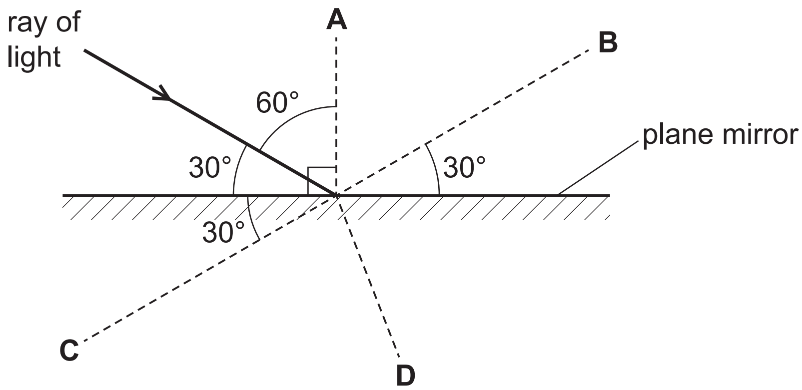
B speed = amplitude × wavelength

C speed = frequency × wavelength

D speed = $\dfrac{\text{frequency}}{\text{wavelength}}$

22 A ray of light is incident on the surface of a plane mirror, as shown.

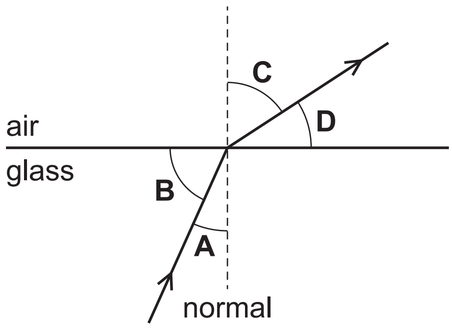
In which position is the reflected ray?



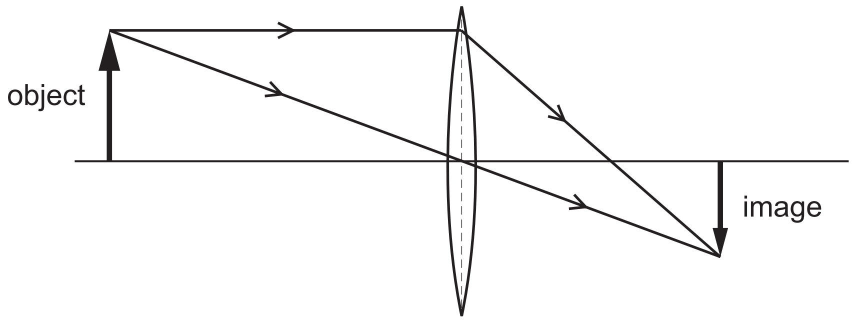
23 The diagram shows a ray of light passing from glass to air.

The dotted line labelled 'normal' is a line drawn perpendicular to the surface of the glass.

What is the angle of refraction?



24 The diagram shows how a converging lens forms an image of an object.



Which description of the image formed by the lens is correct?

A real, inverted and diminished

B real, upright and enlarged

C virtual, inverted and diminished

D virtual, upright and enlarged

25 Which row describes the image viewed in a plane mirror?

	type of image	image size
A	real	enlarged
B	real	same size
C	virtual	enlarged
D	virtual	same size

26 The table gives the range of frequencies of three types of electromagnetic radiation.

type of radiation	range of frequencies / Hz
radio	3×10^{4} to 3×10^{9}
visible light	4×10^{14} to 8×10^{14}
X-rays	3×10^{16} to 3×10^{19}

Which type of electromagnetic radiation has a frequency of 4×10^{15} Hz?

A gamma radiation

B infrared radiation

C microwave radiation

D ultraviolet radiation

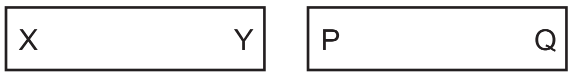
27 An athlete hears a starting pistol fire 1.5 seconds after she sees a puff of smoke from the pistol. The sound and the smoke are made at the same time. The starting pistol is 450 metres away from the athlete.

What is the speed of sound calculated from this observation?

A 150 m/s **B** 300 m/s **C** 330 m/s **D** 675 m/s

28 A plotting compass is placed close to the pole of a magnet with poles marked X and Y. The N pole of the plotting compass points to pole X, as shown.

Pole Y of the magnet is then placed close to another magnet with poles marked P and Q, as shown. The magnets repel each other.



magnets repel

Which row is correct?

	pole X	pole P
A	N	N
B	N	S
C	S	N
D	S	S

29 Why are electrical cables often made from copper?

A Copper contains free electrons and is a good conductor.

B Copper contains free electrons and is a good insulator.

C Copper contains free protons and is a good conductor.

D Copper contains free protons and is a good insulator.

30 There is a current I in a resistor. The potential difference (p.d.) across the resistor is V.

What does the product IV represent?

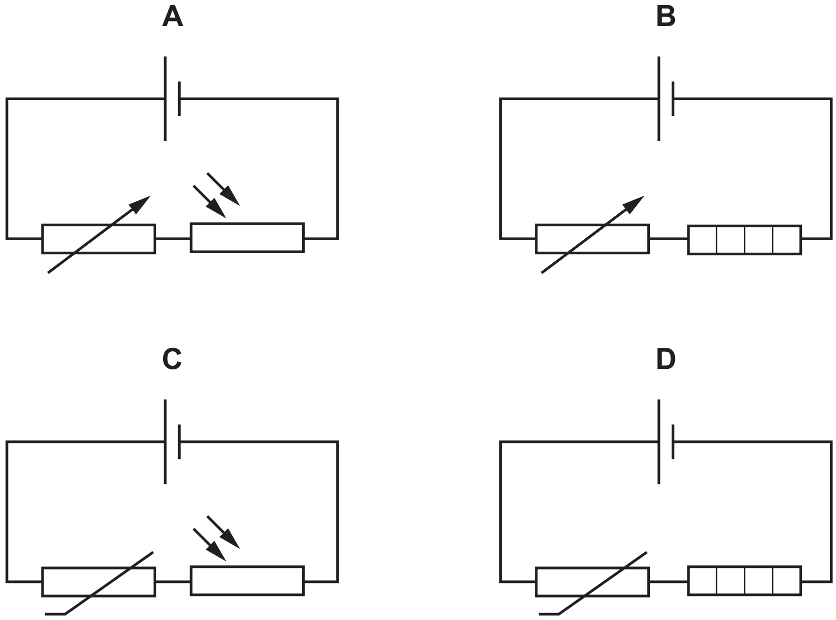
A the charge flowing through the resistor

B the energy transferred by the resistor

C the power dissipated by the resistor

D the resistance of the resistor

31 Which circuit contains a thermistor and a heater?



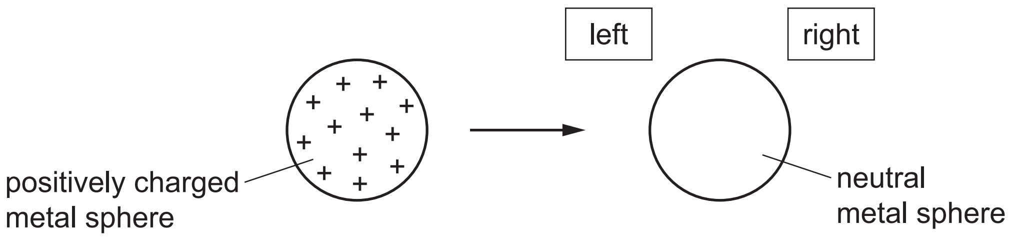
32 Four 2.0 V cells are connected together in series to make a battery, as shown.

What is the combined electromotive force (e.m.f.) of the battery?

A 0.50 V **B** 2.0 V **C** 6.0 V **D** 8.0 V

33 An isolated metal sphere is positively charged.

It is then brought near to another isolated metal sphere that is neutral.



What happens to the charges on the neutral sphere as the positively charged sphere is brought close to it?

A Some positive charges move to the left and some negative charges move to the right.

B Some positive charges move to the right and some negative charges move to the left.

C Some positive charges move to the right, but the negative charges do **not** move.

D The positive charges do **not** move, but some negative charges move to the left.

34 The potential difference (p.d.) across a car headlamp is 12 V. The current in the lamp is 2.5 A.

How much energy is transferred by the lamp in 1.0 hour?

A 1800 J **B** 1800 W **C** 108 000 J **D** 108 000 W

35 A copper wire rests in a uniform magnetic field.

Which change causes an induced electromotive force (e.m.f.) in the wire?

A connecting the wire to a power supply

B connecting the wire to earth

C heating the wire

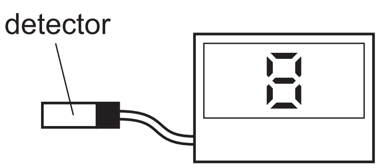
D moving the wire

36 An element has two isotopes.

Which row compares the nucleus of one atom of each isotope?

	the charge on each nucleus	the mass of each nucleus
A	different	different
B	different	the same
C	the same	different
D	the same	the same

37 In a laboratory, a radiation detector is connected to a counter which shows a reading for the count rate due to background radiation.



Which unit is count rate measured in?

A counts / s

B minutes / count

C coulombs / minute

D amperes / s

38 Which statement about alpha, beta and gamma emissions is correct?

A Alpha-particles are lighter than beta-particles.

B Beta-particles are **not** deflected by magnetic fields.

C Gamma rays are more penetrating than alpha-particles and beta-particles.

D Gamma rays are more ionising than alpha-particles and beta-particles.

39 What is an effect of ionising nuclear radiation on living things?

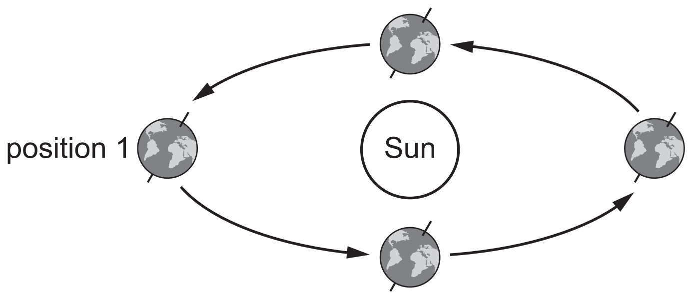
A cell mutation

B electric shock

C internal cooling of body cells

D broken bones

40 The diagram shows the Earth orbiting the Sun.



Which row identifies and explains the correct season in the northern hemisphere on the Earth in position 1?

	season	explanation
A	summer	the northern hemisphere tilts away from the Sun
B	summer	the northern hemisphere tilts towards the Sun
C	winter	the northern hemisphere tilts away from the Sun
D	winter	the northern hemisphere tilts towards the Sun

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PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2025**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	D	**1**
2	B	**1**
3	A	**1**
4	B	**1**
5	D	**1**
6	A	**1**
7	B	**1**
8	D	**1**
9	D	**1**
10	D	**1**
11	D	**1**
12	A	**1**
13	D	**1**
14	A	**1**
15	B	**1**
16	B	**1**
17	B	**1**
18	B	**1**
19	D	**1**
20	B	**1**
21	C	**1**
22	B	**1**
23	C	**1**
24	A	**1**
25	D	**1**
26	D	**1**
27	B	**1**
28	C	**1**

Question	Answer	Marks
29	A	**1**
30	C	**1**
31	D	**1**
32	D	**1**
33	D	**1**
34	C	**1**
35	D	**1**
36	C	**1**
37	A	**1**
38	C	**1**
39	A	**1**
40	B	**1**

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PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2024**

45 minutes

You must answer on the multiple choice answer sheet.

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You will need: Multiple choice answer sheet
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Soft pencil (type B or HB is recommended)

INSTRUCTIONS

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- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

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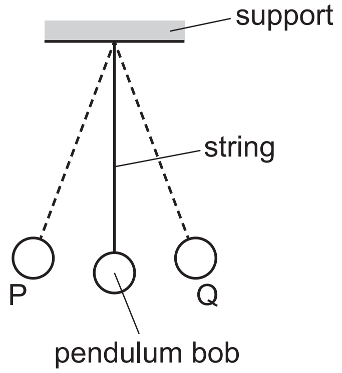
1 The diagram shows a pendulum that oscillates between P and Q.



Which method is used to find the average period of oscillation for the pendulum?

A Measure the time it takes to swing from P to Q.

B Measure the time it takes to swing from P to Q to P 10 times and divide that time by 10.

C Measure the time it takes to swing from P to Q to P 10 times and multiply that time by 10.

D Measure the time it takes to swing from Q to P.

2 A car moves through a measured distance *s* in a known time *t*.

What is the correct equation used to calculate the average speed *v* of the car?

A $v = \frac{s}{t}$ **B** $v = s \times t$ **C** $v = \frac{t}{s}$ **D** $v = s \times t^2$

3 On Mars, the acceleration of free fall is $3.7\,\text{m/s}^2$.

What is the weight of a 2.0 kg mass on Mars?

A 0.54 N **B** 1.9 N **C** 7.4 N **D** 20 N

4 A rocket is travelling vertically upwards. Three vertical forces act on it.

The thrust acts upwards and is equal to 100 000 N.

The weight acts downwards and is equal to 80 000 N.

What is the air resistance force acting on the rocket when it is travelling upwards at a constant speed?

A 20 000 N downwards

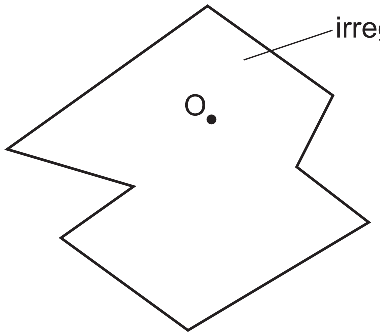
B 20 000 N upwards

C 180 000 N downwards

D 180 000 N upwards

5 An irregular plane object is suspended freely from point O.

The object is displaced and then released. It swings a few times, and comes to rest as shown.



Where is the centre of gravity of the object?

A horizontally to the left of O

B horizontally to the right of O

C vertically above O

D vertically below O

6 Three situations are listed.

1 someone blowing air into a party balloon

2 a crane lifting a block of concrete

3 a pile of books at rest on a shelf

In which situations is work being done?

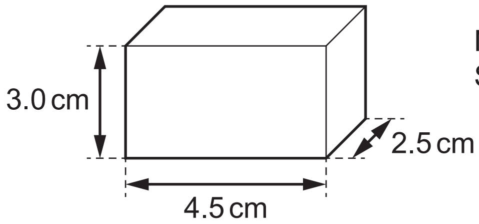
A 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 2 and 3 only

7 Which row describes an advantage and a disadvantage of wind turbines?

	advantage	disadvantage
A	no fuel needed	harmful gases released
B	variable supply	fuel needed
C	no harmful gases released	variable supply
D	constant supply	noisy

8 The diagram shows a rectangular metal block with dimensions 2.5 cm × 3.0 cm × 4.5 cm. The block is resting on the ground.

The block exerts a pressure of 0.23 N / cm^2 on the ground.



What is the weight of the metal block?

A 0.020 N **B** 1.7 N **C** 2.6 N **D** 3.1 N

9 Which row correctly describes the movement of particles in solids and liquids?

	solids	liquids
A	no movement	move around each other
B	no movement	vibration only
C	vibration only	move around each other
D	vibration only	vibration only

10 What is the boiling temperature of water at standard atmospheric pressure?

A –173 K **B** 100 K **C** 273 K **D** 373 K

11 Which row describes how the volume and mass of liquid mercury change as it is heated?

	volume	mass
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

12 In which situation does the store of internal energy of the water increase?

A water at 0 °C freezing to form ice

B water being heated from 25 °C to 40 °C

C steam condensing on a window

D water in a puddle on a breezy day

13 Which statement describes what happens to particles during the evaporation of a liquid?

A Particles become fixed in position.

B Particles escape from the surface.

C Particles move much closer together.

D Particles start to move around at random.

14 Which of the following is **not** an example of convection?

A a spoon handle becoming hot when it is placed in a hot drink

B a thermal up-draught in the atmosphere that glider pilots use

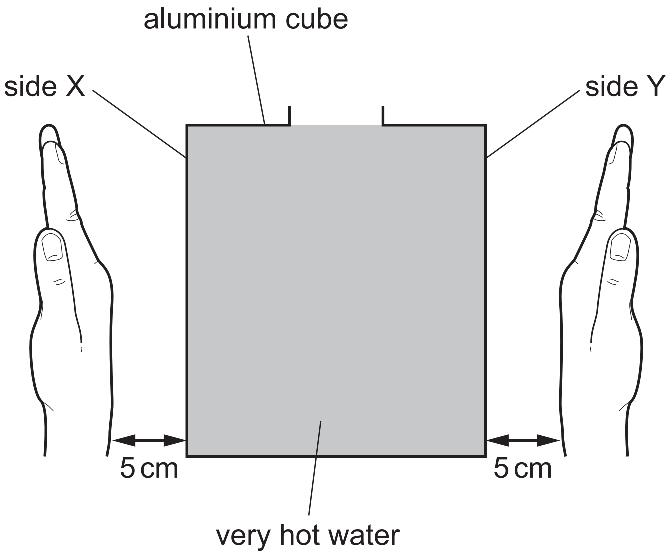
C flames and smoke rising from a fire

D warming up water in a kettle on a gas ring

15 A hollow aluminium cube is filled with very hot water.

Side X of the cube is opposite side Y of the cube. One of these two sides is black and one is white.

A student holds the back of one hand 5 cm from side X, and then immediately holds the back of the other hand 5 cm from side Y.

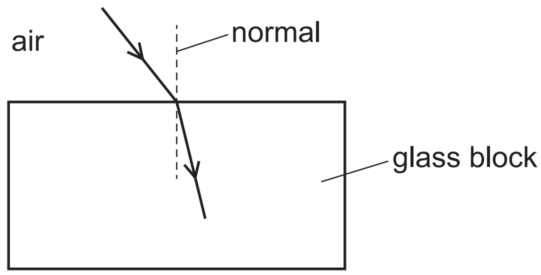


The hand held near side Y feels warmer than the hand held near side X.

Which row identifies the black side and correctly compares the rate of emission of thermal radiation from each side?

	black side	rate of emission of thermal radiation
A	X	greater for X
B	X	the same for X and Y
C	Y	greater for Y
D	Y	the same for X and Y

16 The diagram shows a ray of light travelling from air into a glass block and changing direction so that it is closer to the normal.



What is the name of this process and why does it happen?

	process	why it happens
A	diffraction	speed of the light decreases
B	diffraction	speed of the light increases
C	refraction	speed of the light increases
D	refraction	speed of the light decreases

17 Which row gives an example of a longitudinal wave and describes the direction of the vibrations?

	example of a longitudinal wave	direction of vibrations
A	light wave	at right angles to the direction of propagation
B	light wave	parallel to the direction of propagation
C	sound wave	at right angles to the direction of propagation
D	sound wave	parallel to the direction of propagation

18 Which statement about total internal reflection is correct?

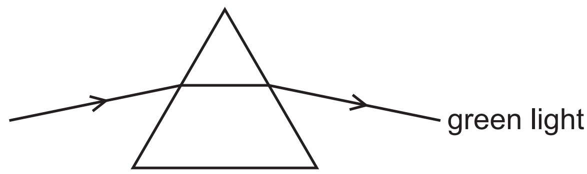
A Total internal reflection occurs when light is incident on a glass to air boundary at an angle of incidence greater than the critical angle.

B Total internal reflection occurs when light is incident on a glass to air boundary at an angle of incidence less than the critical angle.

C Total internal reflection occurs when light is incident on an air to glass boundary at an angle of incidence greater than the critical angle.

D Total internal reflection occurs when light is incident on an air to glass boundary at an angle of incidence less than the critical angle.

19 A ray of green light passes through a glass prism, as shown.



Which colours of light refract as shown in the table?

	refracts more than green	refracts less than green
A	red	blue
B	red	yellow
C	violet	blue
D	violet	yellow

20 Which electromagnetic waves are used for sterilising medical equipment?

A gamma rays

B microwaves

C radio waves

D visible light

21 Which process causes a sound wave to produce an echo?

A diffraction

B dispersion

C reflection

D refraction

22 A student stands 75 m from a wall and makes a short, loud sound by tapping two pieces of wood together.

What is the approximate time between making the sound and the student hearing the echo produced?

A 0.2 s **B** 0.5 s **C** 2 s **D** 5 s

23 Which statement is **always** true?

A A magnetic material attracts a non-magnetic material.

B A magnetic material repels another magnetic material.

C A magnetic material attracts another magnetic material if one of them is magnetised.

D A non-magnetic material repels another non-magnetic material.

24 A polythene rod is charged by friction. The polythene rod becomes negatively charged.

Which statement is correct?

A The rod gains electrons.

B The rod loses electrons.

C The rod gains protons.

D The rod loses protons.

25 What is the unit of electromotive force (e.m.f.)?

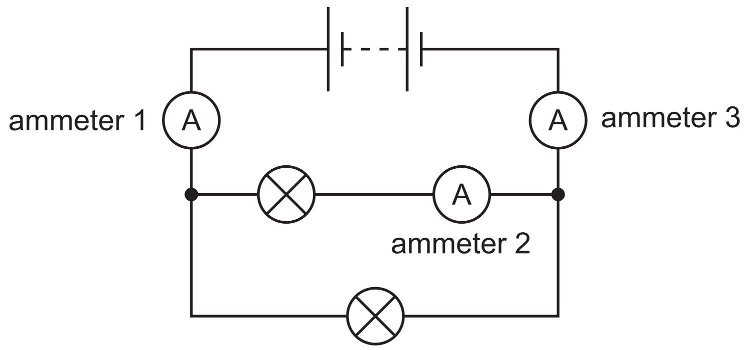
A ampere

B newton

C ohm

D volt

26 The diagram shows a circuit containing two identical lamps and three ammeters.



The current in ammeter 1 is 0.30 A.

Which row gives possible values for the currents in ammeters 2 and 3?

	current in ammeter 2 / A	current in ammeter 3 / A
A	0.15	0.00
B	0.15	0.30
C	0.30	0.00
D	0.30	0.30

27 Into which wire is a switch connected in order to switch a mains appliance on and off safely?

A live

B negative

C neutral

D positive

28 Which device uses the magnetic effect of an electric current?

A heater

B lamp

C relay

D thermistor

29 A transformer has 200 turns on its primary coil and is connected to a 240 V a.c. supply.

The output voltage of the transformer is 60 V a.c.

How many turns are on the secondary coil of the transformer?

A 20 **B** 50 **C** 72 **D** 800

30 Which statement about an atom is correct?

A An atom has positive charge because there are only positive charges in the nucleus.

B An atom has negative charge because there are only negative charges in the nucleus.

C An atom has neutral charge because the positive charge in the nucleus equals the total negative charge of the electrons.

D An atom has neutral charge because the negative charge in the nucleus equals the total positive charge of the electrons.

31 The table gives information about four nuclei.

nucleus	proton number	nucleon number
1	26	59
2	27	59
3	27	60
4	28	60

Which nuclei are isotopes of the same element?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

32 Which apparatus is used to measure background radiation?

A a counter, a detector and a source

B a counter and a detector only

C a counter and a source only

D a detector and a source only

33 Which row correctly identifies the nature of each radioactive emission?

	alpha (α)	beta (β)	gamma (γ)
A	helium nucleus	electromagnetic wave	electron
B	electron	helium nucleus	electromagnetic wave
C	helium nucleus	electron	electromagnetic wave
D	electromagnetic wave	electron	helium nucleus

34 A nucleus spontaneously emits radiation and becomes a nucleus of a different element.

What could the emitted radiation be?

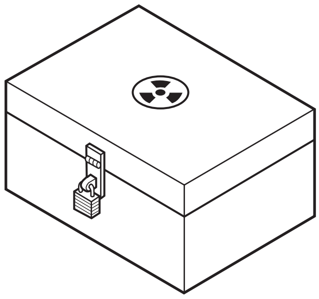
A alpha or beta

B alpha or gamma

C beta or neutron

D gamma or neutron

35 The diagram shows a lead-lined box used for storing a radioactive source.



Why is the inside of the box lined with lead?

A It helps the source to stay radioactive for longer.

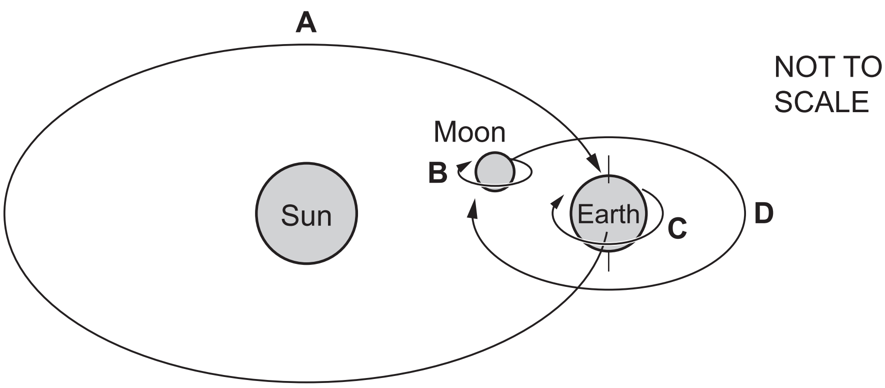
B It makes the box heavier.

C It makes the radioactive source more stable.

D It reduces the amount of radiation that can escape from the box.

36 A student draws a simplified diagram showing the Sun and the different movements of the Moon and the Earth.

Which arrow represents a motion taking 365 days to complete?



37 The table gives information about four of the planets in our Solar System.

Two of these planets are gaseous and two are rocky.

planet	average distance from the Sun / km
1	1.4×10^9
2	2.3×10^8
3	5.8×10^7
4	7.8×10^8

Which two planets are rocky?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

38 Which row shows the most common elements in the Sun and the regions of the electromagnetic spectrum in which most of the Sun's energy is radiated?

	elements	regions
A	carbon and oxygen	gamma and radio waves
B	carbon and oxygen	infrared and visible
C	hydrogen and helium	gamma and radio waves
D	hydrogen and helium	infrared and visible

39 What is the meaning of a light-year?

A the distance light travels in space in one year

B the number of years it takes light to travel from the Sun to the Earth

C the speed that light travels in space

D the time it takes light to travel from the Sun to the nearest star

40 What is the approximate diameter of the Milky Way?

A 100 light-years

B 1000 light-years

C 10 000 light-years

D 100 000 light-years

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Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2024**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	B	**1**
2	A	**1**
3	C	**1**
4	A	**1**
5	D	**1**
6	B	**1**
7	C	**1**
8	C	**1**
9	C	**1**
10	D	**1**
11	D	**1**
12	B	**1**
13	B	**1**
14	A	**1**
15	C	**1**
16	D	**1**
17	D	**1**
18	A	**1**
19	D	**1**
20	A	**1**
21	C	**1**
22	B	**1**
23	C	**1**
24	A	**1**
25	D	**1**
26	B	**1**
27	A	**1**
28	C	**1**

Question	Answer	Marks
29	B	**1**
30	C	**1**
31	C	**1**
32	B	**1**
33	C	**1**
34	A	**1**
35	D	**1**
36	A	**1**
37	C	**1**
38	D	**1**
39	A	**1**
40	D	**1**

Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2023**

45 minutes

You must answer on the multiple choice answer sheet.

1287925071

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

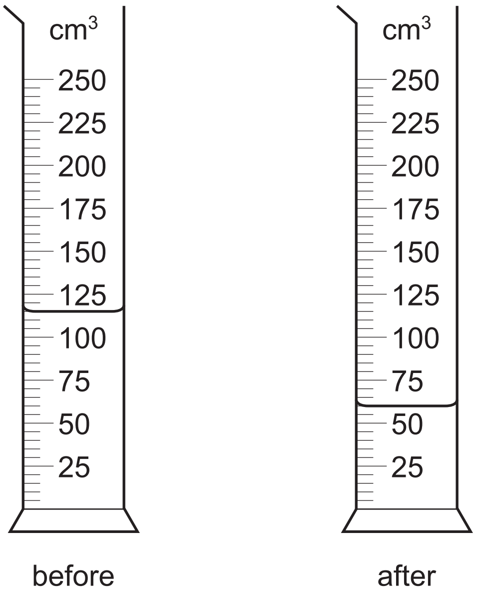
INSTRUCTIONS
- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION
- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

1 A measuring cylinder contains water.

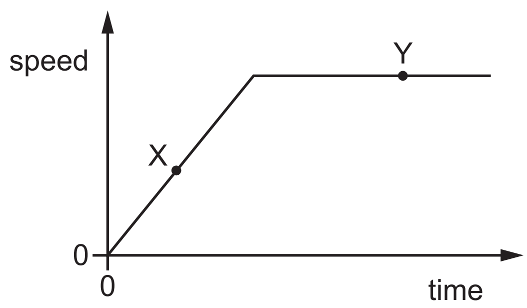
The diagrams show the measuring cylinder before and after some of the water is poured into a beaker.



How much water has been poured into the beaker?

A 51 cm^3 **B** 52 cm^3 **C** 55 cm^3 **D** 63 cm^3

2 The diagram shows the speed–time graph for a car.



Which row describes the motion of the car at point X and at point Y?

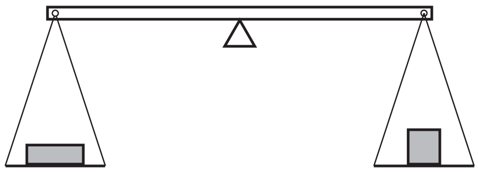
	point X	point Y
A	at rest	moving with constant speed
B	moving with constant speed	at rest
C	moving with changing speed	at rest
D	moving with changing speed	moving with constant speed

3 A ball is dropped in a vacuum from a height of 4.0 m above the surface of Mars. The acceleration of the ball at a height of 2.0 m is 3.8 m/s^2.

What is the acceleration of the ball at a height of 1.0 m above the surface of Mars?

A 1.9 m/s^2 **B** 3.8 m/s^2 **C** 5.7 m/s^2 **D** 7.6 m/s^2

4 Two objects are placed on a balance, one on each side, as shown.



Which properties of the objects can be compared using the balance?

A weight, mass and volume

B weight and mass only

C volume and density

D density only

5 A rectangular swimming pool is 50 m long and 25 m wide.

It contains water at a depth of 2 m.

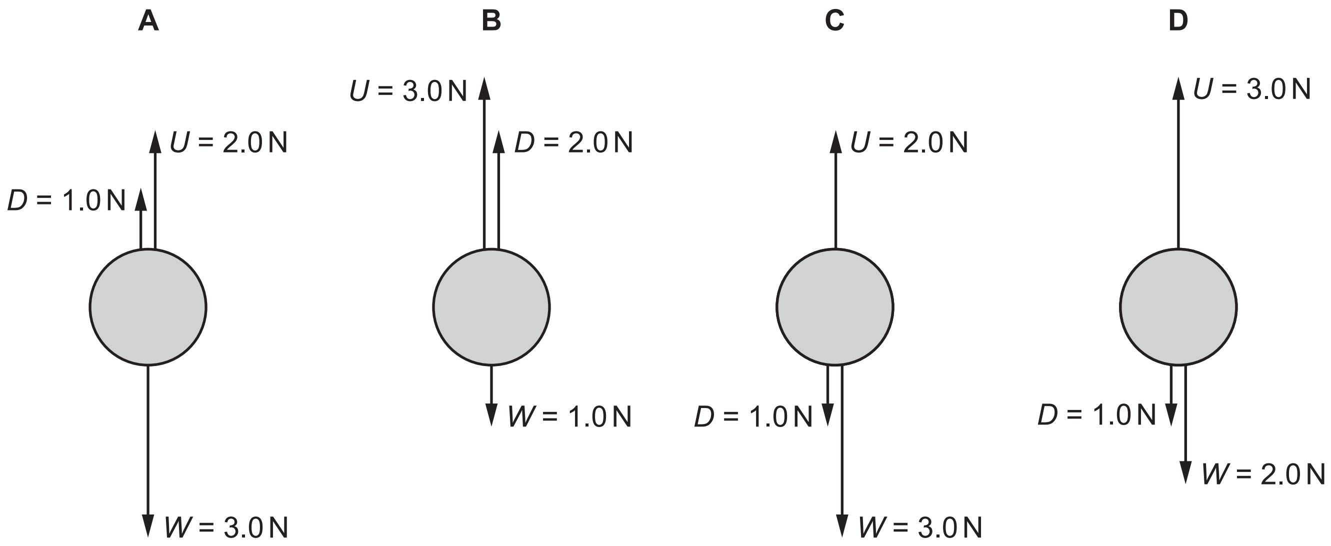
The density of the water is 1000 kg/m^3.

What is the mass of the water in the pool?

A 2.5 kg **B** 2500 kg **C** 77 000 kg **D** 2 500 000 kg

6 An object is rising vertically at constant speed through water. There are three vertical forces acting on it: the weight W, the drag force D, and the upward force U.

Which diagram shows the magnitude and direction of the vertical forces acting on the object?



7 Which force produces heating during contact with a moving object?

A weight

B friction

C electrostatic force

D magnetic force

8 A metre rule is balanced at its midpoint.

It remains balanced when a 3.0 N load is hung from the 40 cm mark and a second load is hung from the 80 cm mark.

What is the weight of the second load?

A 1.0 N **B** 1.5 N **C** 3.0 N **D** 9.0 N

9 A student measures the length of a spring. She then attaches different weights to the spring. She measures the length of the spring for each weight.

The table shows her results.

weight / N	length / mm
0	520
1.0	524
2.0	528
3.0	533
4.0	537
5.0	540

What is the extension of the spring with a weight of 3.0 N attached to it?

A 4 mm **B** 5 mm **C** 12 mm **D** 13 mm

10 Which power station produces the greatest atmospheric pollution for each unit of energy generated?

A a gas fired power station

B a hydroelectric power station

C a nuclear power station

D a wind farm

11 A 500 N weight is raised through a height of 5 cm.

How much work is done by the force?

A 25 J **B** 100 J **C** 2500 J **D** 10 000 J

12 What is the unit of power?

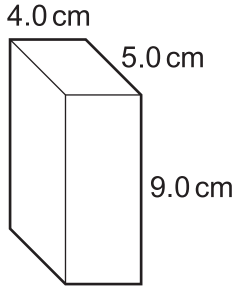
A joule

B newton

C volt

D watt

13 The diagram shows a rectangular block of weight 16 N. It is resting on a flat surface.



What is the pressure at the base of the block due to its weight?

A 0.089 N / cm^2

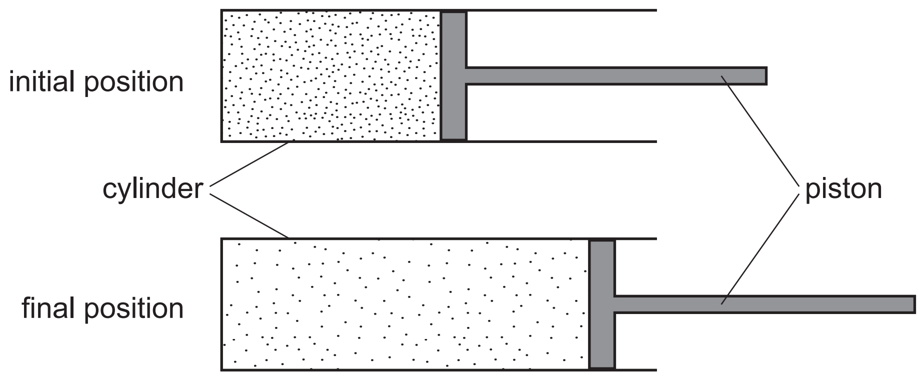
B 0.36 N / cm^2

C 0.80 N / cm^2

D 320 N / cm^2

14 A piston traps a mass of gas inside a cylinder. Initially, the piston is halfway along the length of the cylinder.

The piston is now moved towards the open end of the cylinder. The temperature of the gas remains constant.



How are the density and the pressure of the gas affected by moving the piston?

	density	pressure
A	decreases	decreases
B	decreases	unchanged
C	increases	decreases
D	increases	unchanged

15 Which statement describes what happens to air particles when the air is heated?

A The particles move more slowly.

B The particles move more quickly.

C The particles move closer together.

D The particles vibrate faster.

16 What happens when the temperature of a liquid increases?

A The mass of the liquid increases, making the liquid less dense.

B The mass of the liquid increases, making the liquid more dense.

C The volume of the liquid increases, making the liquid less dense.

D The volume of the liquid increases, making the liquid more dense.

17 In which states of matter can thermal energy be transferred by convection?

A in gases, liquids and solids

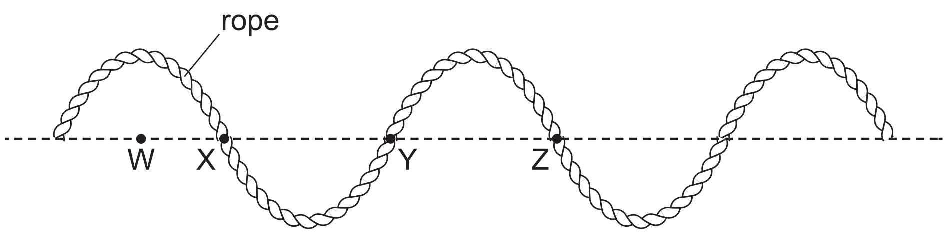
B in gases and liquids only

C in gases and solids only

D in liquids only

18 A transverse wave moves along a rope.

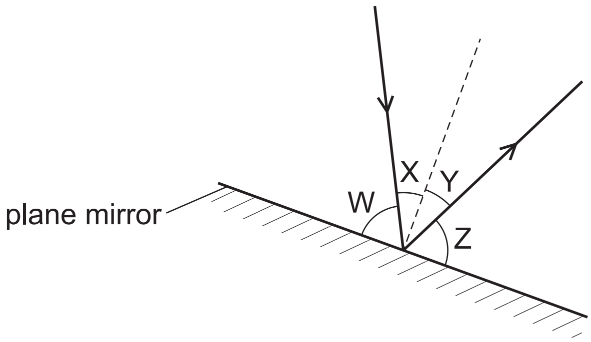
The diagram shows the position of the rope at one particular time.



Which two labelled points are one wavelength apart?

A W and X **B** W and Z **C** X and Z **D** Y and Z

19 The diagram shows a ray of light being reflected from a plane mirror.



Which row identifies the angles of incidence and reflection?

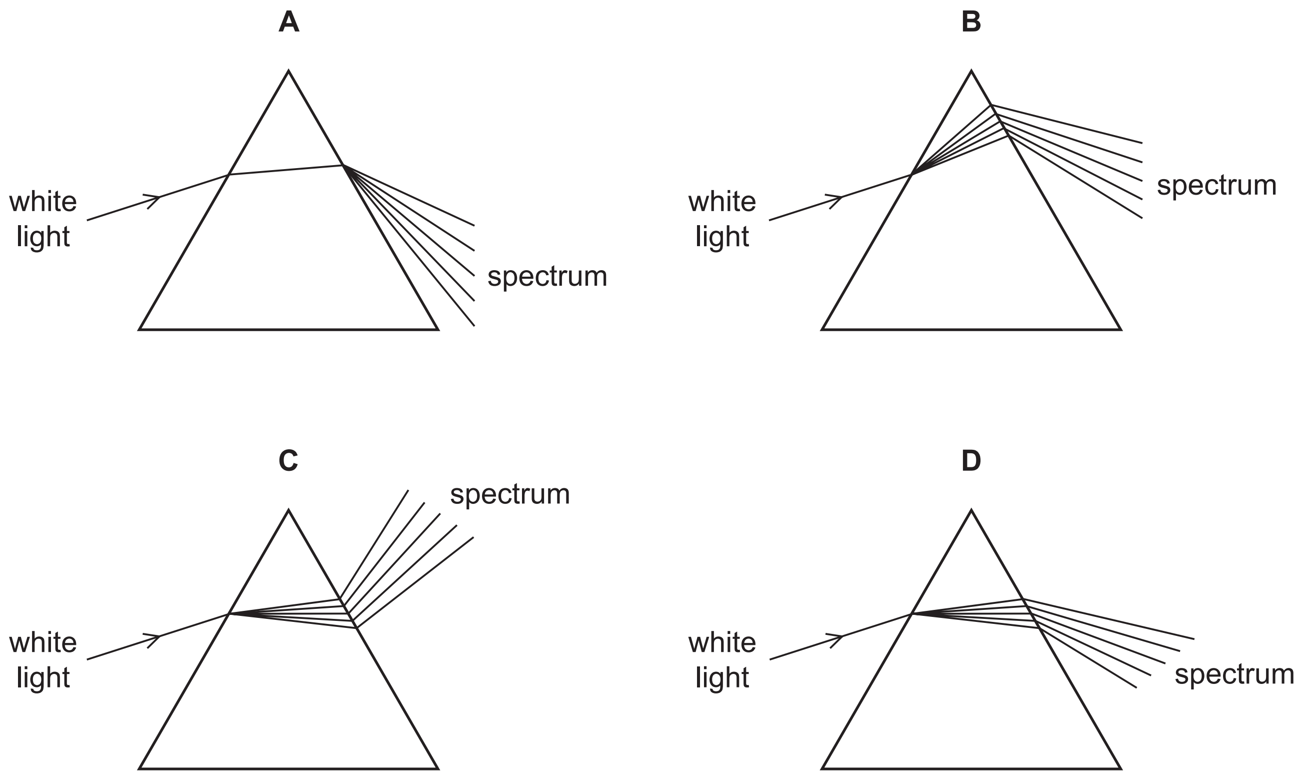
	angle of incidence	angle of reflection
A	W	Y
B	W	Z
C	X	Y
D	X	Z

20 Blue light has a typical wavelength of 5.0×10^{-7} m and frequency of 0.60×10^{15} Hz.

Which row gives a typical wavelength and frequency for red light?

	wavelength	frequency
A	2.4×10^{-7} m	0.40×10^{15} Hz
B	2.4×10^{-7} m	1.3×10^{15} Hz
C	6.9×10^{-7} m	0.40×10^{15} Hz
D	6.9×10^{-7} m	1.3×10^{15} Hz

21 Which diagram shows what happens when a ray of white light passes through a prism?

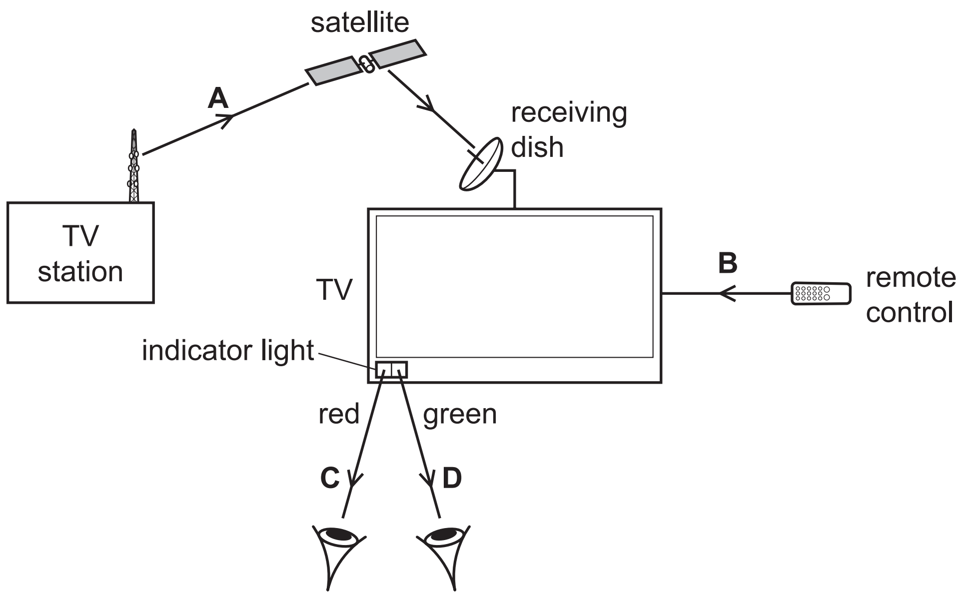


22 A television (TV) station transmits a signal to a television receiving dish.

The television has an on / off indicator light.

The television is switched on by a remote control which changes the indicator light from red to green.

Which electromagnetic wave used in these actions has the longest wavelength?

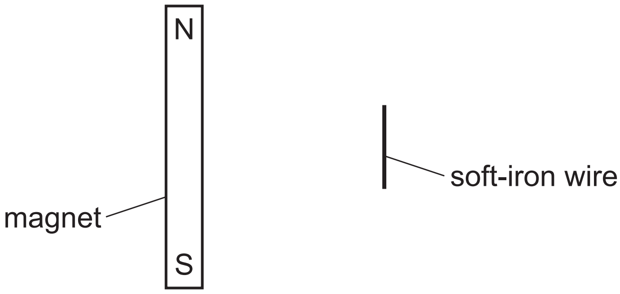


23 A sound wave has a wavelength of 0.024 m.

What is the frequency of this sound wave and is it audible to humans?

	frequency / Hz	audible to humans
A	140	yes
B	140	no
C	14 000	yes
D	14 000	no

24 The diagram shows a bar magnet at rest on a smooth horizontal surface. A length of soft-iron wire is held parallel to the magnet.



The wire is released.

What happens?

A The wire moves away from the magnet.

B The wire moves towards the magnet.

C The wire's centre stays in its present position and the wire rotates through 90° in a clockwise direction.

D The wire's centre stays in its present position and the wire rotates through 90° in an anticlockwise direction.

25 A man walks across a carpet. He becomes negatively charged by friction with the carpet.

What happens as he touches a metal object connected to the Earth?

A The man gains electrons.

B The man loses electrons.

C The man gains protons.

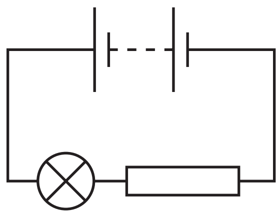
D The man loses protons.

26 A laboratory has a standard wire of known resistance. It also has other wires, made from the same material as the standard wire, but of different lengths and diameters.

Which wire would definitely have a resistance of less than the standard wire?

	wire length	wire diameter
A	longer	larger
B	longer	smaller
C	shorter	larger
D	shorter	smaller

27 The diagram shows a circuit.



Which energy transfers occur?

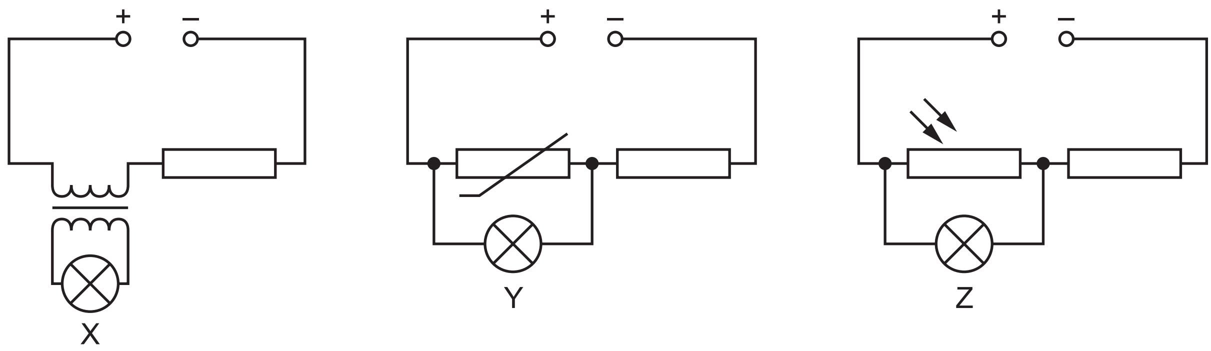
A battery → lamp → surrounding air

B battery → surrounding air → battery

C lamp → surrounding air → battery

D surrounding air → lamp → battery

28 X, Y and Z are lamps.



In which lamps is there a current?

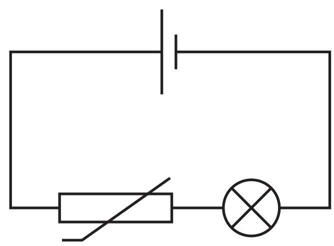
	X	Y	Z
A	✓	✗	✓
B	✓	✗	✗
C	✗	✓	✓
D	✗	✓	✗

key

✓ = current in lamp

✗ = no current in lamp

29 The diagram shows a circuit.



Which change causes the bulb in the circuit to become brighter?

A a decrease in light intensity

B a decrease in temperature

C an increase in light intensity

D an increase in temperature

30 The current in a kettle is 10 A and the kettle is protected by a 13 A fuse.

The owner of the kettle replaces the 13 A fuse with a 3 A fuse.

What happens when the kettle is switched on?

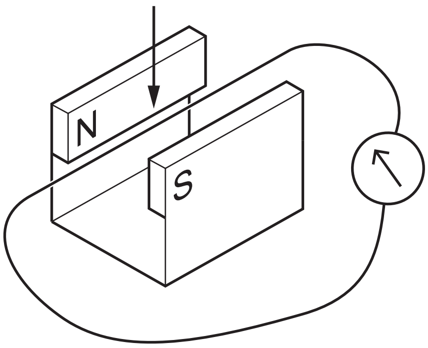
A The fuse melts and the kettle might be damaged.

B The fuse melts and the kettle is undamaged.

C The fuse does not melt and the kettle works correctly.

D The fuse does not melt but the kettle fails to work.

31 A wire is moved down in a direction perpendicular to the magnetic field.



Three changes are suggested.

1 The speed of the movement of the wire is increased.

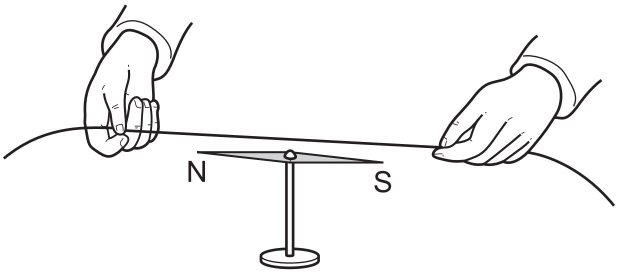
2 The magnetic field strength is decreased.

3 The direction of the magnetic field is reversed.

Which changes increase the electromotive force (e.m.f.) induced in the wire?

A 1 and 3 **B** 1 only **C** 2 and 3 **D** 3 only

32 In an experiment, a wire is held above a compass needle as shown.



An electric current is switched on in the wire and the compass needle is deflected.

Which row explains why this happens and then describes what happens when the current is reversed?

	why this happens	what happens when the current is reversed
A	there is a magnetic field inside the wire	the compass needle deflects in the opposite direction
B	there is a magnetic field inside the wire	the compass needle remains deflected in the same direction
C	there is a magnetic field around the wire	the compass needle deflects in the opposite direction
D	there is a magnetic field around the wire	the compass needle remains deflected in the same direction

33 Over time, the strength of the magnets in an electric motor decreases.

Which row describes **two** ways to keep the motor running at its original speed?

	current in the coil	number of turns on the coil
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

34 A rechargeable battery contains lithium.

The lithium exists as positive lithium ions.

How does an ion of lithium differ from an atom of lithium?

A The ion has fewer electrons orbiting the nucleus.

B The ion has more electrons orbiting the nucleus.

C The nucleus of the ion has less charge.

D The nucleus of the ion is more positively charged.

35 An iron nuclide is represented by the symbol shown.

$$^{56}_{26}\mathrm{Fe}$$

Which statements about a nucleus of this iron nuclide are correct?

1 The nucleus contains 56 neutrons.

2 The nucleon number is 30.

3 The proton number is 26.

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 only

36 What is an artificial source of background radiation?

A X-ray machines in hospitals

B radon gas from rocks

C cosmic rays from the Sun

D plants and other living things

37 A sample of a radioactive isotope has an initial rate of emission of 128 counts per minute and a half-life of 4 days.

How long will it take for the rate of emission to fall to 32 counts per minute?

A 2 days **B** 4 days **C** 8 days **D** 12 days

38 Approximately how long does it take for the Moon to make one complete orbit of the Earth?

A 24 hours

B 1 month

C 1 season

D 1 year

39 The Sun has a mass of 2.0×10^{30} kg.

Which element accounts for most of this mass?

A carbon

B hydrogen

C oxygen

D uranium

40 The nearest star to the Sun is Proxima Centauri at a distance of 4.2 light years.

Which statements are correct?

1 Telescope images of Proxima Centauri show it as it was 4.2 years ago.

2 If a spacecraft near Proxima Centauri sent a radio message to the Earth, it would take 4.2 years to arrive.

3 Proxima Centauri is outside the Milky Way galaxy.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

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PHYSICS | **0625/12**

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Question	Answer	Marks
1	C	**1**
2	D	**1**
3	B	**1**
4	B	**1**
5	D	**1**
6	D	**1**
7	B	**1**
8	A	**1**
9	D	**1**
10	A	**1**
11	A	**1**
12	D	**1**
13	C	**1**
14	A	**1**
15	B	**1**
16	C	**1**
17	B	**1**
18	C	**1**
19	C	**1**
20	C	**1**
21	D	**1**
22	A	**1**
23	C	**1**
24	B	**1**
25	B	**1**
26	C	**1**
27	A	**1**
28	C	**1**

Question	Answer	Marks
29	D	**1**
30	B	**1**
31	B	**1**
32	C	**1**
33	D	**1**
34	A	**1**
35	D	**1**
36	A	**1**
37	C	**1**
38	B	**1**
39	B	**1**
40	B	**1**

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PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2022**

45 minutes

You must answer on the multiple choice answer sheet.

* 8 1 7 7 7 5 8 0 4 5 *

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

1 A student investigates a pendulum.

He measures the time for the pendulum to complete 20 oscillations.

He repeats the experiment three more times.

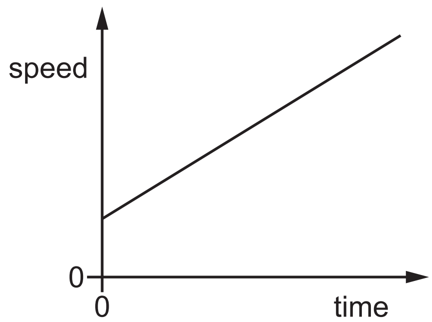
The readings are shown.

experiment	time for 20 oscillations / s
1	17.6
2	19.8
3	17.6
4	18.6

What is the average period of the pendulum?

A 0.88 s **B** 0.92 s **C** 17.6 s **D** 18.4 s

2 The diagram shows a speed–time graph for a moving object.



Which statement describes the motion of the object?

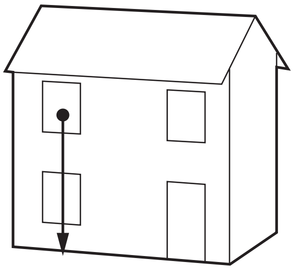
A The speed of the object is increasing with constant acceleration.

B The speed of the object is increasing with an acceleration that is not constant.

C The speed of the object is decreasing with constant deceleration.

D The speed of the object is decreasing with a deceleration that is not constant.

3 A tennis ball falls from the upstairs window of a house.



What can be said about the acceleration of the ball if air resistance is ignored?

A It depends on the density of the ball.

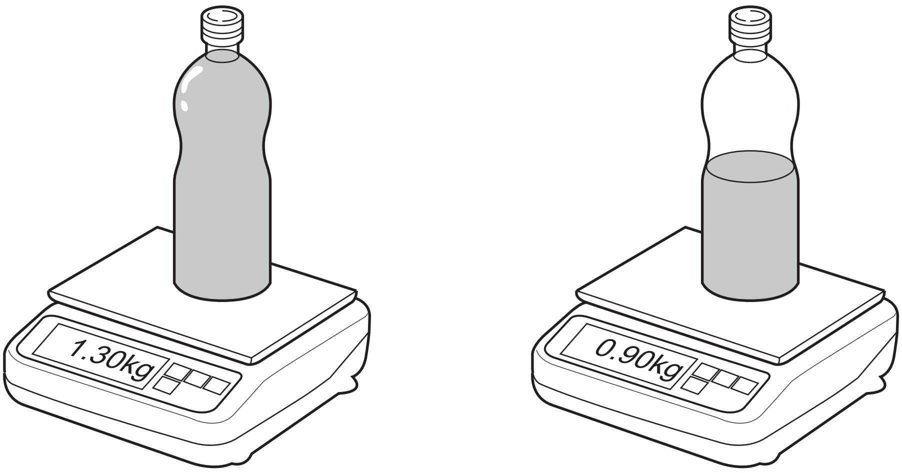
B It depends on the mass of the ball.

C It increases as the ball falls.

D It stays the same as the ball falls.

4 The mass of a full bottle of cooking oil is 1.30 kg.

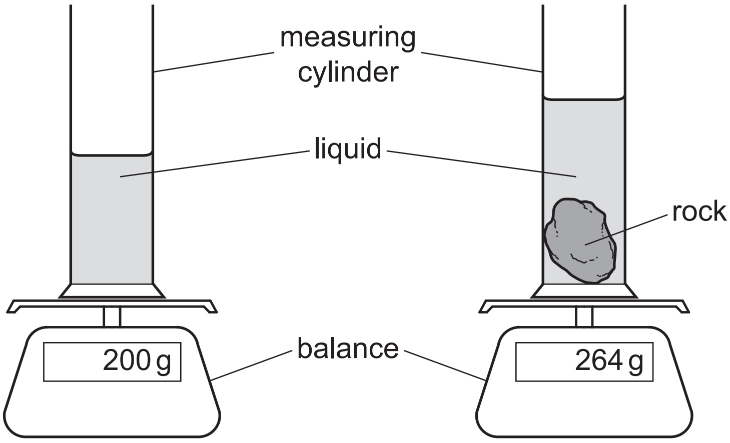
When exactly half of the oil has been used, the mass of the bottle plus the remaining oil is 0.90 kg.



What is the mass of the empty bottle?

A 0.40 kg **B** 0.50 kg **C** 0.65 kg **D** 0.80 kg

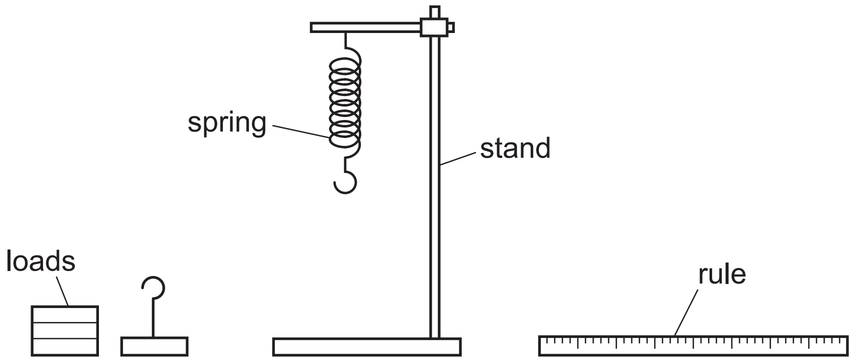
5 A student carries out an experiment to find the density of a rock.



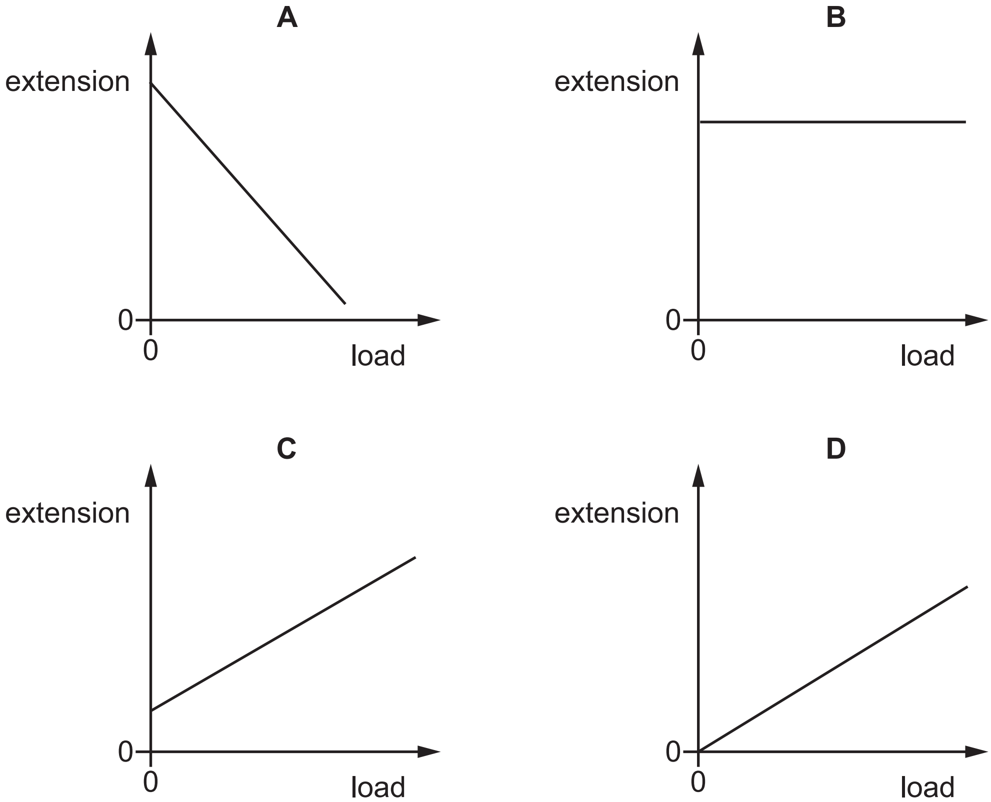
Which two quantities does the student need to make to determine the density of the rock?

	quantity 1	quantity 2
A	increase in mass	increase in volume of liquid
B	final mass	increase in depth of liquid
C	increase in mass	increase in depth of liquid
D	final mass	increase in volume of liquid

6 A spring is suspended from a stand. Loads are added and the extensions of the spring are measured.



Which graph shows the result of plotting extension against load?



7 Which are examples of friction?

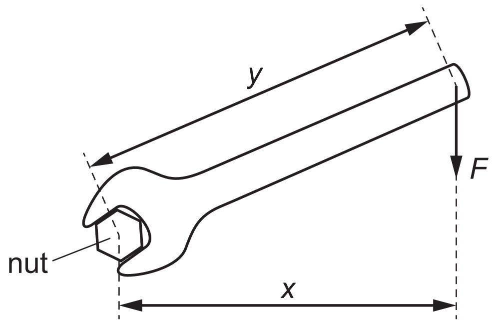
1 tension in a spring

2 air resistance

3 weight

A 1 only **B** 1 and 3 **C** 2 only **D** 2 and 3

8 The diagram shows a nut being turned with a spanner by applying a force F to the spanner.



Which equation gives the moment of the force F about the centre of the nut?

A moment = Fx

B moment = Fy

C moment = $\frac{F}{x}$

D moment = $\frac{F}{y}$

9 A child pushes a toy car along a horizontal surface and then releases it.

As the car slows down, what is the main energy transfer?

A from chemical to thermal

B from chemical to kinetic

C from kinetic to gravitational (potential)

D from kinetic to thermal

10 Energy resources are used to produce electricity.

Which resource is non-renewable?

A hydroelectric

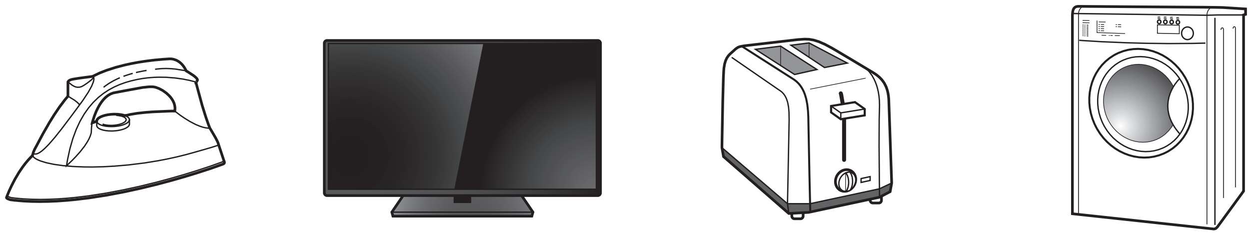
B nuclear fission

C waves

D wind

11 The diagrams show four appliances and their power ratings.

Which appliance transfers the most energy per second?



A	**B**	**C**	**D**
iron 1.0 kW	TV 150 W	toaster 800 W	washing machine 3.0 kW

12 An object is at rest on a horizontal surface.

Which equation is used to calculate the pressure that the object exerts?

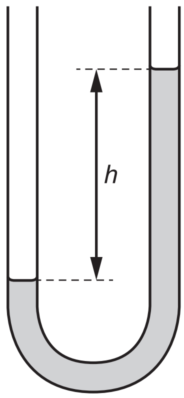
A $\dfrac{\text{mass of the object}}{\text{area of contact}}$

B $\dfrac{\text{weight of the object}}{\text{area of contact}}$

C mass of the object × area of contact

D weight of the object × area of contact

13 Which device is shown?



A barometer

B galvanometer

C manometer

D newton meter

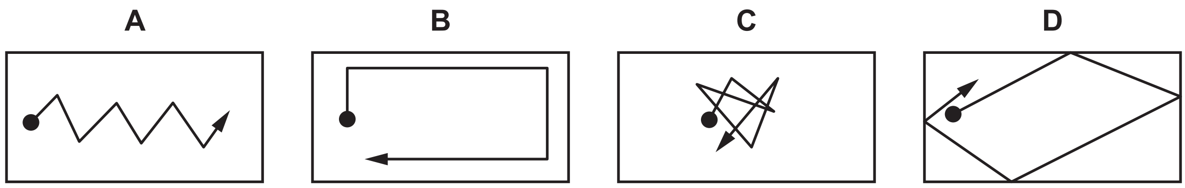
14 Which properties does a liquid have?

	definite shape	can be compressed easily
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key
✓ = has this property
✗ = does not have this property

15 In the diagrams, the black circle represents a smoke particle in air.

Which diagram shows a likely path that the particle takes because of Brownian motion?



16 Which name is given to the change in volume of a gas when it is heated at constant pressure?

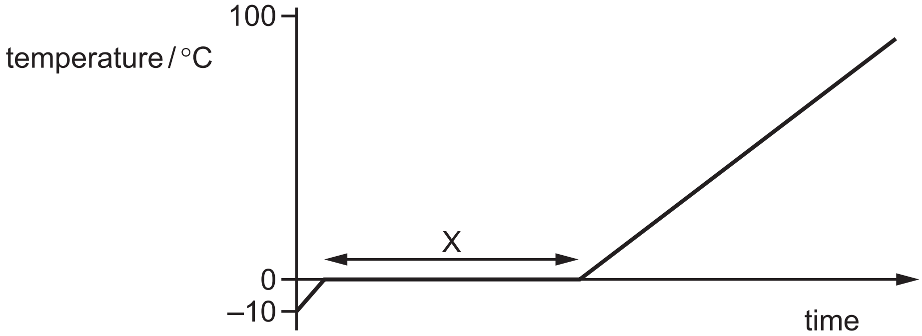
A thermal capacity

B thermal conduction

C thermal energy

D thermal expansion

17 Some ice is slowly heated and its temperature is measured. A graph is plotted of temperature against time.



Which row describes what happens to the thermal energy and to the temperature in section X?

	thermal energy	temperature of ice
A	gained by ice	rises
B	gained by ice	stays the same
C	not gained by ice	rises
D	not gained by ice	stays the same

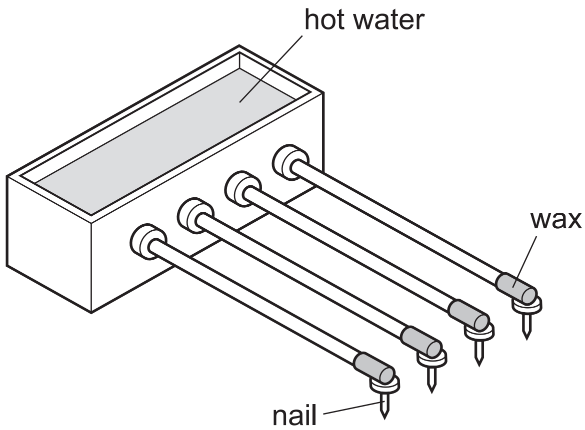
18 The melting point of a substance is –78 °C and its boiling point is 23 °C.

Which row gives the correct state of matter of the substance at the given temperatures?

	state at temperature of 0 °C	state at temperature of 100 °C
A	solid	liquid
B	solid	gas
C	liquid	solid
D	liquid	gas

19 A student sets up an experiment to find out how well different metals conduct thermal energy.

One end of a rod of each metal is in hot water and the other end has a small nail attached to it by wax.



The rods have the same thickness.

Which conditions should be satisfied in order to make this a valid test?

	same size nails	hot water kept at constant temperature	same length rods
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key
✓ = condition applies
✗ = condition does not apply

20 Two similar metal containers, P and Q, each contain water at 90 °C. Both containers are the same size and both are sealed.

The water in container P cools more quickly than the water in container Q.

Which statement is correct?

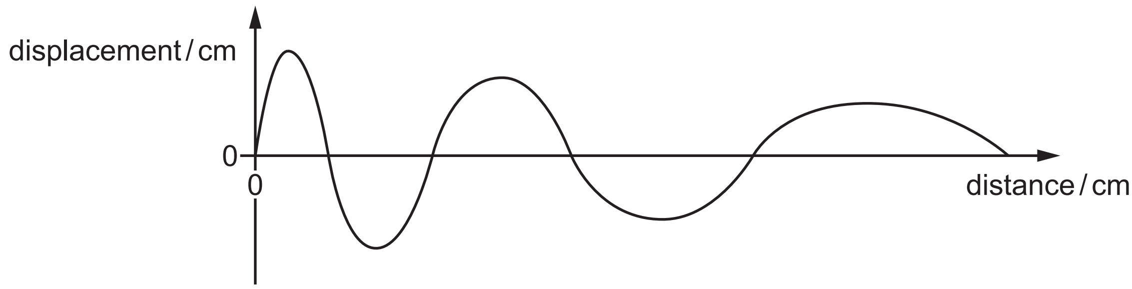
A P contains more water than Q.

B P has a shinier surface than Q.

C P is painted a darker colour than Q.

D P is surrounded by a vacuum and Q is surrounded by air.

21 A wave has the appearance shown.



How do the properties of the wave change as the distance from the origin increases?

	amplitude	wavelength
A	decreases	increases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

22 A tank contains water. Ripples are produced on the surface of the water. Refraction is observed.

What causes the ripples to refract?

A The cold water in the tank is replaced by warm water.

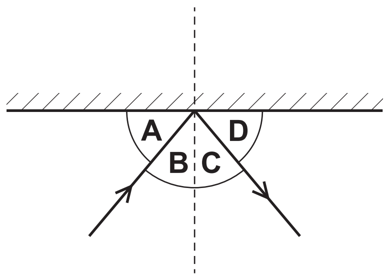
B The ripples change speed as they move from deep to shallow water.

C The ripples hit the wall of the tank.

D The ripples pass through a narrow gap.

23 The diagram shows a plane mirror and a ray of light reflected from it.

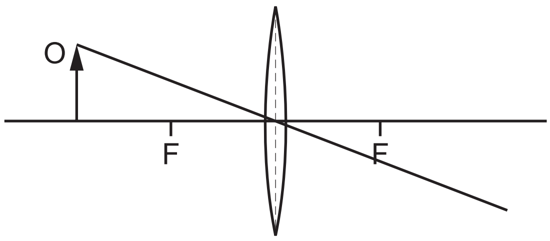
Which angle is the angle of incidence?



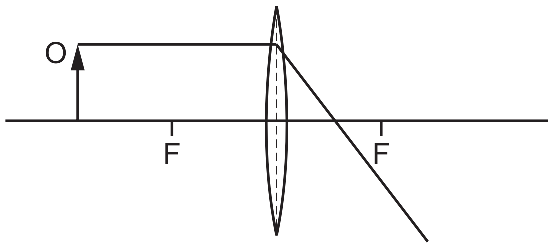
24 The diagrams each show a ray of light from an object O passing through a thin converging lens.

The principal focuses in each diagram are labelled F.

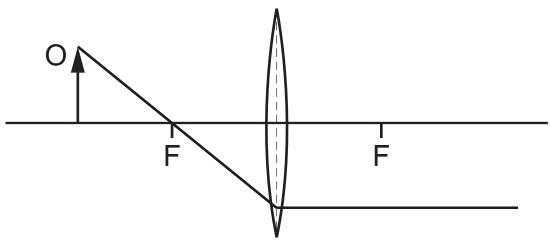
1



2



3



Which diagrams are correct?

A 1 and 2 **B** 2 only **C** 1 and 3 **D** 3 only

25 Which row is correct?

	frequency of infrared waves compared to microwaves	use of infrared waves
A	greater	radiant heater
B	greater	satellite television
C	lower	radiant heater
D	lower	satellite television

26 A student investigates sound waves from a loudspeaker.

The frequency of the sound wave is 25 000 Hz. The student has normal hearing but she cannot hear the sound.

What should she do if she wants to hear a sound from the loudspeaker?

A decrease the amplitude

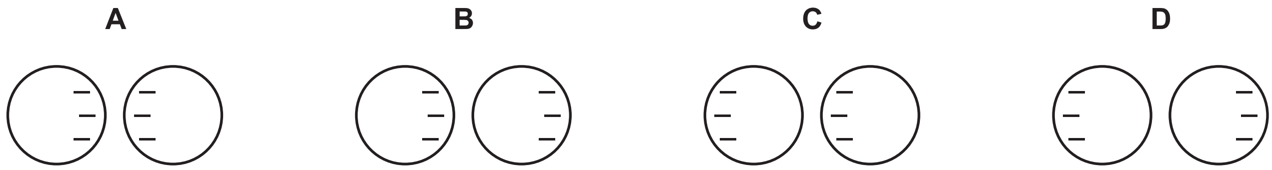
B decrease the frequency

C increase the amplitude

D increase the frequency

27 Two isolated metal spheres are both negatively charged. The spheres are brought close together but do not touch.

Which diagram shows the charge distribution on the spheres?



28 When a plastic comb is placed next to a small piece of aluminium foil hanging from a nylon thread, the foil is repelled by the comb.

Why is this?

A The comb is charged and the foil is uncharged.

B The comb is uncharged and the foil is charged.

C The comb and the foil have charges of opposite sign.

D The comb and the foil have charges of the same sign.

29 The diagram shows a piece of metal resistance wire.

Which wire, made of the same metal, has a smaller resistance?

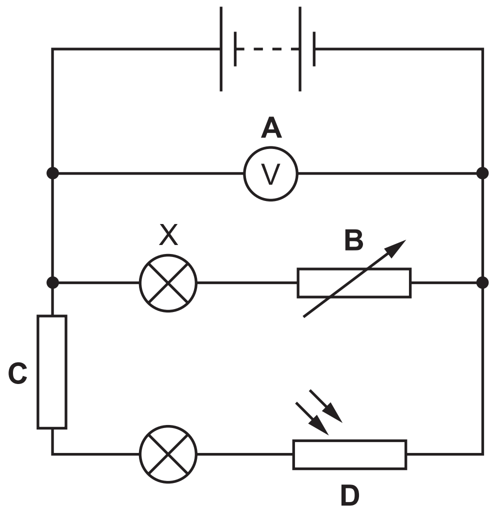
A a wire of the same length with a larger diameter

B a wire of the same length with a smaller diameter

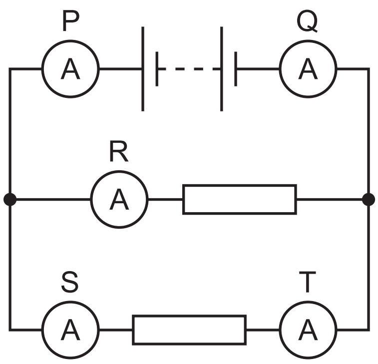
C a wire of greater length with the same diameter

D a wire of greater length with a smaller diameter

30 Which labelled component in the circuit shown controls the brightness of lamp X?



31 A circuit includes a battery, two identical resistors and five ammeters, P, Q, R, S and T.



Which statement about the readings on the ammeters is **not** correct?

A P has a greater reading than Q.

B P has a greater reading than R.

C P has a greater reading than S.

D P has a greater reading than T.

32 An electrician sets up a potential divider circuit in a fridge so that when the fridge door is open and light from the room enters the fridge, a warning light turns on.

Which component does the electrician need for the potential divider in addition to a variable resistor?

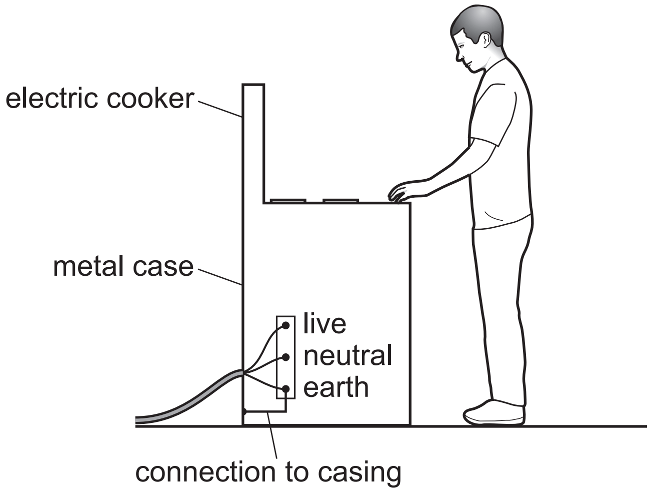
A light-dependent resistor

B relay

C thermistor

D motor

33 A simple wiring diagram for an electric cooker is shown.



Why is there a wire connecting the metal case of the cooker to earth?

A It improves the efficiency of the cooker.

B It prevents the metal case from becoming too hot when the cooker is left on.

C It reduces the risk of an electric shock if the live wire touches the metal case.

D The electric cooker will not switch on without it.

34 Three statements about a.c. and d.c. currents are given.

1 A d.c. current is in one direction only whilst an a.c. current repeatedly changes direction.

2 d.c. is the abbreviation for direct current and a.c. is the abbreviation for amplitude current.

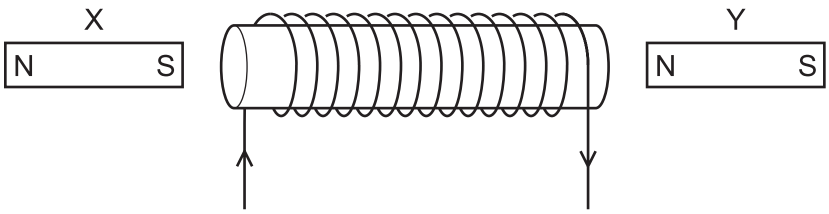
3 An a.c. current is in one direction only whilst a d.c. current repeatedly changes direction.

Which statements are correct?

A 1 and 2 **B** 1 only **C** 2 and 3 **D** 3 only

35 Two magnets are placed near a current-carrying coil.

The diagram shows this experimental arrangement and the current direction in the coil.



Which statement is correct?

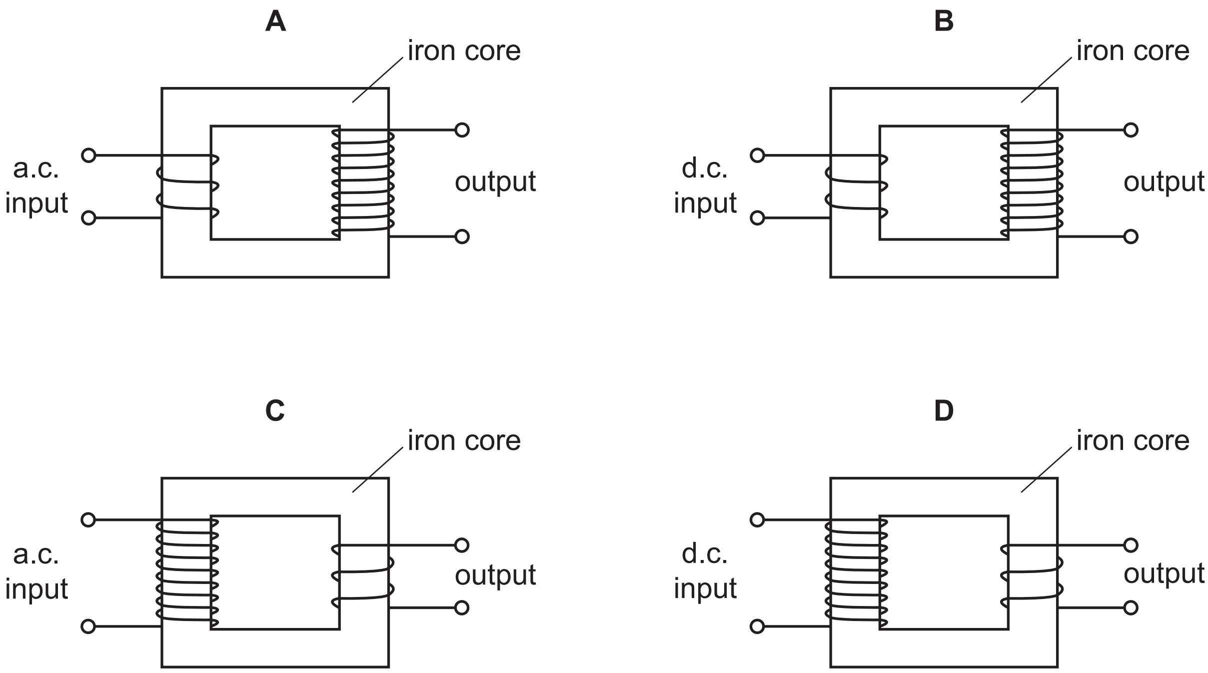
A Both X and Y are attracted to the coil.

B Both X and Y are repelled by the coil.

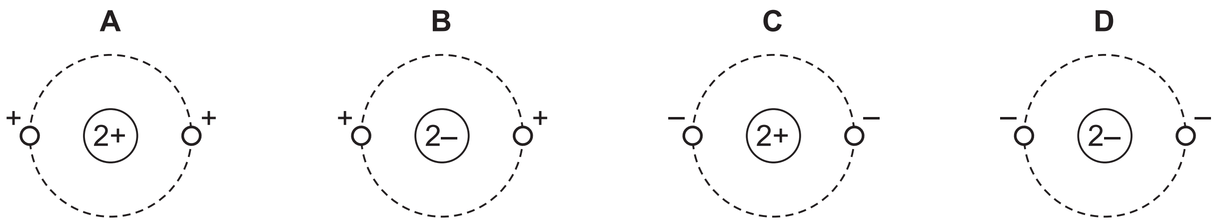
C X is attracted to the coil and Y is repelled.

D X is repelled by the coil and Y is attracted.

36 Which arrangement can be used to step up a voltage?



37 Which diagram shows the structure of an atom containing a nucleus and two orbiting electrons?



38 A nuclide has the symbol $^{14}_{6}C$.

Which statement about all atoms of this nuclide is correct?

A There are 6 protons in the nucleus.

B There are 14 neutrons in the nucleus.

C There are 6 electrons in the nucleus.

D There are 20 nucleons in the nucleus.

39 The table compares the penetrating abilities and ionising effects of α-particles and of γ-radiation.

Which row is correct?

	least penetrating	most ionising
A	α	α
B	α	γ
C	γ	α
D	γ	γ

40 When a radioactive isotope is set up close to a counter, a count rate of 38 000 counts/s is obtained. The table shows the count rate from the isotope over a three-year period.

time / years	$\dfrac{\text{count rate}}{\text{counts / s}}$
0	38 000
1	26 000
2	17 000
3	12 000

What is the half-life of the isotope?

A less than 1 year

B more than 1 year but less than 2 years

C more than 2 years but less than 3 years

D more than 3 years

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Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2022**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

Question	Answer	Marks
1	B	**1**
2	A	**1**
3	D	**1**
4	B	**1**
5	A	**1**
6	D	**1**
7	C	**1**
8	A	**1**
9	D	**1**
10	B	**1**
11	D	**1**
12	B	**1**
13	C	**1**
14	D	**1**
15	C	**1**
16	D	**1**
17	B	**1**
18	D	**1**
19	C	**1**
20	C	**1**
21	A	**1**
22	B	**1**
23	B	**1**
24	C	**1**
25	A	**1**
26	B	**1**
27	D	**1**
28	D	**1**

Question	Answer	Marks
29	A	**1**
30	B	**1**
31	A	**1**
32	A	**1**
33	C	**1**
34	B	**1**
35	B	**1**
36	A	**1**
37	C	**1**
38	A	**1**
39	A	**1**
40	B	**1**

Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2021**

45 minutes

You must answer on the multiple choice answer sheet.

* 9 2 8 8 5 2 6 9 8 4 *

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

IB21 03_0625_12/3RP

1 A student has a measuring cylinder containing water and also has a balance.

Which of these could she use to find the volume of a small metal sphere?

She has no other apparatus.

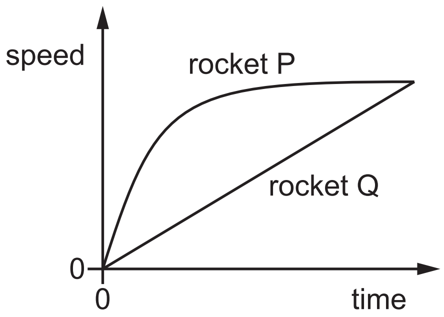
A either the measuring cylinder containing water or the balance

B the measuring cylinder containing water only

C the balance only

D neither the measuring cylinder nor the balance

2 Two rockets are launched at the same time from the surface of the Earth. The graph shows how the speeds of the rockets change with time.



Which statement about the rockets is correct?

A Both rockets travel the same distance.

B Rocket P accelerates and then decelerates.

C Rocket P travels further than rocket Q.

D Rocket Q has zero acceleration.

3 A train begins a journey from a station and travels 60 km in a time of 20 minutes.

What is the average speed of the train?

A 3.0 m / s **B** 5.0 m / s **C** 50 m / s **D** 60 m / s

4 What is weight?

A a frictional force

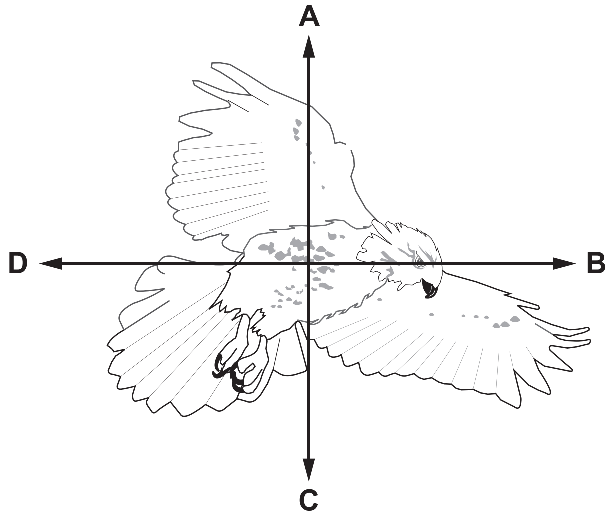
B a magnetic force

C a gravitational force

D an electromagnetic force

5 The diagram shows a bird in flight. The bird is flying in a horizontal direction to the right.

In which direction does air resistance act on the bird?



6 A small bottle has a mass of 20 g when empty. The volume of the bottle is $10\,cm^3$.

When full of liquid, the total mass is 150 g.

What is the density of the liquid?

A $0.50\,g/cm^3$ **B** $2.0\,g/cm^3$ **C** $13\,g/cm^3$ **D** $15\,g/cm^3$

7 A rocket is travelling vertically upwards. Three vertical forces act on it.

The thrust acts upwards and is equal to 100 000 N.

The weight acts downwards and is equal to 80 000 N.

What is the air resistance force acting on the rocket when it is travelling upwards at constant speed?

A 20 000 N downwards

B 20 000 N upwards

C 180 000 N downwards

D 180 000 N upwards

8 Which energy source boils water to make steam in power stations?

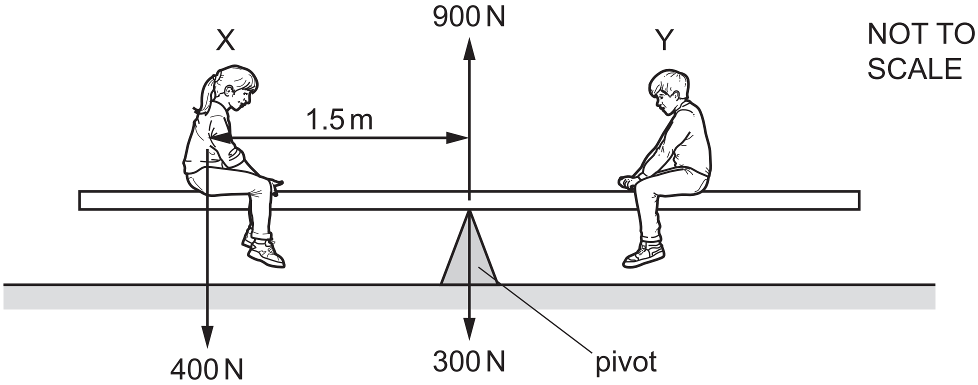
A energy from tides

B energy from waves

C hydroelectric energy

D nuclear energy

9 Two people X and Y sit on a see-saw, as shown.



X weighs 400 N and sits 1.5 m from the pivot.

The weight of the see-saw is 300 N and acts through the pivot.

When the see-saw is balanced, the pivot pushes up on the see-saw with a force of 900 N.

What is the weight of person Y and how far from the pivot is he sitting?

	weight of Y / N	distance from pivot / m
A	200	1.5
B	200	3.0
C	400	1.5
D	400	3.0

10 A horizontal force pulls a box along a horizontal surface.

The box gains 30 J of kinetic energy and 10 J of thermal energy is produced by the friction between the box and the surface.

How much work is done by the force?

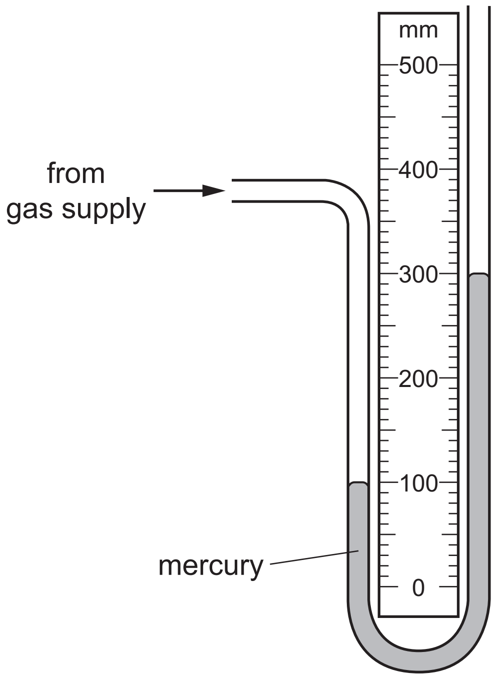
A 10 J **B** 20 J **C** 30 J **D** 40 J

11 The table shows the work done and the time taken by four machines.

Which machine is the most powerful?

	work done / kJ	time taken / s
A	100	10
B	100	20
C	200	10
D	200	20

12 The diagram shows a manometer connected to a gas supply.



What is the pressure of the gas supply?

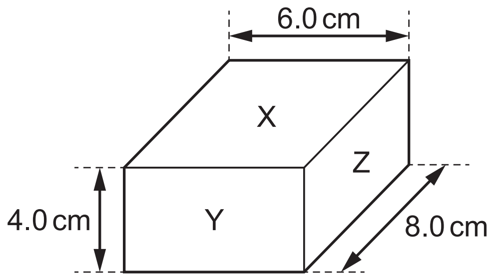
A 100 mm Hg above atmospheric pressure

B 100 mm Hg below atmospheric pressure

C 200 mm Hg above atmospheric pressure

D 200 mm Hg below atmospheric pressure

13 The diagram shows a box of dimensions 6.0 cm × 8.0 cm × 4.0 cm.



The box rests on a flat horizontal surface.

On which face must the box rest in order to exert the least pressure?

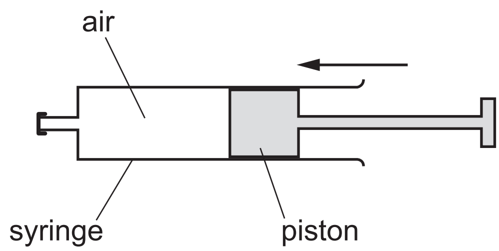
A face X

B face Y

C face Z

D The pressure is the same for all the faces.

14 Air in a sealed syringe is slowly compressed by moving the piston. The temperature of the air stays the same.



Which statement about the air is correct?

A The pressure of the air decreases because its molecules now travel more slowly.

B The pressure of the air decreases because the area of the syringe walls is now smaller.

C The pressure of the air increases because its molecules now hit the syringe walls more frequently.

D The pressure of the air increases because its molecules now travel more quickly.

15 In an experiment, smoke particles are suspended in air and viewed through a microscope.

The smoke particles move about with short random movements.

Which of the following statements is correct?

A Air particles have large masses compared to smoke particles and they move in one direction only.

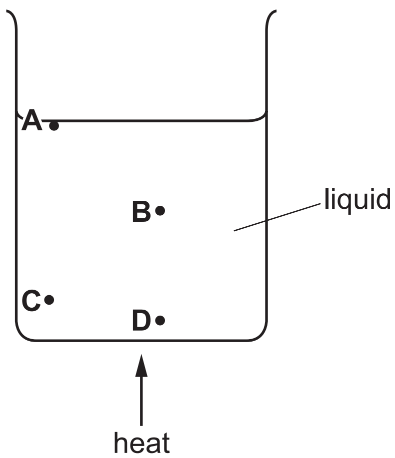
B Air particles have large masses compared to smoke particles and they move in random directions.

C Air particles move at high speeds compared to smoke particles and they move in one direction only.

D Air particles move at high speeds compared to smoke particles and they move in random directions.

16 A beaker of liquid is heated slowly so that the liquid evaporates.

During evaporation, from where do the more energetic molecules leave the liquid?



17 Four samples of materials with the same dimensions are tested.

Which material gives the highest rate of thermal conduction?

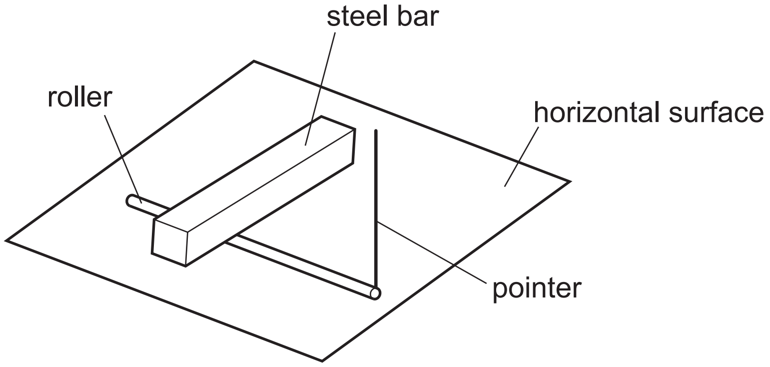
A rubber

B copper

C plastic

D steel

18 The diagram shows a simple type of thermometer.



The roller rests on a horizontal surface and the steel bar rests on top of the roller.

When the temperature changes, the pointer rotates.

Which physical property is being used to measure temperature?

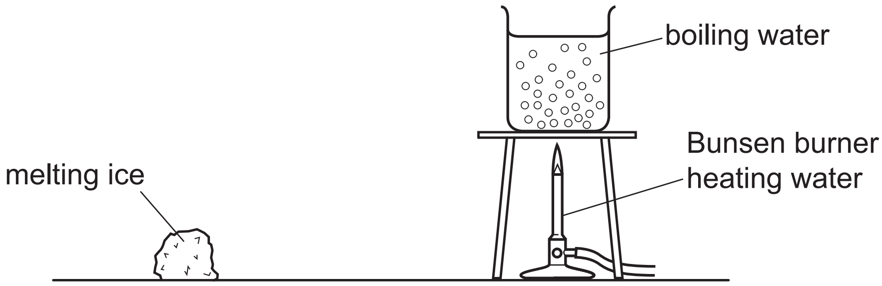
A the length of the pointer

B the length of the roller

C the length of the steel bar

D the thickness of the roller

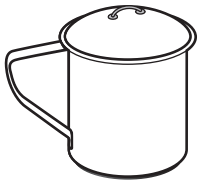
19 A piece of melting ice at 0 °C and a beaker of boiling water are both in a laboratory. The laboratory is at 20 °C.



What is happening to the temperature of the melting ice and what is happening to the temperature of the boiling water?

	temperature of melting ice	temperature of boiling water
A	constant	constant
B	constant	increasing
C	increasing	constant
D	increasing	increasing

20 Four metal mugs with lids each contain the same volume of hot water at the same temperature.



The mugs are made from the same metal and are the same shape and size, but each has its outside surface painted with a different combination of texture and colour.

Which mug radiates thermal energy at the fastest rate?

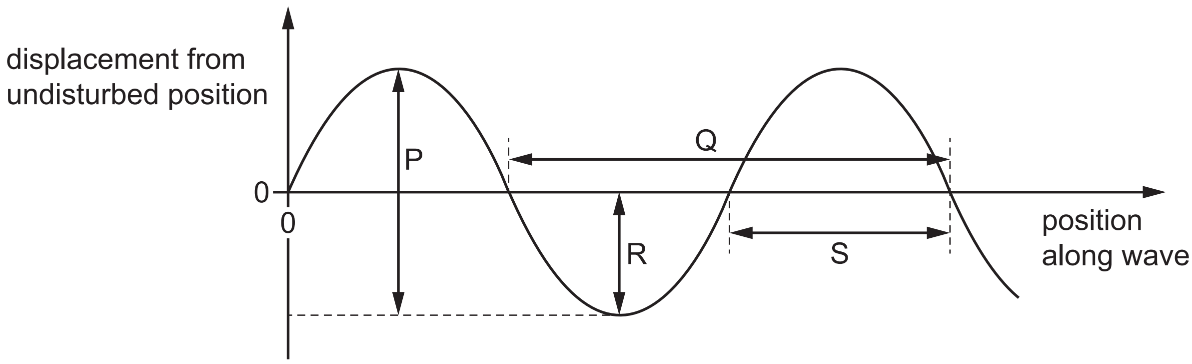
A the mug painted shiny white

B the mug painted dull white

C the mug painted shiny black

D the mug painted dull black

21 The diagram shows a wave.



Which dimensions describe the properties of the wave?

A P = amplitude, S = speed

B P = speed, Q = wavelength

C R = amplitude, Q = wavelength

D R = amplitude, S = wavelength

22 What causes the change in direction when light travels from air into glass?

A The amplitude of the light changes.

B The colour of the light changes.

C The frequency of the light changes.

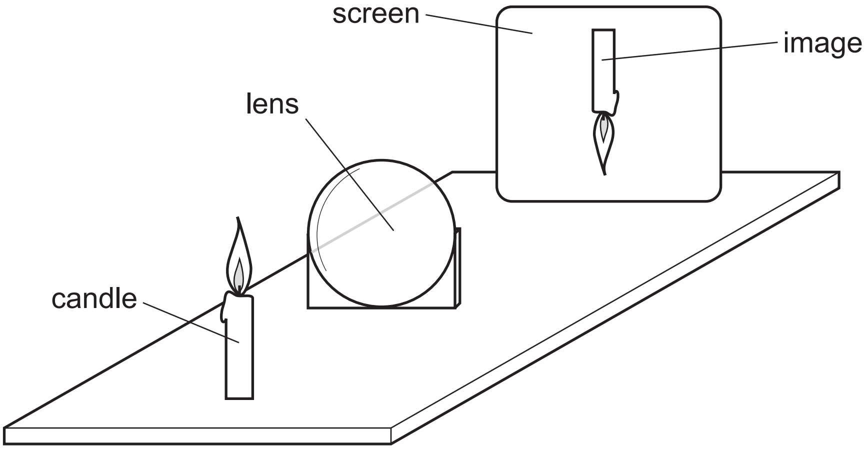
D The speed of the light changes.

23 Light from a torch is incident on a plane mirror. The angle of incidence is 38°.

What is the angle of reflection?

A 38° **B** 52° **C** 76° **D** 142°

24 A thin converging lens is used to produce a sharp image of a candle.



Various sharp images are produced on the screen by moving the lens and the screen backwards and forwards.

Which statement is **always** correct?

A The image is at the principal focus (focal point) of the lens.

B The image is bigger than the object.

C The image is closer to the lens than the object.

D The image is inverted.

25 Thermal radiation is part of the electromagnetic spectrum.

What is the name of this region of the spectrum?

A gamma rays

B infrared rays

C ultraviolet rays

D visible light rays

26 Which process causes a sound wave to produce an echo?

A diffraction

B dispersion

C reflection

D refraction

27 A quiet sound is produced by a loudspeaker. The pitch of the sound remains constant but the loudness of the sound is increased.

Which property of the sound wave is increased?

A amplitude

B frequency

C speed

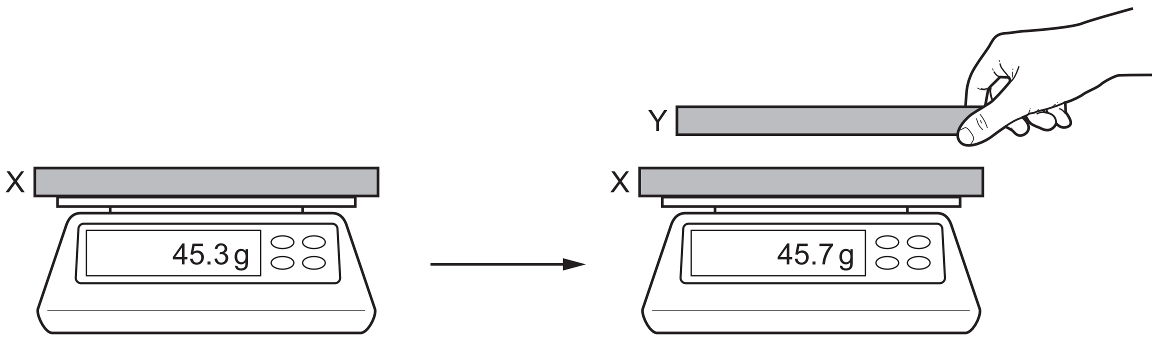
D wavelength

28 A soft-iron bar is not magnetised. It is held close to the N pole of a magnet and then to the S pole of the same magnet.

What will be the result?

	N pole	S pole
A	attracts	attracts
B	attracts	repels
C	repels	attracts
D	repels	repels

29 A charged rod X is placed on a balance and another rod Y is brought close to it, as shown.



Which combination of charges would cause the change in the balance reading shown?

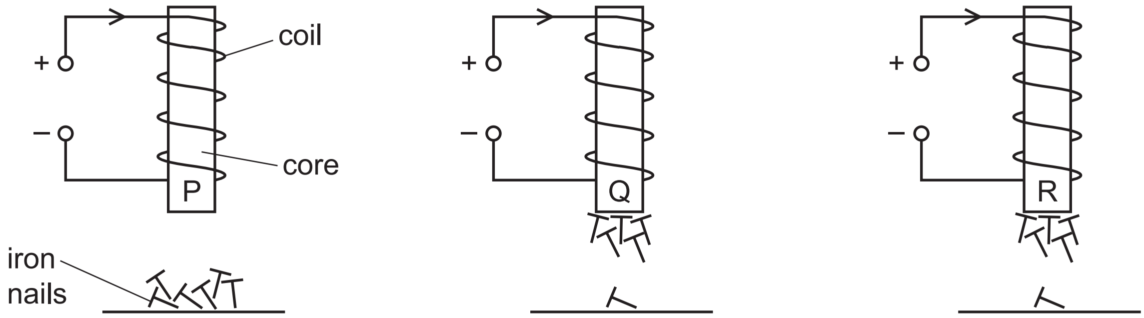
	X	Y
A	negative charge	negative charge
B	negative charge	positive charge
C	negative charge	no charge
D	positive charge	no charge

30 Three cores of different metals P, Q and R are placed inside identical coils of wire.

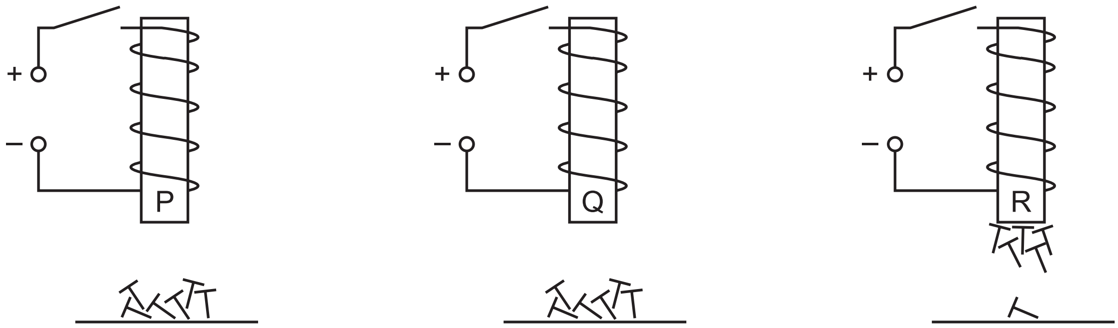
At least one of the metals is non-magnetic.

The cores are held above some iron nails.

The three diagrams show what happens when there is a current in the coils.



The three diagrams below show what happens when the current is then switched off.



Which core metals are magnetic?

A P only **B** R only **C** P and Q **D** Q and R

31 A length of metal wire is used as a resistor in a circuit.

Which change to the wire results in an increase in current in the circuit?

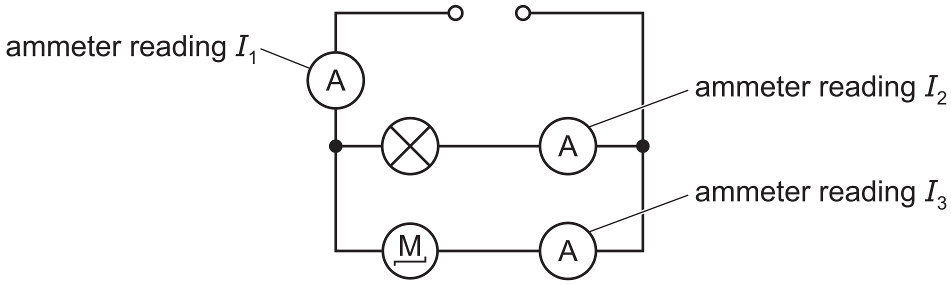
A connecting a second identical length of wire in parallel with the original wire

B connecting a second identical length of wire in series with the original wire

C decreasing the thickness of the wire

D increasing the length of the wire

32 The diagram shows a circuit containing a power supply, a lamp and a motor. Three ammeters measure the current at different points in the circuit. The readings on the ammeters are I_1, I_2 and I_3.



Which statement about the ammeter readings is correct?

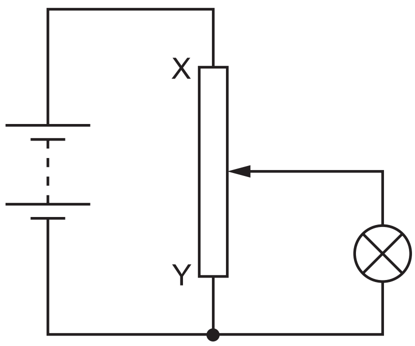
A All three readings are the same.

B I_1 is greater than I_2 and is greater than I_3.

C I_2 is greater than I_1 and is greater than I_3.

D I_3 is greater than I_1 and is greater than I_2.

33 A student designs a circuit to use as a dimmer switch for a lamp.



What happens to the brightness of the lamp and the potential difference (p.d.) across the lamp, when the slider is moved from X to Y?

	brightness of lamp	p.d. across the lamp
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

34 Which electrical component is connected in series with an electric circuit to protect it from damage by a very large current?

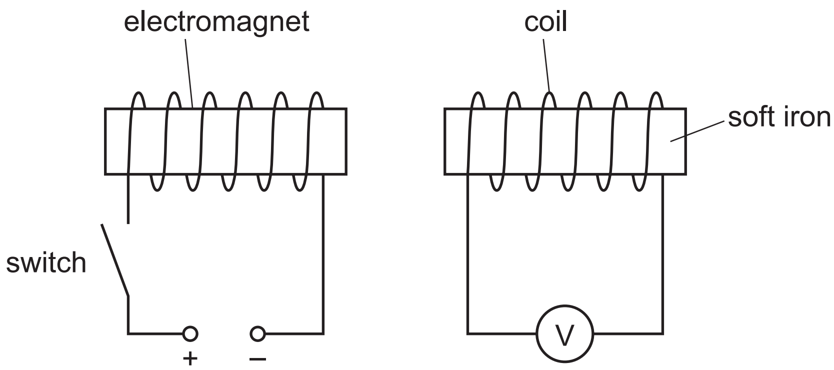
A earth wire

B fuse

C relay

D thermistor

35 The diagram shows an electromagnet near a coil of wire connected to a voltmeter. The reading on the voltmeter is zero.



The switch is closed. The electromagnet magnetises quickly.

What happens to the reading on the voltmeter?

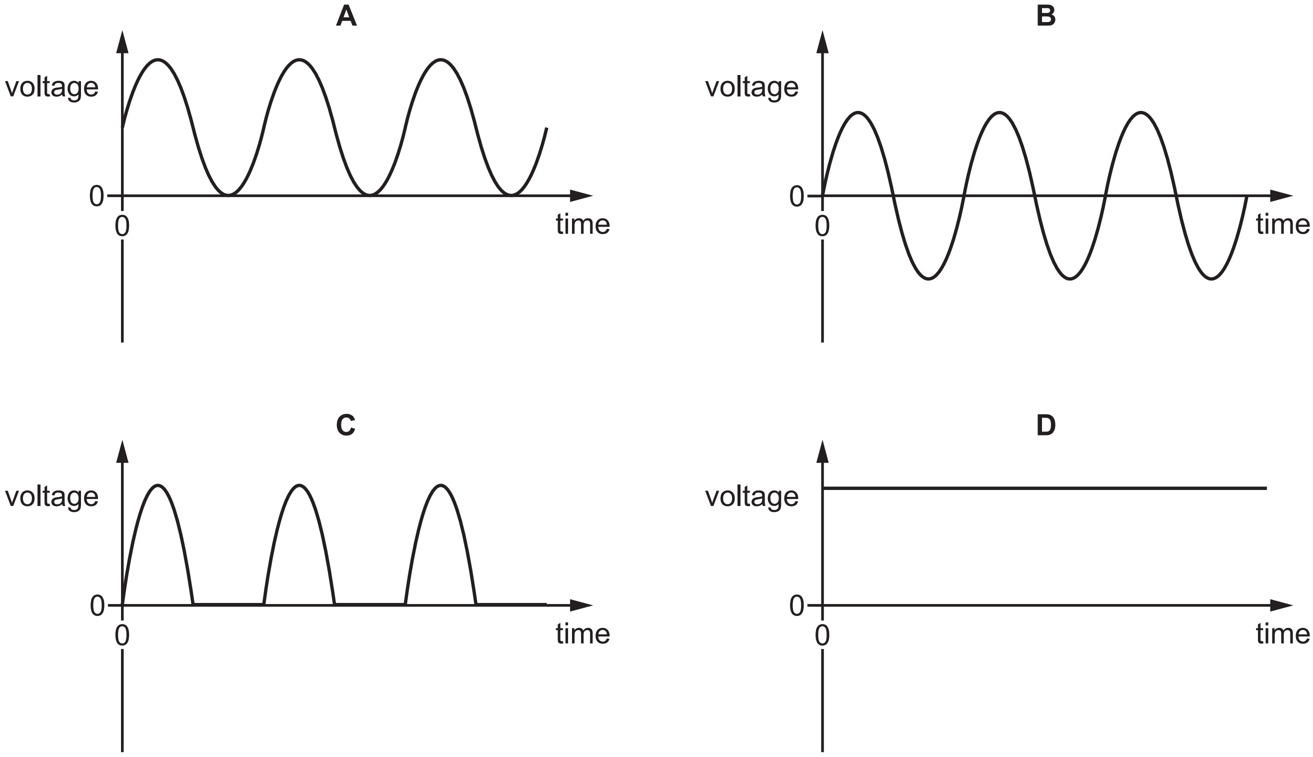
A It keeps increasing.

B It quickly increases and stays at maximum.

C It quickly increases and then decreases.

D It stays on zero.

36 Which diagram represents the voltage output of a simple a.c. generator?



37 Three students are describing the structure of an atom.

student 1 All the positively charged particles are in the nucleus.

student 2 Positive electrons are in the nucleus.

student 3 Negative electrons orbit around the nucleus.

Which students are making a correct statement?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

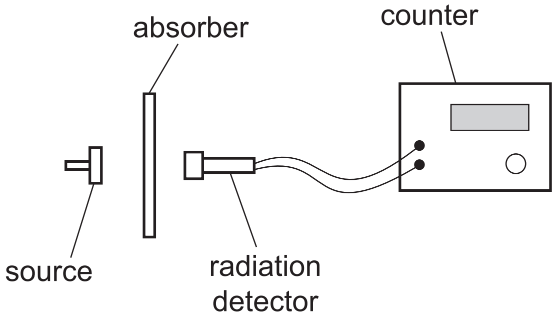
38 The symbol below describes a nuclide.

$${}^{9}_{4}\mathrm{Be}$$

Which row is correct?

	proton number (*Z*)	nucleon number (*A*)
A	4	5
B	4	9
C	9	4
D	9	5

39 The diagram shows a radioactivity experiment.



When a piece of paper is used as the absorber, the count rate drops to the background count rate.

Which radiation is the source emitting?

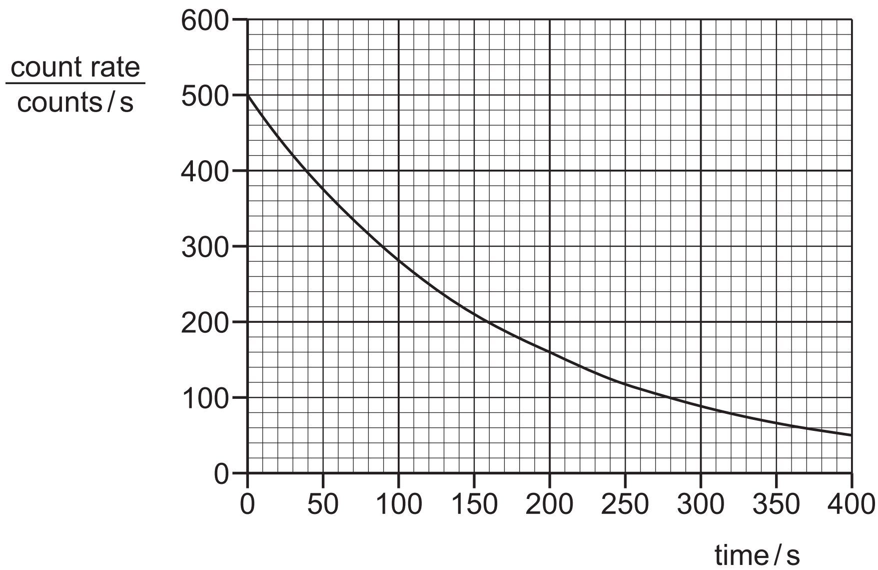
A α-radiation only

B β-radiation only

C γ-radiation only

D α-radiation, β-radiation and γ-radiation

40 The graph shows how the count rate from a radioactive sample changes with time.



What is the half-life for this sample?

A 90 s **B** 120 s **C** 200 s **D** 400 s

Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **March 2021**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the March 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

Question	Answer	Marks
1	B	**1**
2	C	**1**
3	C	**1**
4	C	**1**
5	D	**1**
6	C	**1**
7	A	**1**
8	D	**1**
9	B	**1**
10	D	**1**
11	C	**1**
12	C	**1**
13	A	**1**
14	C	**1**
15	D	**1**
16	A	**1**
17	B	**1**
18	C	**1**
19	A	**1**
20	D	**1**
21	C	**1**
22	D	**1**
23	A	**1**
24	D	**1**
25	B	**1**
26	C	**1**
27	A	**1**
28	A	**1**

Question	Answer	Marks
29	A	**1**
30	D	**1**
31	A	**1**
32	B	**1**
33	A	**1**
34	B	**1**
35	C	**1**
36	B	**1**
37	C	**1**
38	B	**1**
39	A	**1**
40	B	**1**

Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2020**

45 minutes

You must answer on the multiple choice answer sheet.

6993319596

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

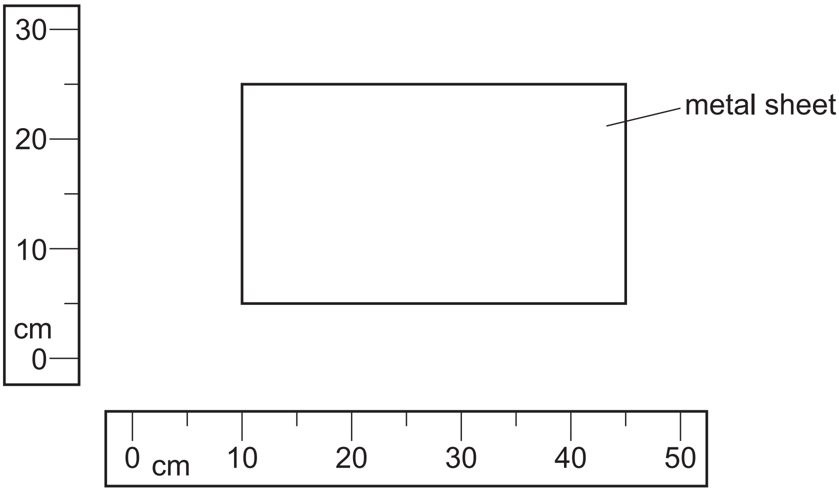
- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = $10\,m/s^2$).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **20** pages. Blank pages are indicated.

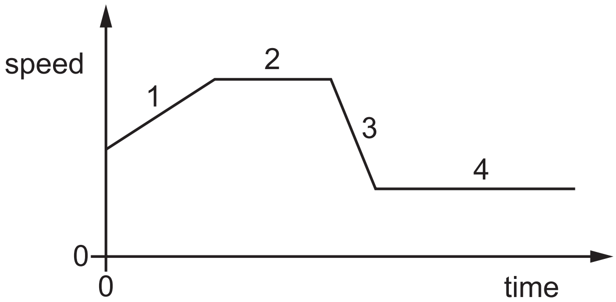
1 The diagram shows a rectangular metal sheet close to two rulers.



What is the area of the metal sheet?

A $700\,cm^2$ **B** $875\,cm^2$ **C** $900\,cm^2$ **D** $1125\,cm^2$

2 The graph shows how the speed of a car changes with time over part of a journey.



Which section of the graph shows acceleration and which section of the graph shows deceleration?

	acceleration	deceleration
A	1	2
B	1	3
C	2	4
D	3	1

3 A steel ball is dropped from the top floor of a building. Air resistance can be ignored.

Which statement describes the motion of the ball?

A The ball falls with constant acceleration.

B The ball falls with constant speed.

C The ball falls with decreasing speed.

D The ball falls with increasing acceleration.

4 Diagram 1 shows a piece of flexible material that contains many pockets of air. Diagram 2 shows the same piece of flexible material after it has been compressed so that its volume decreases.

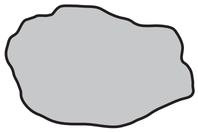


diagram 1
(before compression)

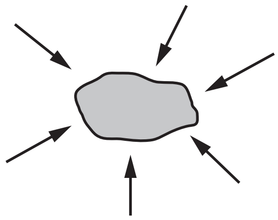


diagram 2
(after compression)

What happens to the mass and to the weight of the flexible material when it is compressed?

	mass	weight
A	increases	increases
B	increases	no change
C	no change	increases
D	no change	no change

5 On the Moon, the gravitational field strength *g* is 1.6 N / kg.

An object has a mass of 2.0 kg.

What is the weight of the object on the Moon?

A 0 N **B** 1.3 N **C** 3.2 N **D** 20.0 N

6 A measuring cylinder contains 40 cm^3 of water.

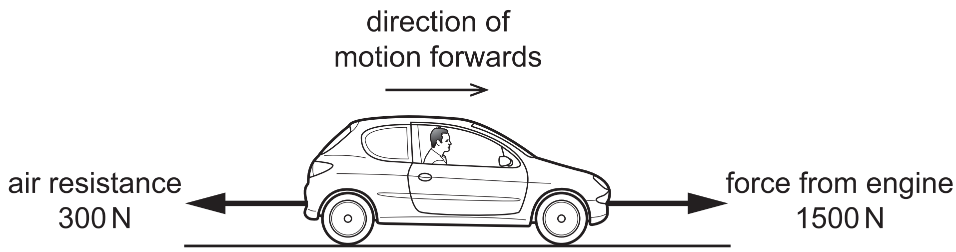
A solid metal ball is dropped into the water and the water level rises to 56 cm^3.

The mass of the ball is 80 g.

What is the density of the metal from which the ball is made?

A 0.20 g/cm^3 **B** 1.4 g/cm^3 **C** 2.0 g/cm^3 **D** 5.0 g/cm^3

7 A car travels along a horizontal road at constant speed. Three horizontal forces act on the car. The diagram shows two of these forces.



What is the size and the direction of the third horizontal force acting on the car?

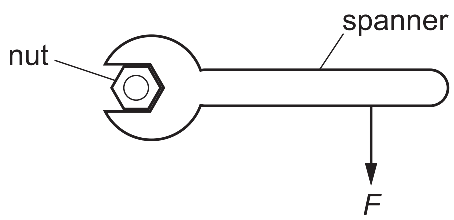
A 1200 N backwards

B 1200 N forwards

C 1800 N backwards

D 1800 N forwards

8 A force *F* is applied to a spanner, as shown.



Which action increases the moment of *F* about the centre of the nut?

A apply the force *F* to the end of the spanner handle

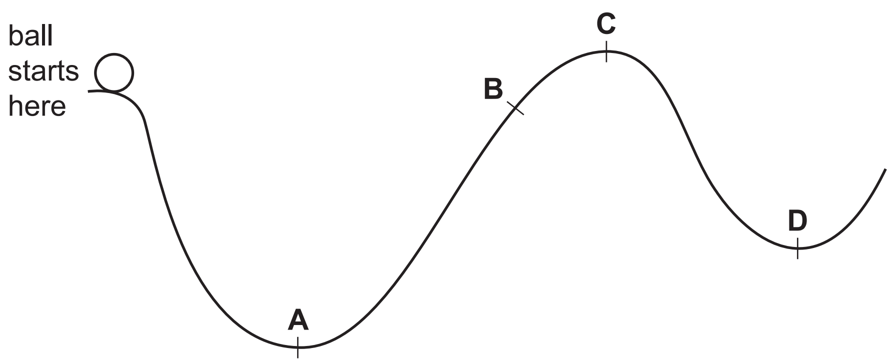
B apply the force *F* parallel to the spanner handle

C spray oil on the nut

D use a shorter spanner

9 A ball is released from rest and rolls down a track from the position shown.

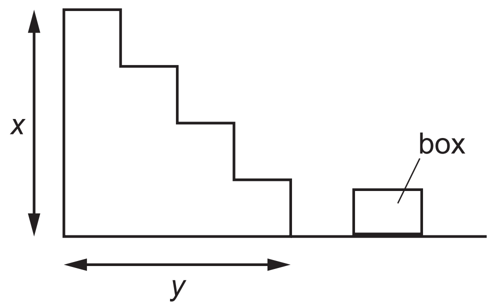
What is the furthest position that it is possible for the ball to reach?



10 Which row describes an advantage and a disadvantage of wind turbines?

	advantage	disadvantage
A	no fuel needed	harmful gases released
B	variable supply	fuel needed
C	no harmful gases released	variable supply
D	constant supply	noisy

11 A box of mass *m* and weight *W* is carried up some stairs of total height *x* and total width *y*.



On which quantities does the work done against gravity on the box depend?

A *m* and *y* **B** *W* and *x* **C** *W* and *y* **D** *x* and *y*

12 Diagram 1 shows a tube sealed at one end and partly immersed in mercury. The tube has a diameter d. The top of the mercury in the tube is a height h above the mercury reservoir.

diagram 1

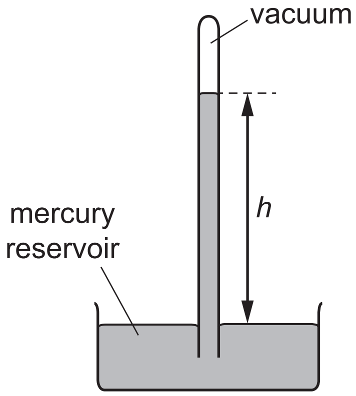
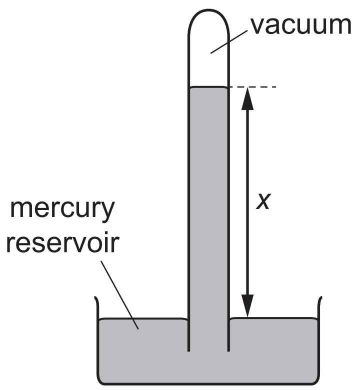


Diagram 2 shows a similar arrangement with a tube that has a diameter $2d$.

diagram 2



What is the relationship between h and x?

A $x = 2h$

B $x = h$

C $x = \frac{h}{2}$

D $x = \frac{h}{4}$

13 A skier is standing still on a flat area of snow.

The weight of the skier is 550 N. The total area of his skis in contact with the ground is 0.015 m^2.

What is the pressure exerted on the ground by the skier?

A 0.83 N / m^2 **B** 8.3 N / m^2 **C** 3700 N / m^2 **D** 37 000 N / m^2

14 The molecules of a substance in a particular state of matter move freely with random motion. The average speed of the molecules is increasing.

What is being described?

A a gas being heated

B a liquid evaporating

C a solid being heated

D a solid melting

15 A student measures the mass of warm water in an open container over two minutes. The container is kept at a constant temperature. The results are in the table.

time / minutes	mass / g
0.0	33.9
0.5	30.6
1.0	27.6
1.5	24.9
2.0	22.5

Why does the mass of the water change?

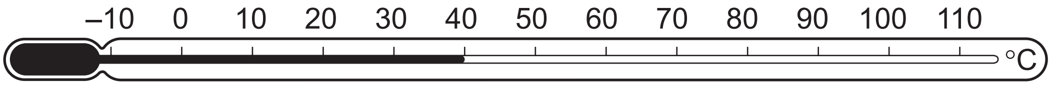
A The water evaporates.

B The water freezes.

C The water condenses.

D The water boils.

16 Which points are the fixed points of the liquid-in-glass thermometer shown?



A the beginning and end points of the column of liquid

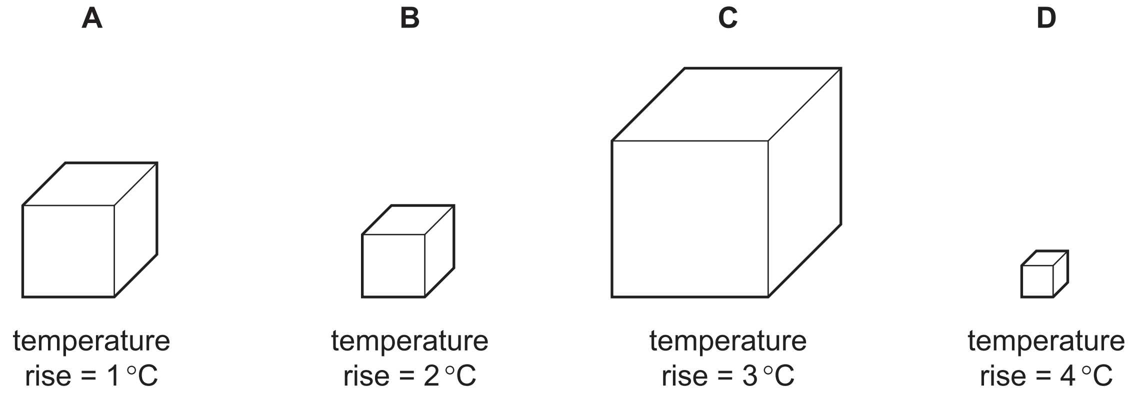
B the points marked –10 °C and 110 °C

C the points marked 0 °C and 100 °C

D the top and bottom points of the thermometer bulb

17 Four different metal blocks are given the same quantity of thermal energy.

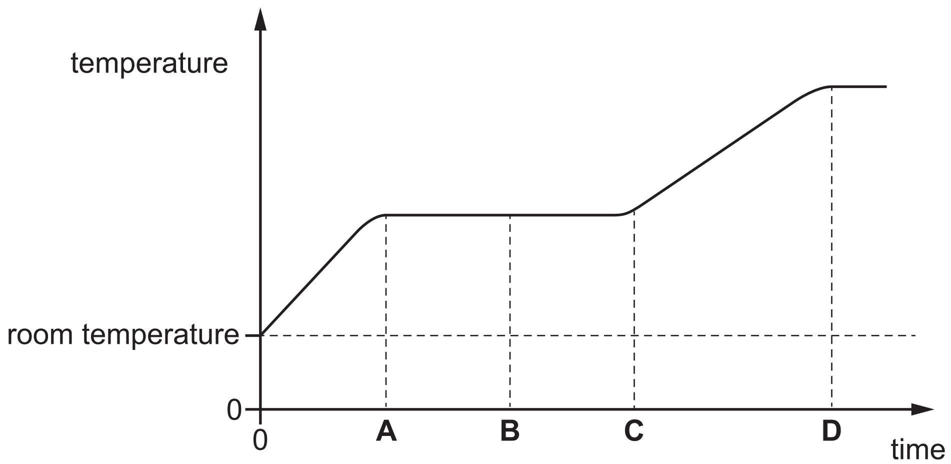
Which block has the greatest thermal capacity?



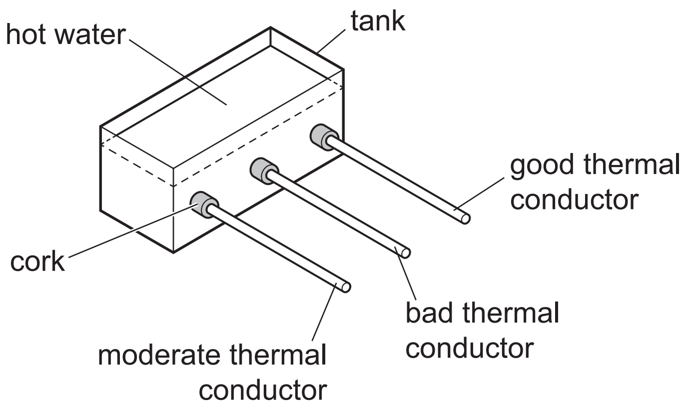
18 A solid is heated from room temperature.

The graph shows how its temperature changes with time as it is heated constantly.

At which time has it just become completely liquid?



19 Rods of the same shape and size are inserted through corks into a tank of hot water. Each rod is covered with a layer of solid wax that has a low melting point. After a period of time, some wax melts.



On which rod will the wax melt first?

A all at the same time

B good thermal conductor

C bad thermal conductor

D moderate thermal conductor

20 A boy jumps into an indoor swimming pool. He notices that the water appears to get colder as he goes deeper underwater. This is due to convection.

Which statement is correct?

A Cold water is more dense than warm water so it sinks to the bottom of the pool.

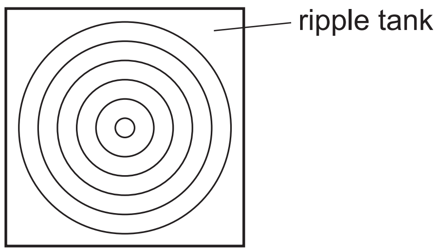
B Warm water is more dense than cold water so it rises to the surface of the pool.

C The molecules in cold water have more kinetic energy than the molecules in warm water so they move to the bottom of the pool faster.

D The molecules in warm water are closer together than the molecules in cold water so they rise to the surface of the pool.

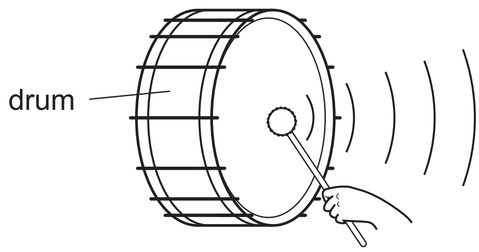
21 The diagrams show examples of wave motion.

1



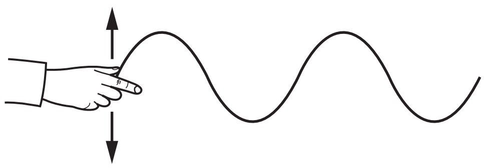
waves on water

2



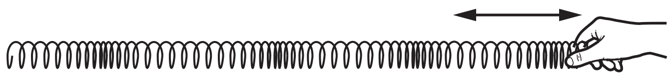
waves in air

3



waves on a rope

4



waves in a spring

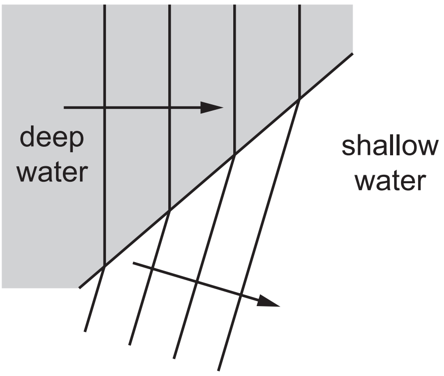
Which waves are longitudinal?

A 1 only **B** 2 and 3 only **C** 2, 3 and 4 **D** 2 and 4 only

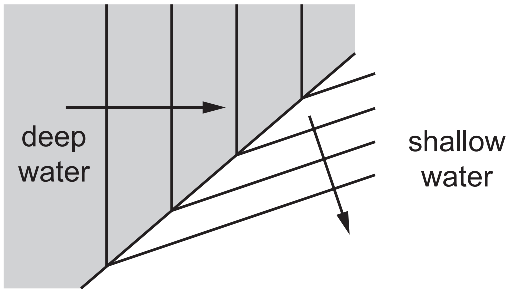
22 Waves on the surface of water travel from deep to shallow water.

Which diagram shows the correct path of the waves in the shallow water?

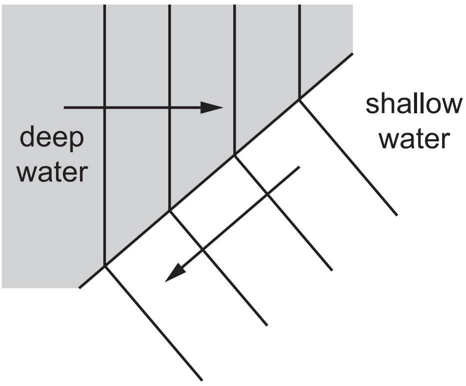
A



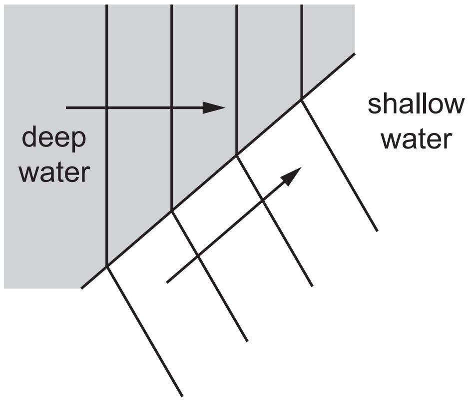
B



C

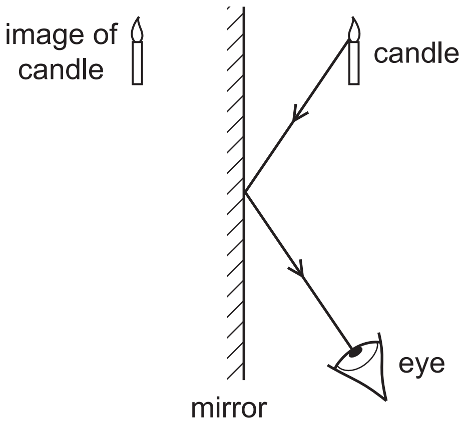


D

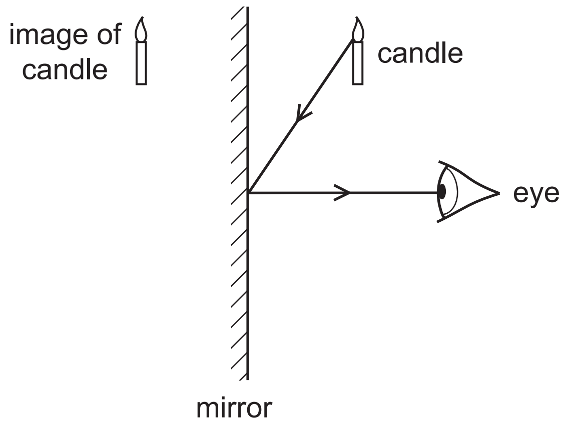


23 Which diagram shows how the light from a candle is reflected by a mirror, and shows the position of the image formed?

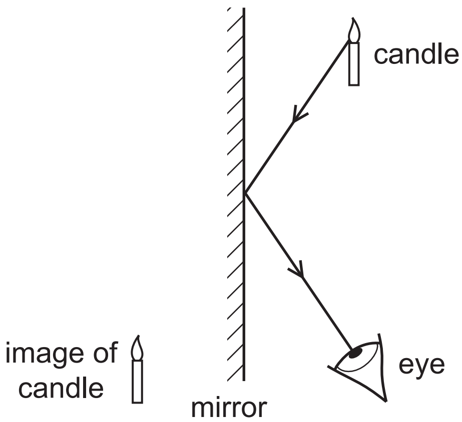
A



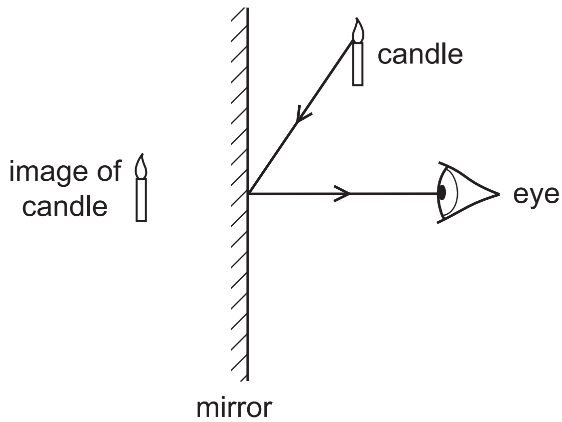
B



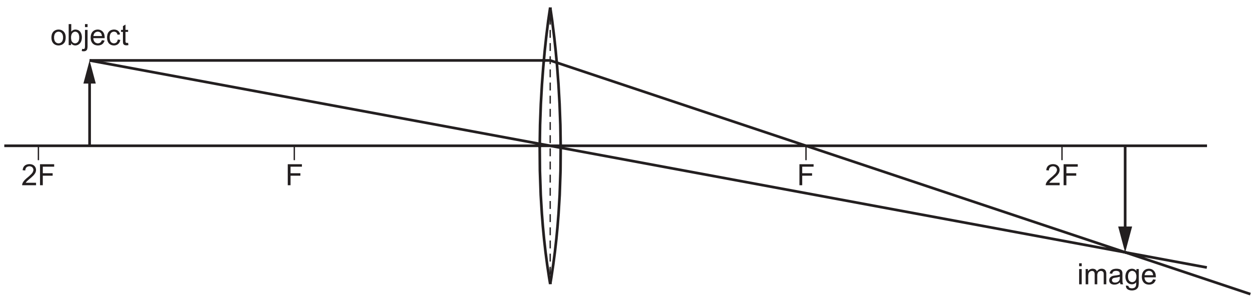
C



D



24 The diagram shows an image being formed by a converging lens.



Which description of the image formed is correct?

A enlarged and inverted

B enlarged and upright

C diminished and inverted

D diminished and upright

25 The horn on a ship makes a sound. The captain on the ship hears an echo from a cliff 4.0 s later.

The speed of sound is 340 m / s.

How far away is the cliff from the ship?

A 170 m **B** 340 m **C** 680 m **D** 1360 m

26 Bats produce ultrasound waves to navigate.

What is a possible frequency range for these waves?

A 0–20 Hz

B 20 Hz–2000 Hz

C 2 kHz–20 kHz

D 20 kHz–120 kHz

27 The table shows the forces that exist between magnetic poles.

Which row is correct?

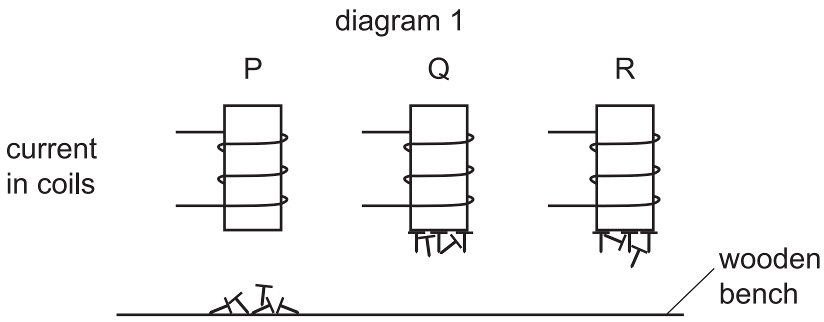
	N pole and N pole	N pole and S pole
A	attraction	attraction
B	attraction	repulsion
C	repulsion	attraction
D	repulsion	repulsion

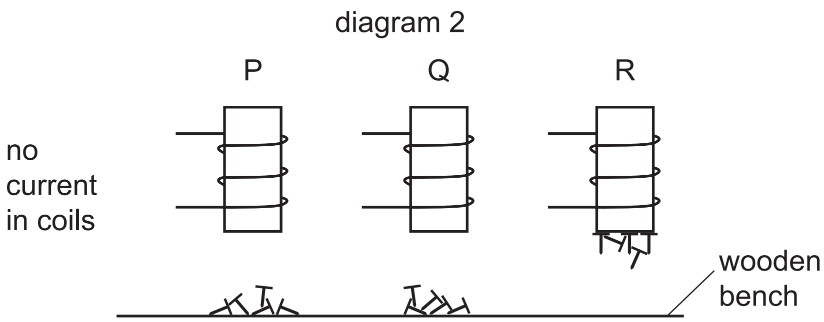
28 The diagrams show three different metal rods P, Q and R, inside coils of wire.

Small iron nails are placed on a wooden bench under the rods.

Diagram 1 shows the situation when there are electric currents in the wires.

Diagram 2 shows the situation when the currents are switched off.





Which row correctly identifies the metal rods?

	P	Q	R
A	copper	soft iron	steel
B	soft iron	copper	steel
C	steel	soft iron	copper
D	copper	steel	soft iron

29 A polythene rod becomes negatively charged when it is rubbed with a cloth.

Which statement explains this?

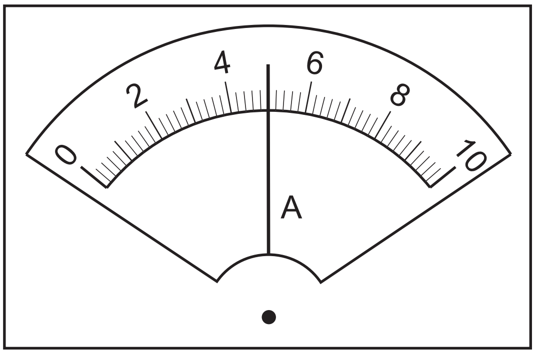
A The rod gains electrons.

B The rod loses electrons.

C The rod gains protons.

D The rod loses protons.

30 The diagram shows a reading on an ammeter.

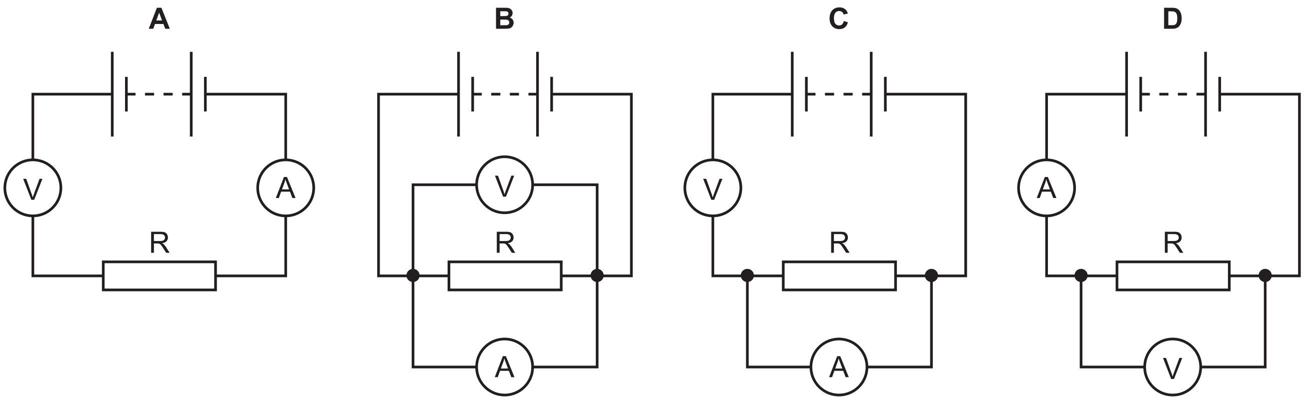


What is the reading?

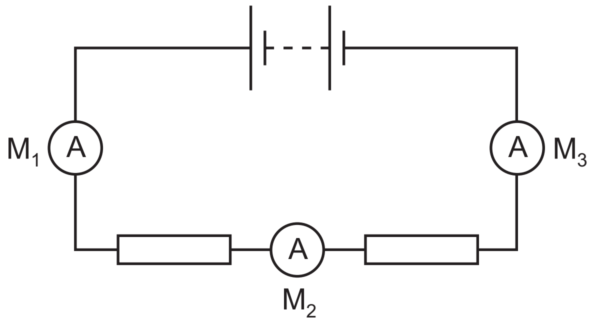
A 0.45 A **B** 0.50 A **C** 4.5 A **D** 5.0 A

31 A student is to determine the resistance of resistor R. She uses a circuit including a voltmeter and an ammeter.

Which circuit should be used?



32 The diagram shows a battery connected to two resistors. Three ammeters M_1, M_2 and M_3 are connected in the circuit.



Ammeter M_1 reads 1.0 A.

What are the readings on M_2 and on M_3?

	reading on M_2 / A	reading on M_3 / A
A	0.0	0.0
B	0.5	0.5
C	0.5	1.0
D	1.0	1.0

33 Lamps in a circuit are connected in parallel.

What is the advantage of this?

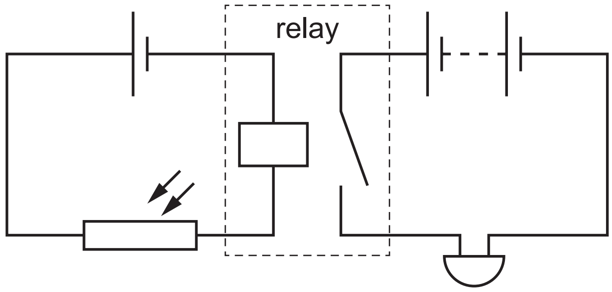
A If one lamp breaks, the others remain lit.

B Less current is taken from the power supply.

C The lamps use less power than if they were connected in series.

D The potential difference across each lamp is less than that of the power supply.

34 The diagram shows a relay being used to control a buzzer.



What is the function of the circuit?

A The buzzer switches off when the circuit becomes cold.

B The buzzer switches off when the circuit becomes light.

C The buzzer switches on when the circuit becomes cold.

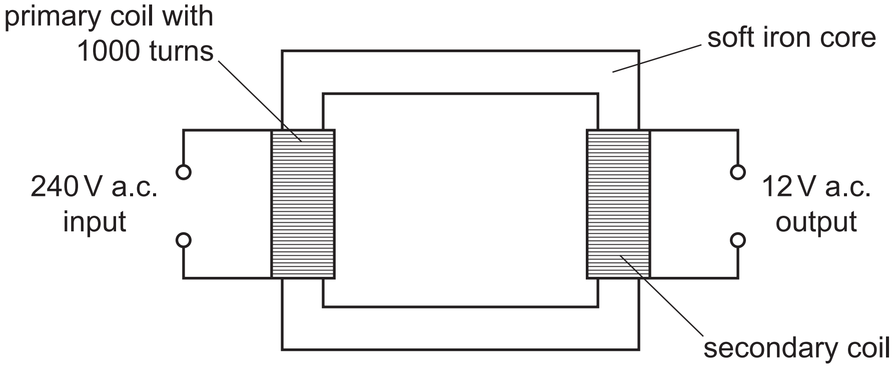
D The buzzer switches on when the circuit becomes light.

35 There is a 5 A fuse in the circuit of an appliance.

What is the purpose of the fuse and in which wire is it connected?

	purpose	wire in which it is connected
A	breaks the circuit if the current is greater than 5 A	earth
B	breaks the circuit if the current is greater than 5 A	live
C	breaks the circuit if the current is less than 5 A	earth
D	breaks the circuit if the current is less than 5 A	live

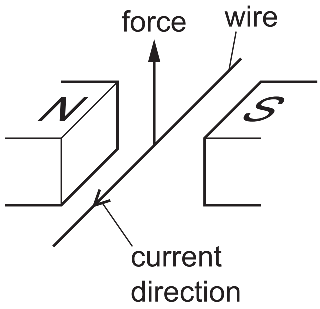
36 The diagram shows a transformer that has an output voltage of 12 V.



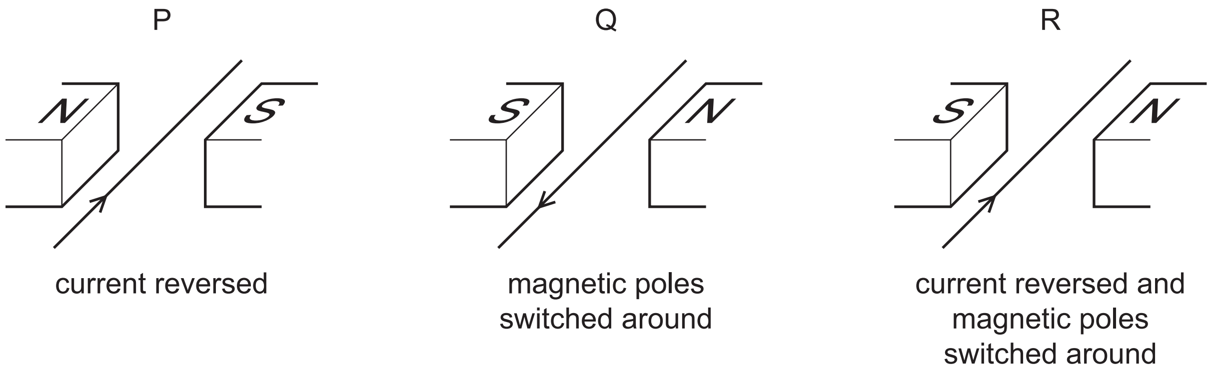
How many turns of wire are in the secondary coil?

A 12 **B** 20 **C** 50 **D** 20 000

37 A current-carrying conductor is placed between two magnetic poles. The current causes a force to act on the wire.



Three other arrangements P, Q and R, of the wire and magnetic poles are set up as shown.



Which arrangements will cause a force in the opposite direction to that shown in the top diagram?

A P, Q and R **B** P and Q only **C** P only **D** R only

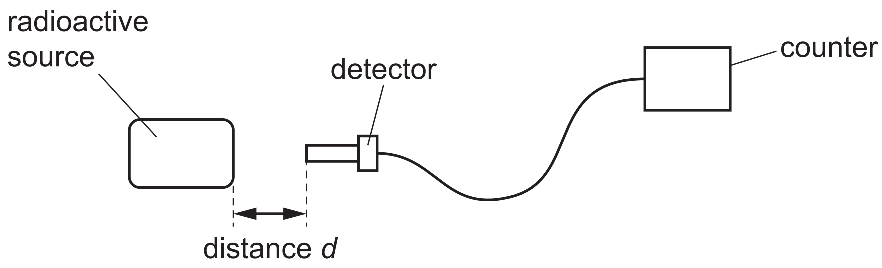
38 The symbol for a radioactive nuclide of carbon is ${}^{14}_{6}C$.

How many neutrons are in its nucleus?

A 6 **B** 8 **C** 14 **D** 20

39 A student measures the rate at which ionising radiation is emitted from a radioactive substance.

He places a detector at different distances from the radioactive source.



The table shows how the reading on the counter varies with distance *d*.

The readings on the counter are corrected for background radiation.

distance *d* / cm	0	2	4	6
counter reading / counts per minute	1250	115	0	0

Which type of ionising radiation is being emitted by the substance?

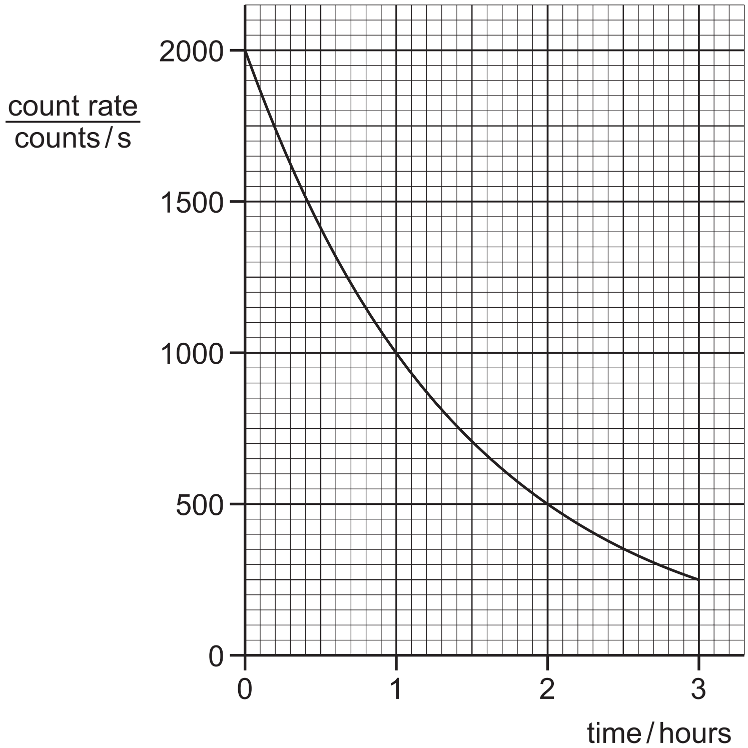
A α-particles

B β-particles

C γ-rays

D X-rays

40 The graph shows the count rate from a radioactive source over a period of time.



What is the half-life of the source?

A 0.5 hour **B** 1.0 hour **C** 1.5 hours **D** 3.0 hours

Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **March 2020**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the March 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

Question	Answer	Marks
1	A	**1**
2	B	**1**
3	A	**1**
4	D	**1**
5	C	**1**
6	D	**1**
7	A	**1**
8	A	**1**
9	B	**1**
10	C	**1**
11	B	**1**
12	B	**1**
13	D	**1**
14	A	**1**
15	A	**1**
16	C	**1**
17	A	**1**
18	C	**1**
19	B	**1**
20	A	**1**
21	D	**1**
22	A	**1**
23	A	**1**
24	A	**1**
25	C	**1**
26	D	**1**
27	C	**1**
28	A	**1**

Question	Answer	Marks
29	A	**1**
30	D	**1**
31	D	**1**
32	D	**1**
33	A	**1**
34	D	**1**
35	B	**1**
36	C	**1**
37	B	**1**
38	B	**1**
39	A	**1**
40	B	**1**

Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2019**

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

* 2 3 1 8 1 4 7 3 8 5 *

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Electronic calculators may be used.
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m / s^2).

This document consists of **16** printed pages.

1 Water drips from a tap into a measuring cylinder.

The table shows the volume of water in the cylinder every minute for four minutes.

time / minutes	volume of water / cm^3
0	0
1.0	27
2.0	57
3.0	83
4.0	112

What is the average volume of water collected in the cylinder each minute?

A $22\,cm^3$ **B** $28\,cm^3$ **C** $56\,cm^3$ **D** $57\,cm^3$

2 An object is moving with uniform deceleration.

Which statement describes its motion?

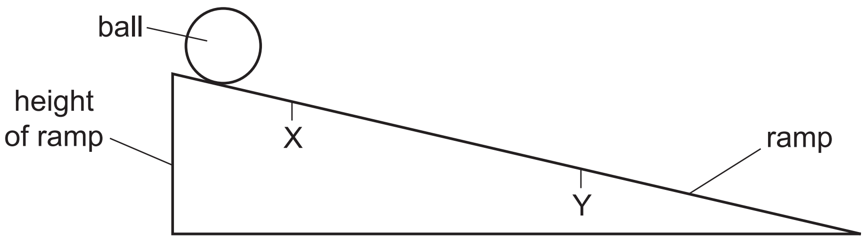
A Its rate of change of speed is decreasing.

B Its speed is constant.

C Its speed is decreasing.

D Its speed is increasing.

3 A ball rolls down a ramp. The time it takes to move from X to Y is measured.



Which other quantity must be measured in order to calculate the average speed of the ball between point X and point Y?

A angle of slope

B diameter of ball

C distance between X and Y

D height of ramp

4 An astronaut in a space station orbits above the Earth.

In the space station, the acceleration due to gravity is $7.5\,m/s^2$.

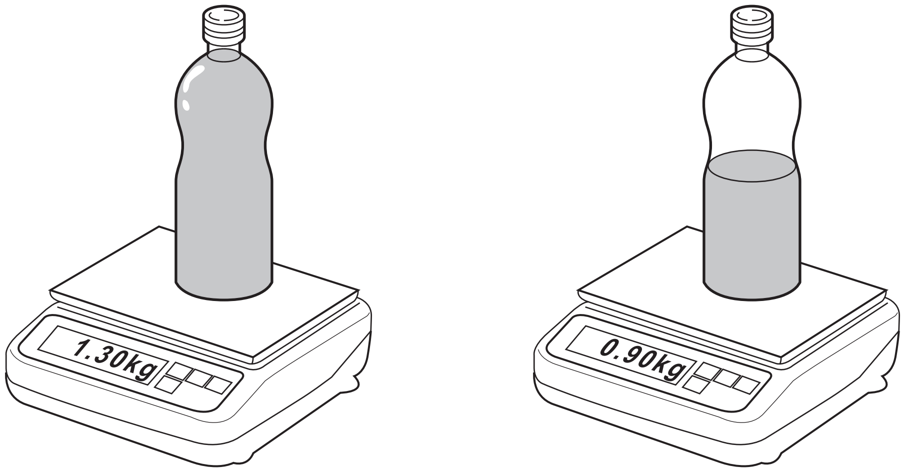
On Earth, the acceleration due to gravity is $10\,m/s^2$.

Which statement about the astronaut's mass and weight in the space station is correct?

	mass of astronaut	weight of astronaut
A	same as on the Earth	less than on the Earth
B	same as on the Earth	greater than on the Earth
C	lower than on the Earth	less than on the Earth
D	lower than on the Earth	greater than on the Earth

5 The mass of a full bottle of cooking oil is 1.30 kg.

When exactly half of the oil has been used, the mass of the bottle plus the remaining oil is 0.90 kg.



What is the mass of the empty bottle?

A 0.40 kg **B** 0.50 kg **C** 0.65 kg **D** 0.80 kg

6 A solid ball has a volume of $4.0\,cm^3$. The density of the ball is $1.6\,g/cm^3$.

What is the mass of the ball?

A 0.4 g **B** 2.5 g **C** 4.0 g **D** 6.4 g

7 Which property of an object **cannot** be changed by applying forces?

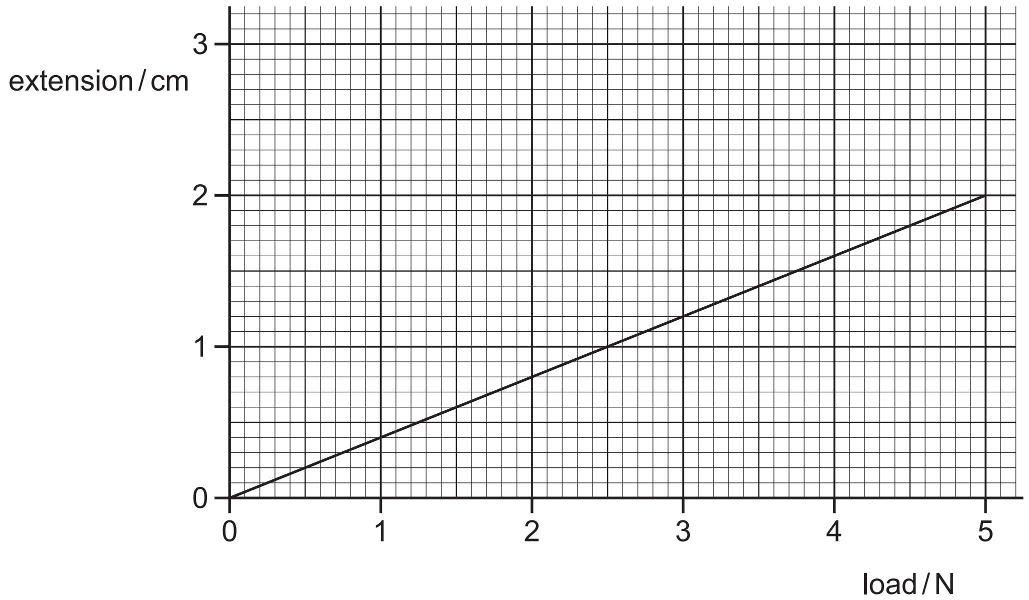
A mass

B shape

C speed

D volume

8 The extension / load graph for a spring is shown. The unstretched length of the spring is 15.0 cm.

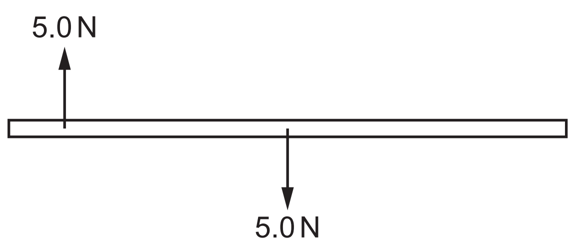


When an object of unknown weight is suspended on the spring, the length of the spring is 16.4 cm.

What is the weight of the object?

A 0.55 N **B** 0.67 N **C** 3.5 N **D** 4.1 N

9 The diagram shows a wooden beam with two forces acting on it.



Which way will the beam move?

A accelerate up the page

B accelerate down the page

C turn anticlockwise

D turn clockwise

10 Which source of energy uses the production of steam to generate electrical energy?

A hydroelectric

B nuclear

C tides

D waves

11 What is the unit of power?

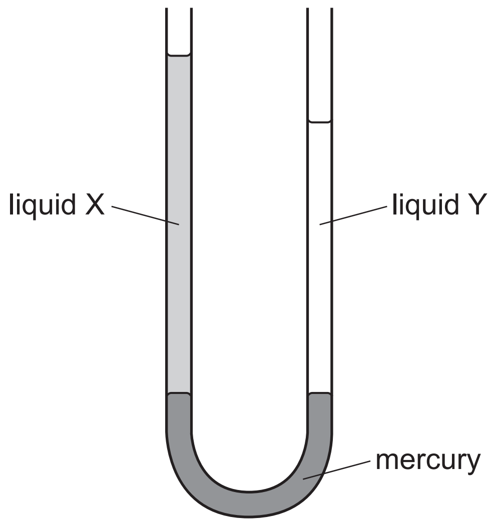
A joule

B newton

C pascal

D watt

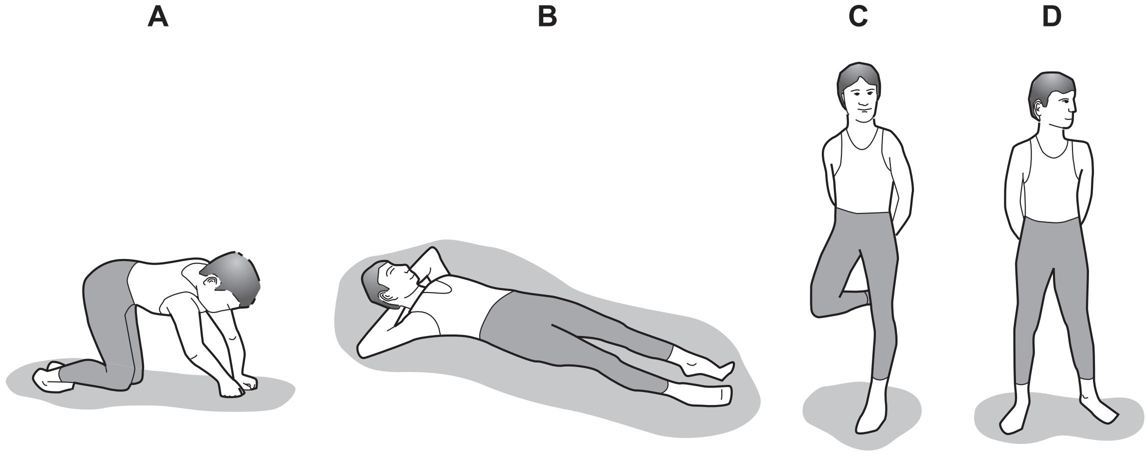
12 The diagram shows a U-tube manometer containing three liquids: mercury, liquid X and liquid Y. Neither liquid X or liquid Y mixes with mercury.



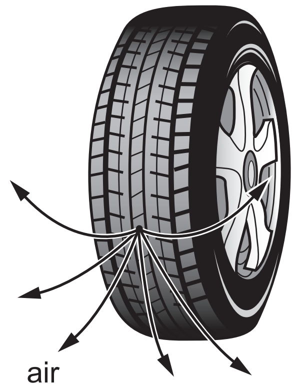
Which row compares the pressure exerted by liquid X and by liquid Y on the mercury, and the density of liquid X and the density of liquid Y?

	pressure exerted by X and by Y on the mercury	densities of X and of Y
A	pressure of X is greater than Y	density of X is greater than Y
B	pressure of Y is greater than X	density of Y is greater than X
C	pressure of X and of Y is the same	density of X is greater than Y
D	pressure of X and of Y is the same	density of Y is greater than X

13 Which diagram shows an athlete exerting least pressure on the ground?



14 A car tyre runs over a nail which makes a hole in it. The air in the tyre leaks out.



Why does the air leave the tyre?

A The pressure inside the tyre is greater than the pressure outside.

B The pressure inside the tyre is less than the pressure outside.

C The temperature inside the tyre is greater than the temperature outside.

D The temperature inside the tyre is less than the temperature outside.

15 On a cold day, a metal measuring tape graduated in millimetres is used to measure the distance between two fence posts. The measuring tape reads 3.000 m.

On a much hotter day, the metal measuring tape is used to measure the length of the same distance again. The metal measuring tape has a higher temperature than the ground. The temperature of the ground remains constant.

Which statement is correct?

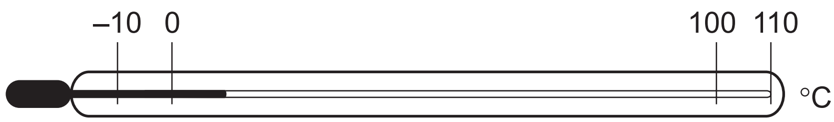
A The measuring tape reads less than 3.000 m because the graduations are closer together.

B The measuring tape reads less than 3.000 m because the graduations are further apart.

C The measuring tape reads more than 3.000 m because the graduations are closer together.

D The measuring tape reads more than 3.000 m because the graduations are further apart.

16 A thermometer has graduations which start at –10 °C and end at 110 °C.



What is the lower fixed point and what is the upper fixed point of the Celsius scale?

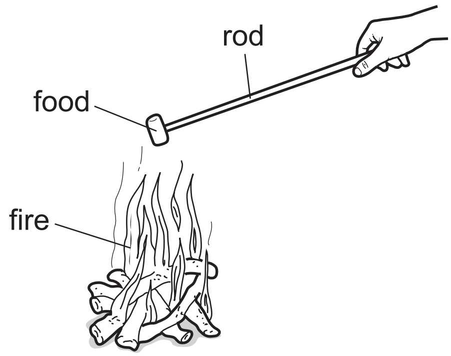
	lower fixed point / °C	upper fixed point / °C
A	–10	100
B	–10	110
C	0	100
D	0	110

17 Which row describes the process of melting?

	initial state	final state	change in temperature?
A	liquid	gas	yes
B	liquid	solid	no
C	solid	gas	yes
D	solid	liquid	no

18 Four campers are warming their food on a fire.

They use different rods, each of the same dimensions, to hold their food near the fire.



Which material is the best choice to prevent their hands from getting too hot?

A aluminium

B copper

C steel

D wood

19 A beaker of water is heated and thermal energy travels through the water by convection.

What happens to the density of the water when it is heated and how does the water move?

A The density decreases and the heated water moves downwards.

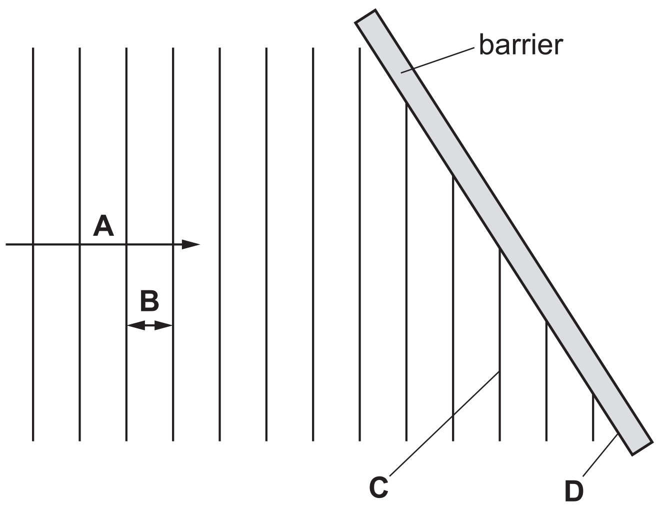
B The density decreases and the heated water moves upwards.

C The density increases and the heated water moves downwards.

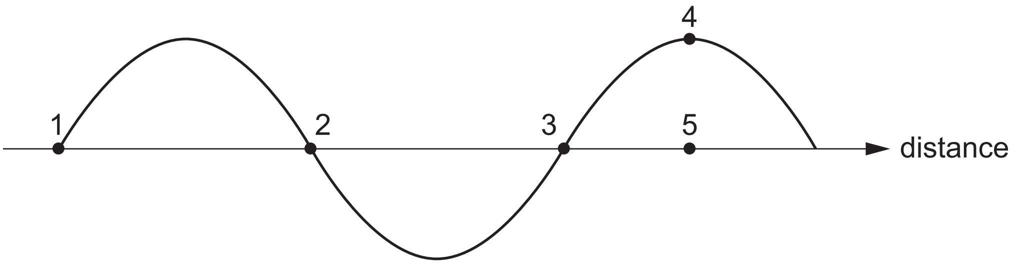
D The density increases and the heated water moves upwards.

20 The diagram shows a wave before it reflects from a barrier.

Which labelled section of the diagram represents a wavefront?



21 The diagram shows a transverse wave.



Which distance is equal to one wavelength?

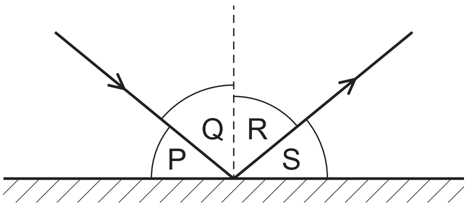
A the distance between points 1 and 2

B the distance between points 1 and 3

C the distance between points 2 and 3

D the distance between points 4 and 5

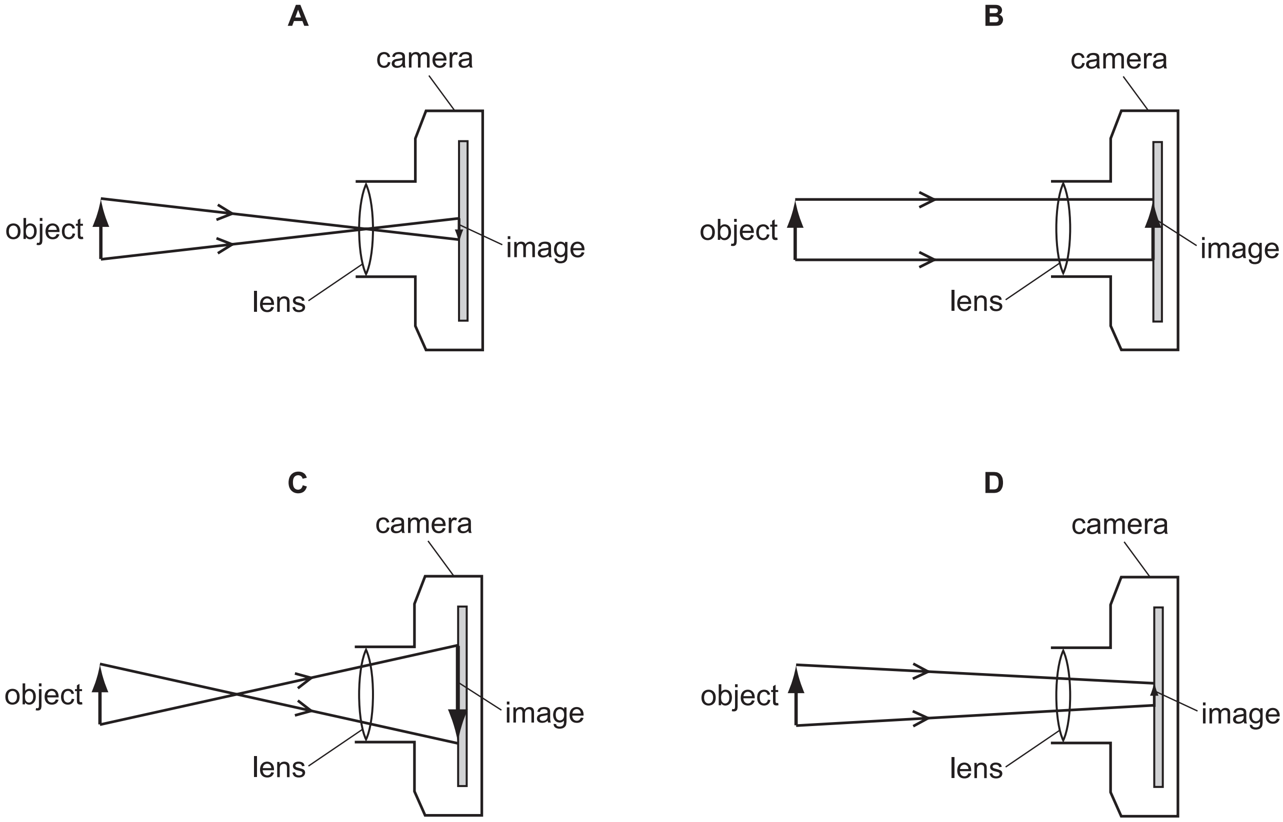
22 A ray of light is reflected by a plane mirror.



Which row shows the angle of incidence and the angle of reflection?

	angle of incidence	angle of reflection
A	P	Q
B	P	S
C	Q	R
D	R	S

23 Which diagram correctly represents rays of light passing through a converging lens in a camera?



24 Which statement about microwaves is correct?

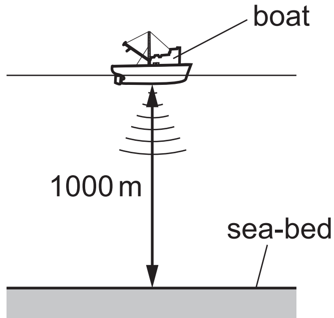
A Microwaves are longitudinal waves.

B The frequencies of microwaves are greater than the frequencies of visible light.

C The speed of microwaves in a vacuum is equal to the speed of visible light in a vacuum.

D The wavelengths of microwaves are smaller than the wavelengths of infra-red.

25 A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from the sea-bed. The sound reaches the boat again after 1.3 s. The sea-bed is 1000 m below the boat.



Using this information, what is the speed of sound in the water?

A 770 m/s **B** 1300 m/s **C** 1500 m/s **D** 2600 m/s

26 Which action will **not** magnetise a rod?

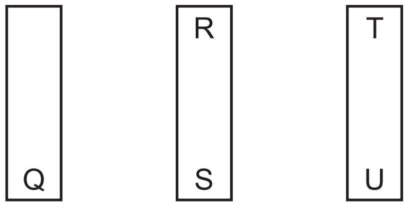
A placing a copper rod inside a coil carrying a direct current

B stroking a steel rod with a permanent magnet

C hammering a steel rod aligned with the Earth's magnetic field

D placing a soft-iron rod close to a permanent magnet

27 The ends of three metal rods are tested by holding end Q of rod 1 close to the others in turn.



rod 1 rod 2 rod 3

The results are as follows.

End Q attracts end R.

End Q attracts end S.

End Q attracts end T.

End Q repels end U.

Which of the metal rods is a magnet?

A rod 1 only

B rod 1 and rod 2

C rod 1 and rod 3

D rod 3 only

28 Which statement about a voltmeter is correct?

A A voltmeter measures the current in a component and is connected in series with the component.

B A voltmeter measures the current in a component and is connected in parallel with the component.

C A voltmeter measures the potential difference (p.d.) across a component and is connected in series with the component.

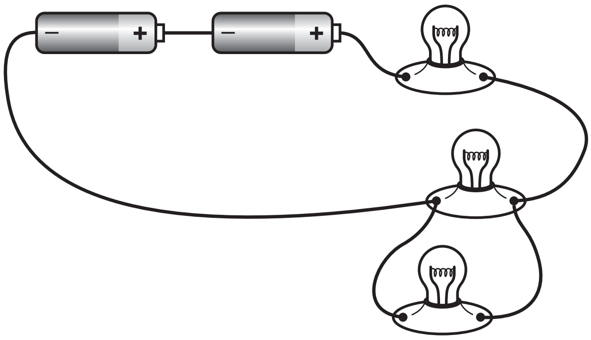
D A voltmeter measures the potential difference (p.d.) across a component and is connected in parallel with the component.

29 Four wires are made from the same metal.

Which wire has the lowest resistance?

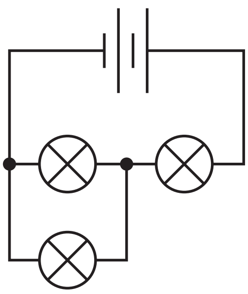
	length of wire / cm	diameter of wire / mm
A	20	0.20
B	20	0.40
C	40	0.20
D	40	0.40

30 A student sets up a circuit containing a battery of two cells and three lamps, as shown.

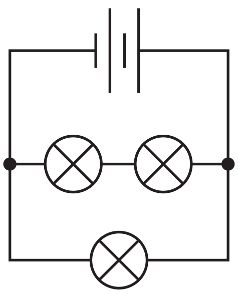


What is the circuit diagram for this arrangement?

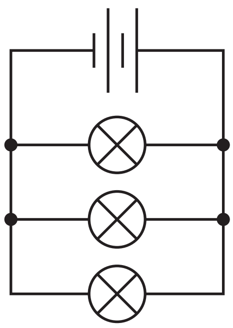
A



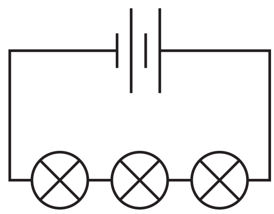
B



C



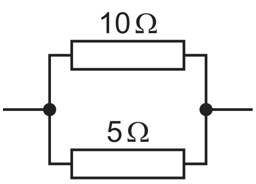
D



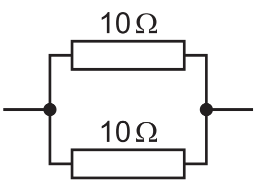
31 The diagrams show different resistor arrangements.

Which arrangement has the smallest combined resistance?

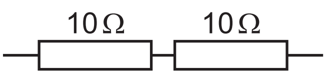
A



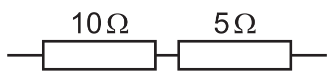
B



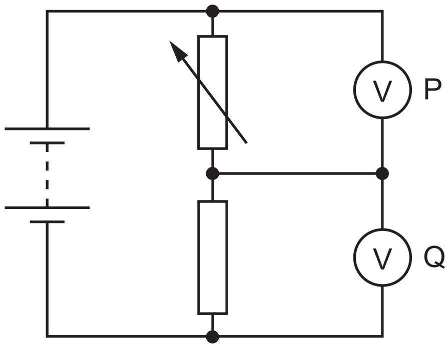
C



D



32 The diagram shows a potential divider circuit.



The resistance of the variable resistor is increased.

Which row shows what happens to the reading on voltmeter P and on voltmeter Q?

	reading on voltmeter P	reading on voltmeter Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

33 The current in a kettle is 10 A and the kettle is protected by a 13 A fuse.

The owner of the kettle replaces the 13 A fuse with a 3 A fuse.

What happens when the kettle is switched on?

A The fuse melts and the kettle might be damaged.

B The fuse melts and the kettle is undamaged.

C The fuse does not melt and the kettle works correctly.

D The fuse does not melt but the kettle fails to work.

34 A student wishes to demonstrate electromagnetic induction.

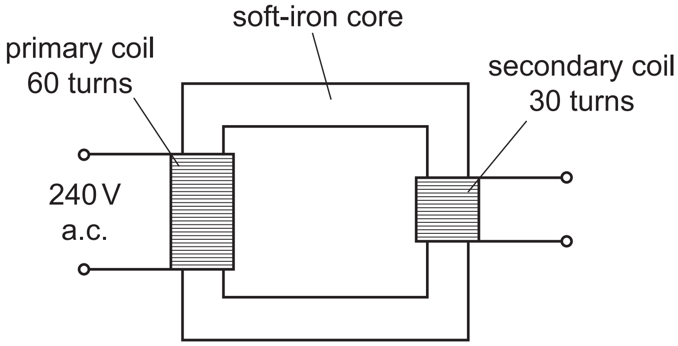
He has a magnet and connecting wires.

Which other apparatus does he need?

	voltmeter	battery
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key
✓ = needed
✗ = not needed

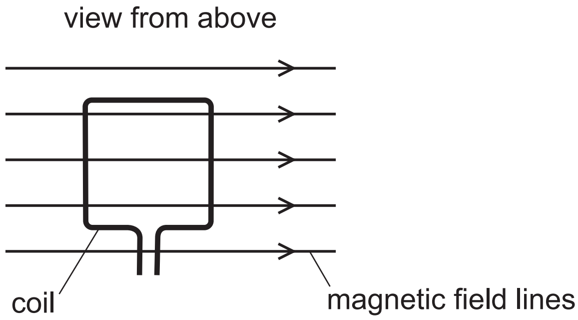
35 The diagram shows a transformer connected to a 240 V a.c. supply.



What is the potential difference across the secondary coil of the transformer?

A 30 V **B** 120 V **C** 240 V **D** 480 V

36 A current-carrying coil is placed in a magnetic field.



Which effect does the coil experience?

A a change in shape

B a change in weight

C a resultant force

D a turning effect

37 Which statement about the structure of an atom is correct?

A Negative electrons surround a neutral nucleus.

B Negative electrons surround a positive nucleus.

C Positive electrons surround a neutral nucleus.

D Positive electrons surround a negative nucleus.

38 What are isotopes of an element?

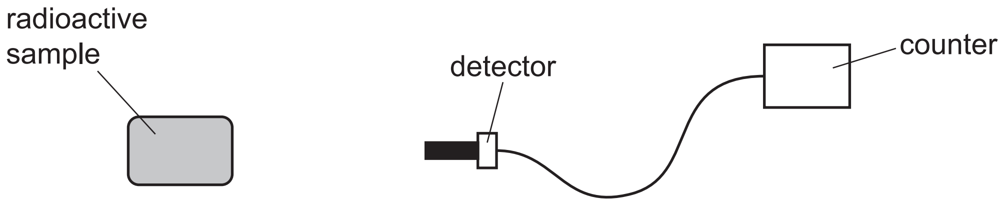
A atoms of a different element with a different number of neutrons

B atoms of a different element with a different number of protons

C atoms of the same element with a different number of neutrons

D atoms of the same element with a different number of protons

39 A student measures the level of radiation emitted by a radioactive sample.



The table shows the readings she records on the counter over a short period of time.

counter reading / counts per minute	106	96	98	100

The sample is removed and the counter then shows a background radiation reading of 4 counts per minute.

What is the best estimate for the average count rate due to the radioactive sample?

A 96 counts per minute

B 98 counts per minute

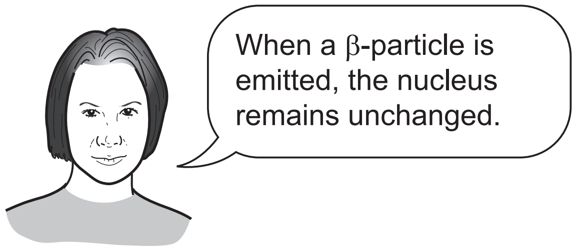
C 100 counts per minute

D 104 counts per minute

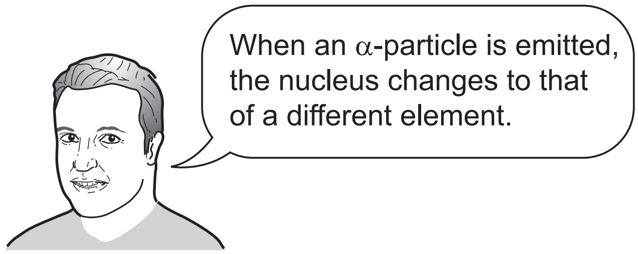
40 Four students are discussing ideas about radioactive decay.

Which student's statement is correct?

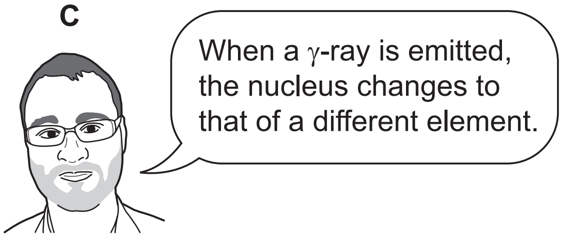
A



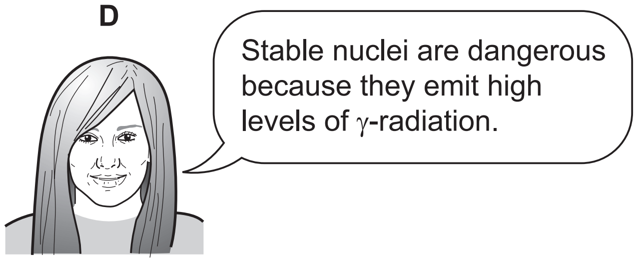
B



C



D



Cambridge IGCSE

Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **March 2019**

MARK SCHEME

Maximum Mark: 40

Published

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Cambridge Assessment International Education

Question	Answer	Marks
1	B	**1**
2	C	**1**
3	C	**1**
4	A	**1**
5	B	**1**
6	D	**1**
7	A	**1**
8	C	**1**
9	D	**1**
10	B	**1**
11	D	**1**
12	D	**1**
13	B	**1**
14	A	**1**
15	B	**1**
16	C	**1**
17	D	**1**
18	D	**1**
19	B	**1**
20	C	**1**
21	B	**1**
22	C	**1**
23	A	**1**
24	C	**1**
25	C	**1**
26	A	**1**
27	C	**1**
28	D	**1**

Question	Answer	Marks
29	B	**1**
30	A	**1**
31	A	**1**
32	C	**1**
33	B	**1**
34	B	**1**
35	B	**1**
36	D	**1**
37	B	**1**
38	C	**1**
39	A	**1**
40	B	**1**

Cambridge IGCSE

Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

* 8 2 0 8 2 2 2 0 4 7 *

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2018**

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Electronic calculators may be used.
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m / s^2).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

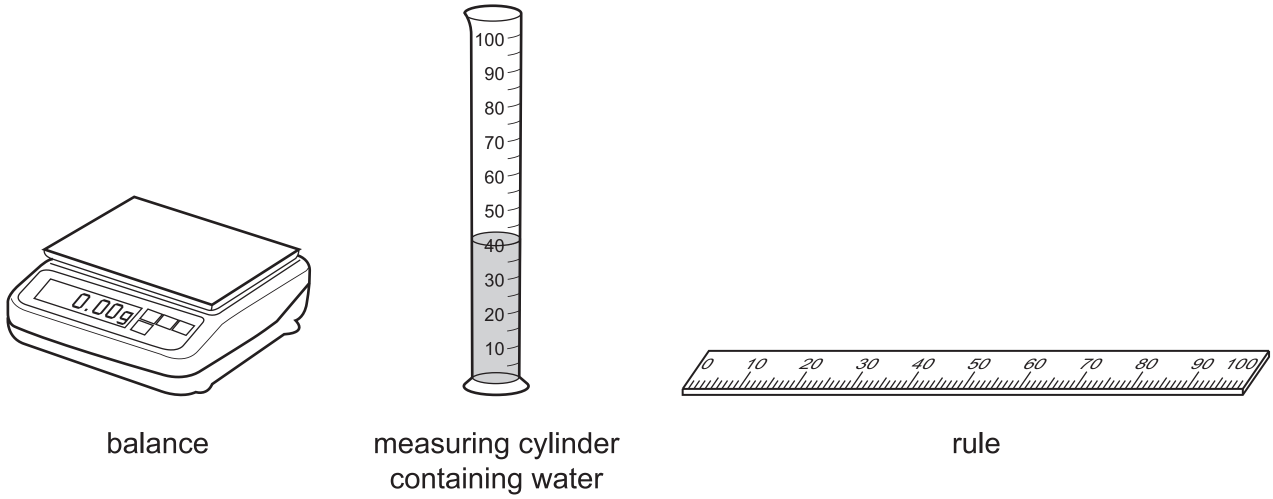
This document consists of **18** printed pages and **2** blank pages.

IB18 03_0625_12/3RP

CAMBRIDGE
International Examinations

1 A student is asked to find the volume of a small irregularly-shaped piece of rock.

He has the following apparatus available.



Which apparatus must the student use to find the volume of the small piece of rock?

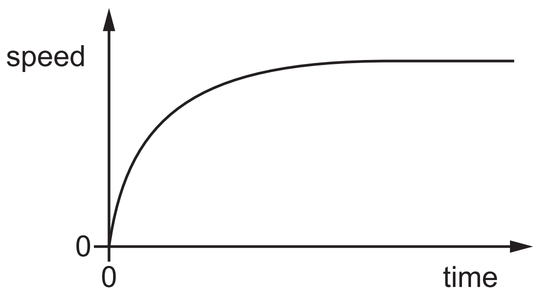
A balance and rule

B rule only

C balance and measuring cylinder

D measuring cylinder only

2 The speed-time graph represents the motion of a car travelling along a straight level road.



Which statement describes the motion of the car?

A It accelerates and reaches a constant speed.

B It accelerates and then stops moving.

C It decelerates and then reaches a constant speed.

D It decelerates and then stops moving.

3 An athlete runs 300 metres up a hill at a steady speed of 3.0 m/s.

She then immediately runs the same distance down the hill at a steady speed of 6.0 m/s.

What is her average speed for the 600 metre run?

A 2.0 m/s **B** 3.0 m/s **C** 4.0 m/s **D** 4.5 m/s

4 Which statement is correct?

A Mass is a gravitational force but weight is not a gravitational force.

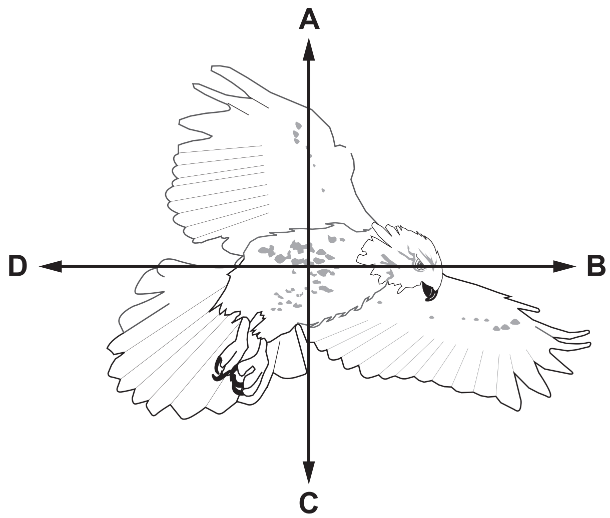
B Mass is not a gravitational force and weight is not a gravitational force.

C The unit of mass is the kilogram and the unit of weight is the newton.

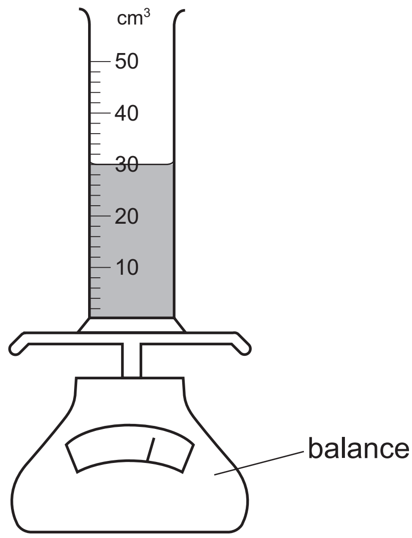
D The unit of mass is the newton and the unit of weight is the kilogram.

5 The diagram shows a bird in flight. The bird is flying in a horizontal direction to the right.

In which direction does air resistance act on the bird?



6 A measuring cylinder contains 30 cm^3 of a liquid.



Some more of the liquid is added until the liquid level reaches the 50 cm^3 mark.

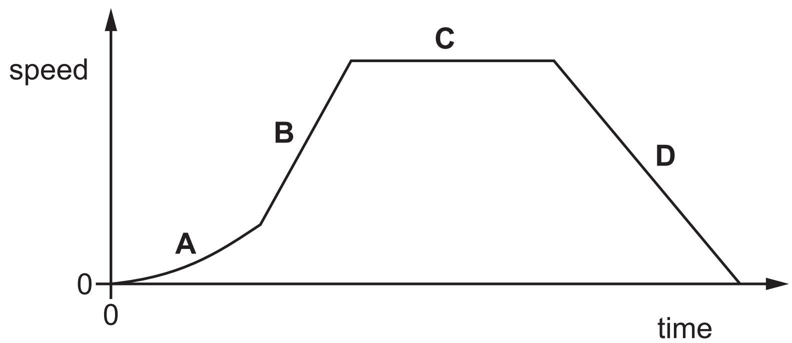
The reading on the balance increases by 30 g.

What is the density of the liquid?

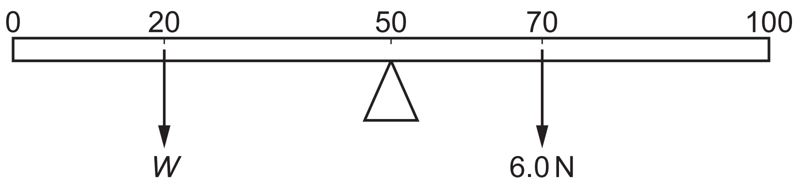
A 0.60 g / cm^3 **B** 0.67 g / cm^3 **C** 1.5 g / cm^3 **D** 1.7 g / cm^3

7 A car is travelling along a straight horizontal road. The speed-time graph is shown.

In which labelled part of the journey is the resultant force on the car zero?



8 The diagram shows a uniform metre rule balanced at its mid-point.

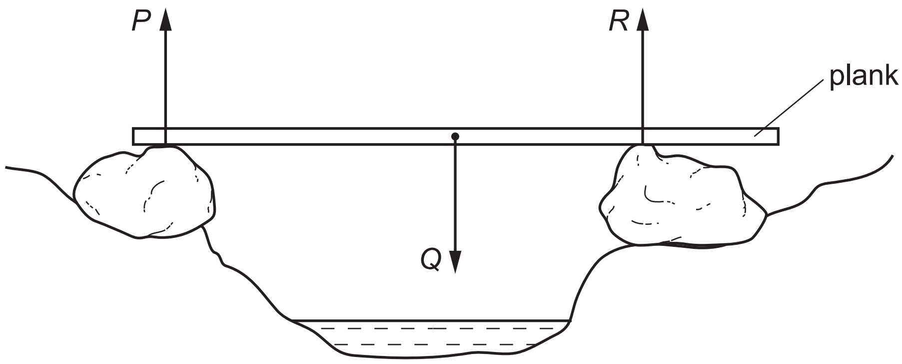


What is the weight W placed at the 20 cm mark?

A 0.25 N **B** 4.0 N **C** 9.0 N **D** 21 N

9 A wooden plank rests in equilibrium on two rocks on opposite sides of a narrow stream.

Three forces *P*, *Q* and *R* act on the plank.



How are the sizes of the forces related?

A $P + Q = R$

B $P + R = Q$

C $P = Q = R$

D $P = Q + R$

10 A ball is at rest at the top of a hill. It rolls down the hill. At the bottom of the hill the ball hits a wall and stops.

Which energy changes occur?

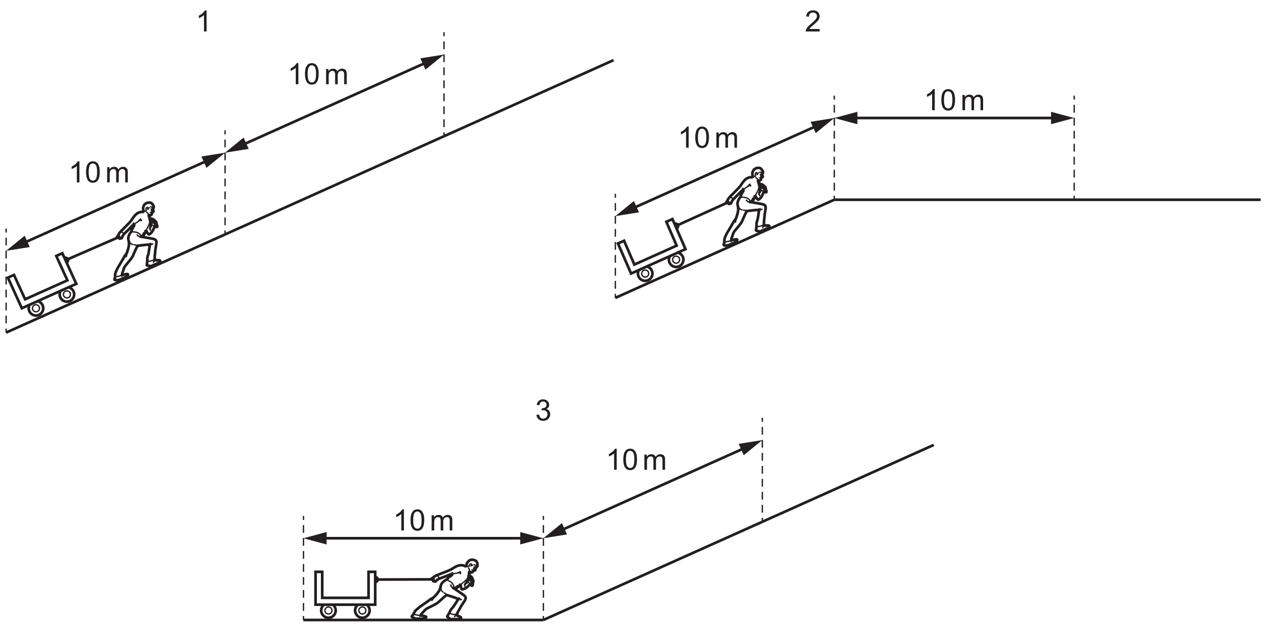
A gravitational potential energy → internal energy → kinetic energy

B gravitational potential energy → kinetic energy → internal energy

C kinetic energy → gravitational potential energy → internal energy

D kinetic energy → internal energy → gravitational potential energy

11 A man pulls a truck a distance of 20 m, as shown in the three diagrams.



Which statement concerning the work done against gravity is correct?

A Most work is done in 1.

B No work is done in 3.

C The work done in all three is equal.

D The work done in 1 and 2 is equal.

12 Which source of energy is used in a nuclear power station to generate electrical energy?

A different types of atom regrouping

B heavy nuclei splitting

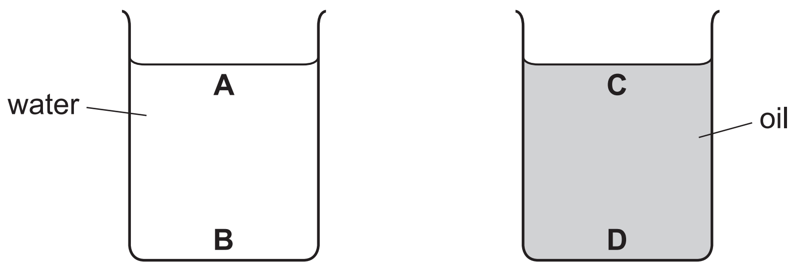
C radioactive isotopes decaying

D radioactive atoms emitting β-particles

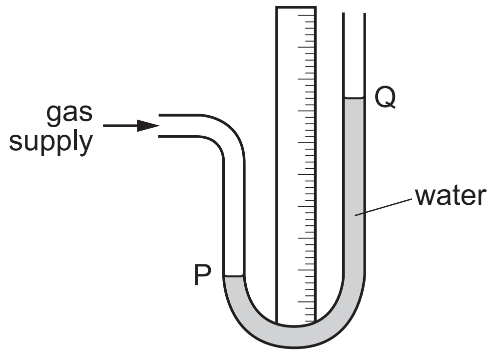
13 Two beakers are filled to the same depth, one with water and one with oil.

The density of water is 1000 kg / m^3 and the density of oil is 920 kg / m^3.

In which position is the pressure the greatest?



14 A water manometer is connected to a gas supply.

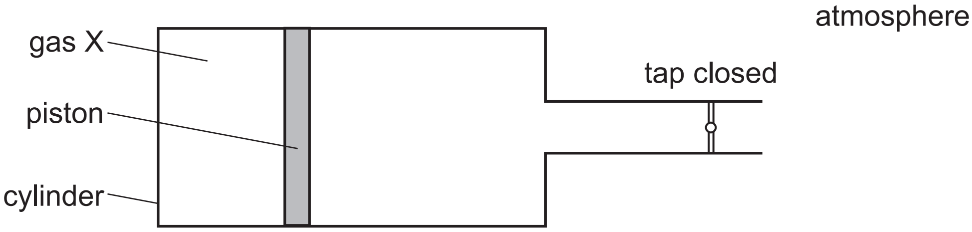


There is a gas leak and the pressure of the gas supply falls.

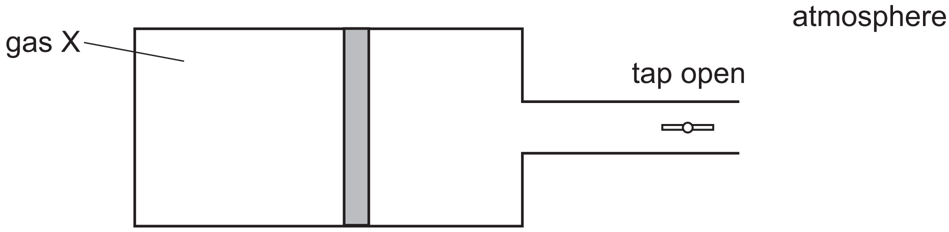
What happens to the water level at P and what happens to the water level at Q?

	water level at P	water level at Q
A	falls	falls
B	falls	rises
C	rises	falls
D	rises	rises

15 A cylinder with a tap contains a fixed mass of gas X. The gas is contained by a piston which can move freely towards or away from the tap.



When the tap is opened, the piston moves slightly to the right, towards the tap.



What can be deduced about the pressure of gas X?

	before opening tap	after opening tap
A	less than atmospheric pressure	more than atmospheric pressure
B	same as atmospheric pressure	more than atmospheric pressure
C	more than atmospheric pressure	less than atmospheric pressure
D	more than atmospheric pressure	same as atmospheric pressure

16 Which quantity does **not** change when there is an increase in temperature?

A the density of a steel block

B the diameter of the hole in a metal nut

C the length of an iron rod

D the mass of a metal coin

17 A thermometer has a low thermal capacity.

Why is this an advantage?

A The thermometer does not absorb much thermal energy to raise its own temperature.

B The thermometer does not conduct much thermal energy to the surroundings.

C The thermometer does not melt when it gets hot.

D The thermometer does not radiate much thermal energy to the surroundings.

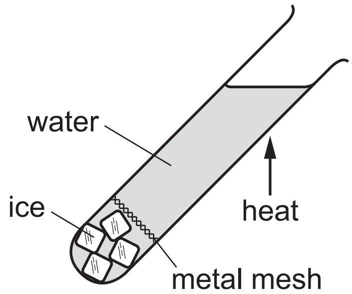
18 In an experiment, a thermometer is placed in a test-tube of hot liquid. The temperature reading of the liquid is recorded every half minute. The table shows the results.

time / minutes	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5
temperature / °C	73	65	59	55	55	55	51	48	45	42	40	38	36	35	34	33

What is the melting point of the substance?

A 0 °C **B** 33 °C **C** 55 °C **D** 73 °C

19 A teacher demonstrates an experiment to a class. A boiling tube is filled with water and some ice cubes are trapped at the bottom of the tube. The teacher then heats the boiling tube in the position shown until the water at the top boils.



The ice does not melt.

What does this demonstrate?

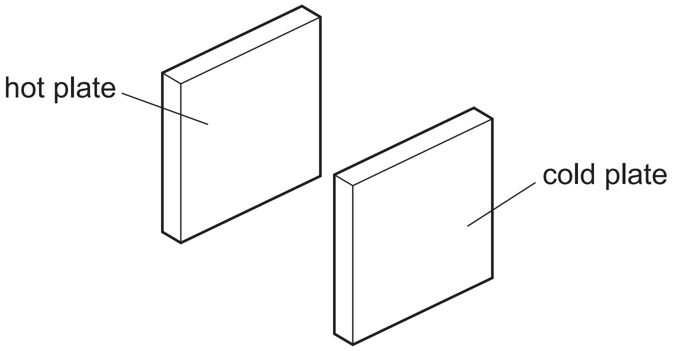
A Water is a good conductor of thermal energy.

B Water is a good convector of thermal energy.

C Water is a poor conductor of thermal energy.

D Water is a poor convector of thermal energy.

20 A hot metal plate is placed near to a cold metal plate. Infra-red radiation transfers thermal energy between the two plates.



Which choice of colour causes the temperature of the cold plate to increase most slowly?

	hot plate	cold plate
A	matt black	matt black
B	matt black	shiny white
C	shiny white	matt black
D	shiny white	shiny white

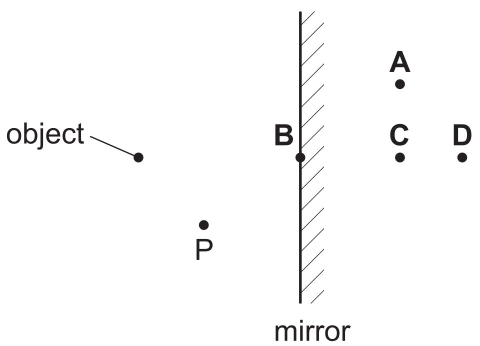
21 Which row shows an example of a transverse wave and an example of a longitudinal wave?

	transverse	longitudinal
A	light	radio
B	radio	sound
C	sound	water
D	water	light

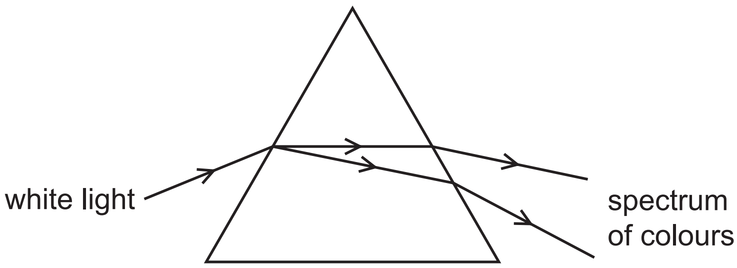
22 An object is placed before a plane mirror as shown.

A student views the image of the object in the mirror from point P.

Where does she see the image?



23 Light passes through a prism as shown. The light changes direction and produces a spectrum of colours.



Which term is used to describe the production of the spectrum of colours?

A diffraction

B dispersion

C refraction

D total internal reflection

24 Which type of waves are produced by a television remote controller?

A infra-red waves

B radio waves

C ultraviolet waves

D visible light

25 Both the amplitude and the frequency of a sound wave decrease.

What happens to the sound heard?

A The sound is louder and has a higher pitch.

B The sound is louder and has a lower pitch.

C The sound is quieter and has a higher pitch.

D The sound is quieter and has a lower pitch.

26 An electromagnet is used to remove a splinter from an eye.

What material is the splinter made from?

A aluminium

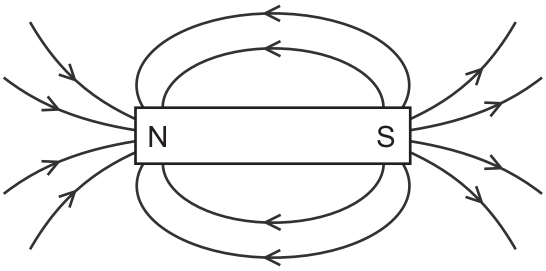
B glass

C iron

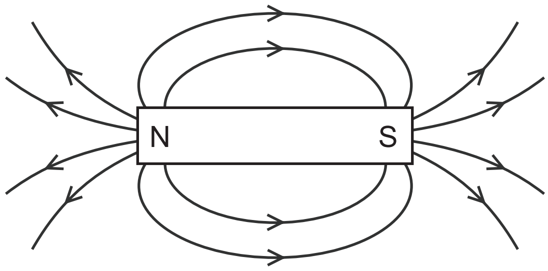
D wood

27 Which diagram shows the magnetic field around a bar magnet?

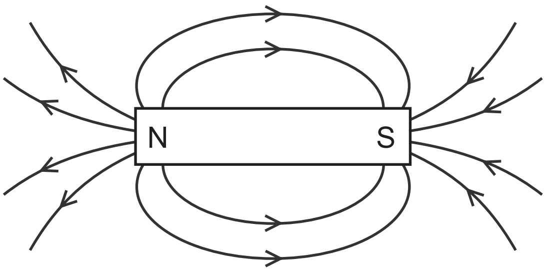
A



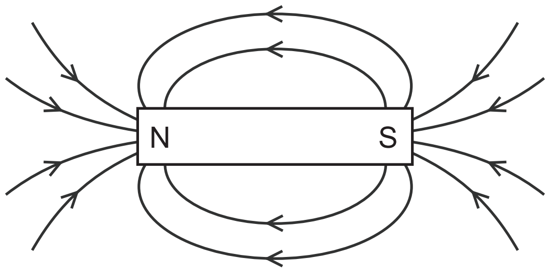
B



C



D



28 A metal conductor is connected to a battery.

Which statement describes the current in the metal conductor?

A It is a flow of electrons from the negative to the positive terminal.

B It is a flow of electrons from the positive to the negative terminal.

C It is a flow of protons from the negative to the positive terminal.

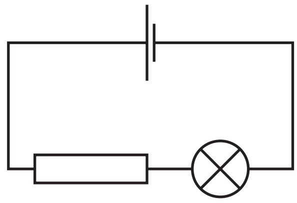
D It is a flow of protons from the positive to the negative terminal.

29 A student sets up the circuit shown to measure three quantities.

1 the current in the circuit

2 the electromotive force (e.m.f.) of the cell

3 the potential difference across the lamp



How many ammeters and how many voltmeters are needed?

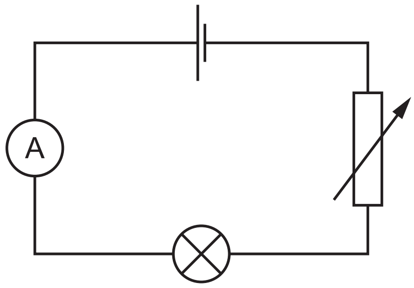
	ammeters	voltmeters
A	0	3
B	1	2
C	2	1
D	3	0

30 The current in a car headlamp is 3.0 A when connected to a 12 V battery.

What is the resistance of the lamp when it is lit?

A 0.25 Ω **B** 4.0 Ω **C** 15 Ω **D** 36 Ω

31 The diagram shows a circuit containing a cell, an ammeter, a lamp and a variable resistor.



The resistance of the variable resistor is increased.

What happens to the ammeter reading and what happens to the brightness of the lamp?

	ammeter reading	lamp brightness
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

32 Diagram 1 shows two identical lamps connected in series with a cell. Diagram 2 shows the same two lamps connected in parallel with an identical cell.

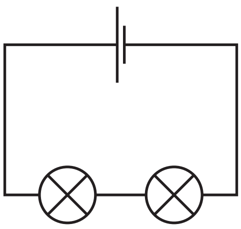


diagram 1

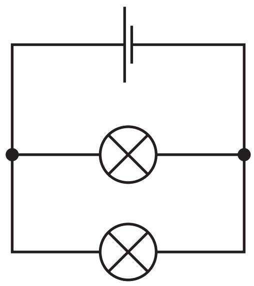


diagram 2

Which statement is correct?

A The cell in diagram 1 can supply current to the lamps for longer than the cell in diagram 2.

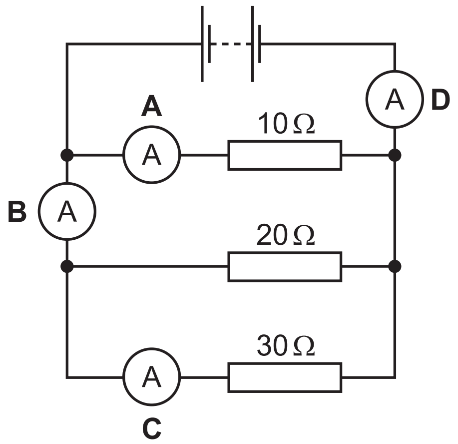
B The current in the cell in diagram 1 is greater than the current in the cell in diagram 2.

C The lamps in diagram 1 are brighter than the lamps in diagram 2.

D The voltage across each lamp in diagram 1 is greater than the voltage in diagram 2.

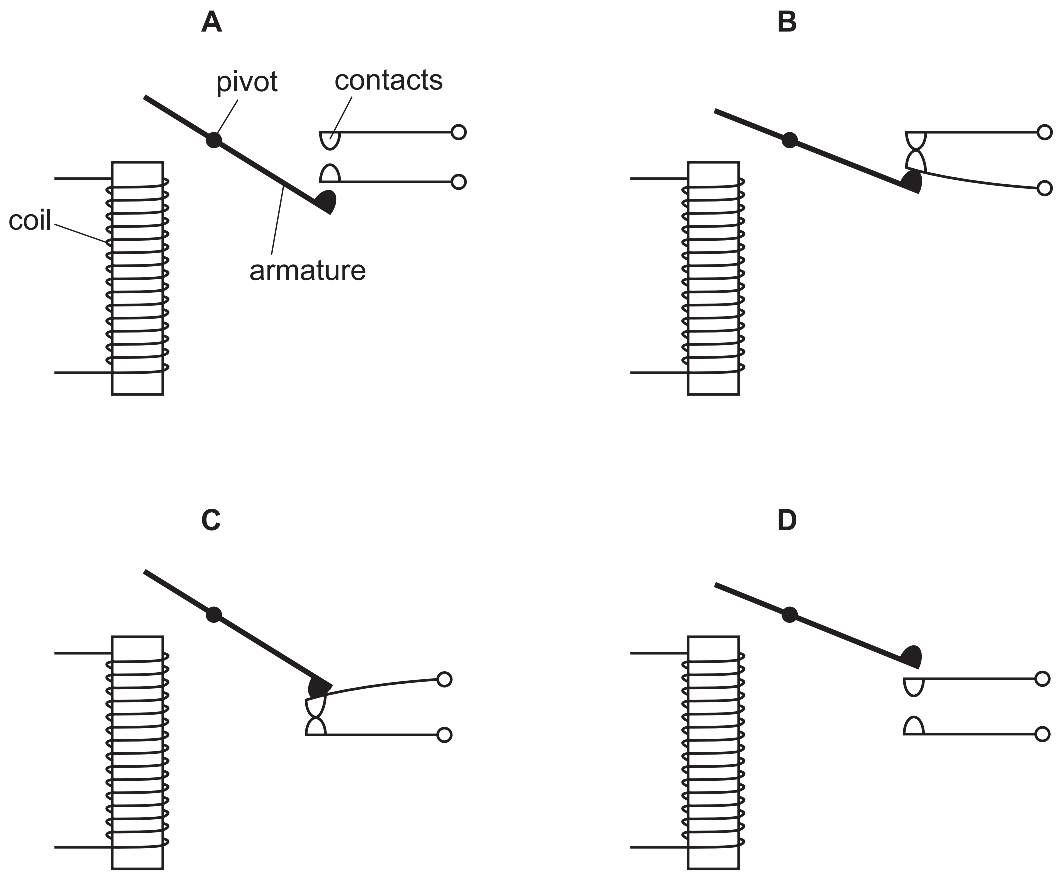
33 A circuit contains four ammeters and three resistors with different values.

Which ammeter shows the largest reading?

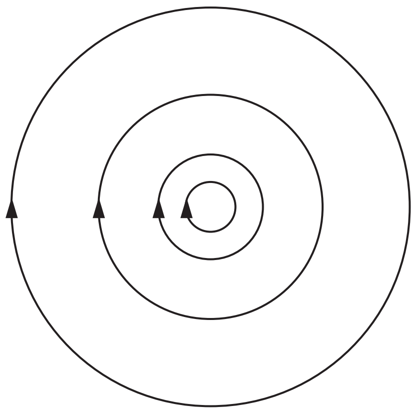


34 The diagram shows the arrangement of the coil, armature and contacts of a relay.

Which diagram shows this arrangement when there is no current in the coil?



35 The diagram represents a magnetic field.



What causes this field?

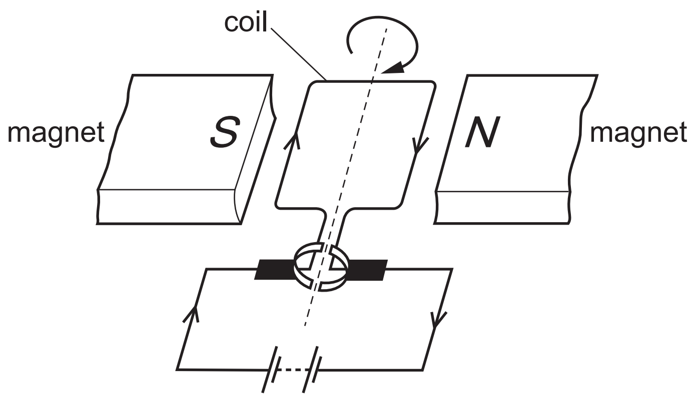
A a bar magnet

B a solenoid carrying a current

C a straight wire carrying a current

D two north poles close together

36 The diagram shows a simple d.c. electric motor which is rotating.



Which change makes the motor rotate more quickly?

A increasing the number of turns on the coil

B removing the magnets

C reversing the battery

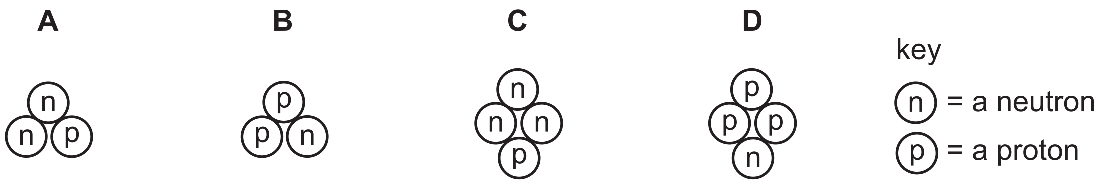
D reversing the polarity of the magnets

37 The notation for an isotope of sodium is $^{23}_{11}\text{Na}$.

Which row gives the composition of a neutral atom of this isotope of sodium?

	number of protons	number of neutrons	number of electrons
A	11	12	11
B	11	12	12
C	11	23	11
D	12	11	12

38 Which diagram represents a nucleus of $^{3}_{1}\text{H}$?



39 When measuring the emissions from a radioactive rock brought into the laboratory, a teacher mentions that background radiation must be taken into account.

What is this background radiation?

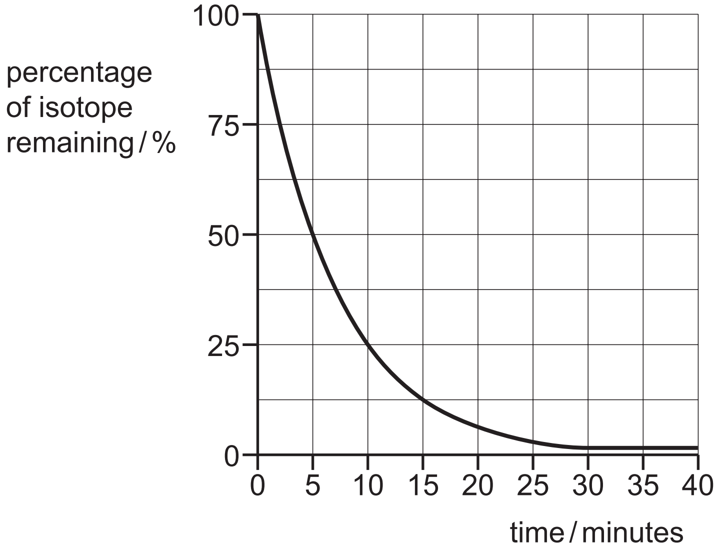
A infra-red radiation from warm objects in the laboratory

B infra-red radiation from the Sun

C ionising radiation from the radioactive rock brought into the laboratory

D ionising radiation in the laboratory when the radioactive rock is not present

40 The graph shows the percentage of a radioactive isotope that remains as time passes.



A student measures the initial rate of emission of this isotope as 40 000 emissions per second.

How long does it take for the rate of emission to fall to 5000 emissions per second?

A 5 minutes **B** 15 minutes **C** 20 minutes **D** 40 minutes

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Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **March 2018**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	D	**1**
2	A	**1**
3	C	**1**
4	C	**1**
5	D	**1**
6	C	**1**
7	C	**1**
8	B	**1**
9	B	**1**
10	B	**1**
11	A	**1**
12	B	**1**
13	B	**1**
14	C	**1**
15	D	**1**
16	D	**1**
17	A	**1**
18	C	**1**
19	C	**1**
20	D	**1**
21	B	**1**
22	D	**1**
23	B	**1**
24	A	**1**
25	D	**1**
26	C	**1**
27	C	**1**
28	A	**1**

Question	Answer	Marks
29	B	**1**
30	B	**1**
31	A	**1**
32	A	**1**
33	D	**1**
34	A	**1**
35	C	**1**
36	A	**1**
37	A	**1**
38	A	**1**
39	D	**1**
40	B	**1**

Cambridge IGCSE

Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **February/March 2017**

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

9265727117

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

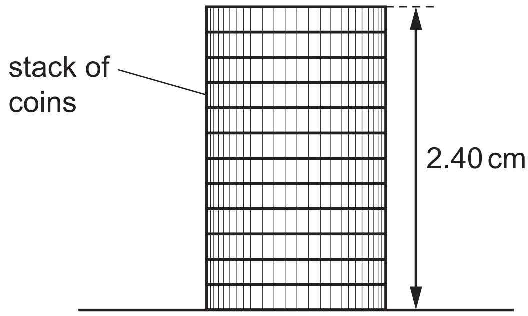
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Electronic calculators may be used.
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m / s^2).

This document consists of **19** printed pages and **1** blank page.

IB17 03_0625_12/3RP

CAMBRIDGE International Examinations

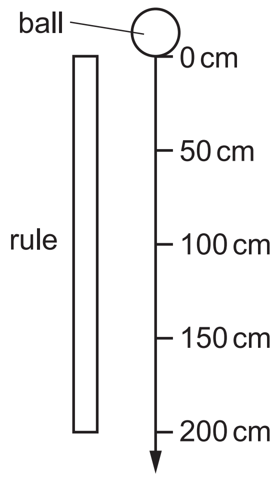
1 The diagram shows the height of a stack of identical coins.



What is the thickness of one coin?

A 0.20 mm **B** 2.0 mm **C** 0.24 cm **D** 2.0 cm

2 In a laboratory, a ball is dropped in a vacuum and falls 200 cm.



Which statement describes the acceleration of the ball?

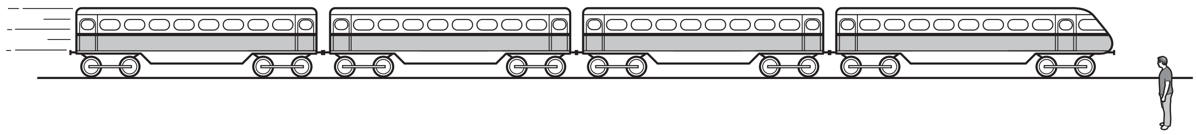
A It is greater at 10 cm than at 200 cm.

B It is greatest at 200 cm.

C It is smaller at 50 cm than at 100 cm.

D It is the same value at 50 cm as at 150 cm.

3 A man stands by a railway track.



A train travelling at 40 m / s takes 2.0 s to pass the man.

What is the length of the train?

A 20 m **B** 38 m **C** 40 m **D** 80 m

4 Which statement about the masses and weights of objects on the Earth is correct?

A A balance can only be used to compare weights, not masses.

B Heavy objects always have more mass than light ones.

C Large objects always have more mass than small ones.

D Mass is a force but weight is not.

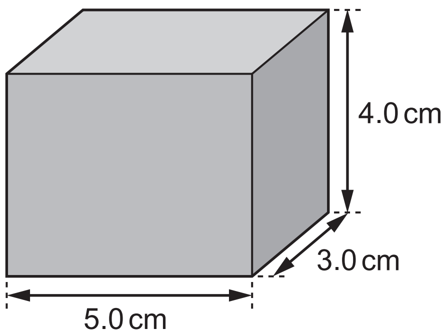
5 A stone has a weight of 5.7 N.

The gravitational field strength *g* is 10 N / kg.

What is the mass of the stone?

A 0.57 kg **B** 5.7 kg **C** 57 kg **D** 570 kg

6 The block of metal shown has a mass of 240 g.



What is the density of the metal?

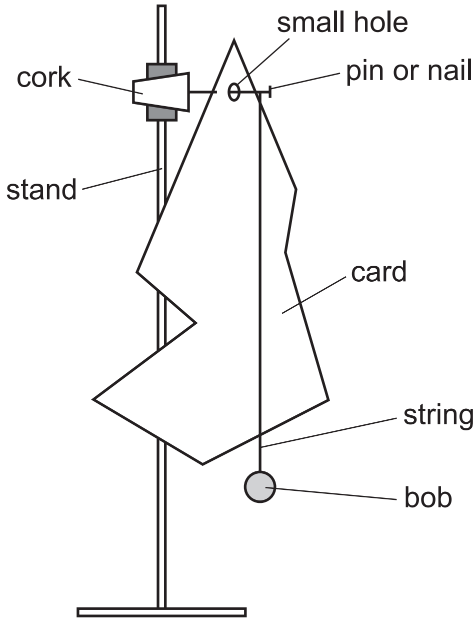
A 0.25 g / cm^3 **B** 4.0 g / cm^3 **C** 16 g / cm^3 **D** 14 400 g / cm^3

7 A car engine causes a forward force of 100 kN to act on the car. The total resistive force on the car is 20 kN.

What is the resultant force on the car?

A 5.0 kN **B** 60 kN **C** 80 kN **D** 120 kN

8 A student sets up the apparatus shown in the diagram to find the centre of mass of the card.



The student makes sure that the card, the string and the bob are all at rest.

What should the student do next?

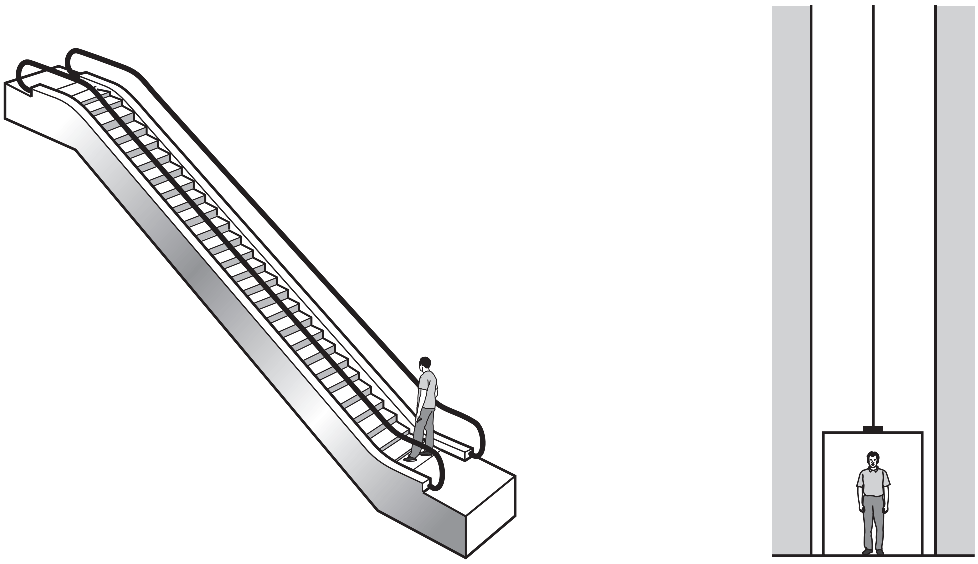
A Mark a horizontal line on the card level with the middle of the string.

B Mark the line of the string on the card.

C Pull the bob on the string to one side and release it.

D Replace the bob with a heavier bob.

9 A man can either take an escalator or a lift to travel up between two floors in a hotel.



The escalator takes 20 seconds to carry the man between the two floors. The useful work done against gravity is *W*. The useful power developed is *P*.

The lift takes 30 seconds to carry the same man between the same two floors.

How much useful work against gravity is done by the lift, and how much useful power is developed by the lift?

	useful work done against gravity by lift	useful power developed by lift
A	more than *W*	less than *P*
B	more than *W*	*P*
C	*W*	less than *P*
D	*W*	*P*

10 The engine of a motor vehicle develops a large amount of power.

Which statement is correct?

A The driving force acting on the vehicle must be large.

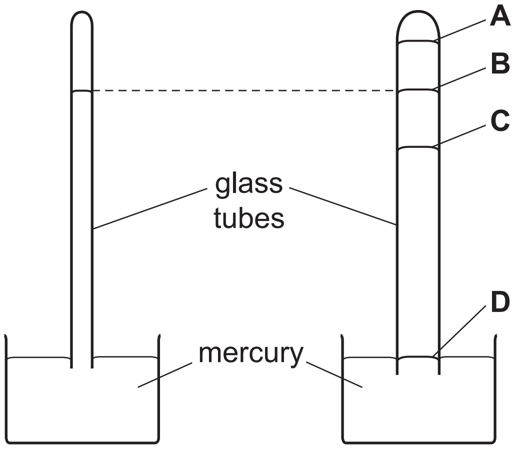
B The engine must have a very large volume.

C The engine must transfer large amounts of energy each second.

D The vehicle must be very fast.

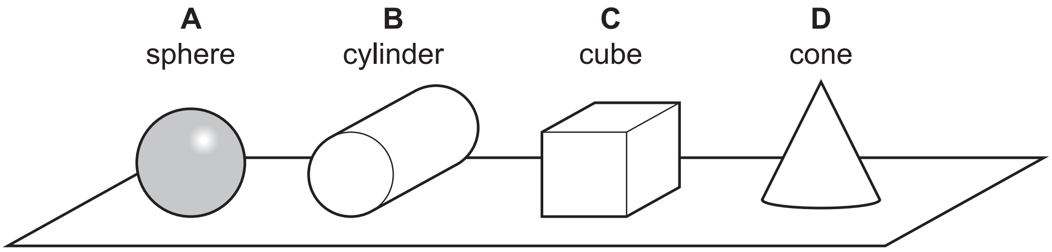
11 The diagrams show two mercury barometers. The right-hand diagram shows a tube of larger diameter. There is a vacuum above the mercury in both tubes.

Which labelled position on the right-hand tube shows the mercury level in the right-hand tube?



12 The diagram shows four solid objects resting on a horizontal surface. The objects all have the **same** weight, and are drawn to the same scale.

Which object exerts the least pressure on the surface?



13 On a hot day, the pressure of the air in a car tyre is greater than on a cold day.

Why is the pressure greater on a hot day?

A The air molecules strike each other more frequently.

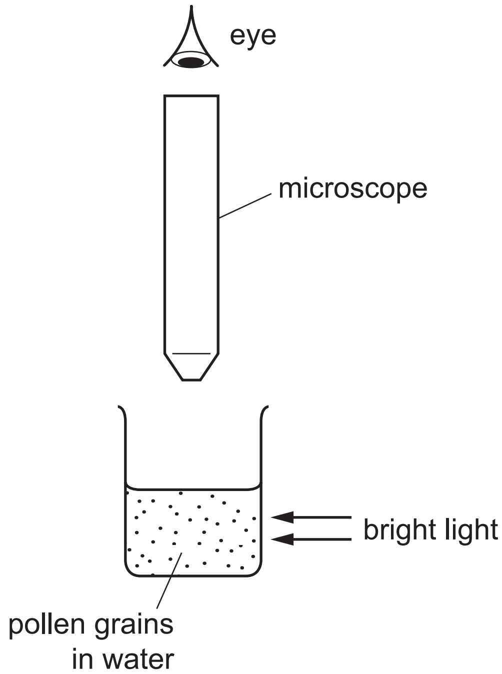
B The air molecules strike each other with greater force.

C The air molecules strike the tyre walls more frequently.

D The number of air molecules in the tyre increases.

14 Very small pollen grains are suspended in water. A bright light shines from the side.

When looked at through a microscope, small specks of light are seen to be moving in a random, jerky manner.



What are the moving specks of light?

A pollen grains being hit by other pollen grains

B pollen grains being hit by water molecules

C water molecules being hit by other water molecules

D water molecules being hit by pollen grains

15 When a thermometer is calibrated, the fixed points are marked.

What are fixed points?

A all the marks on the temperature scale which cannot be removed

B all the marks of the temperature scale

C the lowest and highest temperatures shown on the thermometer

D two temperatures of known value which are easily reproduced

16 Which statement defines the thermal capacity (heat capacity) of a solid body?

A the energy needed to melt the body without a change in temperature

B the energy needed to raise the temperature of the body by one degree Celsius

C the increase in the volume of the body when its temperature is raised by one degree Celsius

D the total amount of internal energy in the body

17 A substance can exist in three different states: solid, liquid or gas.

Each of the two statements below describes a change of state.

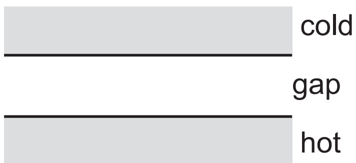
change 1 Molecules move closer together but continue to travel throughout the substance.

change 2 Molecules stop travelling throughout the substance and just vibrate about fixed positions.

Which changes of state do these statements describe?

	change 1	change 2
A	condensation	melting
B	condensation	solidification
C	solidification	condensation
D	solidification	melting

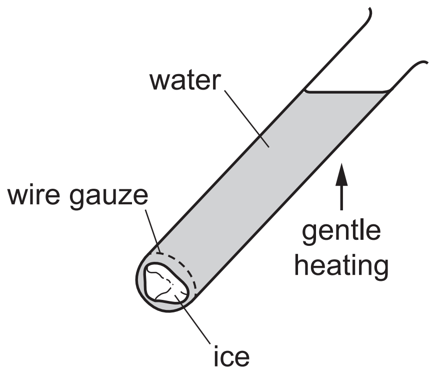
18 The diagram shows the gap between a hot surface and a cold surface. The gap can contain air, solid iron, a vacuum or liquid water.



Which row shows whether heat can be transferred between the surfaces by conduction and convection?

		conduction	convection
A	air (gas)	yes	no
B	iron (solid)	yes	no
C	vacuum	no	yes
D	water (liquid)	yes	no

19 Ice is trapped by a piece of wire gauze at the bottom of a tube containing water. The water at the top of the tube boils before the ice at the bottom of the tube melts.



Why does this happen?

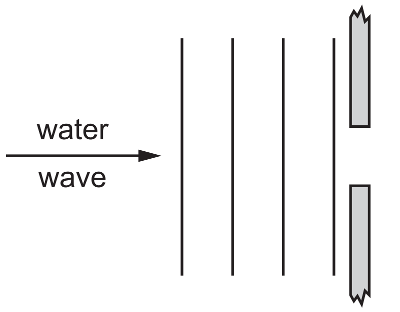
A Convection currents are circulating throughout the water.

B Ice is a poor emitter of thermal energy.

C Water is a poor conductor of thermal energy.

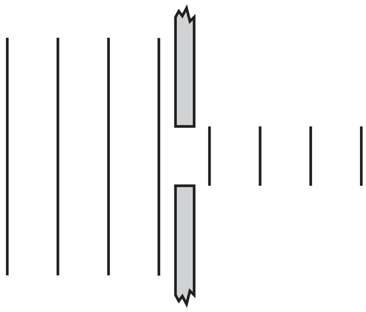
D Wire gauze is a good conductor of thermal energy.

20 In a shallow tank, a water wave moves towards a barrier with a narrow gap.



Which diagram shows the wave beyond the barrier?

A



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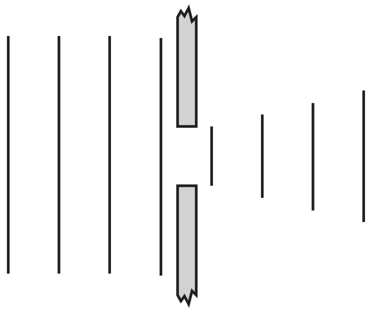

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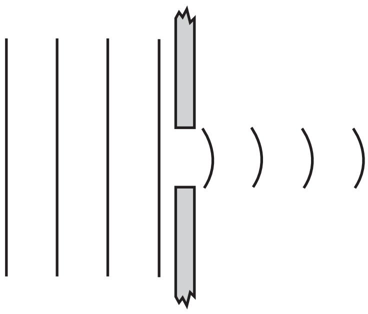
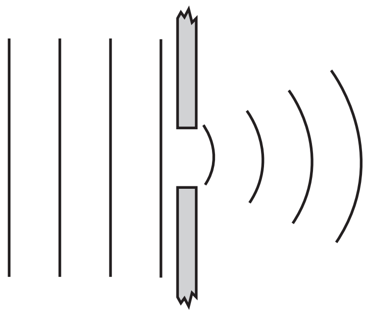
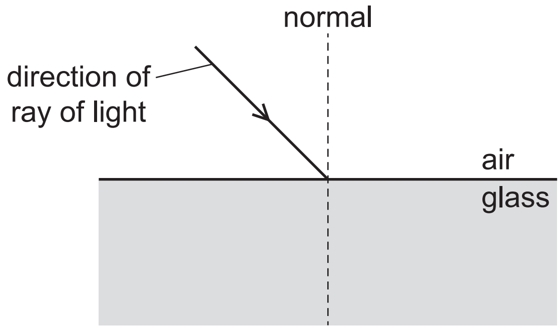


D



21 A ray of light is incident on the surface of a glass block.

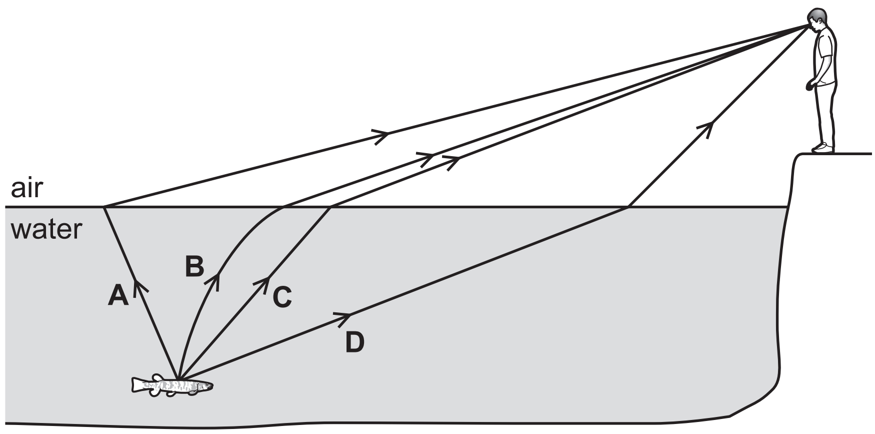


Which row describes how the speed and the direction of the ray of light change when it enters the glass?

	speed in glass	direction in glass
A	decreases	closer to the normal
B	decreases	further from the normal
C	increases	closer to the normal
D	increases	further from the normal

22 A boy sees a fish in a lake.

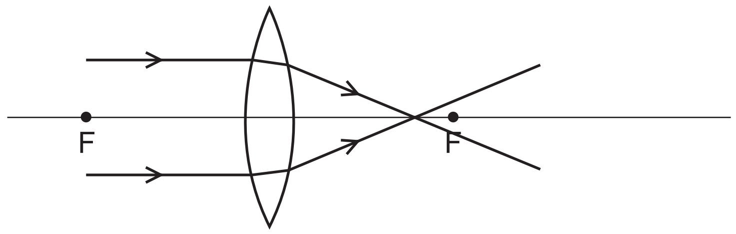
Which labelled path is taken by the light travelling from the fish to the boy's eye?



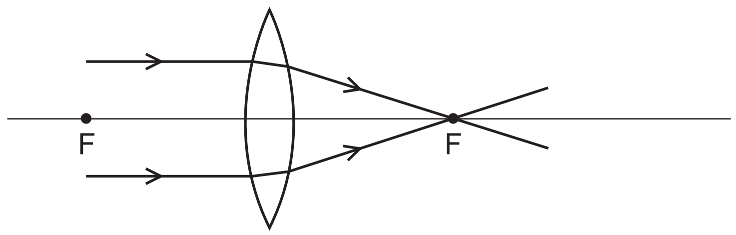
23 The points labelled F are the principal foci of a lens. A beam of parallel light is incident on the lens.

Which diagram shows the path of the light after it passes through the lens?

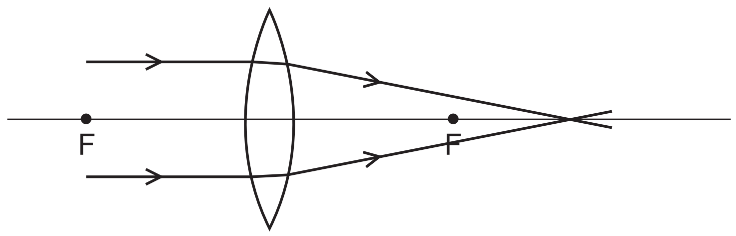
A



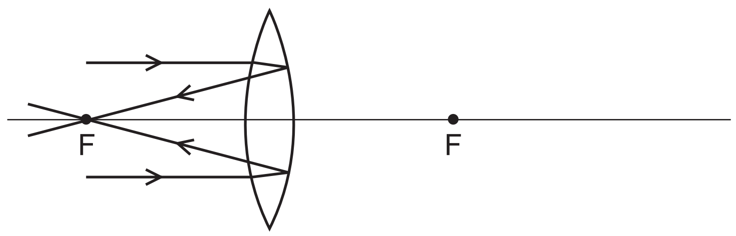
B



C



D



24 What is the main type of radiation given out by a campfire?

A infra-red

B microwave

C ultraviolet

D X-ray

25 A student stands a few hundred metres away from a wall and shouts. He hears a faint echo.

Which statement is correct?

A The sound waves returning are quiet because they have a reduced frequency.

B The sound waves returning are quiet because they have a reduced wavelength.

C The sound waves returning to the student are longitudinal.

D The sound waves returning to the student are transverse.

26 Which statement about ultrasound is correct?

A It is produced by a rapidly vibrating source.

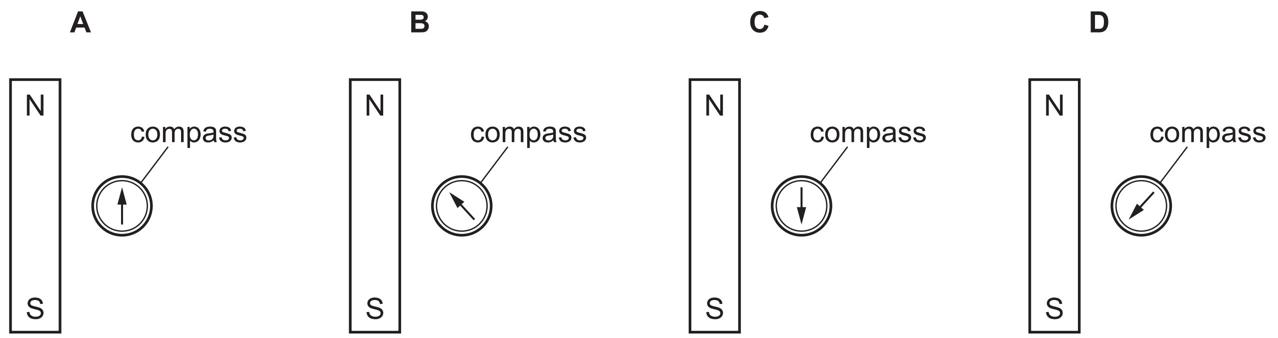
B It is uncomfortable to human ears.

C Its frequency must be greater than 300 kHz.

D It travels the fastest in a vacuum.

27 A small compass is placed close to a strong bar magnet, the same distance from each end.

Which diagram shows the direction in which the compass needle points?



28 A student wishes to magnetise two steel bars X and Y.

She places bar X in a coil connected to a d.c. power supply.

She places bar Y parallel to a magnetic field and hammers it.

Which bars become magnetised?

A X and Y

B X only

C Y only

D neither X nor Y

29 Which particles move in a metal to cause an electric current?

A electrons

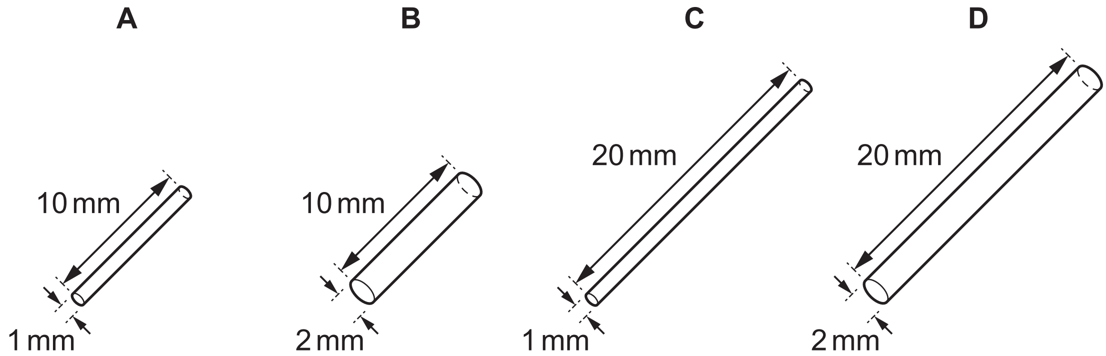
B neutrons

C nucleons

D protons

30 The diagrams represent four copper wires.

Which wire has the greatest resistance?



31 An air-conditioning unit, a lamp and an electric fire all receive electrical energy from the mains supply.

From which of these devices is all this energy eventually transferred to the surroundings?

	air-conditioning unit	lamp	electric fire
A	✓	✓	✓
B	✓	✗	✓
C	✗	✓	✓
D	✗	✗	✓

key

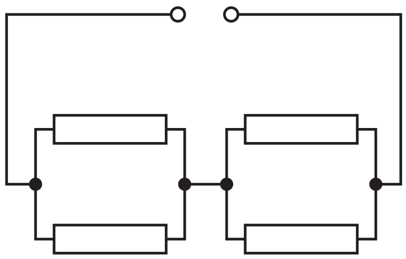
✓ = all energy transferred to surroundings

✗ = not all energy transferred to surroundings

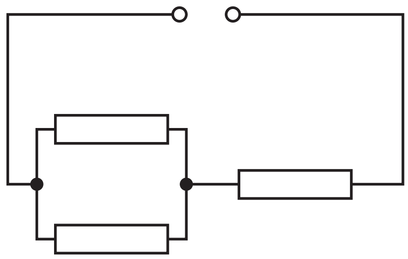
32 In the circuits shown, each of the resistors has a resistance of 1.0 Ω.

Which circuit has the greatest resistance?

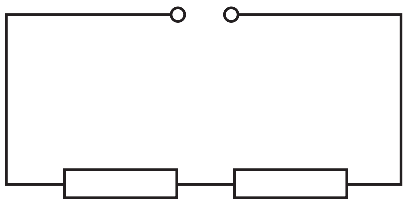
A



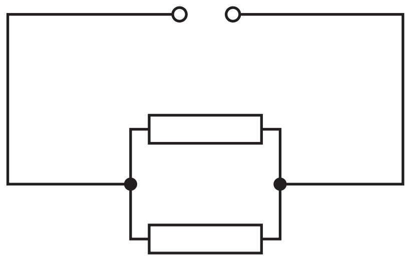
B



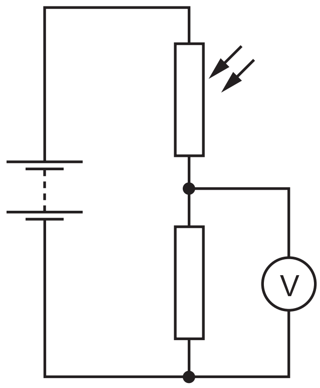
C



D



33 The diagram shows an electric circuit.

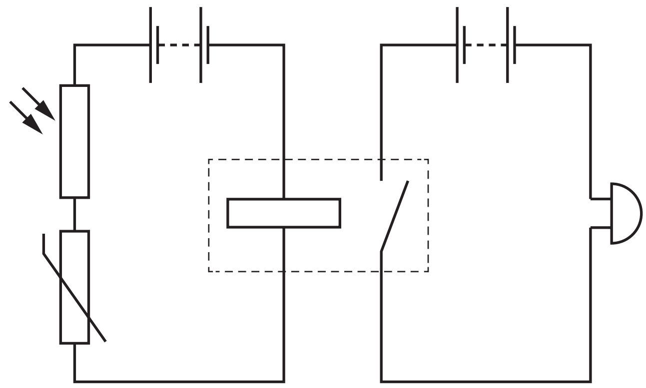


The light falling on the light-dependent resistor (LDR) increases in brightness.

What happens to the resistance of the LDR, the current in the fixed resistor and the reading on the voltmeter?

	resistance of LDR	current in fixed resistor	reading on voltmeter
A	decreases	increases	decreases
B	decreases	increases	increases
C	increases	decreases	decreases
D	increases	decreases	increases

34 The diagram shows two linked circuits to control when a bell is switched on.



The conditions are altered and only one pair of conditions causes the bell to ring.

Which pair causes the bell to ring?

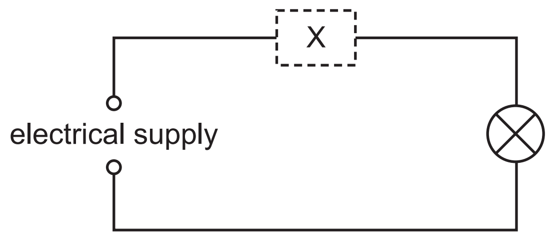
A bright light and high temperature

B bright light and low temperature

C dim light and high temperature

D dim light and low temperature

35 In this circuit, a component at X automatically protects the wiring from overheating if there is a fault.



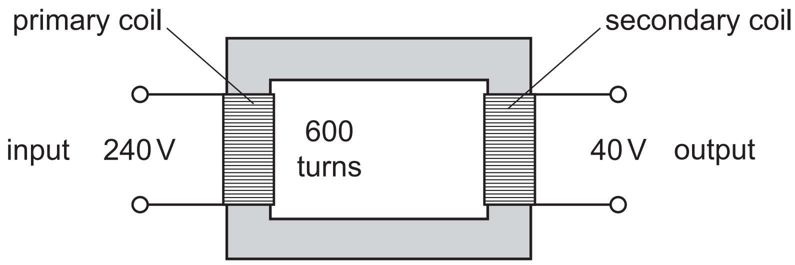
Which row indicates components that are suitable?

	circuit breaker	fuse	switch
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✗

key
✓ = suitable
✗ = not suitable

36 The diagram shows a simple transformer with an input of 240 V and an output of 40 V.

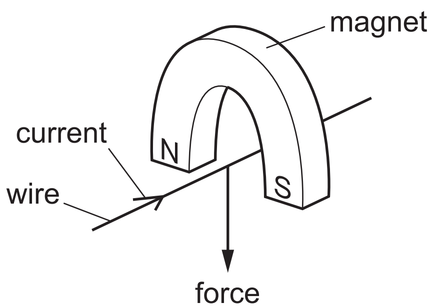
There are 600 turns on the primary coil.



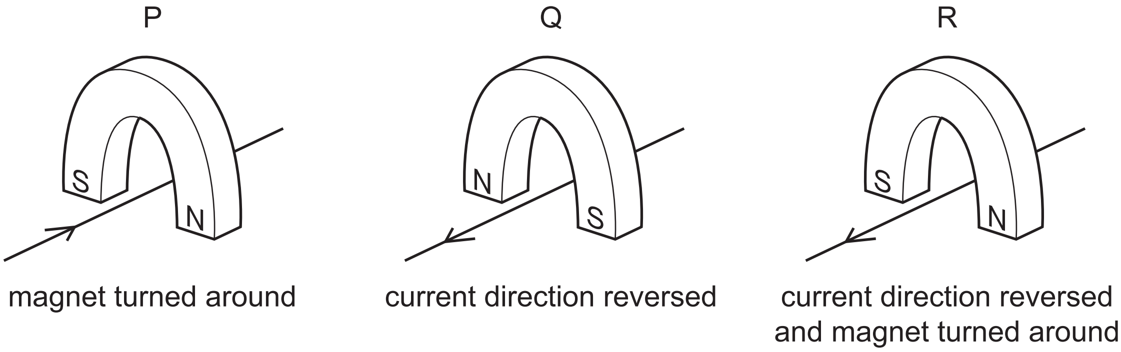
How many turns are there on the secondary coil?

A 100 **B** 320 **C** 400 **D** 3600

37 A wire is placed between the poles of a horseshoe magnet. There is a current in the wire in the direction shown, and this causes a force to act on the wire.



Three other arrangements P, Q and R of the wire and magnet are set up as shown.



Which arrangements cause a force in the same direction as the original arrangement?

A P, Q and R **B** P and Q only **C** P only **D** R only

38 The nuclide notation for an isotope of silver is ${}^{109}_{47}\mathrm{Ag}$.

How many nucleons are in a nucleus of this isotope?

A 47 **B** 62 **C** 109 **D** 156

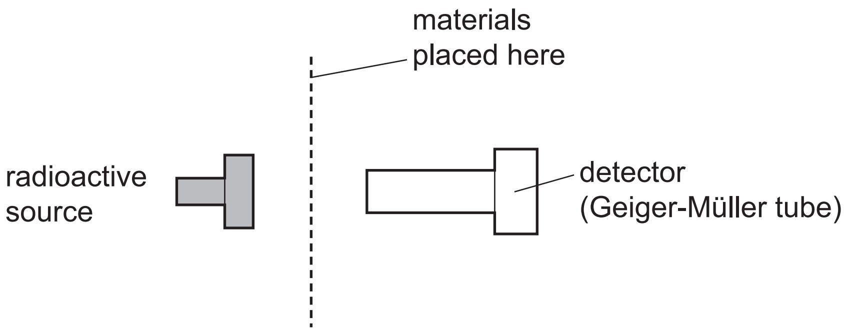
39 A radioactive substance emits radiation at a rate of 600 emissions per second. Four hours later, it emits radiation at a rate of 300 emissions per second.

What is the half-life of the substance and what is the rate of emission after a further four hours?

	half-life / hours	rate of emission after a further four hours / emissions per second
A	2	0
B	2	150
C	4	0
D	4	150

40 A radioactive source is placed near a detector.

The radiation arriving at the detector from the source is measured for 10 minutes with different materials placed between the source and the detector.



material between source and detector	radiation detected / counts
none	5626
sheet of paper	5629
thick sheet of aluminium	2226
thick sheet of lead	255

Which types of radiation are emitted by the source?

A α-particles and γ-rays

B α-particles only

C β-particles and γ-rays

D β-particles only

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Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **March 2017**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the March 2017 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

Question	Answer	Marks
1	B	**1**
2	D	**1**
3	D	**1**
4	B	**1**
5	A	**1**
6	B	**1**
7	C	**1**
8	B	**1**
9	C	**1**
10	C	**1**
11	B	**1**
12	C	**1**
13	C	**1**
14	B	**1**
15	D	**1**
16	B	**1**
17	B	**1**
18	B	**1**
19	C	**1**
20	D	**1**
21	A	**1**
22	C	**1**
23	B	**1**
24	A	**1**
25	C	**1**
26	A	**1**
27	C	**1**
28	A	**1**

Question	Answer	Marks
29	A	**1**
30	C	**1**
31	A	**1**
32	C	**1**
33	B	**1**
34	A	**1**
35	B	**1**
36	A	**1**
37	D	**1**
38	C	**1**
39	D	**1**
40	C	**1**

Cambridge IGCSE

Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS	**0625/12**
Paper 1 Multiple Choice (Core)	**February/March 2016**
	45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

* 7 9 1 7 3 8 8 6 8 9 *

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

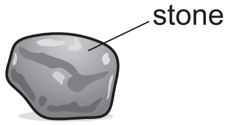
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Electronic calculators may be used.
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m / s^2).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

1 A student wishes to find the volume of a small, irregularly-shaped stone.



A ruler and a measuring cylinder containing some water are available.

Which apparatus is needed?

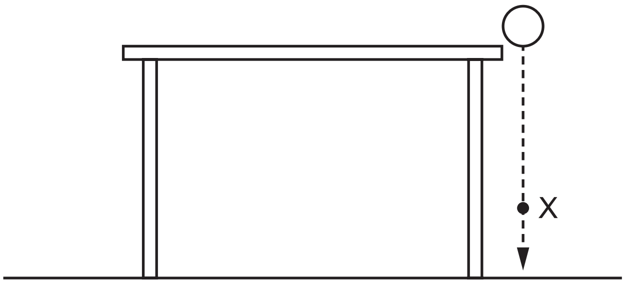
A neither the ruler nor the measuring cylinder

B the measuring cylinder only

C the ruler and the measuring cylinder

D the ruler only

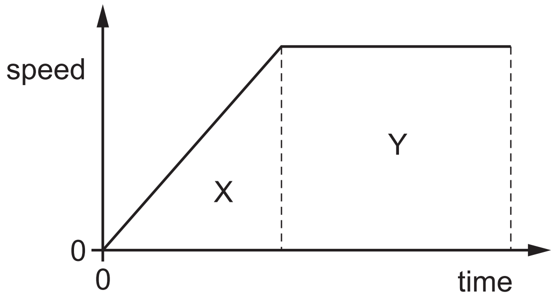
2 A ball is dropped from a table-top. Air resistance may be ignored.



Which row describes the velocity and the acceleration of the ball at point X?

	acceleration	velocity
A	constant	constant
B	constant	increasing
C	increasing	constant
D	increasing	increasing

3 The diagram shows the speed-time graph for a car.



Which area represents the distance travelled while the car is accelerating?

A X **B** X + Y **C** Y **D** Y – X

4 Which quantities are measured in the same unit?

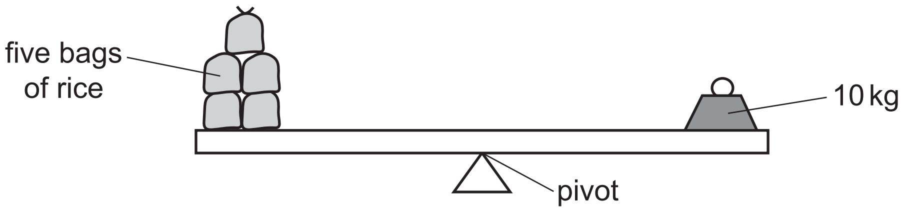
A energy, power and work

B energy and power, but not work

C energy and work, but not power

D power and work, but not energy

5 Five identical bags of rice are balanced on a uniform beam by an object of mass 10 kg.

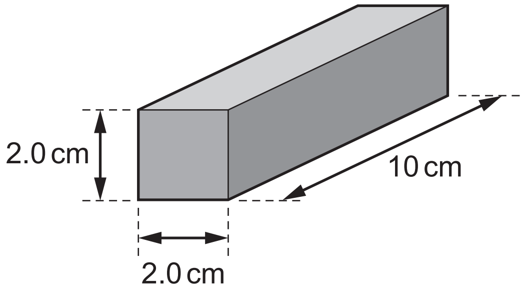


Two more identical bags of rice are added to the other five. The average position of the bags on the beam does not change.

What mass now balances the bags?

A 3.5 kg **B** 7.0 kg **C** 12 kg **D** 14 kg

6 The diagram shows a cuboid block made from a metal of density 2.5 g / cm^3.



What is the mass of the block?

A 8.0 g **B** 16 g **C** 50 g **D** 100 g

7 The diagram shows an unbalanced rod. Two loads X and Y can be moved along the rod.

The rod turns in a clockwise direction as shown.

Which action could make the rod balance?

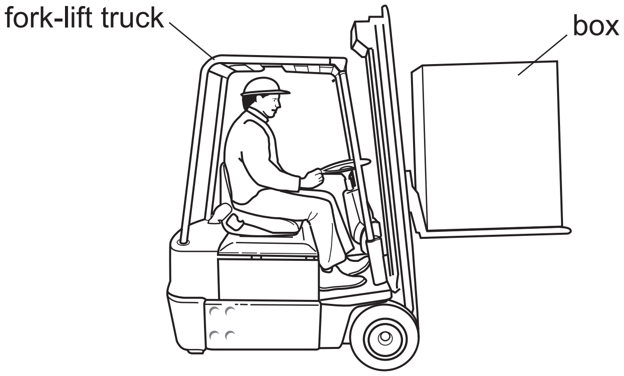
A moving X to the left

B moving X to the right

C moving Y to the right

D moving the pivot to the left

8 A box is being lifted by a fork-lift truck.



The weight of the box is 3000 N.

The force exerted by the fork-lift truck on the box is 3500 N vertically upwards.

What is the resultant vertical force on the box?

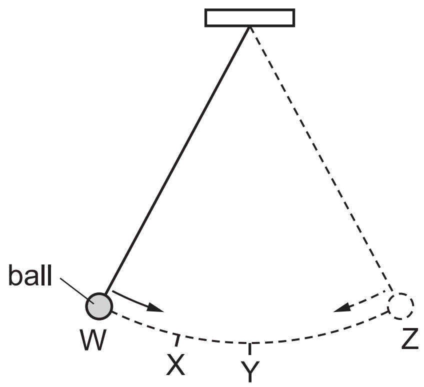
A 500 N downwards

B 500 N upwards

C 6500 N downwards

D 6500 N upwards

9 The diagram shows a ball hanging on a string. The ball swings from point W to point Z and back to point W.



Which statement about the ball is correct?

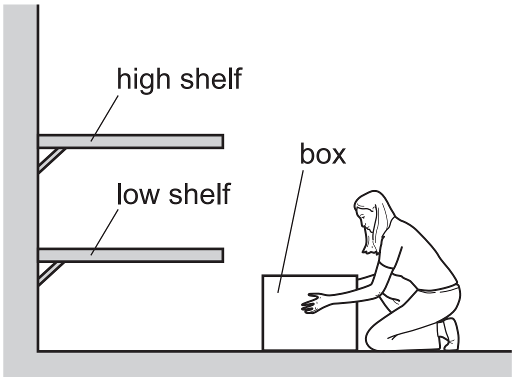
A The kinetic energy of the ball is greatest at point W.

B The kinetic energy of the ball is greatest at point X.

C The kinetic energy of the ball is greatest at point Y.

D The kinetic energy of the ball is the same at all points of the swing.

10 A person in a factory has to lift a box on to a shelf.



Which action involves the person doing the **least** amount of work?

A lifting the box quickly to the high shelf

B lifting the box slowly to the high shelf

C lifting the box to the low shelf first then lifting it to the high shelf

D lifting the box to the low shelf instead of to the high shelf

11 Which is a non-renewable energy resource?

A coal

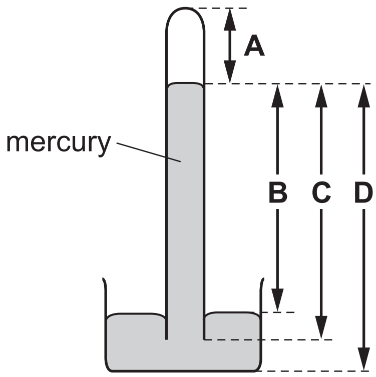
B solar

C tides

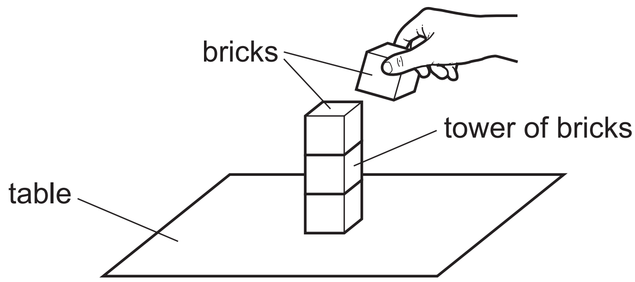
D wind

12 The diagram shows a simple mercury barometer.

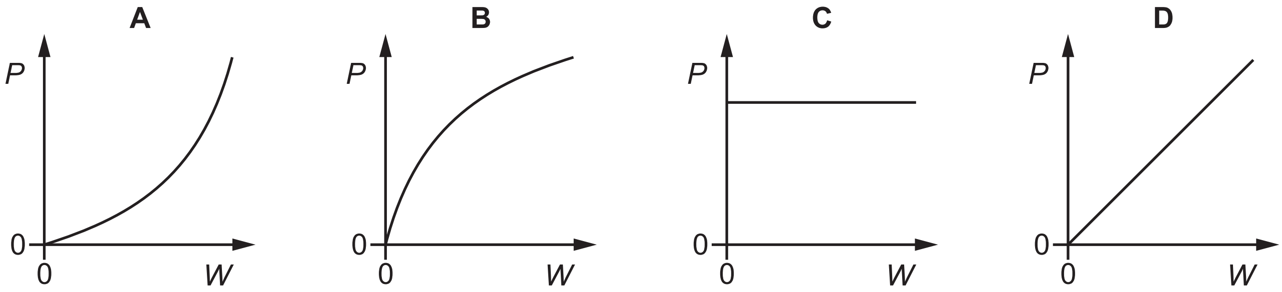
Which labelled length is measured when finding atmospheric pressure?



13 Identical toy bricks are placed one on top of another to make a tower on a table.



Which graph shows the relationship between the pressure P that the tower exerts on the table and the weight W of the tower?

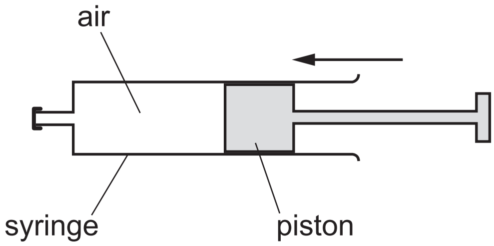


14 The table lists the melting points and the boiling points of four different substances.

Which substance is a liquid at 0 °C?

	melting point / °C	boiling point / °C
A	–219	–183
B	–7	58
C	98	890
D	1083	2582

15 Air in a sealed syringe is slowly compressed by moving the piston. The temperature of the air stays the same.



Which statement about the air is correct?

A The pressure of the air decreases because its molecules now travel more slowly.

B The pressure of the air decreases because the area of the syringe walls is now smaller.

C The pressure of the air increases because its molecules now hit the syringe walls more frequently.

D The pressure of the air increases because its molecules now travel more quickly.

16 Which quantity gives the *thermal capacity* of a solid object?

A the energy lost by radiation from the object in 1.0 s

B the energy needed to melt the object

C the energy needed to raise the temperature of the object by 1.0 °C

D the total amount of thermal energy in the object

17 To mark a temperature scale on a thermometer, standard temperatures known as fixed points are needed.

Which of these is a fixed point on the Celsius scale?

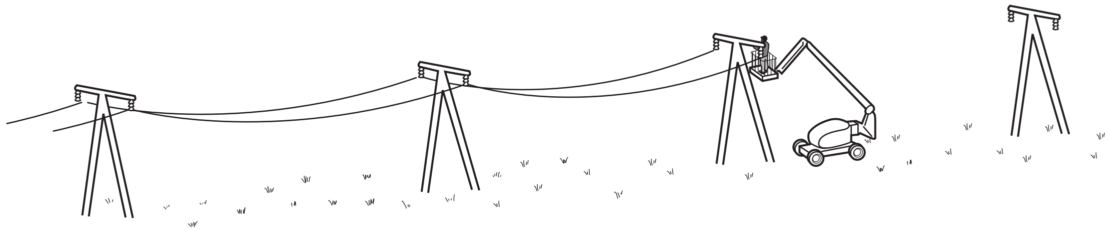
A room temperature

B the temperature inside a freezer

C the temperature of pure melting ice

D the temperature of pure warm water

18 The diagram shows electricity cables being put up on a warm day. The cables are left loose between the poles, as shown in the diagram.



Why are the cables left loose?

A They will contract on cold days.

B They will contract on very warm days.

C They will expand on cold days.

D They will expand on very warm days.

19 A cotton sheet is ironed with a hot electric iron.

How is energy transferred through the metal base of the iron to the sheet?

A by conduction only

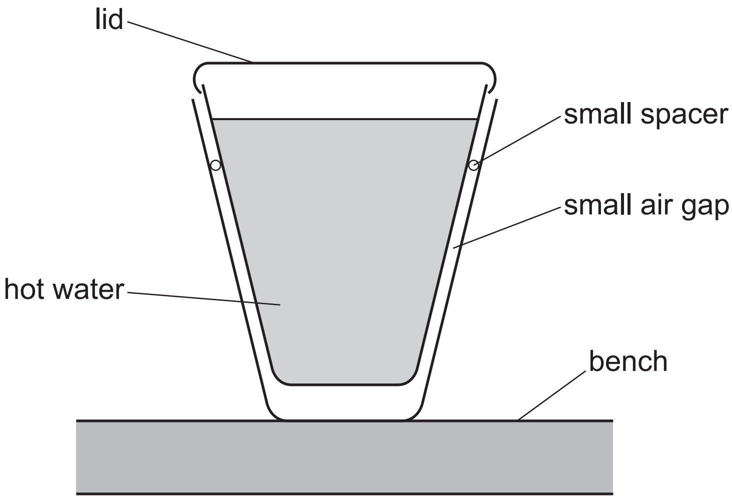
B by convection only

C by radiation only

D by convection and radiation only

20 Two plastic cups are placed one inside the other. A small spacer keeps the two cups separated.

Hot water is poured into the inner cup and a lid is put on top, as shown.



Which statement is correct?

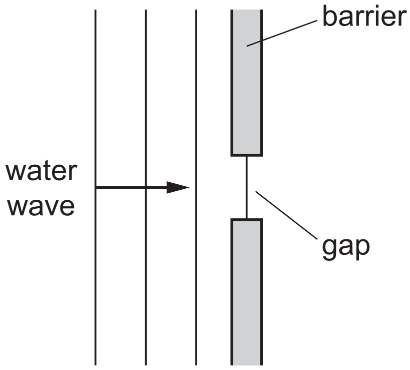
A The bench is heated by convection from the bottom of the outer cup.

B The lid reduces the energy lost by convection.

C There is no thermal conduction through the sides of either cup.

D Thermal radiation is prevented by the small air gap.

21 A water wave in a shallow tank passes through a gap in a barrier.



What happens to the speed and what happens to the wavelength of the wave as it passes through the gap?

	speed	wavelength
A	decreases	decreases
B	decreases	remains constant
C	remains constant	decreases
D	remains constant	remains constant

22 Which is a unit of wavelength?

A hertz

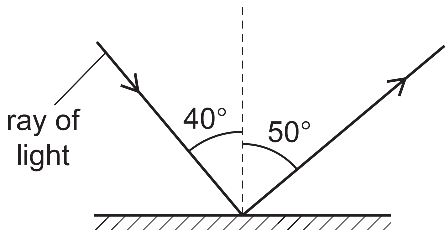
B metre

C metre per second

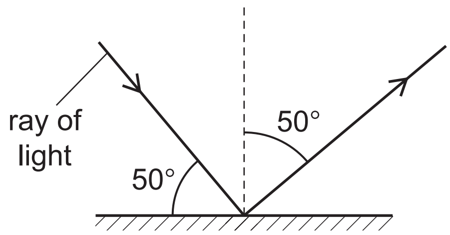
D second

23 Which diagram correctly shows a ray of light reflected by a plane mirror?

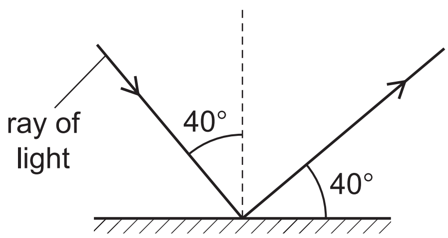
A



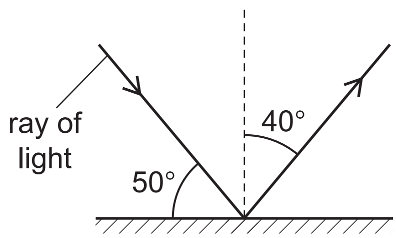
B



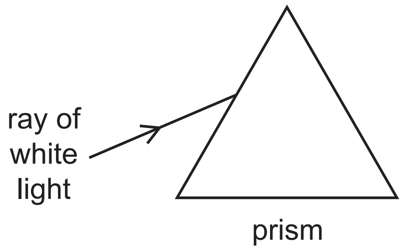
C



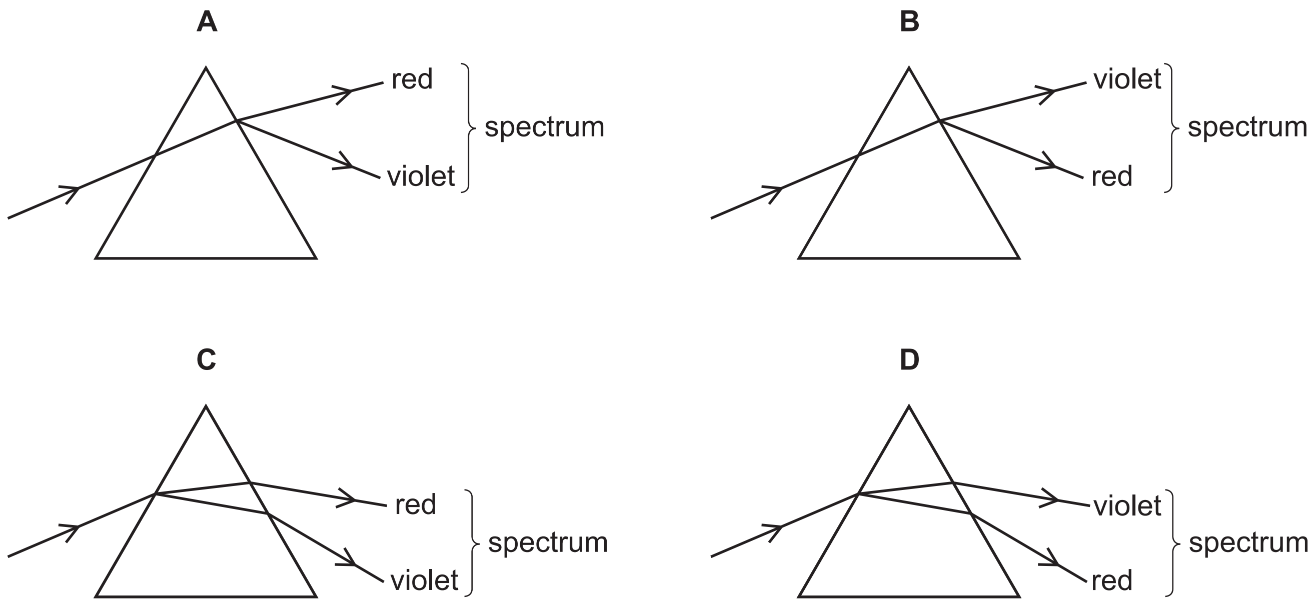
D



24 A ray of white light is incident on a glass prism.



Which ray diagram shows the ray as it passes through the prism and emerges from the opposite side?



25 A sound wave travels from a point X to another point Y.

Which diagram represents the movement of the air molecules, due to the sound wave, in the region between X and Y?

26 Sound wave P has a greater amplitude and a larger wavelength in air than sound wave Q.

How do the loudness and pitch of P compare with the loudness and pitch of Q?

A P is louder and higher in pitch than Q.

B P is louder and lower in pitch than Q.

C P is quieter and higher in pitch than Q.

D P is quieter and lower in pitch than Q.

27 In which pair are both materials magnetic?

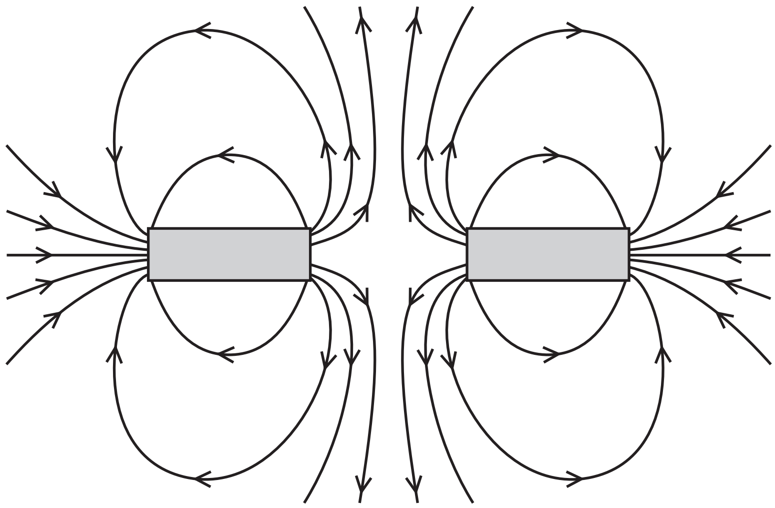
A aluminium and copper

B copper and iron

C iron and steel

D steel and aluminium

28 The diagram shows the magnetic field around two bar magnets.



Which diagram shows the poles of the magnets?

A	N S	N S
B	N S	S N
C	S N	N S
D	S N	S N

29 Which produces an electromotive force (e.m.f.)?

A a battery

B a filament lamp

C a resistor

D a spring balance

30 The table describes four different resistance wires. They are all made from the same metal.

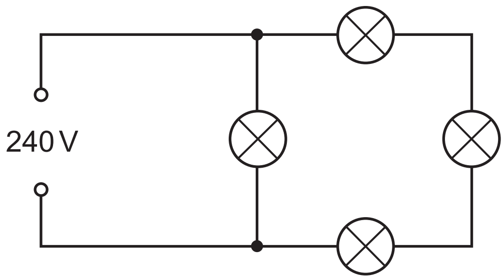
Which wire has the smallest resistance?

	length of wire / m	diameter of wire / mm
A	2.0	1.0
B	2.0	1.5
C	3.0	1.0
D	3.0	1.5

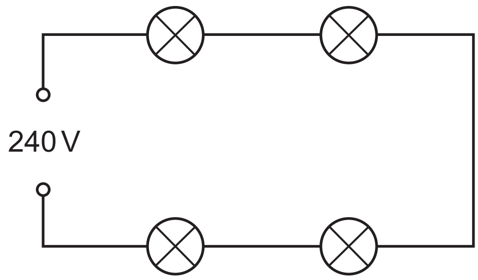
31 Four lamps are each labelled 240 V.

In which circuit do all four lamps have normal brightness?

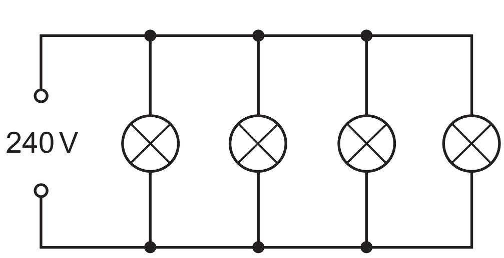
A



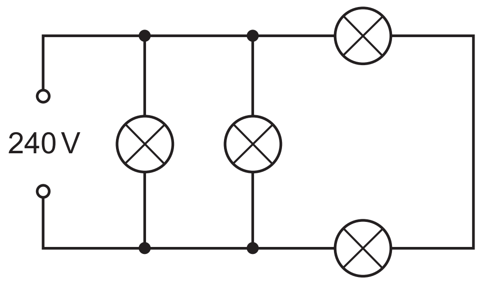
B



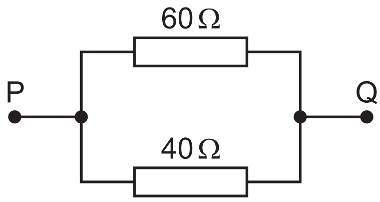
C



D



32 The diagram shows a 40 Ω resistor and a 60 Ω resistor connected in parallel.



What is the total resistance between points P and Q?

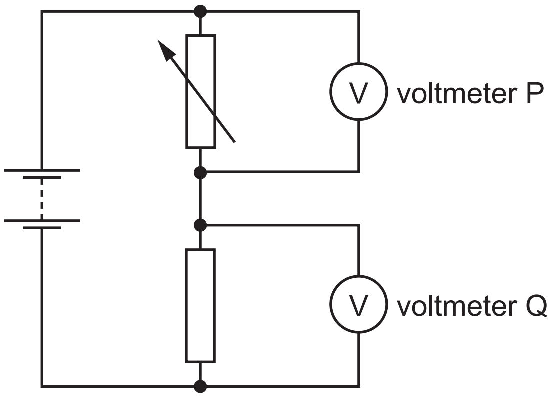
A less than 40 Ω

B 50 Ω

C between 60 Ω and 100 Ω

D 100 Ω

33 The diagram shows a potential divider connected to two voltmeters P and Q.



The resistance of the variable resistor is decreased.

Which row shows what happens to the reading on each voltmeter?

	reading on voltmeter P	reading on voltmeter Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

34 After some building work in a house, a bare (uninsulated) live wire is left protruding from a wall.

What is the greatest hazard?

A a fire

B a fuse blows

C an electric shock

D no current flows

35 A coil carries a current in a magnetic field. The coil experiences a turning effect.

Which device uses this effect?

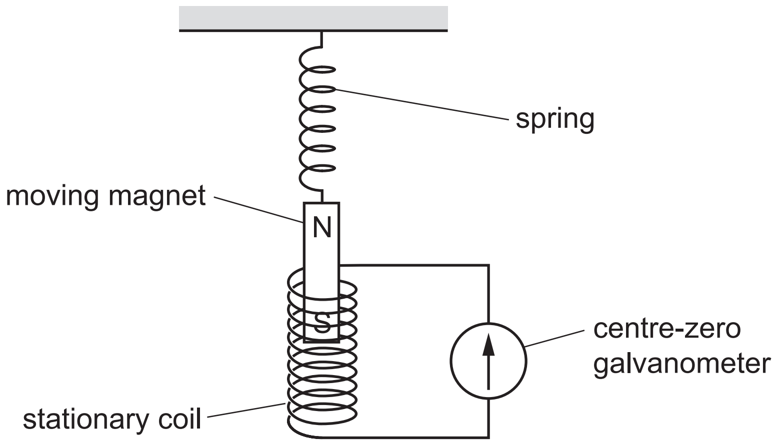
A a d.c. motor

B an electromagnet

C a relay

D a transformer

36 A magnet is suspended from a spring so that it can move freely inside a stationary coil. The coil is connected to a sensitive centre-zero galvanometer.



The magnet repeatedly moves slowly up and down.

What does the galvanometer show?

A a constantly changing reading

B a steady reading to the left

C a steady reading to the right

D a steady reading of zero

37 A nucleus of element X is represented as ${}^{56}_{26}X$.

Which is an isotope of element X?

A ${}^{26}_{56}X$ **B** ${}^{54}_{26}X$ **C** ${}^{56}_{24}X$ **D** ${}^{54}_{28}X$

38 The charge on a proton is *e*.

What is the charge on an electron and what is the charge on a neutron?

	electron	neutron
A	*e*	*e*
B	*e*	0
C	–*e*	–*e*
D	–*e*	0

39 A scientist carries out an experiment using a sealed source which emits β-particles. The range of the β-particles in the air is about 30 cm.

Which precaution is the most effective to protect the scientist from the radiation?

A handling the source with long tongs

B keeping the temperature of the source low

C opening all windows in the laboratory

D washing his hands before leaving the laboratory

40 Which row describes the nature of α-particles and of γ-rays?

	α-particles	γ-rays
A	helium nuclei	electromagnetic radiation
B	helium nuclei	electrons
C	protons	electromagnetic radiation
D	protons	electrons

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CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International General Certificate of Secondary Education

MARK SCHEME for the March 2016 series

0625 PHYSICS

0625/12 Paper 1 (Multiple Choice – Core), maximum raw mark 40

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the March 2016 series for most Cambridge IGCSE® and Cambridge International A and AS Level components.

CAMBRIDGE
International Examinations

Question Number	*Key*	*Question Number*	*Key*
1	**B**	21	**D**
2	**B**	22	**B**
3	**A**	23	**D**
4	**C**	24	**C**
5	**D**	25	**D**
6	**D**	26	**B**
7	**A**	27	**C**
8	**B**	28	**C**
9	**C**	29	**A**
10	**D**	30	**B**
11	**A**	31	**C**
12	**B**	32	**A**
13	**D**	33	**B**
14	**B**	34	**C**
15	**C**	35	**A**
16	**C**	36	**A**
17	**C**	37	**B**
18	**A**	38	**D**
19	**A**	39	**A**
20	**B**	40	**A**

Cambridge IGCSE

Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS **0625/12**

Paper 1 Multiple Choice **February/March 2015**

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

1743635004

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

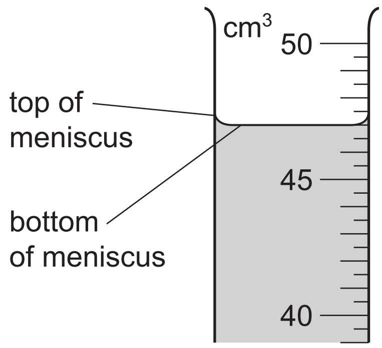
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Electronic calculators may be used.

This document consists of **17** printed pages and **3** blank pages.

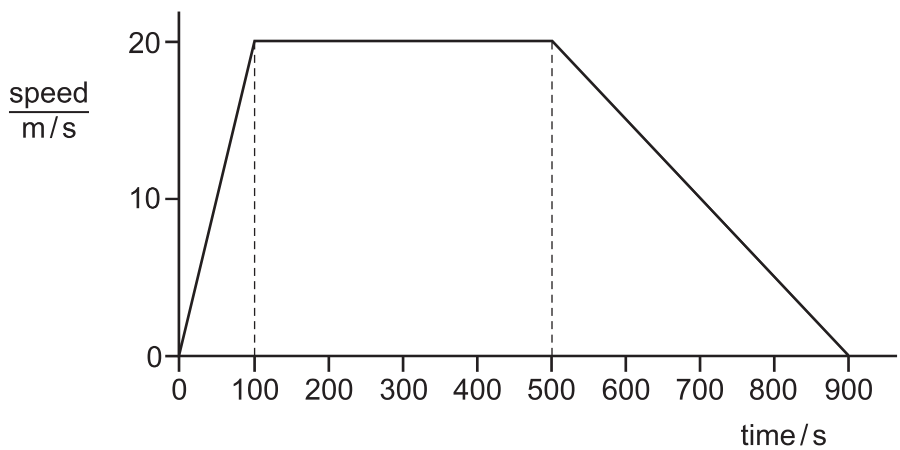
1 A student uses a measuring cylinder to measure the volume of some water. The diagram shows part of the measuring cylinder. The top and bottom of the meniscus are labelled.



What is the volume of the water?

A 47.0 cm^3 **B** 47.5 cm^3 **C** 49.0 cm^3 **D** 49.5 cm^3

2 The graph represents the motion of a train travelling between two stations.



Which statement about the train is correct?

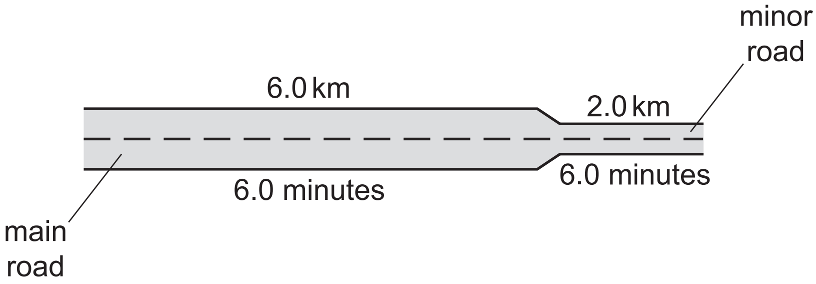
A Its acceleration takes a longer time than its deceleration.

B It travels at constant speed for less than half of its journey time.

C It travels 2000 m in the first 100 s.

D It travels 10 000 m at constant speed.

3 A car travels 6.0 km along a main road in 6.0 minutes. It then travels 2.0 km along a minor road in 6.0 minutes.



Which calculation of average speed for the whole journey is correct?

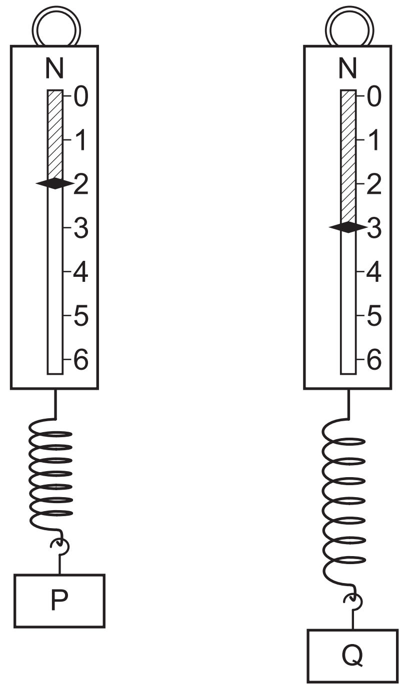
A 8.0 ÷ 12.0 = 0.67 km / minute

B 12.0 ÷ 8.0 = 1.5 km / minute

C 8.0 + 12.0 = 20 km / minute

D 8.0 × 12.0 = 96 km / minute

4 Two metal blocks P and Q have identical dimensions. They hang on identical spring balances.



What can be deduced about P and Q?

A They have different volumes and different weights.

B They have different volumes, but equal masses.

C They have equal volumes and equal weights.

D They have equal volumes, but different masses.

5 A person measures the length, width, height and mass of a metal block with rectangular sides.

Which of these measurements must be used in order to calculate the density of the metal?

A mass only

B height and mass only

C length, width and height only

D length, width, height and mass

6 The diagram shows the only three forces acting on an object.

What is the resultant force on the object?

A 0 N

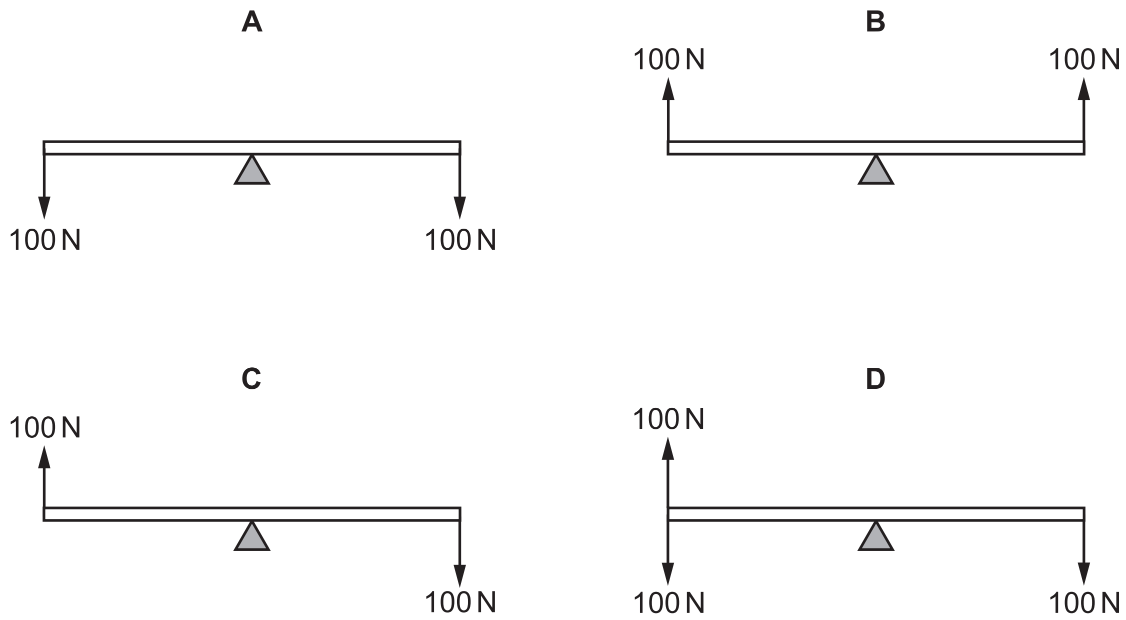
B 5.0 N towards the left

C 5.0 N towards the right

D 10.0 N towards the right

7 A uniform rod rests on a pivot at its centre. The rod is not attached to the pivot. Forces are then applied to the rod in four different ways, as shown. The weight of the rod can be ignored.

Which diagram shows the rod in equilibrium?



8 A helicopter takes off from the ground and rises vertically. It then hovers at a constant height above the ground.

Which sequence of energy changes takes place during the gain in height?

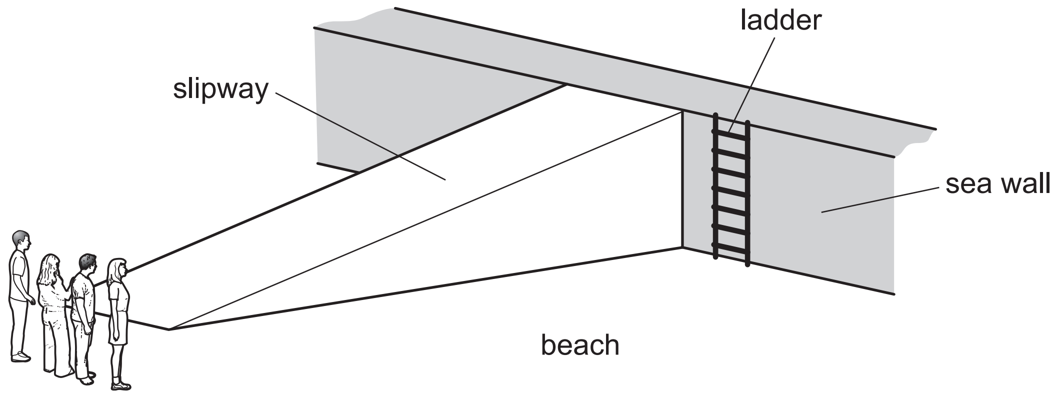
A chemical → gravitational potential → kinetic

B chemical → kinetic → gravitational potential

C gravitational potential → chemical → kinetic

D kinetic → chemical → gravitational potential

9 Four people of equal weight on a beach use different routes to get to the top of a sea wall.



Which person produces the greatest average power?

person	route	time taken / s
A	runs across the beach, then climbs the ladder	8
B	walks across the beach, then climbs the ladder	16
C	runs up the slipway	5
D	walks up the slipway	10

10 A vehicle sinks into soft ground.

The vehicle is changed so that it does not sink as far.

Which change is made?

A a lower centre of mass

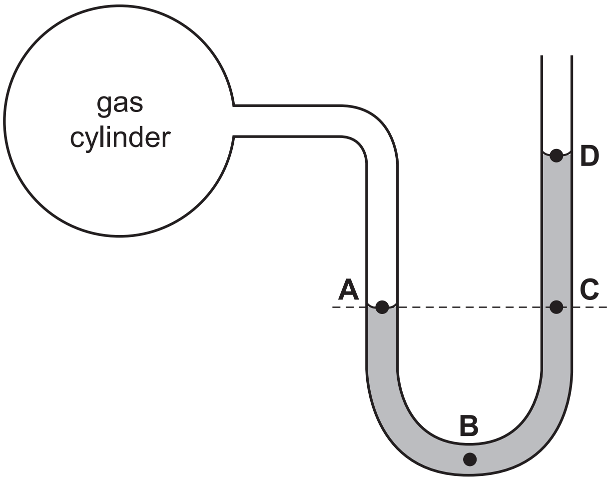
B a more powerful engine

C wheels that are further apart

D wider tyres

11 A manometer is used to measure the pressure of a gas trapped in a cylinder.

At which labelled point on the diagram is the pressure greatest?



12 Extremely small pollen grains in water are viewed through a microscope. The grains are seen to move continually and randomly.

What is the reason for this random movement?

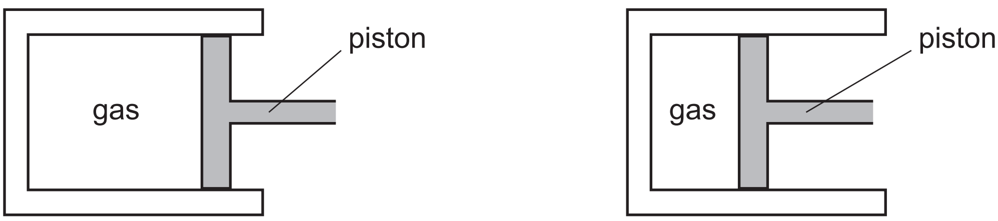
A The grains are moved by randomly moving water molecules.

B The grains are moved by random convection currents in the water.

C The grains are moved by random rays of light reflecting off them.

D The grains are moved by the random motion of their own atoms.

13 A gas is compressed in a sealed cylinder by moving a piston.



initial position

after gas has been compressed

Which row in the table states what happens to the density of the gas and to the pressure of the gas when it is compressed?

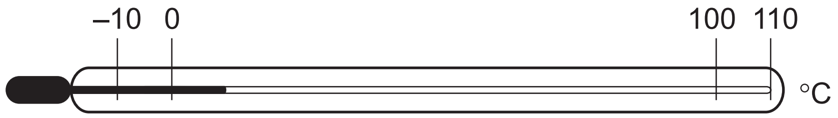
	density	pressure
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

14 When steam condenses it becomes liquid water. When liquid water solidifies it becomes ice.

What happens to the temperature of steam while it is condensing, and what happens to the temperature of water while it is solidifying?

	temperature of steam while it is condensing	temperature of water while it is solidifying
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

15 A thermometer has graduations which start at –10 °C and end at 110 °C.



What is the lower fixed point and what is the upper fixed point of the Celsius scale?

	lower fixed point / °C	upper fixed point / °C
A	–10	100
B	–10	110
C	0	100
D	0	110

16 What is the name of the process of heat transfer using electromagnetic waves?

A conduction

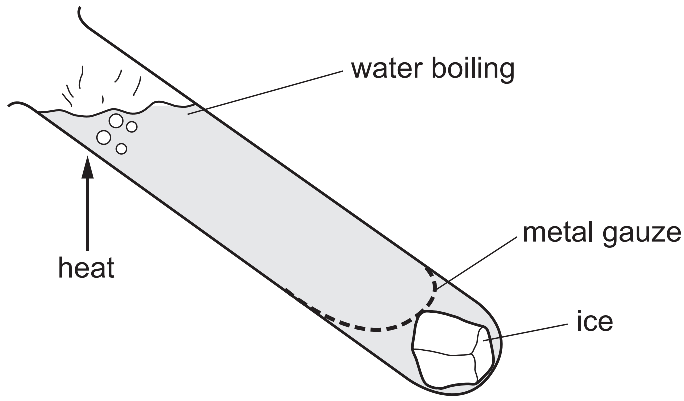
B convection

C evaporation

D radiation

17 Ice is trapped by a metal gauze at the bottom of a tube containing water.

The water is heated strongly at the top, but the ice only melts very slowly.



Why does the ice melt so slowly?

A Heat energy always travels upwards.

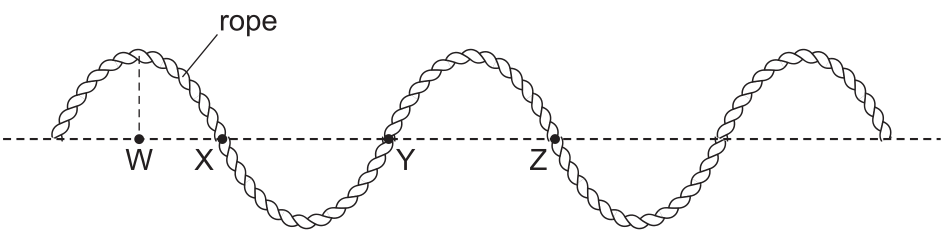
B Hot water is more dense than cold water.

C Metal gauze does not allow heat to pass through.

D Water is a poor conductor of heat.

18 A transverse wave moves along a rope.

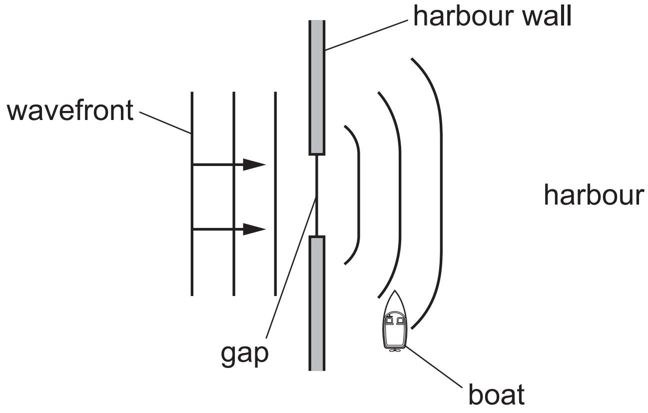
The diagram shows the position of the rope at one particular time.



Which two labelled points are one wavelength apart?

A W and X **B** W and Z **C** X and Z **D** Y and Z

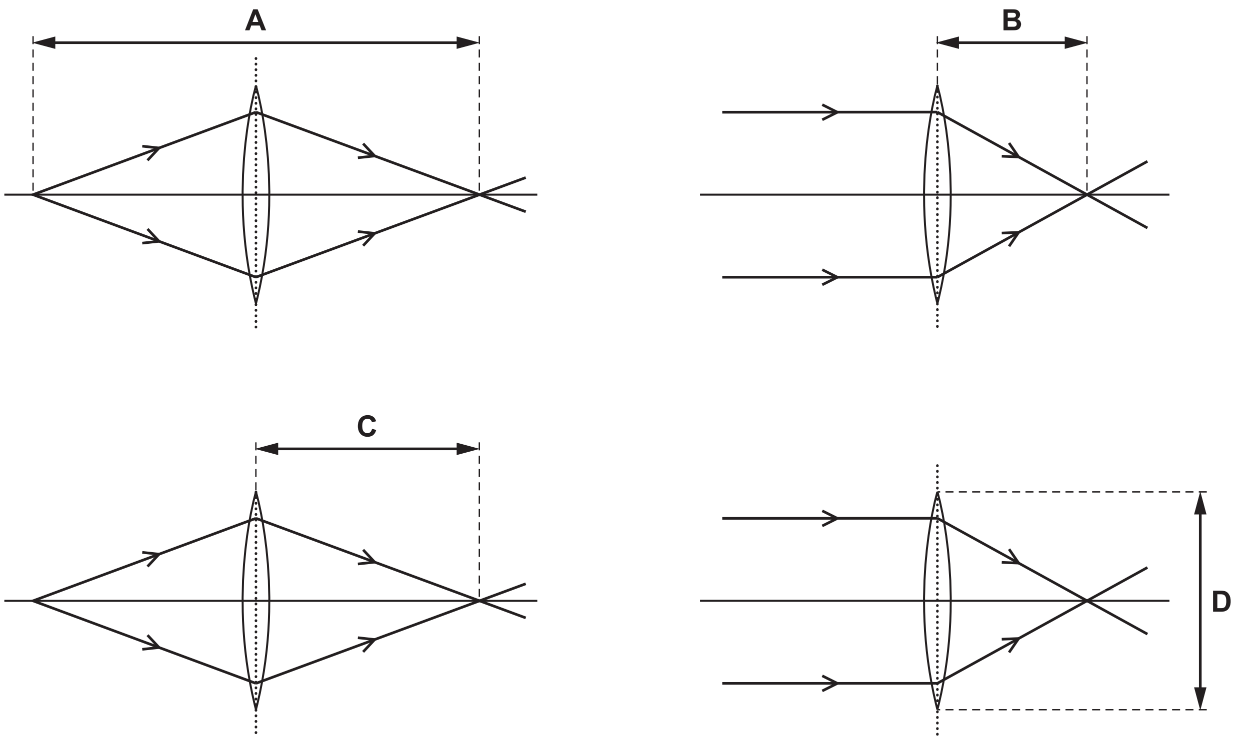
19 The diagram shows a water wave passing through a gap in a harbour wall. The wavefronts curve round the wall and reach a small boat in the harbour.



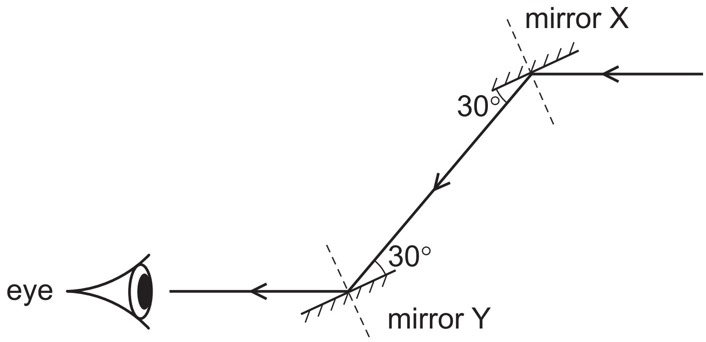
What is the name of this curving effect, and how can the gap be changed so that the wavefronts do not reach the boat?

	name of effect	change to the gap
A	diffraction	make the gap slightly bigger
B	diffraction	make the gap slightly smaller
C	refraction	make the gap slightly bigger
D	refraction	make the gap slightly smaller

20 Which labelled distance is the focal length of the lens?



21 A ray of light is reflected by two parallel plane mirrors X and Y.



Which statement is correct?

A The angle of incidence at mirror X is 30°.

B The angle of incidence at mirror Y is 60°.

C The angle of reflection at mirror X is 120°.

D The angle of reflection at mirror Y is 0°.

22 How do infra-red waves differ from ultraviolet waves?

A Infra-red waves are longitudinal.

B Infra-red waves have a lower speed *in vacuo* (in a vacuum).

C Infra-red waves have lower frequencies.

D Infra-red waves have smaller wavelengths.

23 What is the approximate range of audible sound frequencies for a human with good hearing?

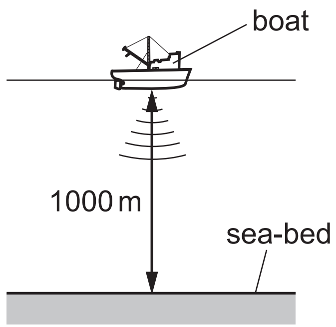
A from 20 Hz to 2000 Hz

B from 20 Hz to 20 000 Hz

C from 200 Hz to 20 000 Hz

D from 200 Hz to 200 000 Hz

24 A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from the sea-bed. The sound reaches the boat again after 1.3 s. The sea-bed is 1000 m below the boat.



Using this information, what is the speed of sound in the water?

A 769 m / s **B** 1300 m / s **C** 1538 m / s **D** 2600 m / s

25 Which metal is suitable to use to make a permanent magnet?

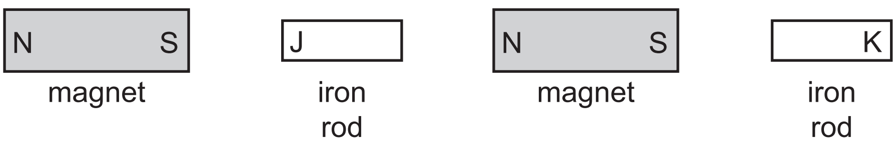
A aluminium

B brass

C iron

D steel

26 The diagram shows two magnets and two iron rods placed in a line.



Which magnetic poles are induced at the ends J and K of the iron rods?

	pole induced at end J	pole induced at end K
A	N	N
B	N	S
C	S	N
D	S	S

27 A teacher wishes to show the production of electrostatic charges.

She holds a rod and rubs it with a cotton cloth. A copper rod, a glass rod, a plastic rod and a steel rod are available.

Which two rods would both be suitable to use?

A a copper rod and a glass rod

B a glass rod and a plastic rod

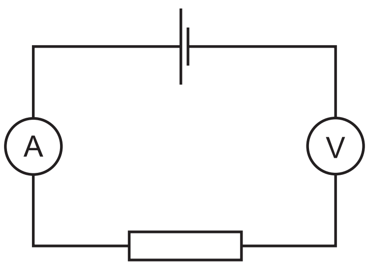
C a plastic rod and a copper rod

D a plastic rod and a steel rod

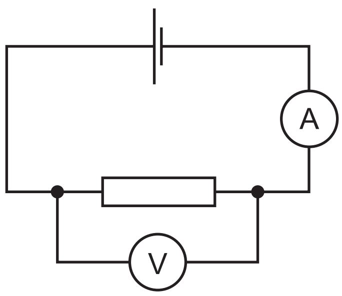
28 A student wishes to determine the resistance of a resistor. She uses an ammeter and a voltmeter in a circuit.

In which circuit are the ammeter and voltmeter connected correctly?

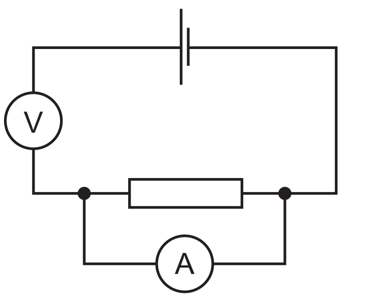
A



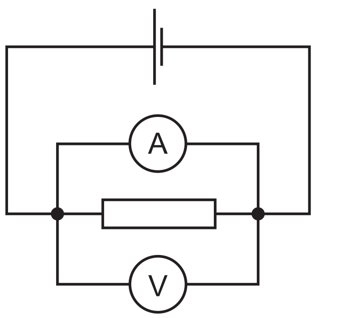
B



C

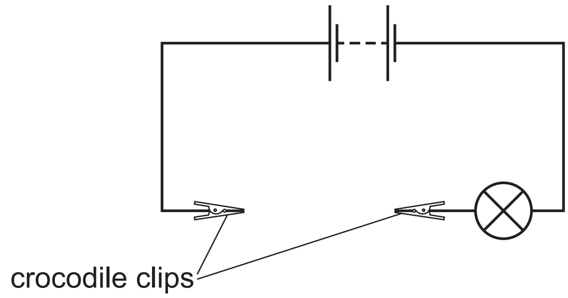


D

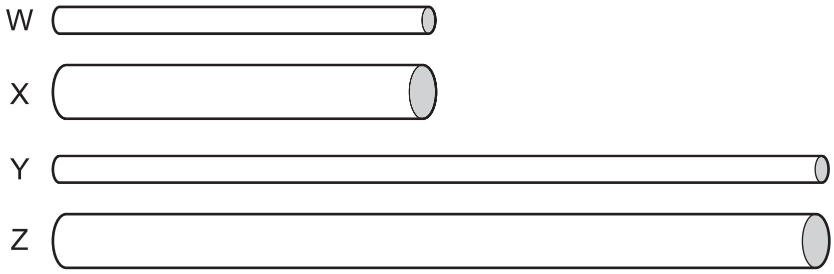


29 A battery is connected to two crocodile clips and a lamp.

There is a gap between the crocodile clips.



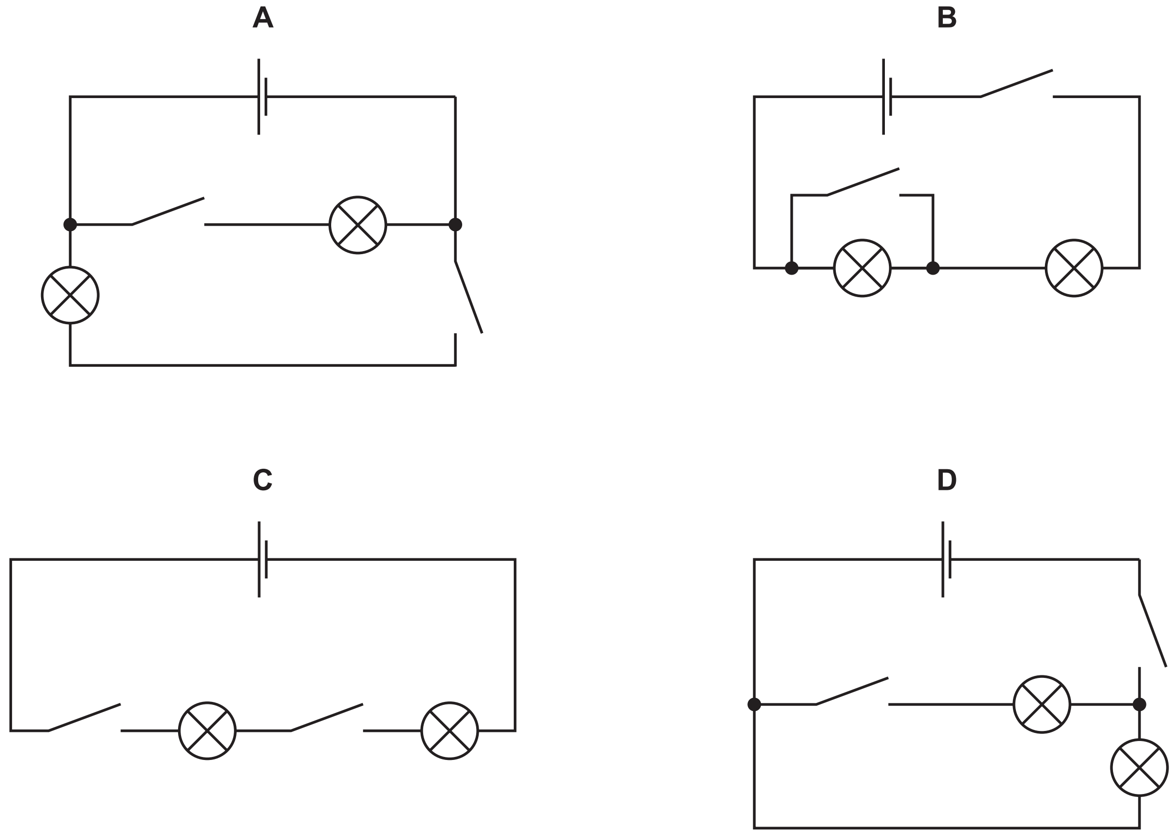
Four cylinders W, X, Y and Z are made of the same metal but have different dimensions. The cylinders are connected in turn, by their ends, between the crocodile clips. The diagrams of the cylinders are all drawn to the same scale.



Which cylinder makes the lamp glow most brightly and which cylinder makes the lamp glow least brightly?

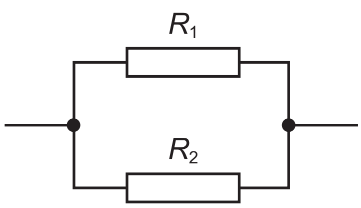
	most brightly	least brightly
A	W	Y
B	W	Z
C	X	Y
D	X	Z

30 In which circuit can the lamps be switched on and off independently?



31 Two resistors, with resistances R_1 and R_2, are connected in parallel.

The resistance R_1 is greater than the resistance R_2.



What is the resistance of the parallel combination?

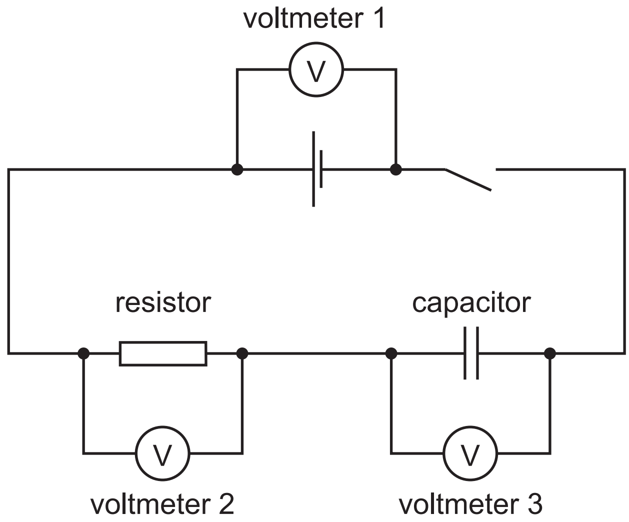
A less than either R_1 or R_2

B equal to R_1

C equal to R_2

D the average of R_1 and R_2

32 In the circuit shown, the switch is closed for a long time, then opened.



Immediately after the switch is opened, which of the voltmeters reads zero?

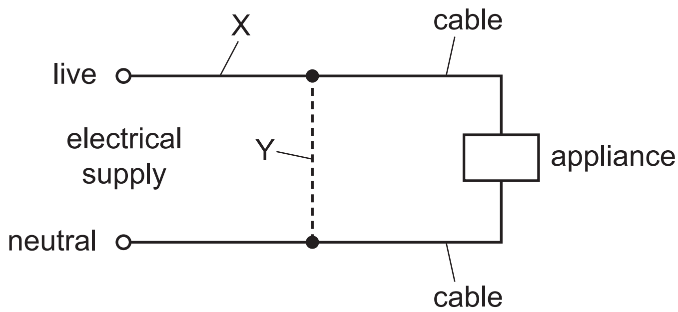
A voltmeter 1 only

B voltmeter 2 only

C voltmeter 3 only

D voltmeter 1, voltmeter 2 and voltmeter 3

33 Either a fuse or a circuit-breaker can be used to protect electrical cables from large currents that could cause overheating.



When a fuse is used, where should it be connected, and when a circuit-breaker is used, where should it be connected?

	position of fuse	position of circuit-breaker
A	X	X
B	X	Y
C	Y	X
D	Y	Y

34 Which device uses slip rings?

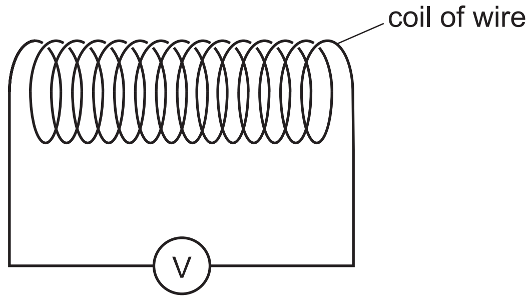
A a d.c. electric motor

B a relay

C a transformer

D an a.c. generator

35 The diagram shows a coil of wire connected to a voltmeter.



A student has a magnet and an unmagnetised iron rod.

How can an e.m.f. be induced across the coil?

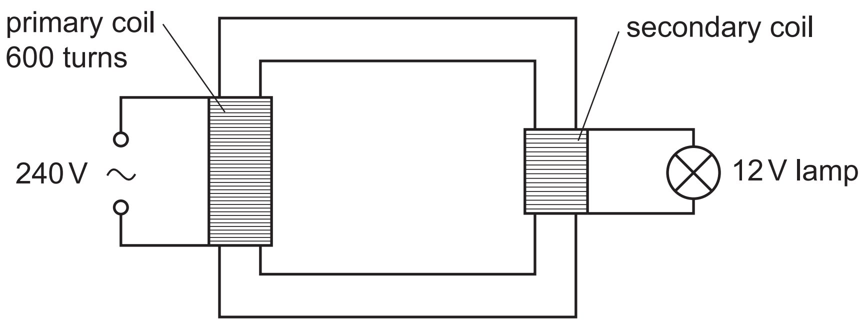
A holding the magnet inside the coil

B holding the iron rod inside the coil

C pushing the magnet into the coil

D pushing the iron rod into the coil

36 A step-down transformer is used to light a 12 V lamp from a 240 V mains supply. The lamp lights at normal brightness. The primary coil has 600 turns.



How many turns are in the secondary coil?

A 12 **B** 20 **C** 30 **D** 50

37 What happens in the process of thermionic emission?

A Complete molecules are emitted from the surface of a hot liquid.

B Single atoms are emitted from the surface of a hot liquid.

C Electrons are emitted from the surface of a hot metal.

D Protons are emitted from the surface of a hot metal.

38 Which row shows the relative ionising effects and penetrating abilities of α-particles and β-particles?

	ionising effect	penetrating ability
A	α greater than β	α greater than β
B	α greater than β	α less than β
C	α less than β	α greater than β
D	α less than β	α less than β

39 A radioactive substance has a half-life of 2 weeks. At the beginning of an investigation, a sample of the substance emits 3000 β-particles per minute.

How many β-particles will it emit per minute after 6 weeks?

A 0 **B** 375 **C** 500 **D** 1500

40 A nuclide has the symbol $^{14}_{6}C$.

How many protons are there in one nucleus of this nuclide?

A 6 **B** 8 **C** 14 **D** 20

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CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International General Certificate of Secondary Education

MARK SCHEME for the March 2015 series

0625 PHYSICS

0625/12 Paper 1 (Multiple Choice), maximum raw mark 40

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the March 2015 series for most Cambridge IGCSE® components.

CAMBRIDGE
International Examinations

Question Number	*Key*	*Question Number*	*Key*
1	**A**	21	**B**
2	**B**	22	**C**
3	**A**	23	**B**
4	**D**	24	**C**
5	**D**	25	**D**
6	**A**	26	**B**
7	**A**	27	**B**
8	**B**	28	**B**
9	**C**	29	**C**
10	**D**	30	**A**
11	**B**	31	**A**
12	**A**	32	**B**
13	**D**	33	**A**
14	**D**	34	**D**
15	**C**	35	**C**
16	**D**	36	**C**
17	**D**	37	**C**
18	**C**	38	**B**
19	**A**	39	**B**
20	**B**	40	**A**